Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-12 14:43:08
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 128
  target_lon_size........................ 256
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_128x256.pdf
  output_dir............................. data/processed_128x256/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 45
  num_layers............................. 5
  num_modes_lat.......................... 48
  num_modes_lon.......................... 96
  mlp_ratio.............................. 4
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.2
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 10
rollout_epoch_start.................... 15
rollout_epoch_end...................... 50
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0005
min_learning_rate...................... 1e-09
batch_size............................. 20
num_epochs............................. 150
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 25
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_128x256/sst_final.nc
checkpoint_dir......................... checkpoints_128x256/
best_model_path........................ checkpoints_128x256/best_model.pt
training_log_path...................... results_128x256/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_128x256/best_model.pt
input_data_path........................ data/processed_128x256/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_128x256/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_128x256/std_params.nc
  trend_params_path..... data/processed_128x256/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

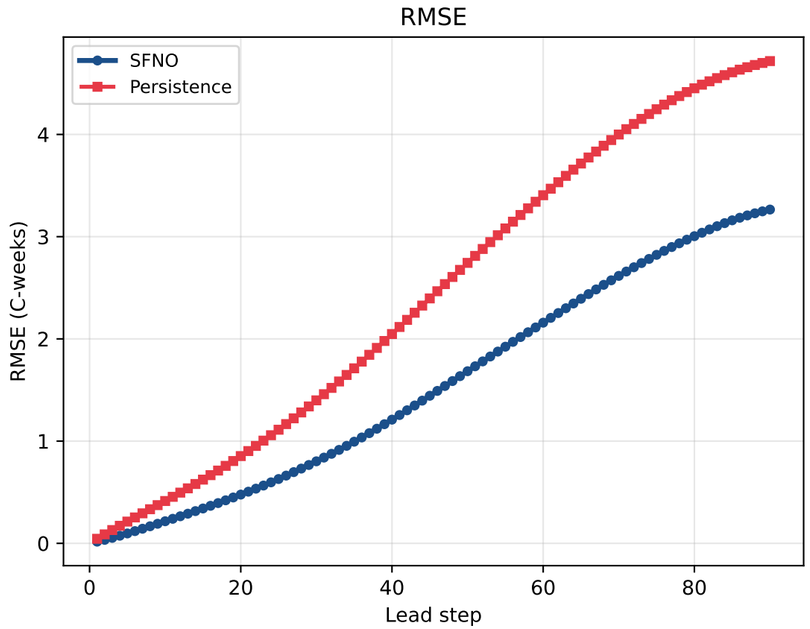
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

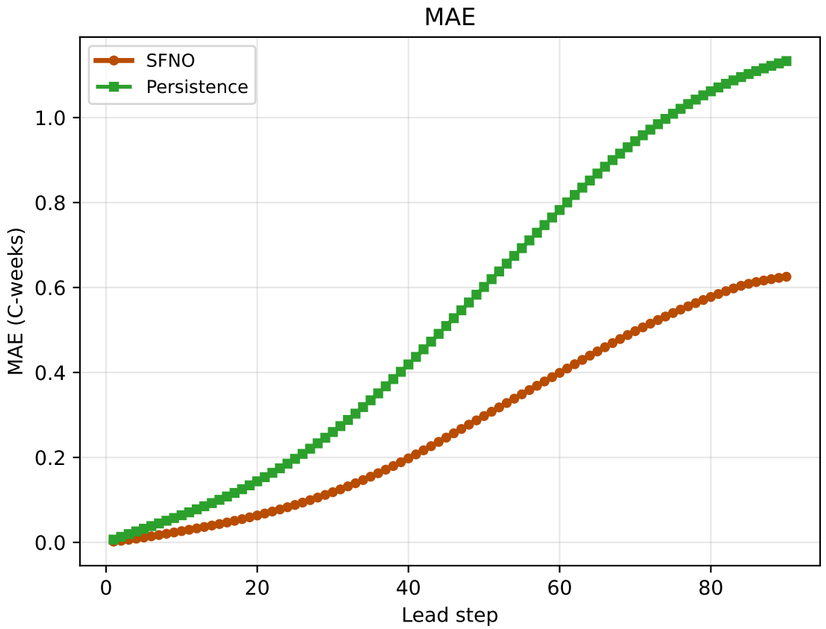
DATASETS
  Test          19 chains x 90 leads x 128 lat x 256 lon
  Val           16 chains x 90 leads x 128 lat x 256 lon
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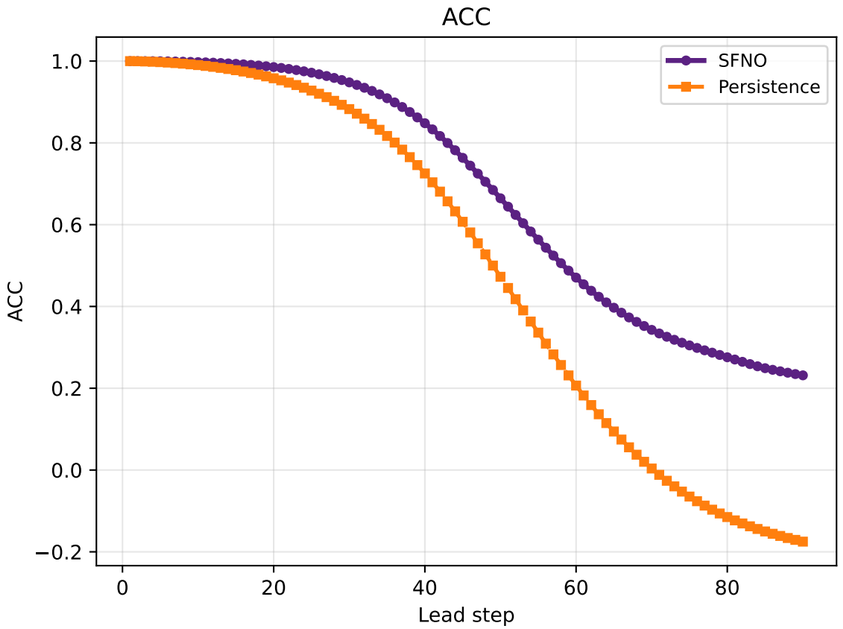
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

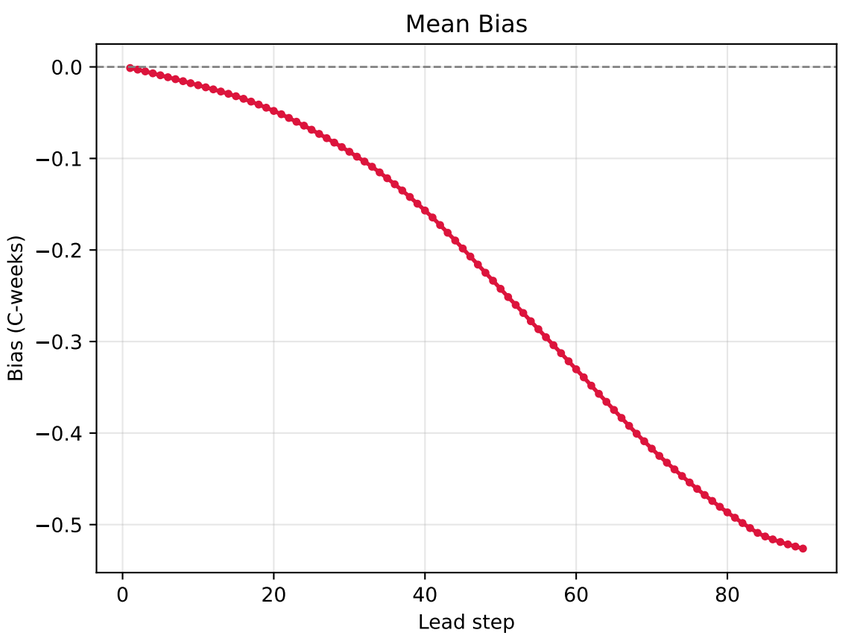
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0168	0.0446	62.3%	0.0018	0.0066	-0.0012	1.0000	0.9999	1.0000
2	0.0343	0.0878	60.9%	0.0038	0.0130	-0.0030	0.9999	0.9996	0.9999
3	0.0537	0.1301	58.7%	0.0061	0.0194	-0.0049	0.9998	0.9990	0.9997
4	0.0746	0.1717	56.5%	0.0087	0.0258	-0.0070	0.9997	0.9983	0.9995
5	0.0968	0.2128	54.5%	0.0114	0.0321	-0.0092	0.9995	0.9974	0.9991
6	0.1197	0.2534	52.8%	0.0143	0.0383	-0.0113	0.9992	0.9964	0.9986
7	0.1430	0.2936	51.3%	0.0172	0.0446	-0.0134	0.9989	0.9951	0.9981
8	0.1670	0.3337	50.0%	0.0203	0.0509	-0.0156	0.9985	0.9937	0.9973
9	0.1914	0.3737	48.8%	0.0234	0.0573	-0.0178	0.9980	0.9920	0.9965
10	0.2161	0.4140	47.8%	0.0265	0.0639	-0.0200	0.9975	0.9902	0.9955
11	0.2402	0.4549	47.2%	0.0296	0.0707	-0.0223	0.9968	0.9882	0.9944
12	0.2649	0.4961	46.6%	0.0329	0.0778	-0.0245	0.9961	0.9859	0.9931
13	0.2900	0.5380	46.1%	0.0362	0.0850	-0.0269	0.9952	0.9834	0.9917
14	0.3154	0.5805	45.7%	0.0396	0.0924	-0.0294	0.9943	0.9807	0.9901
15	0.3410	0.6234	45.3%	0.0431	0.1001	-0.0320	0.9932	0.9777	0.9883
16	0.3671	0.6671	45.0%	0.0468	0.1080	-0.0348	0.9920	0.9744	0.9863
17	0.3938	0.7119	44.7%	0.0507	0.1164	-0.0379	0.9906	0.9709	0.9840
18	0.4210	0.7579	44.5%	0.0547	0.1251	-0.0411	0.9891	0.9669	0.9815
19	0.4486	0.8052	44.3%	0.0589	0.1343	-0.0446	0.9873	0.9627	0.9787
20	0.4769	0.8536	44.1%	0.0633	0.1437	-0.0481	0.9854	0.9580	0.9756
21	0.5054	0.9027	44.0%	0.0679	0.1535	-0.0517	0.9832	0.9529	0.9722
22	0.5349	0.9530	43.9%	0.0726	0.1637	-0.0557	0.9808	0.9474	0.9684
23	0.5653	1.0048	43.7%	0.0775	0.1743	-0.0599	0.9781	0.9414	0.9641
24	0.5968	1.0576	43.6%	0.0827	0.1853	-0.0642	0.9750	0.9349	0.9594
25	0.6292	1.1114	43.4%	0.0881	0.1966	-0.0686	0.9716	0.9278	0.9541
26	0.6627	1.1666	43.2%	0.0937	0.2083	-0.0732	0.9677	0.9201	0.9483
27	0.6968	1.2230	43.0%	0.0995	0.2205	-0.0779	0.9635	0.9118	0.9418
28	0.7313	1.2805	42.9%	0.1054	0.2332	-0.0827	0.9588	0.9028	0.9348
29	0.7665	1.3394	42.8%	0.1117	0.2463	-0.0876	0.9536	0.8930	0.9271
30	0.8024	1.3987	42.6%	0.1181	0.2598	-0.0927	0.9479	0.8826	0.9187
31	0.8388	1.4591	42.5%	0.1247	0.2737	-0.0981	0.9416	0.8713	0.9096
32	0.8757	1.5206	42.4%	0.1318	0.2881	-0.1034	0.9347	0.8592	0.8998
33	0.9141	1.5834	42.3%	0.1391	0.3031	-0.1091	0.9271	0.8461	0.8889
34	0.9538	1.6473	42.1%	0.1467	0.3185	-0.1152	0.9185	0.8320	0.8770
35	0.9944	1.7121	41.9%	0.1546	0.3345	-0.1216	0.9092	0.8169	0.8642
36	1.0356	1.7778	41.7%	0.1628	0.3508	-0.1282	0.8989	0.8007	0.8503
37	1.0779	1.8444	41.6%	0.1711	0.3674	-0.1350	0.8877	0.7835	0.8353
38	1.1212	1.9117	41.4%	0.1798	0.3844	-0.1421	0.8755	0.7651	0.8193
39	1.1654	1.9795	41.1%	0.1886	0.4015	-0.1494	0.8622	0.7456	0.8022
40	1.2101	2.0477	40.9%	0.1977	0.4189	-0.1569	0.8481	0.7252	0.7843
41	1.2552	2.1165	40.7%	0.2071	0.4365	-0.1645	0.8330	0.7035	0.7654
42	1.3016	2.1861	40.5%	0.2167	0.4545	-0.1728	0.8168	0.6808	0.7456
43	1.3486	2.2558	40.2%	0.2265	0.4725	-0.1812	0.7997	0.6571	0.7251
44	1.3960	2.3259	40.0%	0.2365	0.4908	-0.1897	0.7819	0.6325	0.7041
45	1.4436	2.3960	39.8%	0.2466	0.5093	-0.1984	0.7634	0.6070	0.6826
46	1.4912	2.4661	39.5%	0.2567	0.5277	-0.2072	0.7443	0.5810	0.6610
47	1.5393	2.5360	39.3%	0.2669	0.5463	-0.2160	0.7248	0.5544	0.6392
48	1.5876	2.6057	39.1%	0.2770	0.5647	-0.2249	0.7049	0.5273	0.6173
49	1.6356	2.6749	38.9%	0.2872	0.5830	-0.2336	0.6849	0.5000	0.5956
50	1.6838	2.7437	38.6%	0.2974	0.6013	-0.2424	0.6645	0.4726	0.5740
51	1.7321	2.8119	38.4%	0.3075	0.6195	-0.2514	0.6441	0.4452	0.5525
52	1.7799	2.8796	38.2%	0.3178	0.6378	-0.2600	0.6239	0.4177	0.5317
53	1.8279	2.9468	38.0%	0.3281	0.6561	-0.2688	0.6035	0.3904	0.5111
54	1.8759	3.0138	37.8%	0.3383	0.6744	-0.2778	0.5832	0.3631	0.4908
55	1.9236	3.0805	37.6%	0.3484	0.6926	-0.2865	0.5633	0.3360	0.4710
56	1.9710	3.1466	37.4%	0.3585	0.7108	-0.2953	0.5437	0.3092	0.4517
57	2.0181	3.2122	37.2%	0.3686	0.7289	-0.3040	0.5244	0.2827	0.4331
58	2.0650	3.2770	37.0%	0.3788	0.7469	-0.3127	0.5058	0.2568	0.4151
59	2.1118	3.3414	36.8%	0.3889	0.7647	-0.3216	0.4878	0.2314	0.3977
60	2.1587	3.4054	36.6%	0.3990	0.7824	-0.3303	0.4706	0.2065	0.3813
61	2.2059	3.4692	36.4%	0.4092	0.8002	-0.3391	0.4541	0.1823	0.3655
62	2.2530	3.5325	36.2%	0.4194	0.8178	-0.3482	0.4384	0.1589	0.3506
63	2.2999	3.5947	36.0%	0.4295	0.8350	-0.3571	0.4237	0.1364	0.3364
64	2.3464	3.6556	35.8%	0.4396	0.8518	-0.3659	0.4099	0.1149	0.3230
65	2.3925	3.7155	35.6%	0.4495	0.8682	-0.3747	0.3969	0.0943	0.3105
66	2.4384	3.7745	35.4%	0.4594	0.8842	-0.3834	0.3847	0.0746	0.2986
67	2.4839	3.8325	35.2%	0.4691	0.8999	-0.3921	0.3731	0.0556	0.2873
68	2.5289	3.8894	35.0%	0.4787	0.9152	-0.4007	0.3623	0.0375	0.2767
69	2.5731	3.9451	34.8%	0.4881	0.9300	-0.4089	0.3522	0.0202	0.2666
70	2.6164	3.9993	34.6%	0.4972	0.9444	-0.4170	0.3427	0.0037	0.2572
71	2.6589	4.0518	34.4%	0.5061	0.9582	-0.4249	0.3340	-0.0117	0.2484
72	2.7007	4.1026	34.2%	0.5148	0.9716	-0.4323	0.3259	-0.0262	0.2401
73	2.7418	4.1521	34.0%	0.5234	0.9845	-0.4397	0.3185	-0.0399	0.2324
74	2.7822	4.2000	33.8%	0.5317	0.9969	-0.4469	0.3114	-0.0527	0.2251
75	2.8222	4.2466	33.5%	0.5399	1.0090	-0.4540	0.3048	-0.0648	0.2180
76	2.8610	4.2915	33.3%	0.5479	1.0206	-0.4609	0.2986	-0.0762	0.2114
77	2.8985	4.3346	33.1%	0.5557	1.0318	-0.4677	0.2927	-0.0869	0.2051
78	2.9350	4.3756	32.9%	0.5633	1.0423	-0.4741	0.2871	-0.0969	0.1989
79	2.9703	4.4146	32.7%	0.5706	1.0523	-0.4805	0.2814	-0.1062	0.1928
80	3.0048	4.4516	32.5%	0.5777	1.0618	-0.4866	0.2758	-0.1149	0.1866
81	3.0384	4.4868	32.3%	0.5847	1.0709	-0.4925	0.2702	-0.1231	0.1806
82	3.0708	4.5199	32.1%	0.5913	1.0795	-0.4983	0.2646	-0.1306	0.1746
83	3.1022	4.5510	31.8%	0.5978	1.0876	-0.5038	0.2591	-0.1375	0.1685
84	3.1324	4.5803	31.6%	0.6040	1.0953	-0.5090	0.2537	-0.1441	0.1627
85	3.1600	4.6080	31.4%	0.6086	1.1025	-0.5129	0.2491	-0.1502	0.1577
86	3.1851	4.6338	31.3%	0.6128	1.1093	-0.5161	0.2451	-0.1558	0.1534
87	3.2079	4.6576	31.1%	0.6164	1.1158	-0.5190	0.2415	-0.1612	0.1495
88	3.2286	4.6794	31.0%	0.6196	1.1218	-0.5216	0.2380	-0.1662	0.1460
89	3.2477	4.6993	30.9%	0.6226	1.1275	-0.5239	0.2347	-0.1708	0.1427
90	3.2652	4.7176	30.8%	0.6253	1.1330	-0.5262	0.2316	-0.1750	0.1396

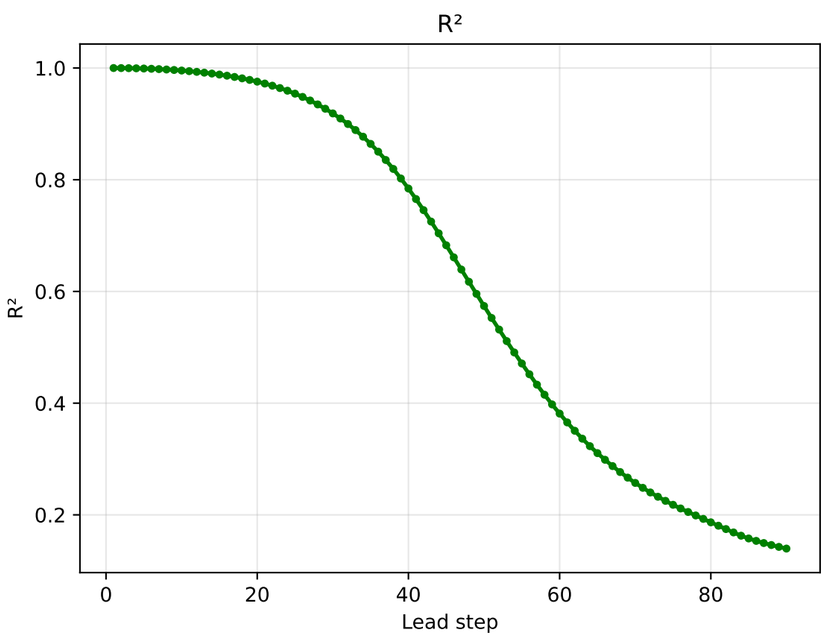
DHW Forecast Skill vs Lead Step - Test

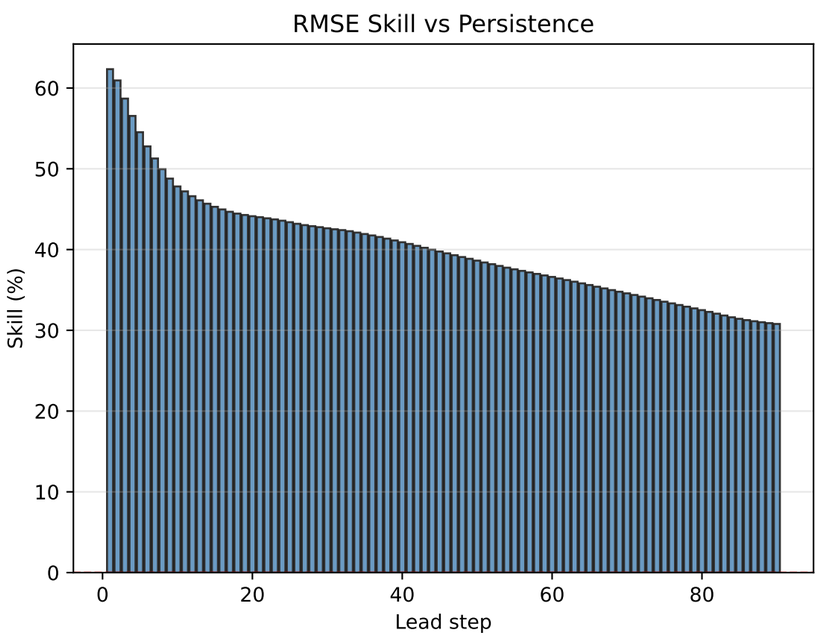


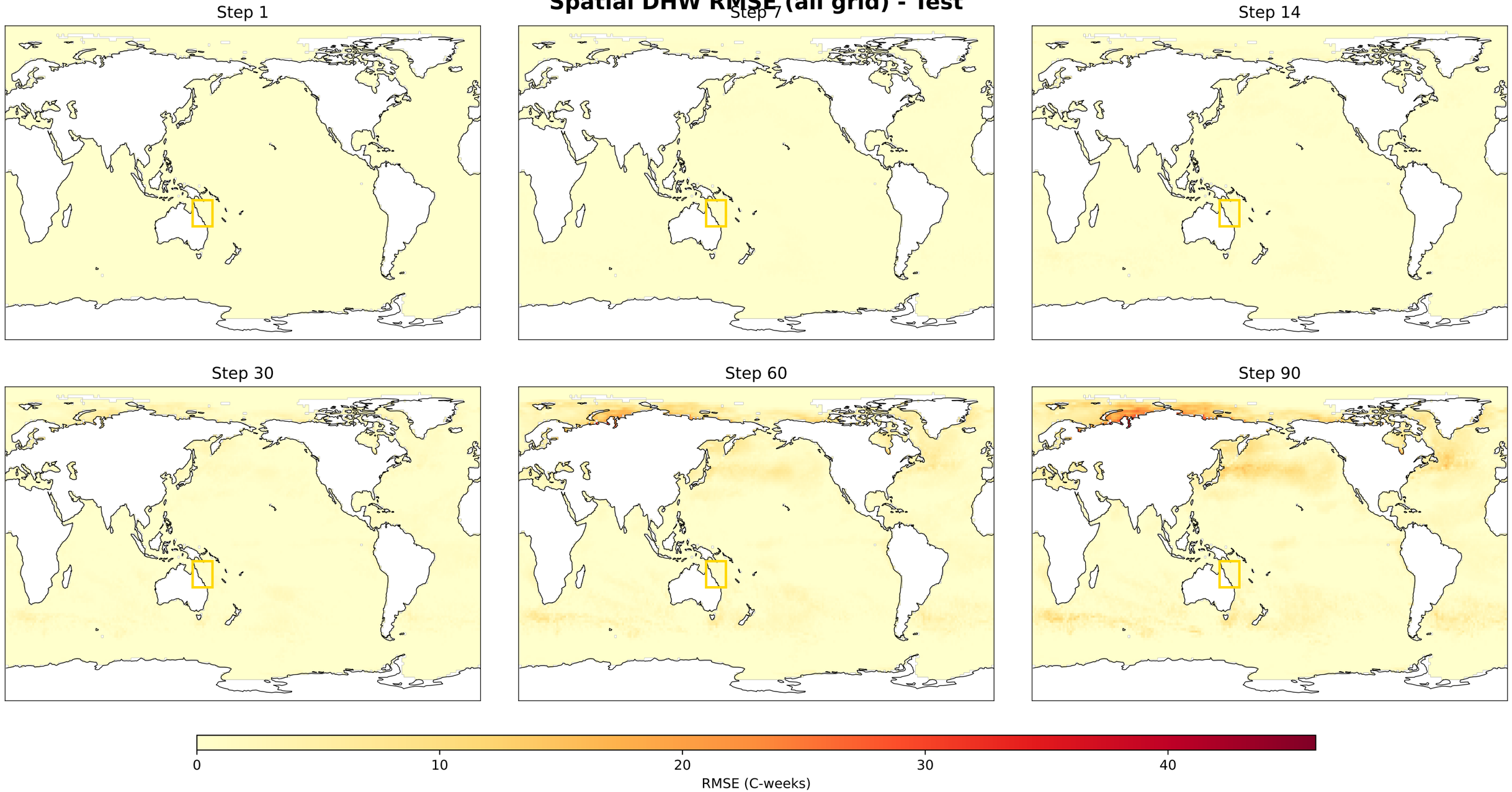


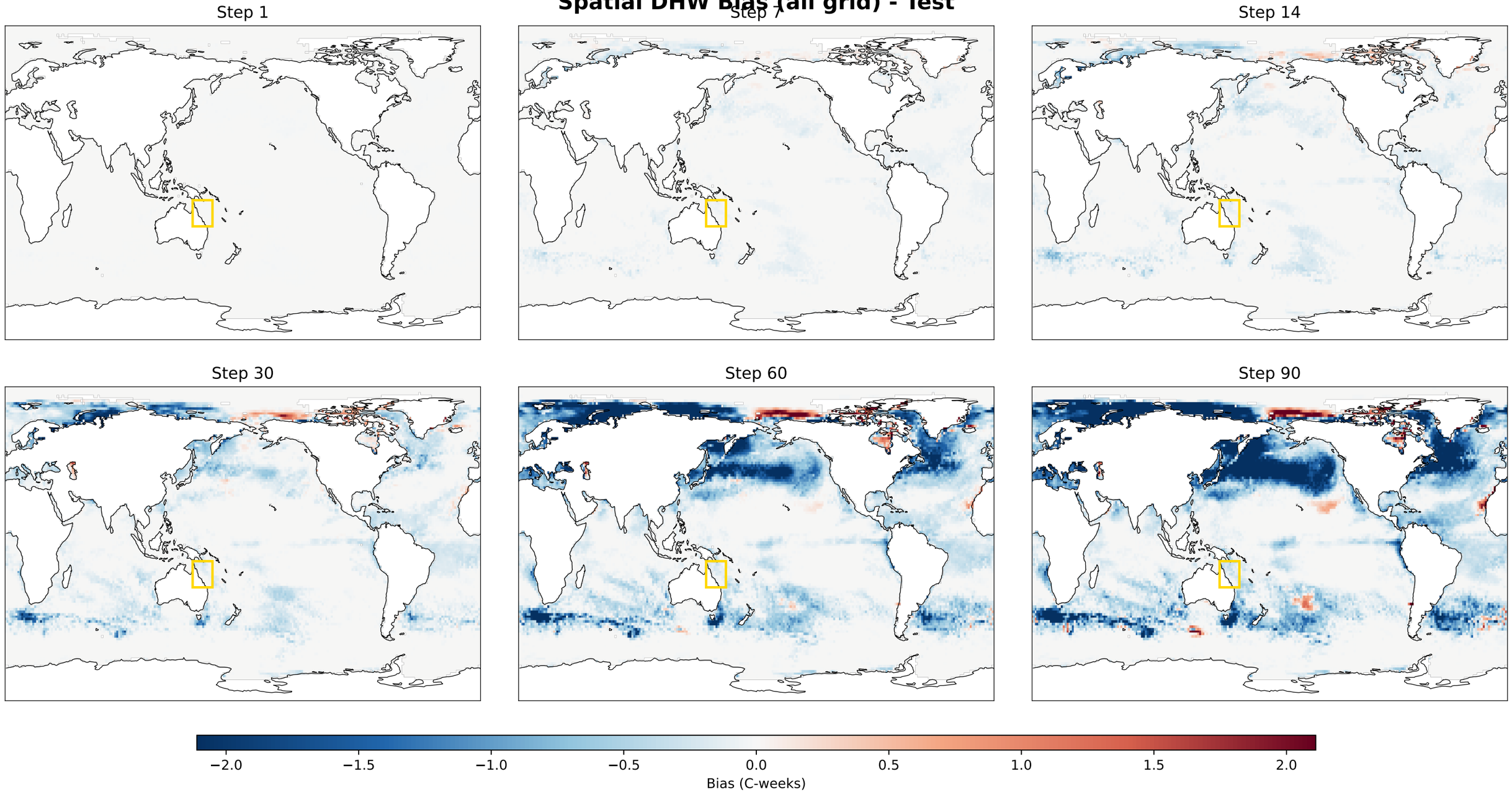


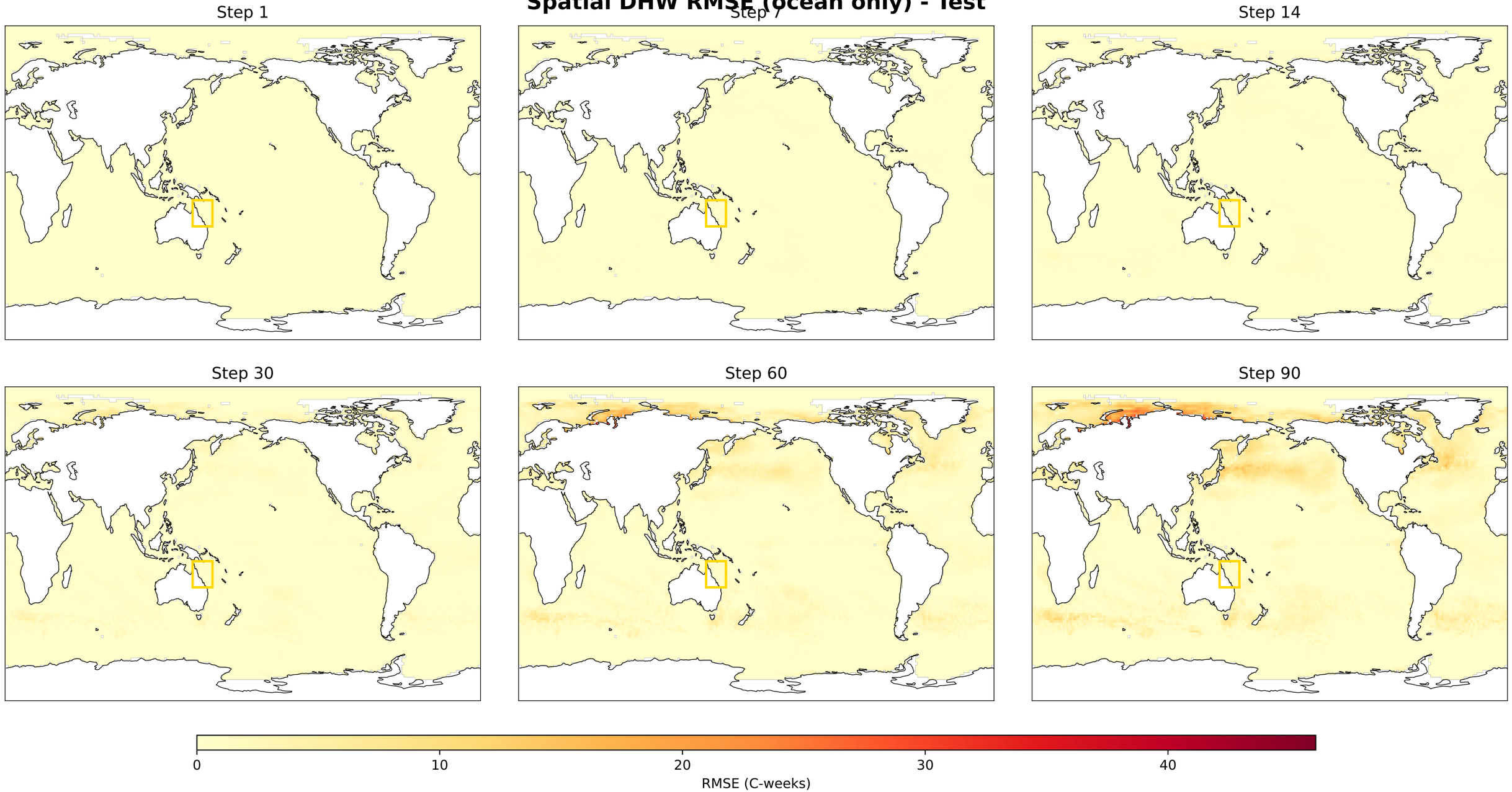


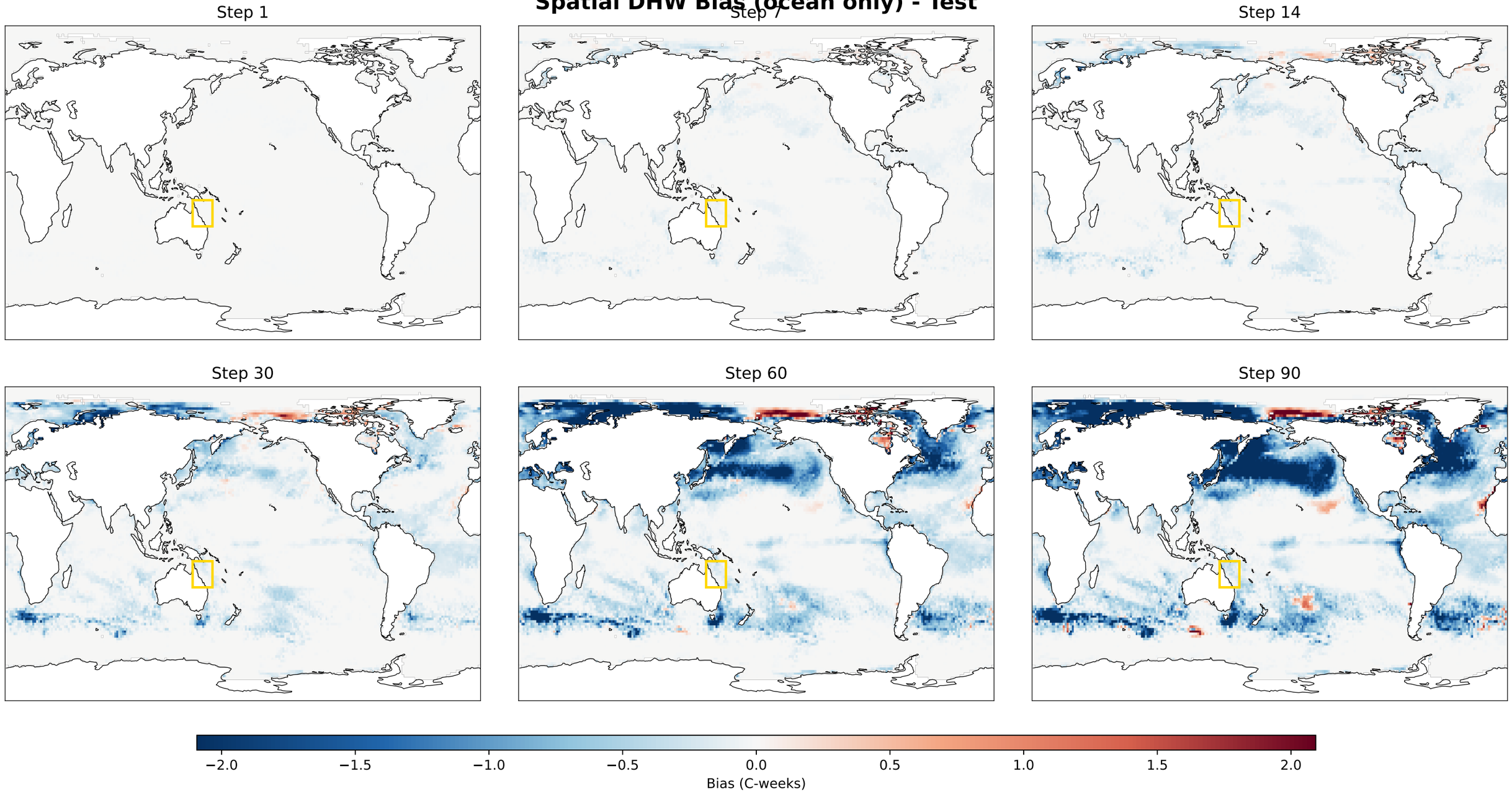





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
RMSE (C-weeks)

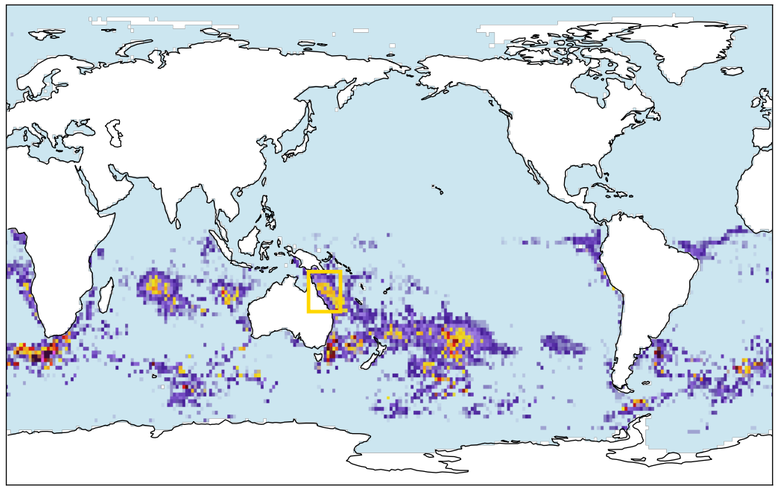

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
RMSE (C-weeks)

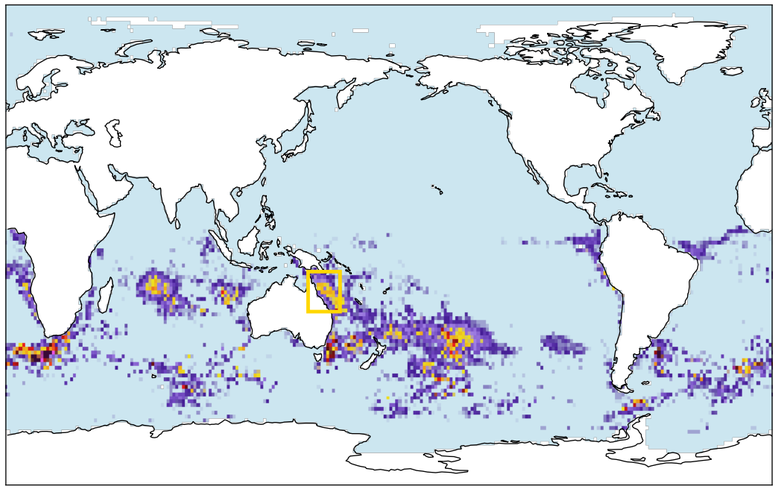

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

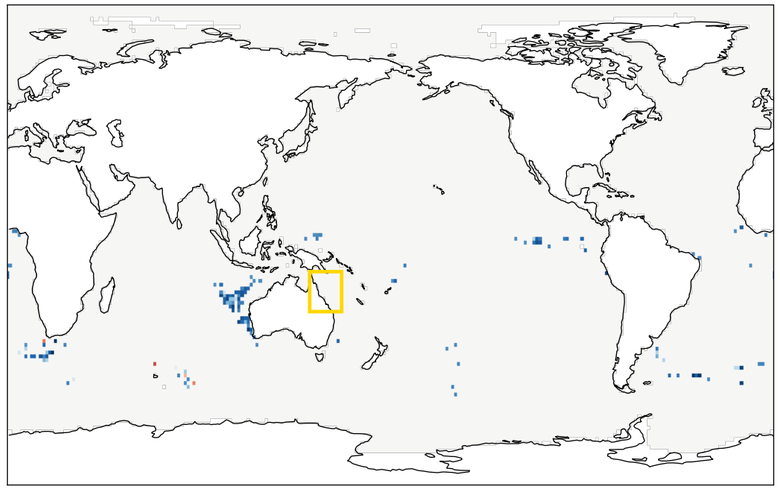
Actual DHW - Step 1



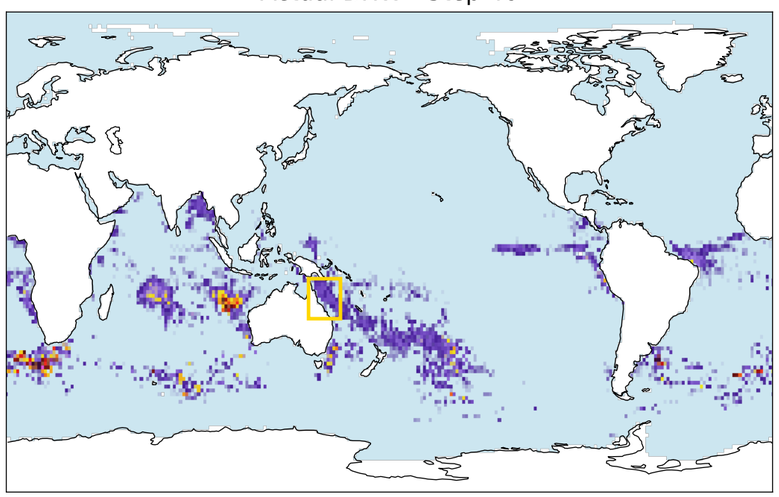
Predicted DHW - Step 1



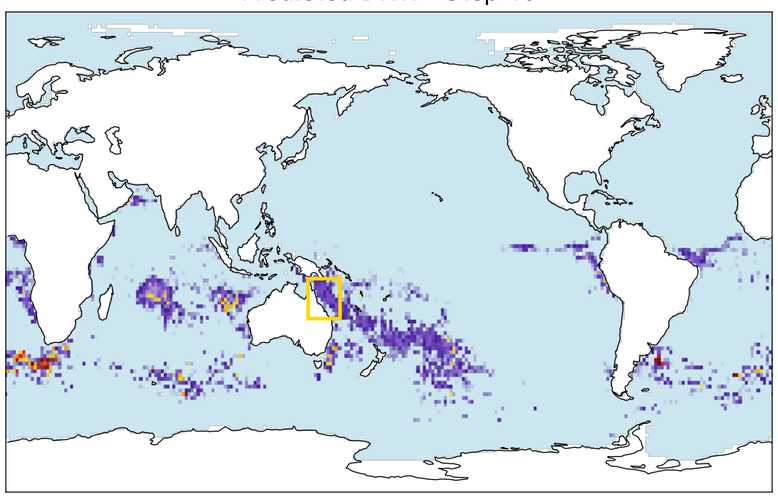
Error (Pred - Actual) - Step 1 [scale ±0.28]



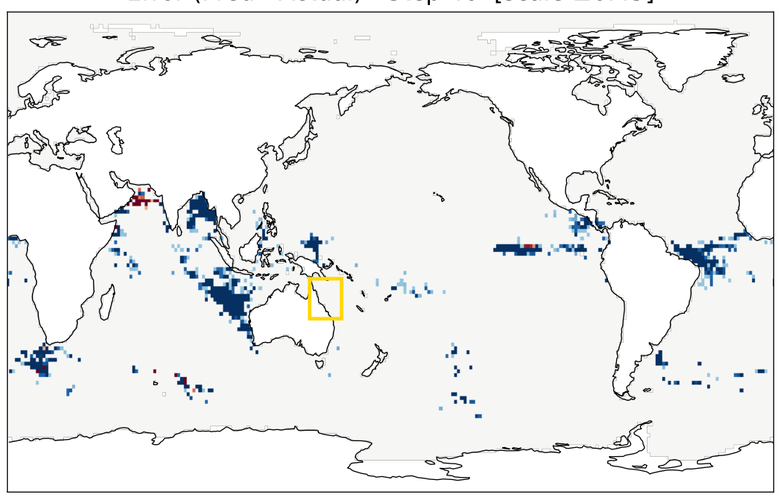
Actual DHW - Step 46



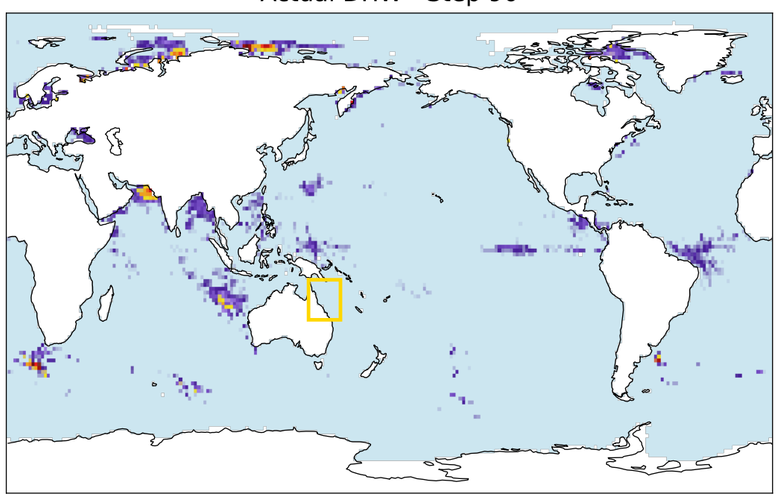
Predicted DHW - Step 46



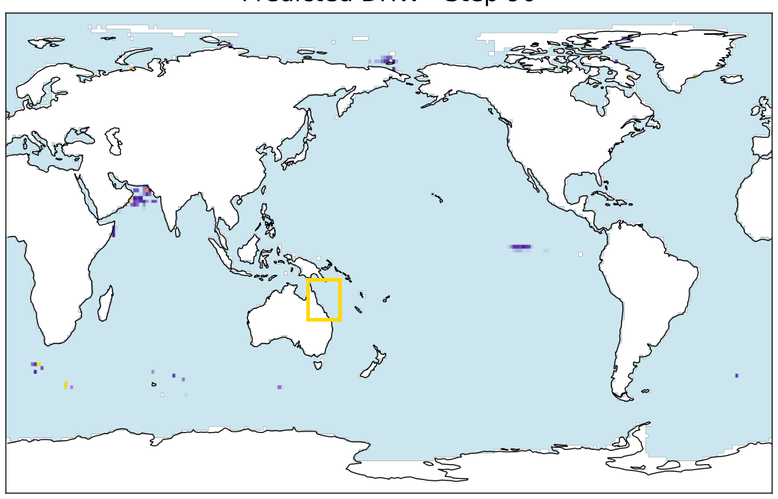
Error (Pred - Actual) - Step 46 [scale ±0.45]



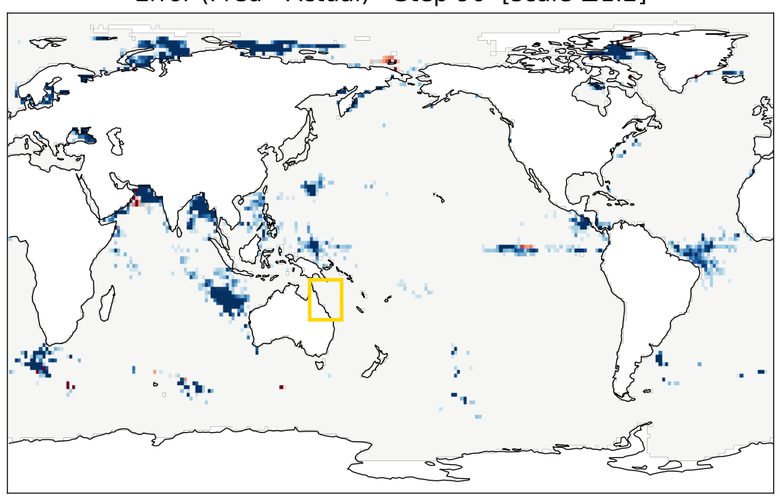
Actual DHW - Step 90

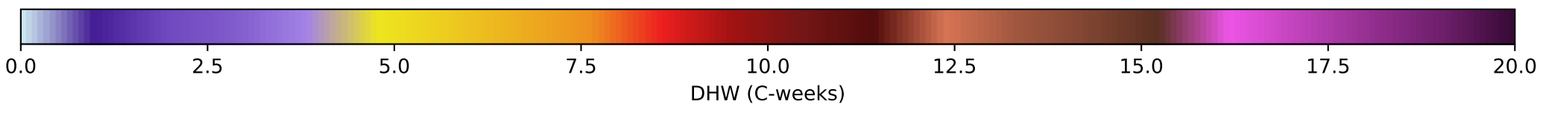


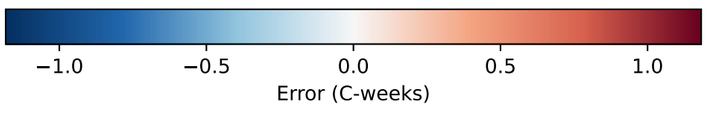
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.2]

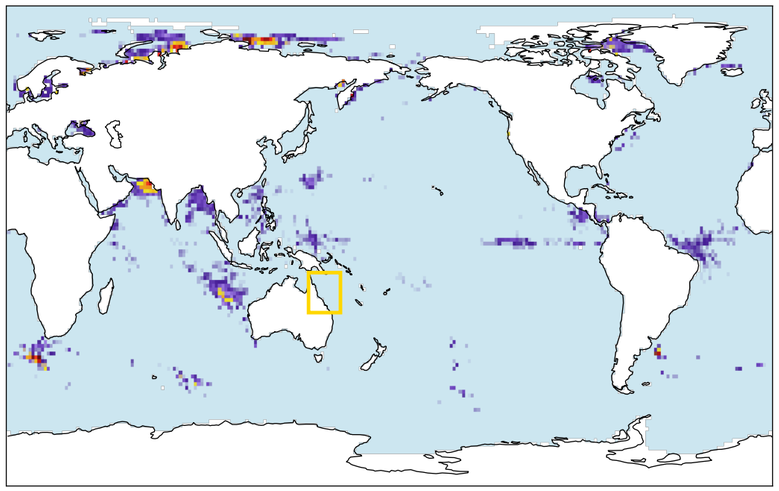




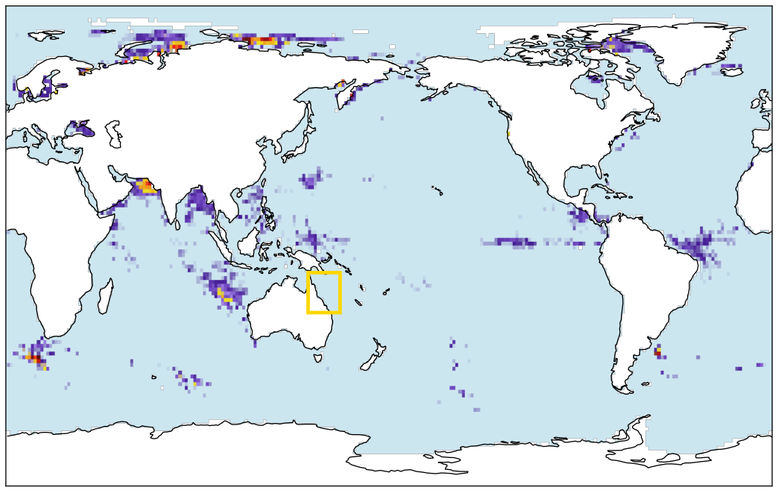


DHW Chain init: 2020-07-09 - Test (C-weeks)

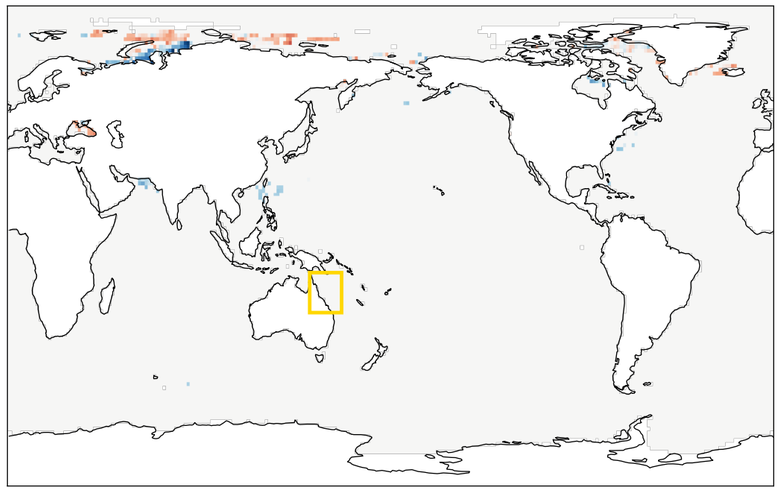
Actual DHW - Step 1



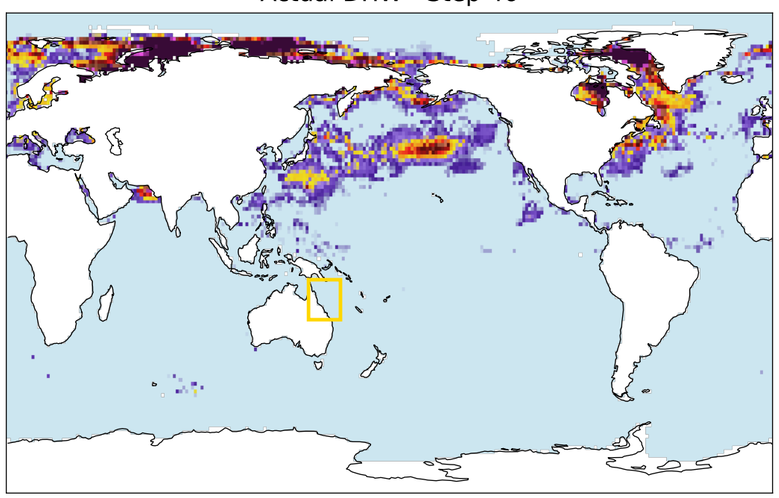
Predicted DHW - Step 1



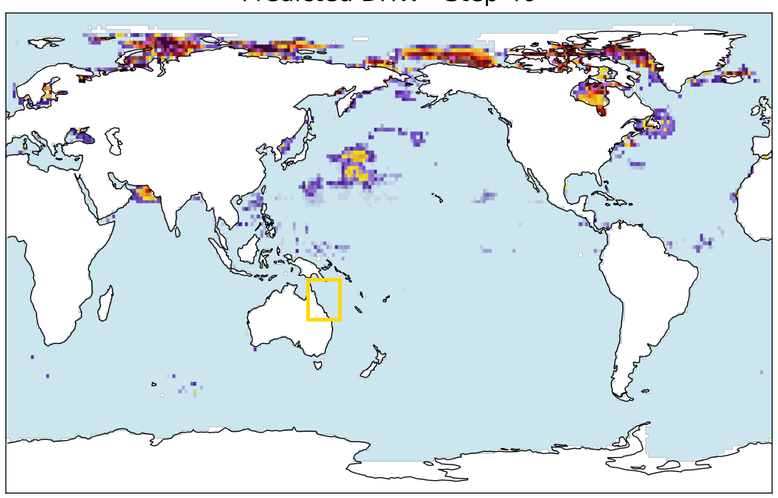
Error (Pred - Actual) - Step 1 [scale ±0.58]



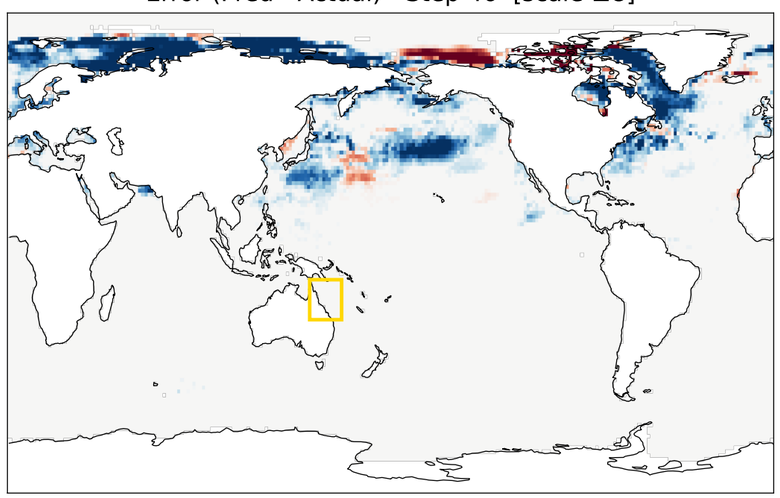
Actual DHW - Step 46



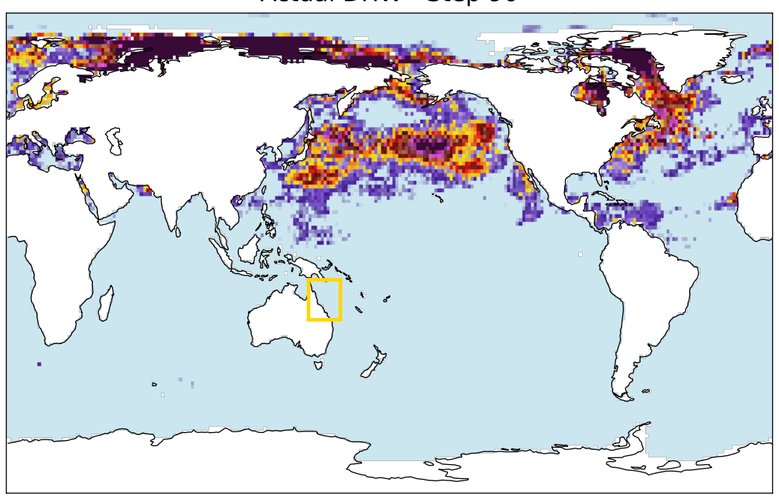
Predicted DHW - Step 46



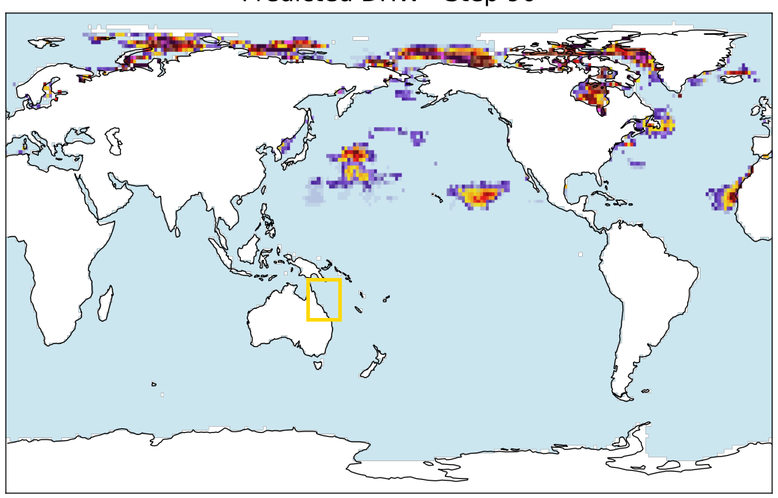
Error (Pred - Actual) - Step 46 [scale ±8]



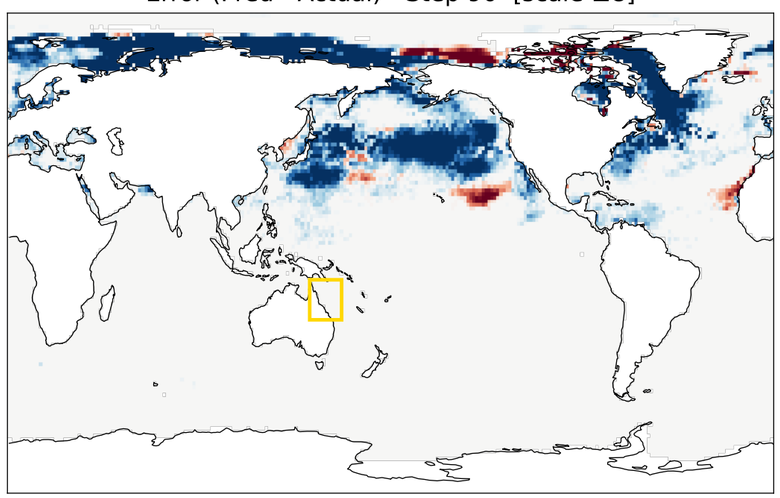
Actual DHW - Step 90

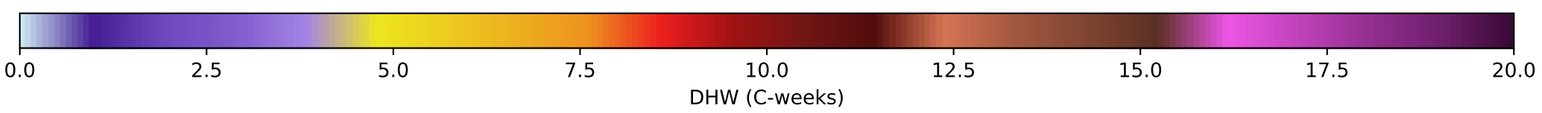


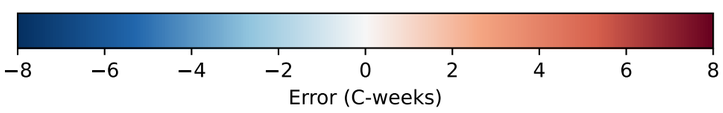
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

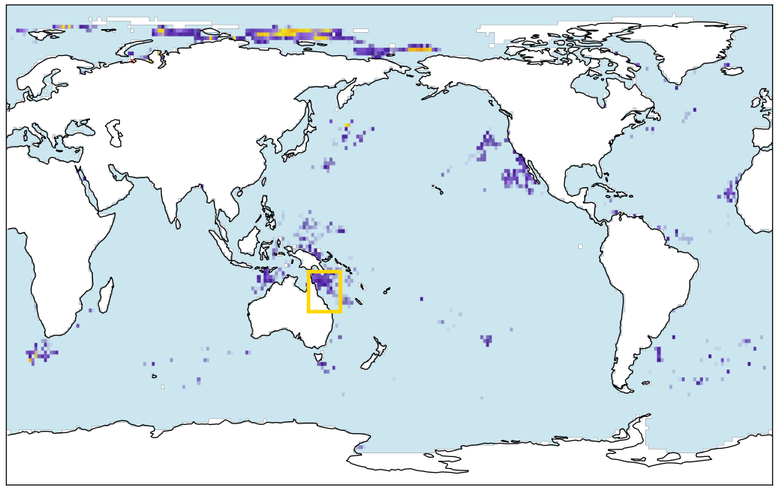




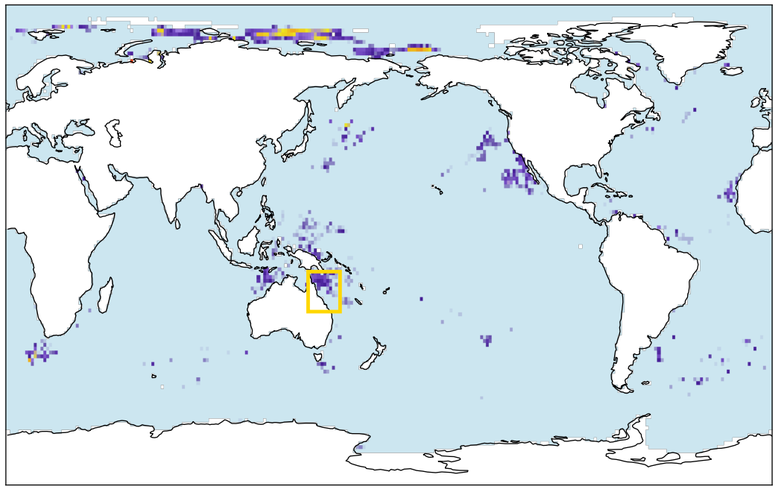


DHW Chain init: 2021-01-06 - Test (C-weeks)

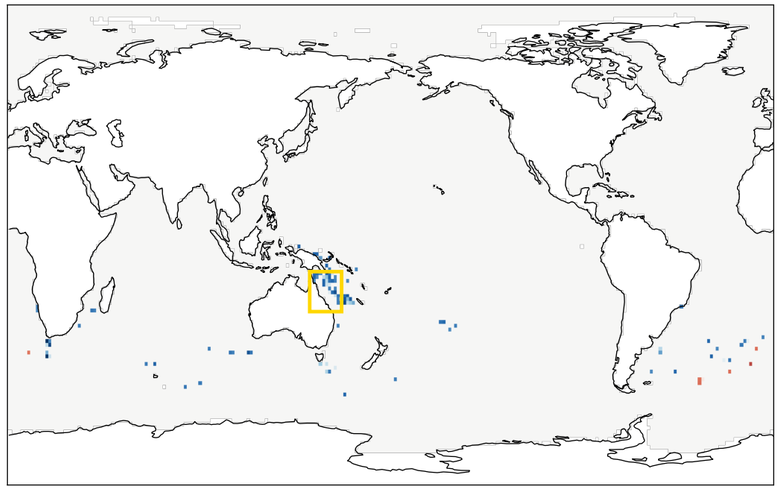
Actual DHW - Step 1



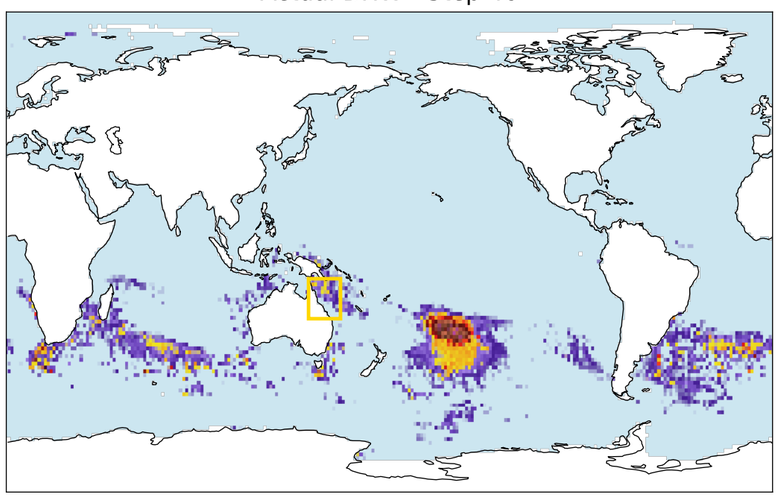
Predicted DHW - Step 1



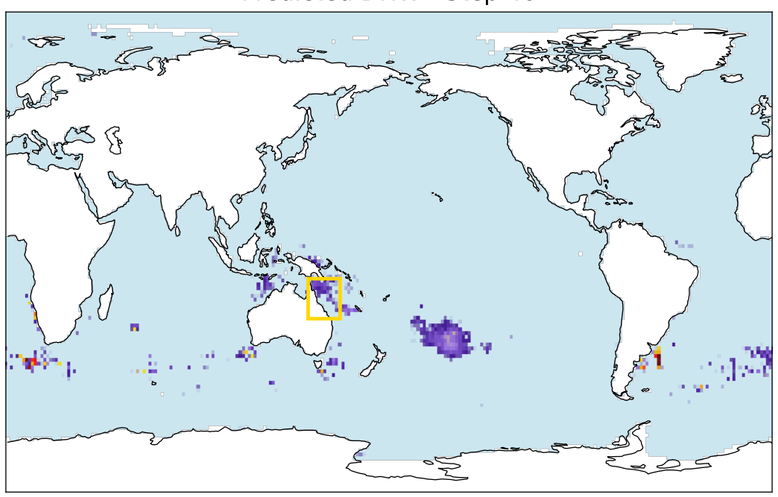
Error (Pred - Actual) - Step 1 [scale ±0.27]



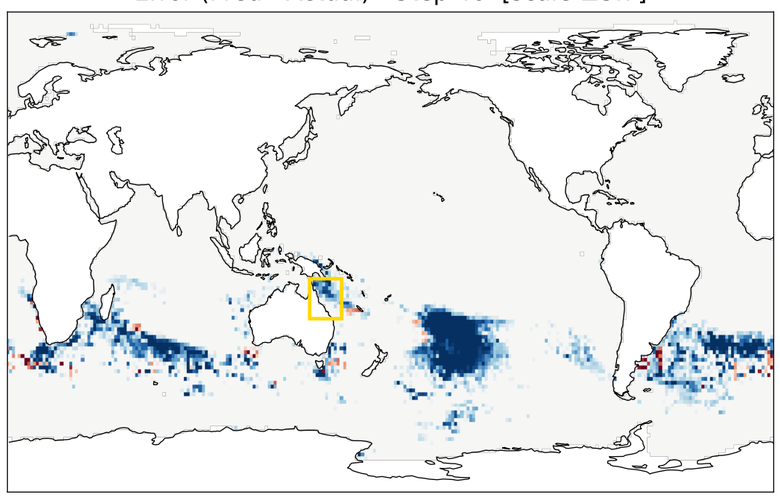
Actual DHW - Step 46



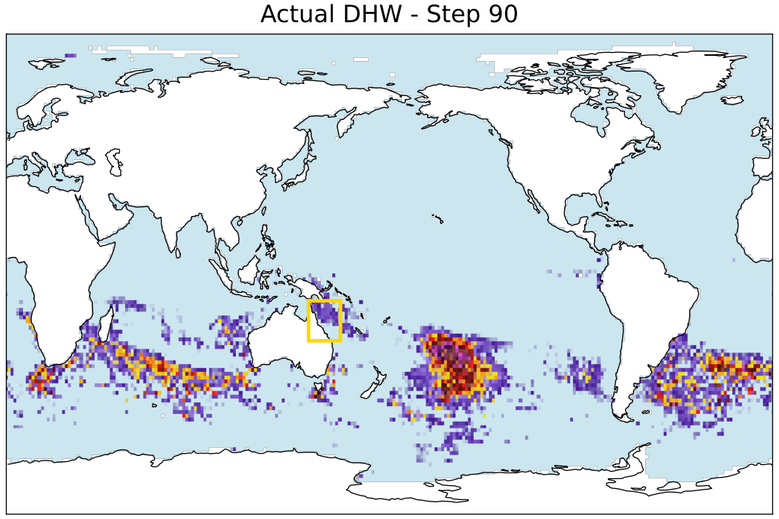
Predicted DHW - Step 46



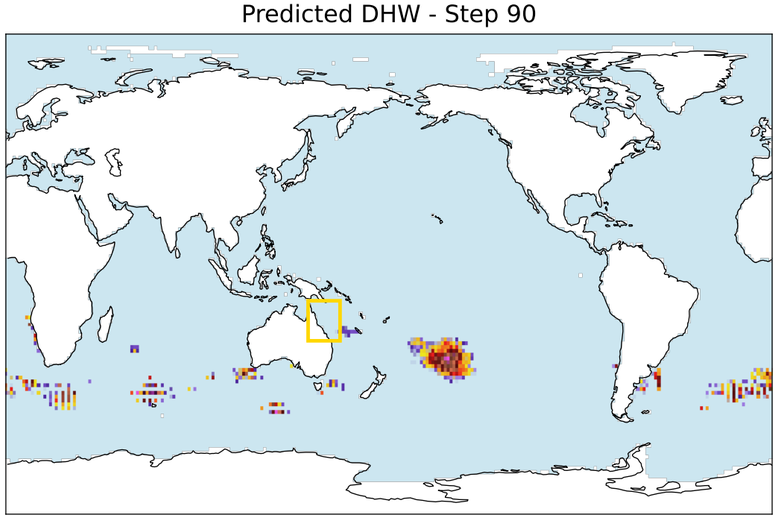
Error (Pred - Actual) - Step 46 [scale ±3.7]



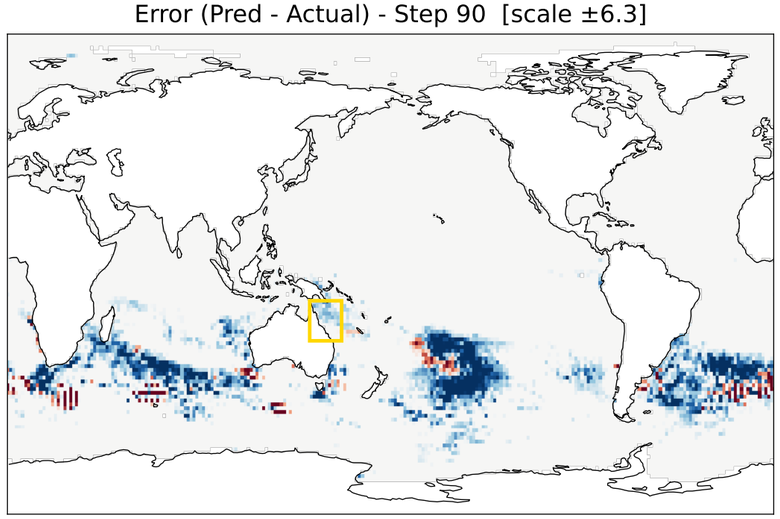
Actual DHW - Step 90

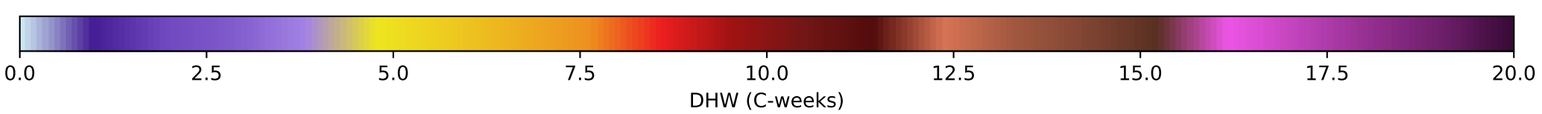


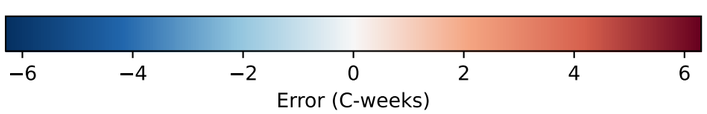
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.3]

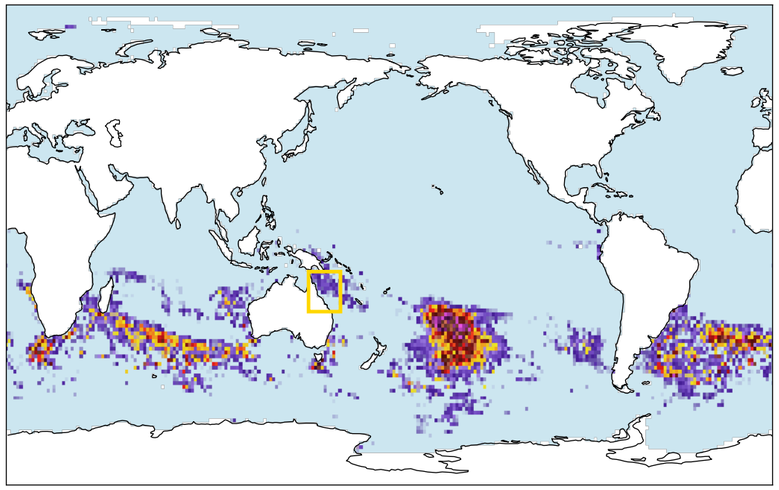




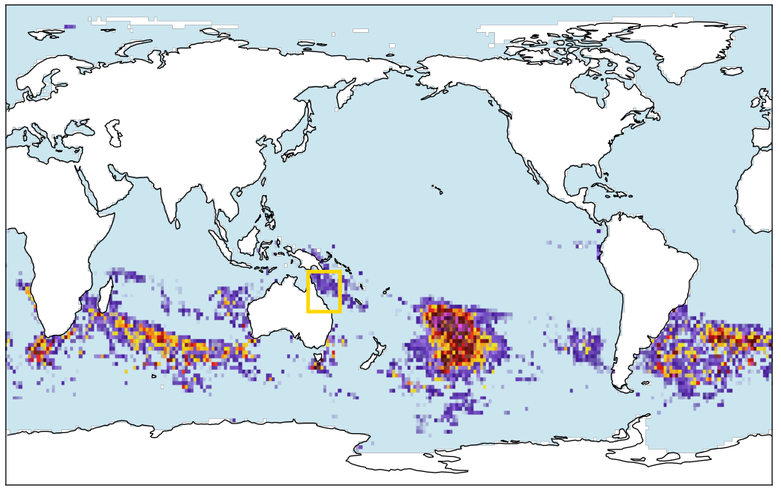


DHW Chain init: 2021-04-06 - Test (C-weeks)

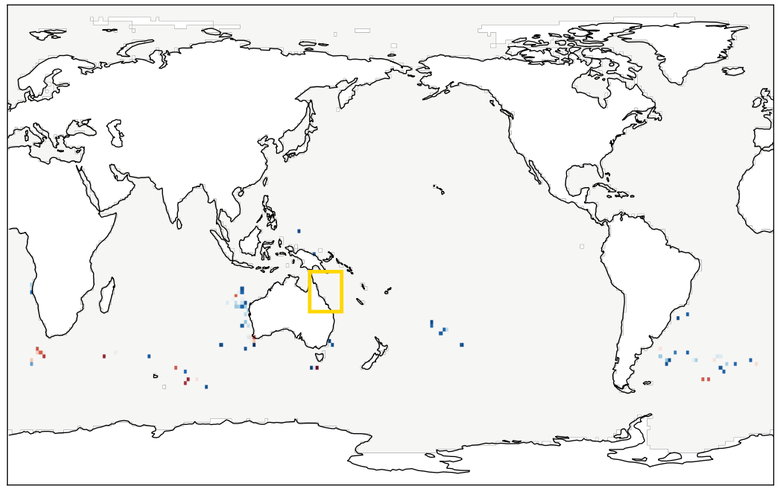
Actual DHW - Step 1



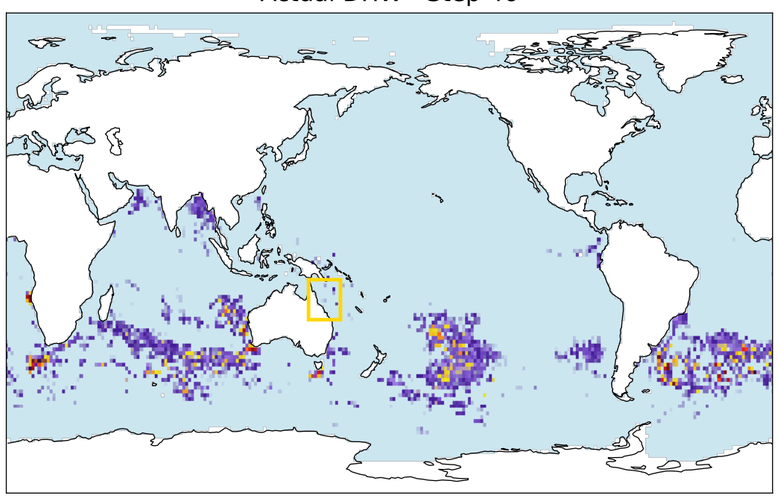
Predicted DHW - Step 1



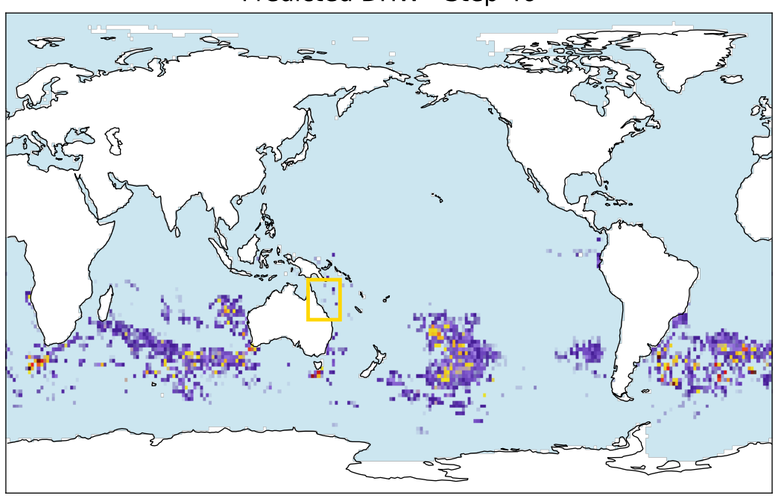
Error (Pred - Actual) - Step 1 [scale ±0.22]



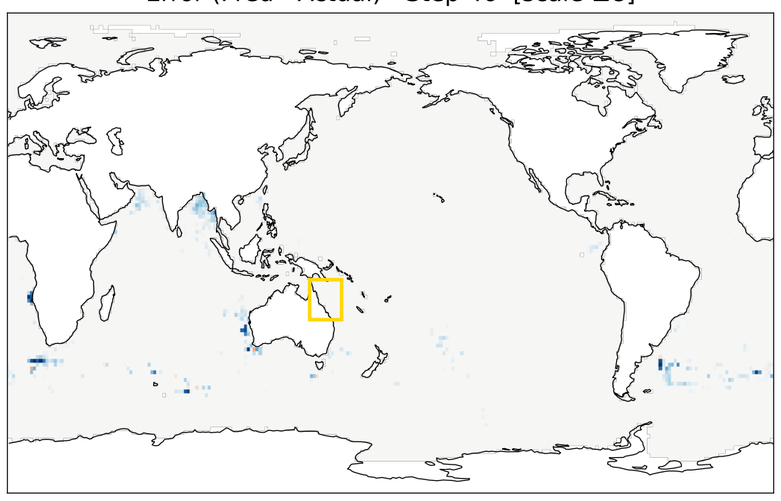
Actual DHW - Step 46



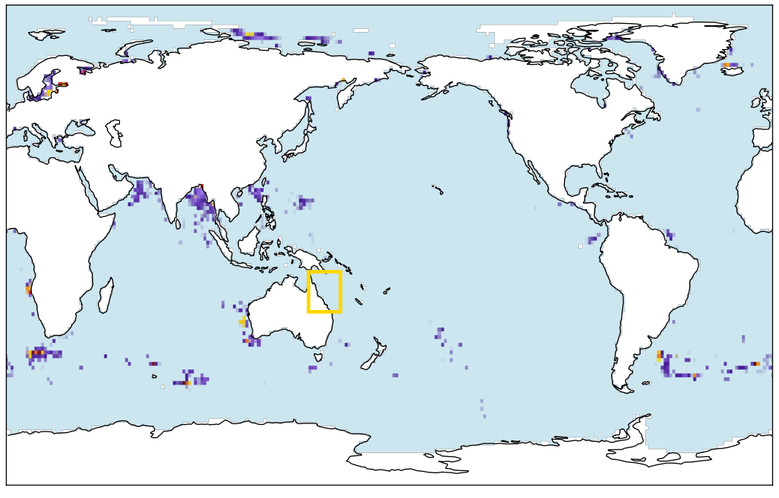
Predicted DHW - Step 46



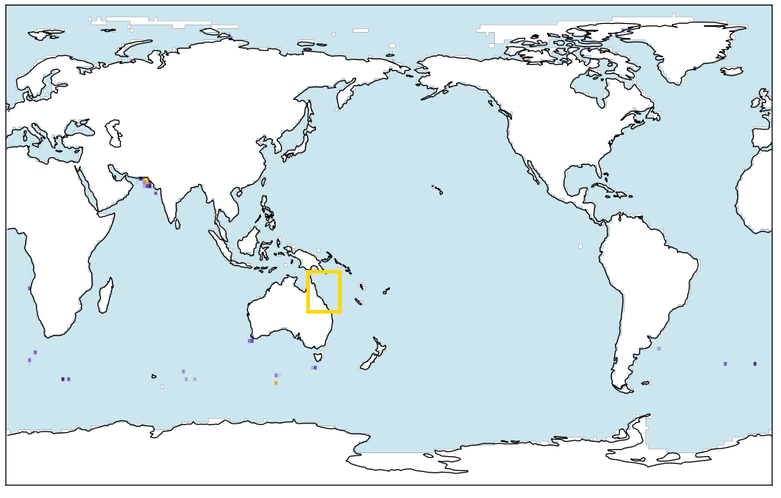
Error (Pred - Actual) - Step 46 [scale ±8]



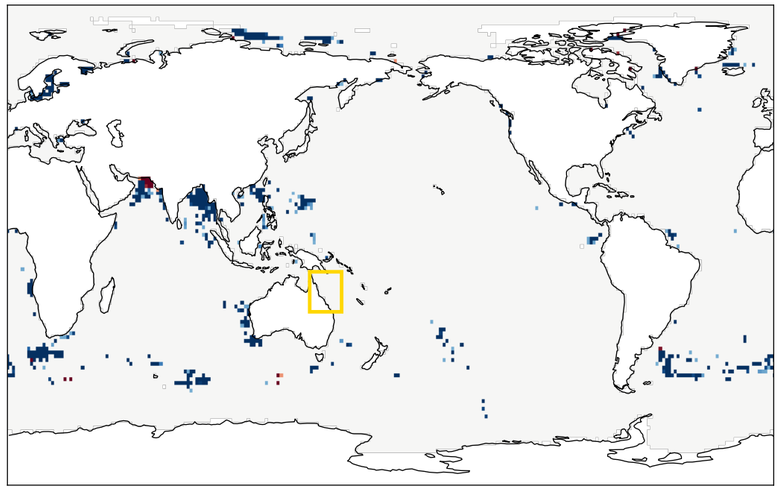
Actual DHW - Step 90

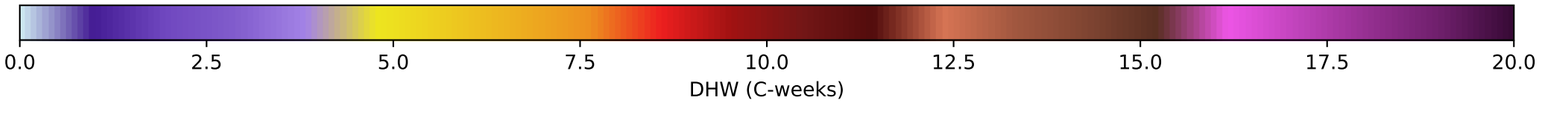


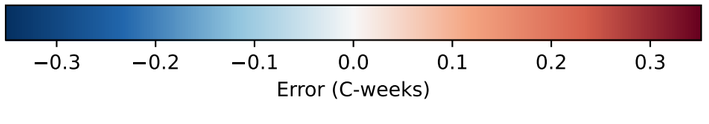
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.35]

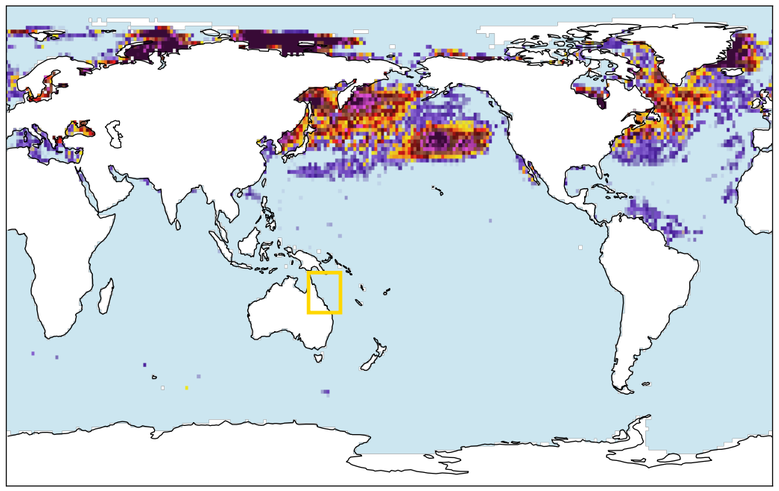




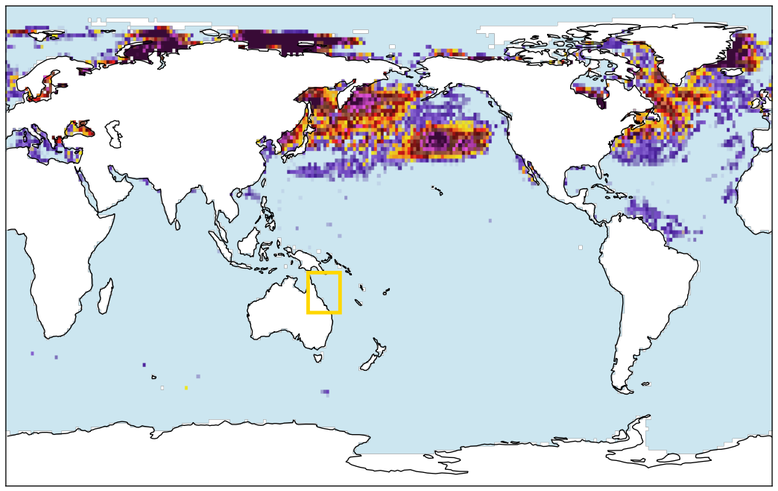


DHW Chain init: 2021-10-03 - Test (C-weeks)

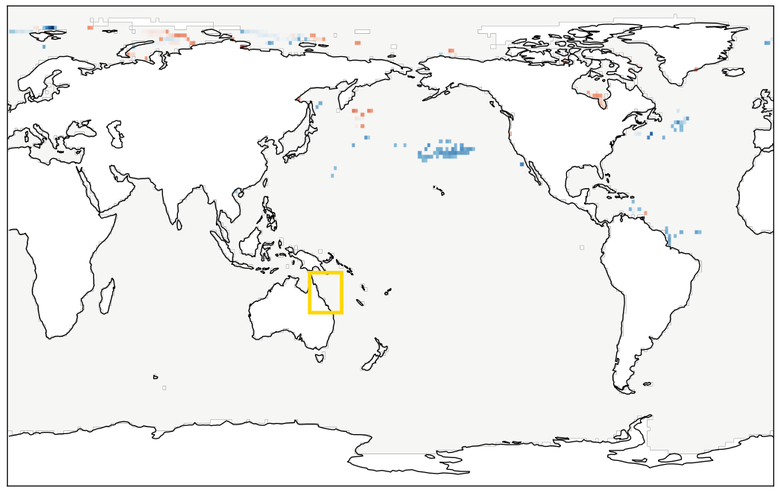
Actual DHW - Step 1



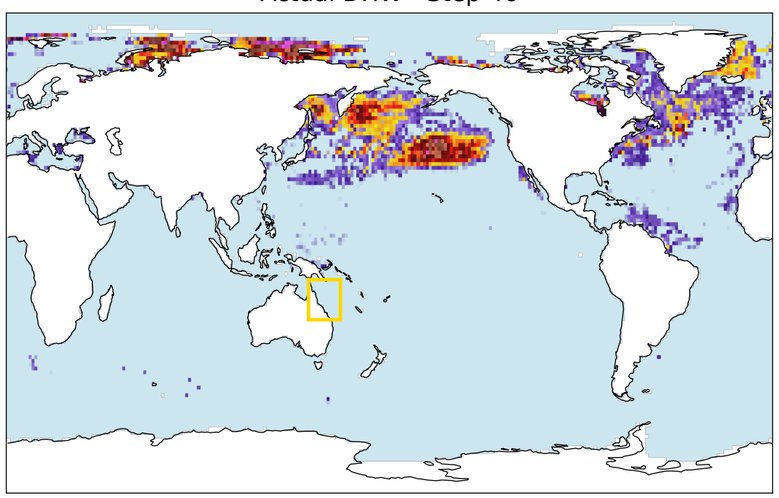
Predicted DHW - Step 1



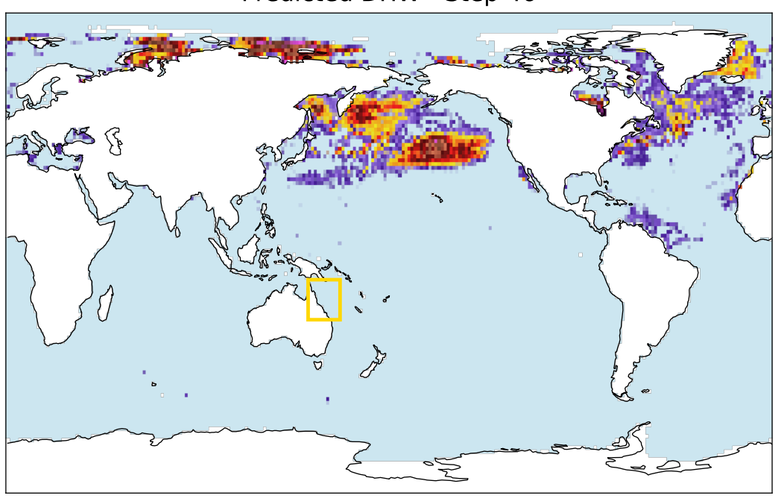
Error (Pred - Actual) - Step 1 [scale ±0.4]



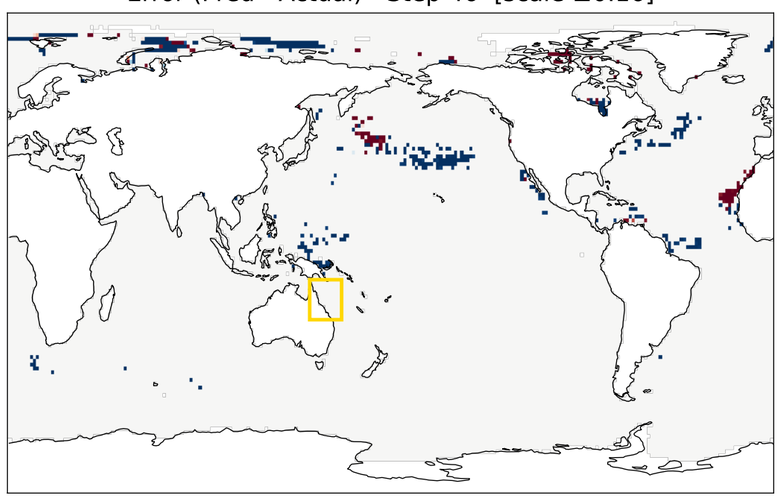
Actual DHW - Step 46



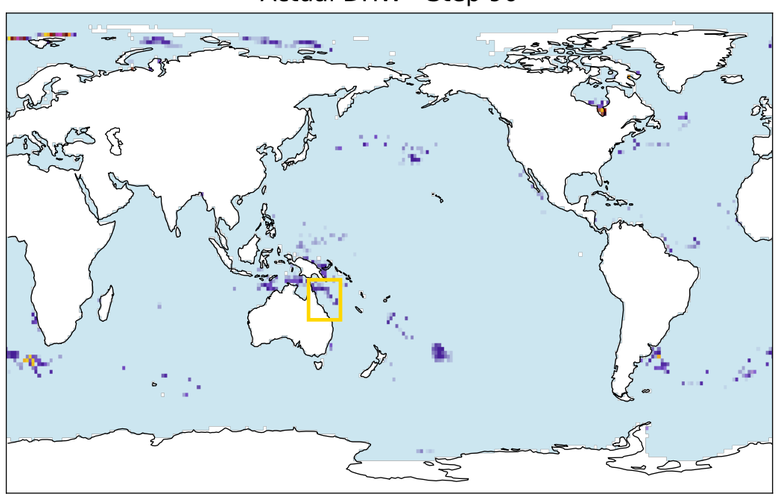
Predicted DHW - Step 46



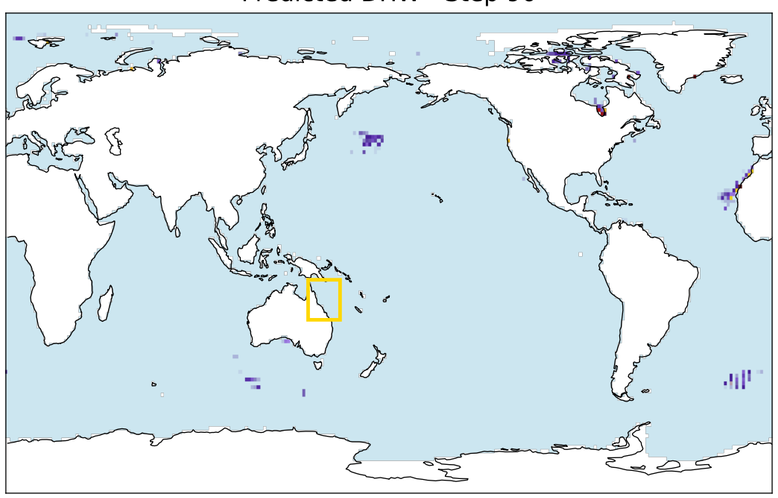
Error (Pred - Actual) - Step 46 [scale ±0.16]



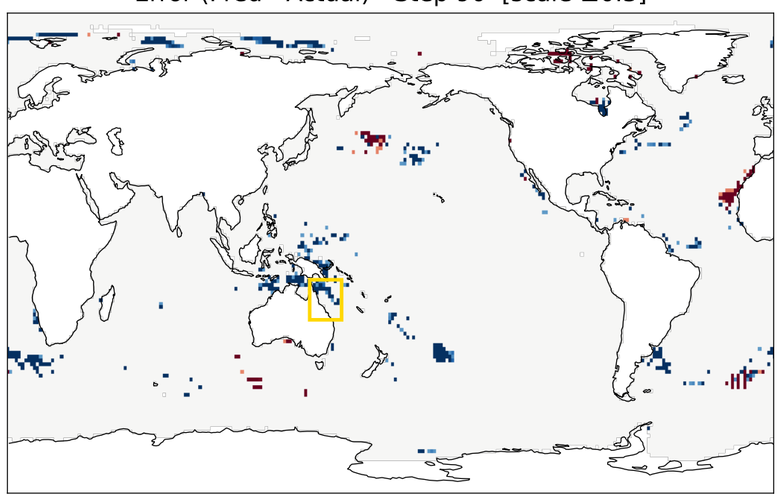
Actual DHW - Step 90

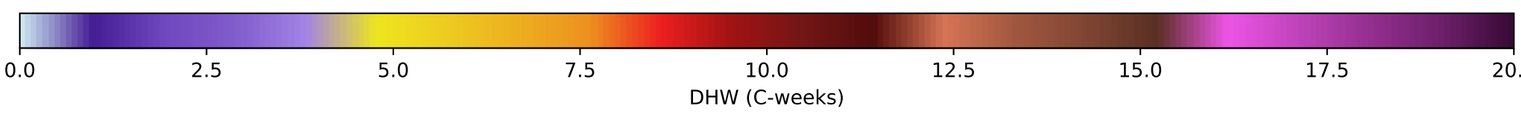


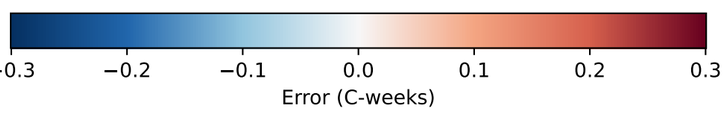
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.3]

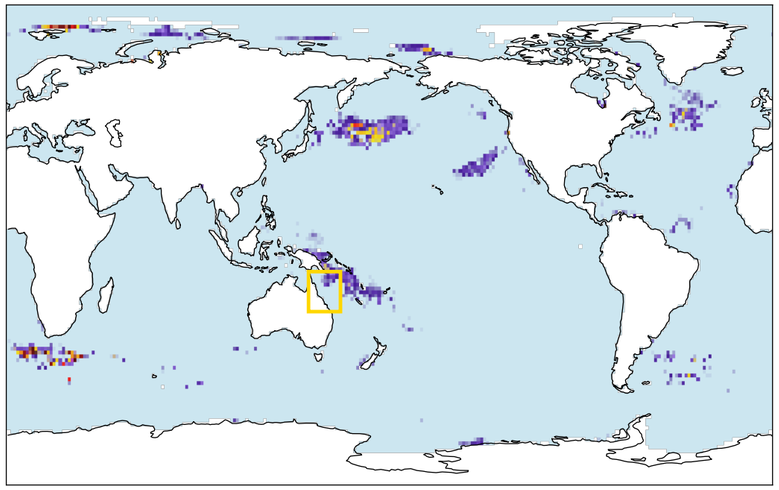




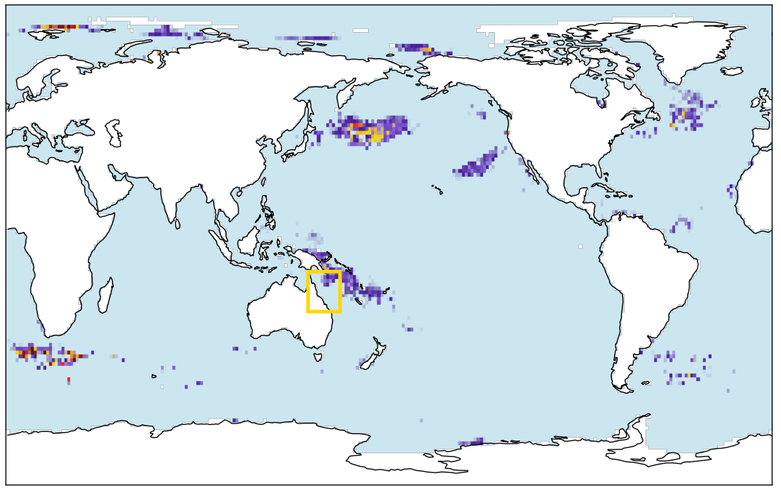


DHW Chain init: 2022-12-27 - Test (C-weeks)

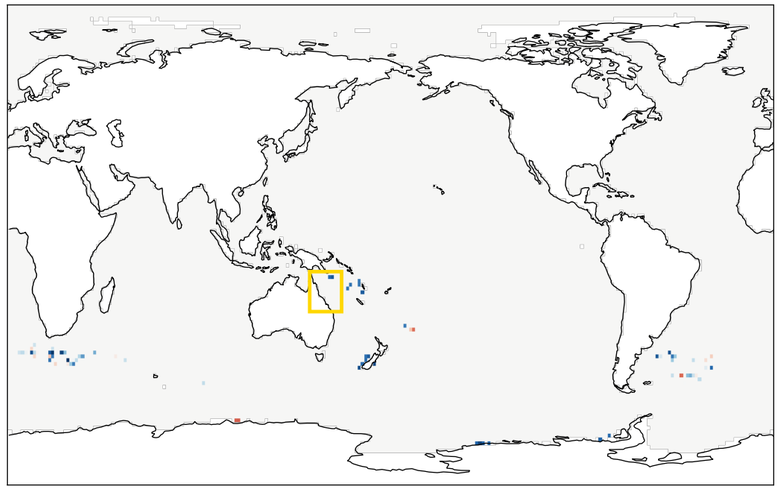
Actual DHW - Step 1



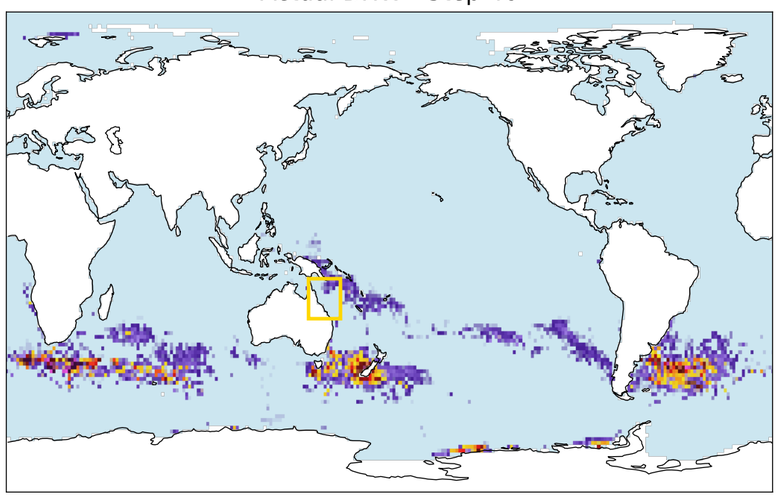
Predicted DHW - Step 1



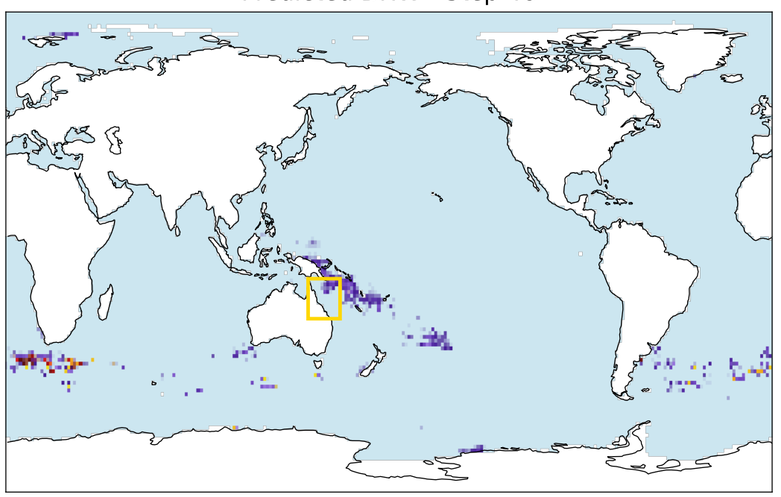
Error (Pred - Actual) - Step 1 [scale ±0.24]



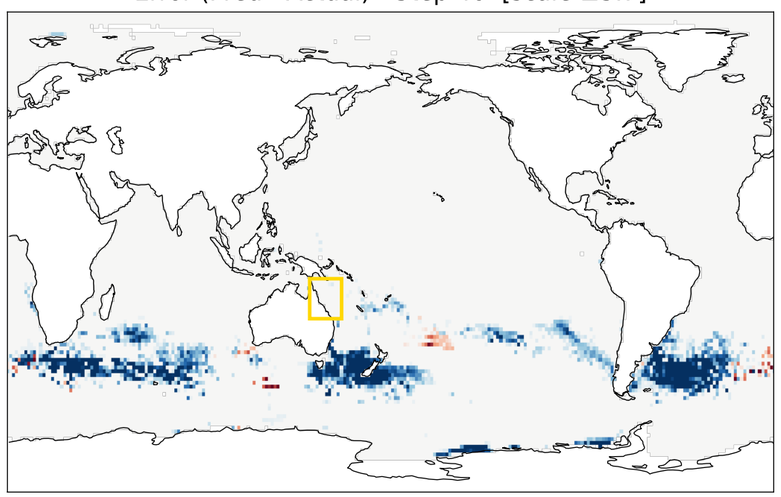
Actual DHW - Step 46



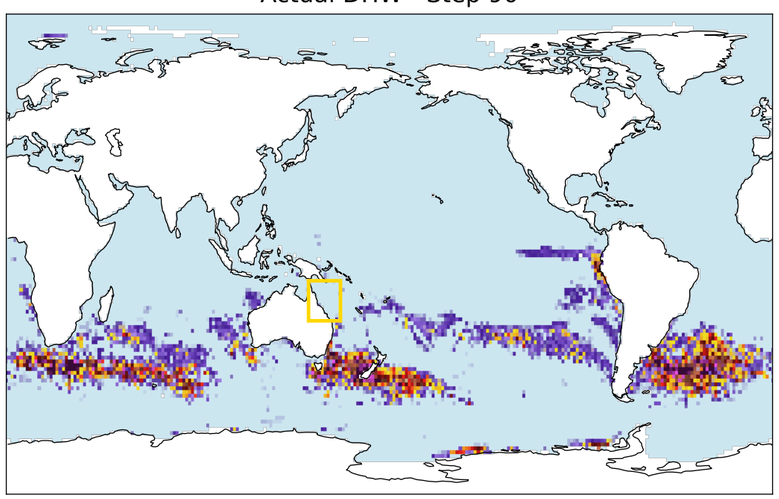
Predicted DHW - Step 46



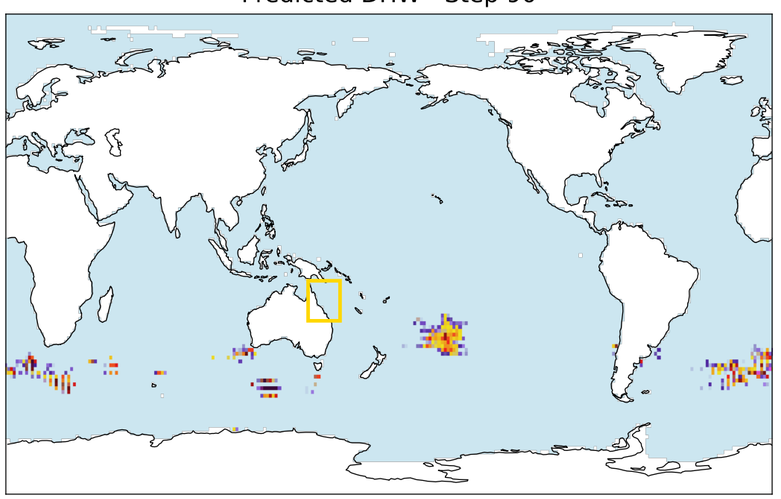
Error (Pred - Actual) - Step 46 [scale ±3.7]



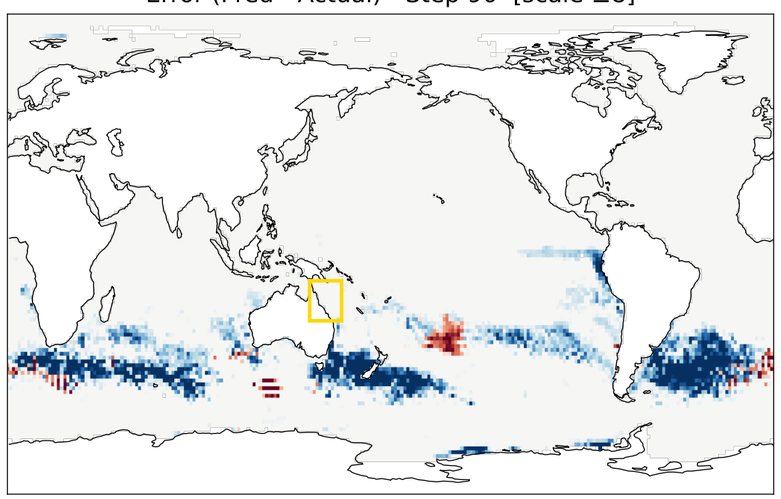
Actual DHW - Step 90

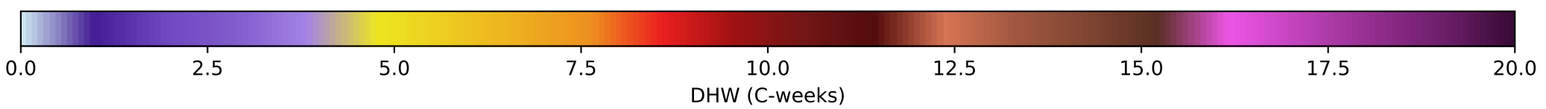


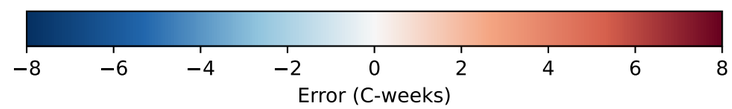
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

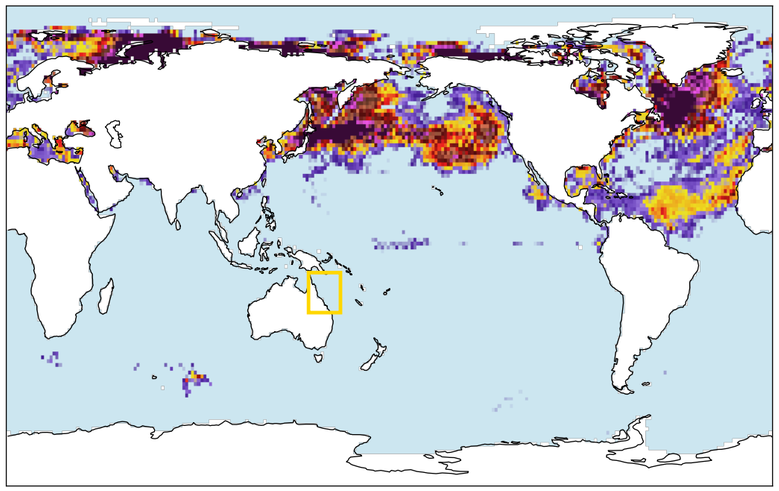




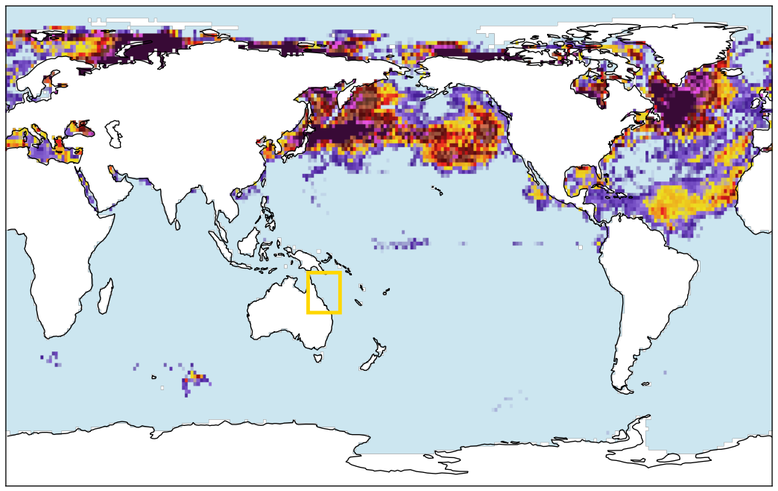


DHW Chain init: 2023-09-23 - Test (C-weeks)

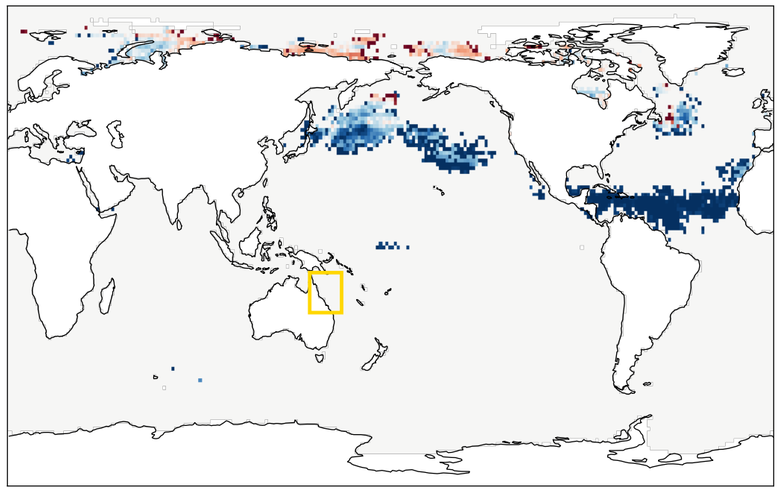
Actual DHW - Step 1



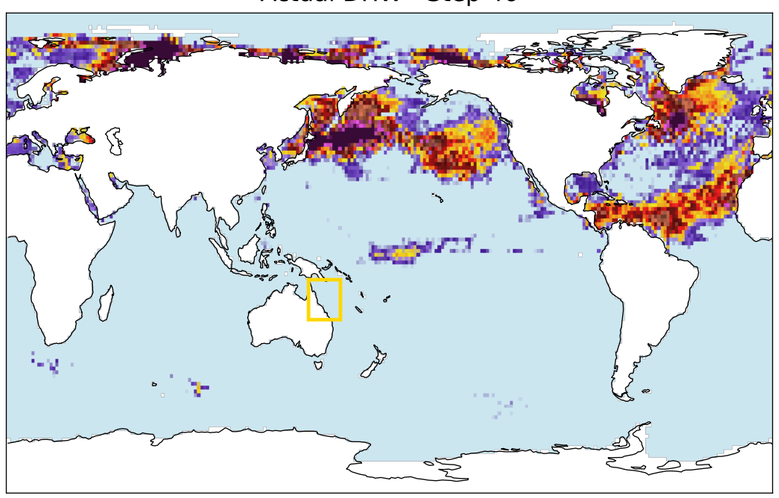
Predicted DHW - Step 1



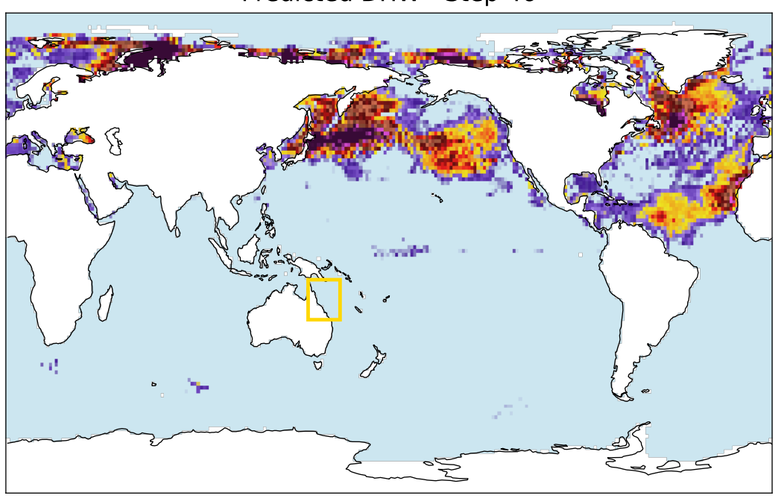
Error (Pred - Actual) - Step 1 [scale ±0.17]



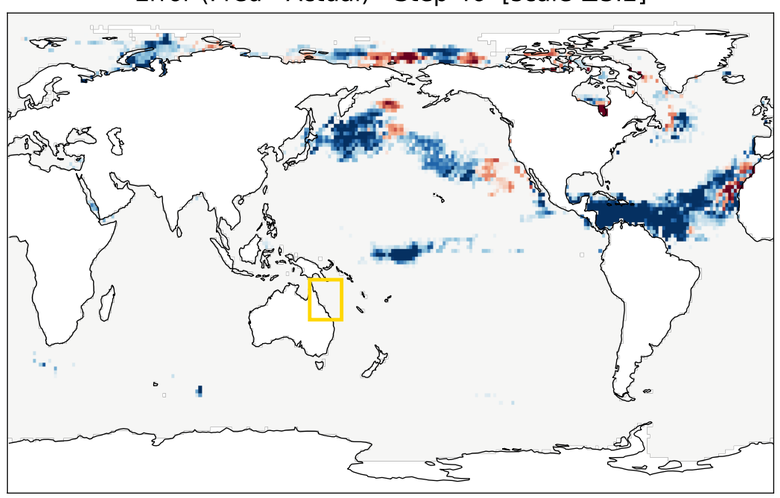
Actual DHW - Step 46



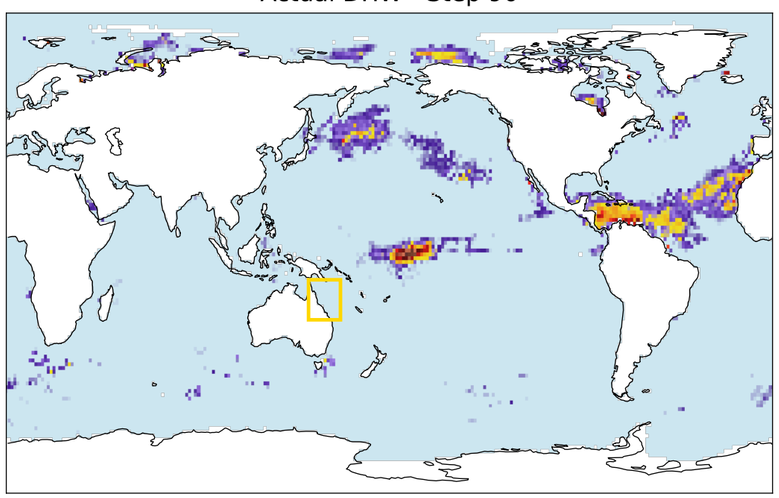
Predicted DHW - Step 46



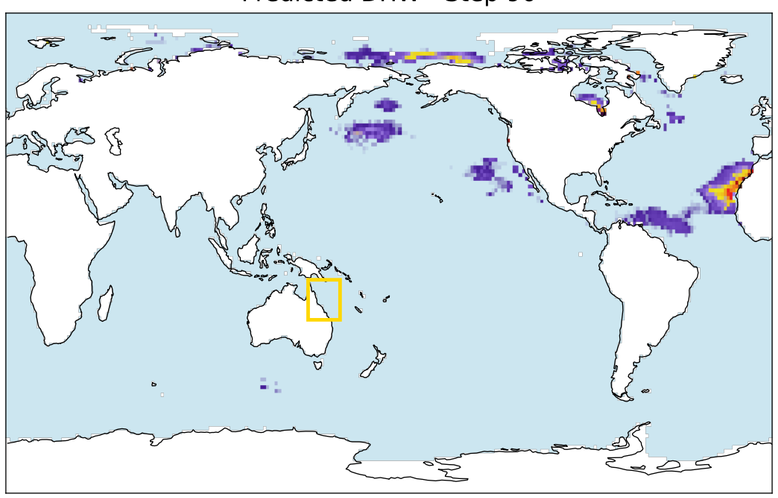
Error (Pred - Actual) - Step 46 [scale ±3.2]



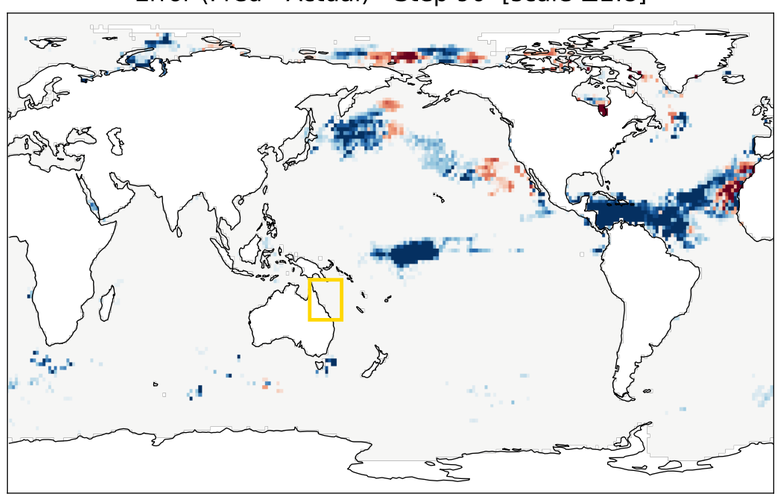
Actual DHW - Step 90

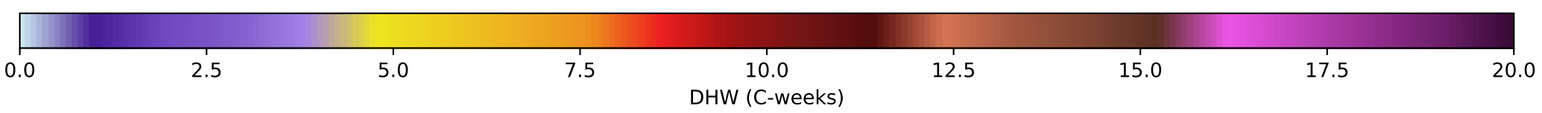


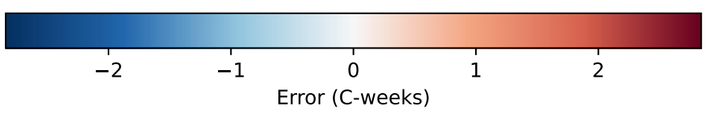
Predicted DHW - Step 90



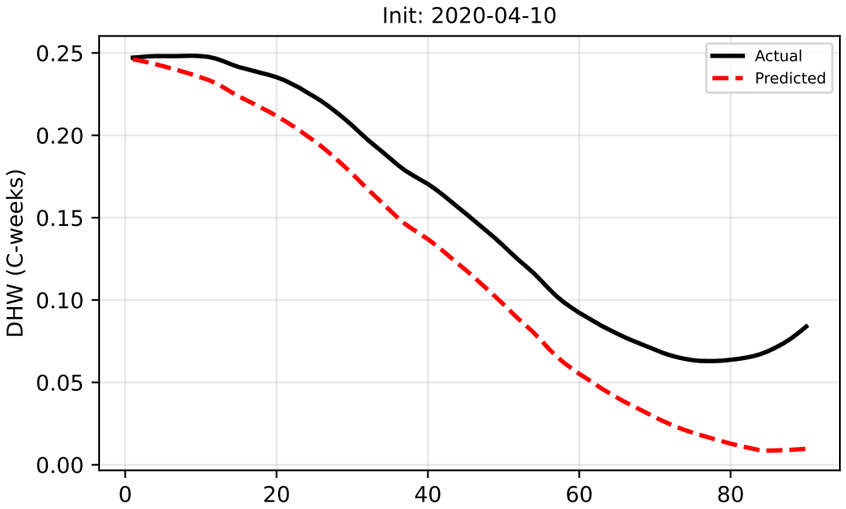
Error (Pred - Actual) - Step 90 [scale ±2.8]

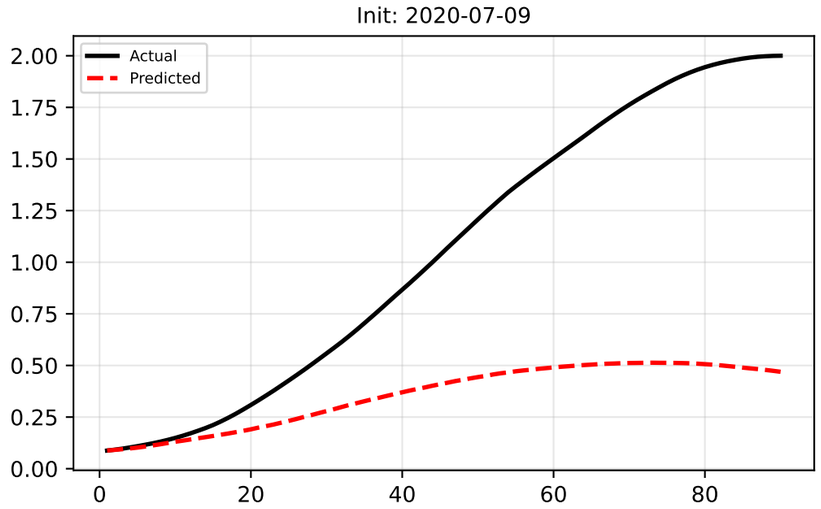


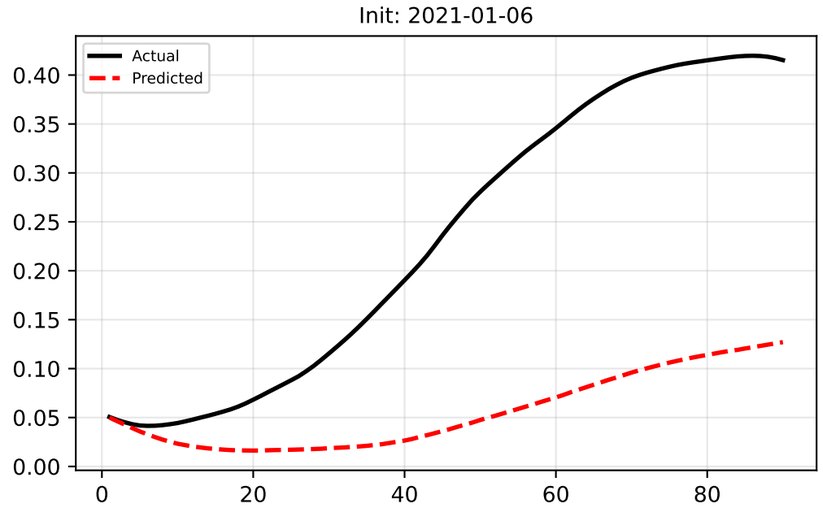


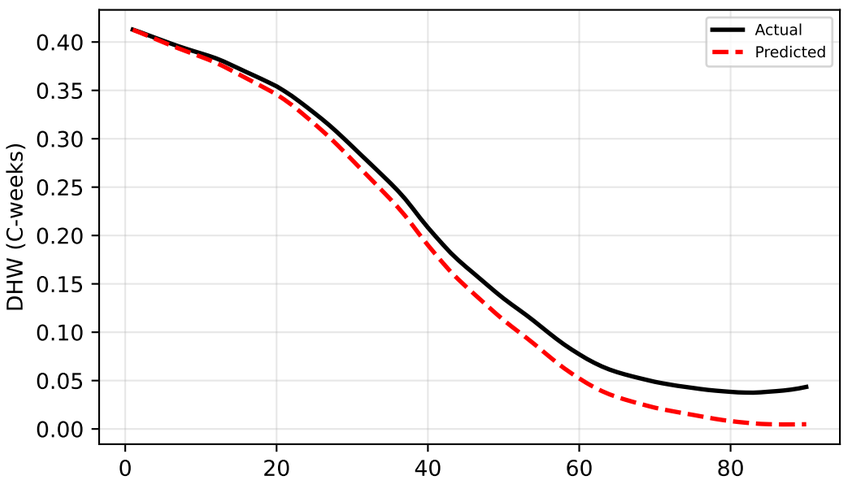


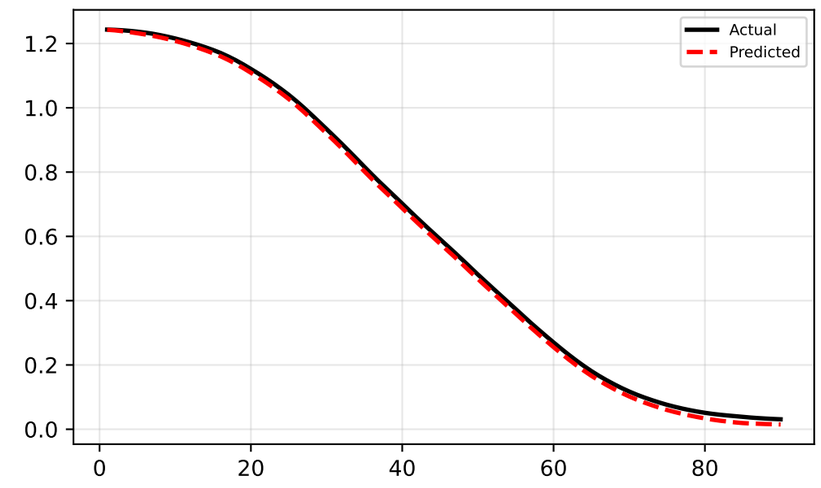
Global-Mean DHW - Test (C-weeks)

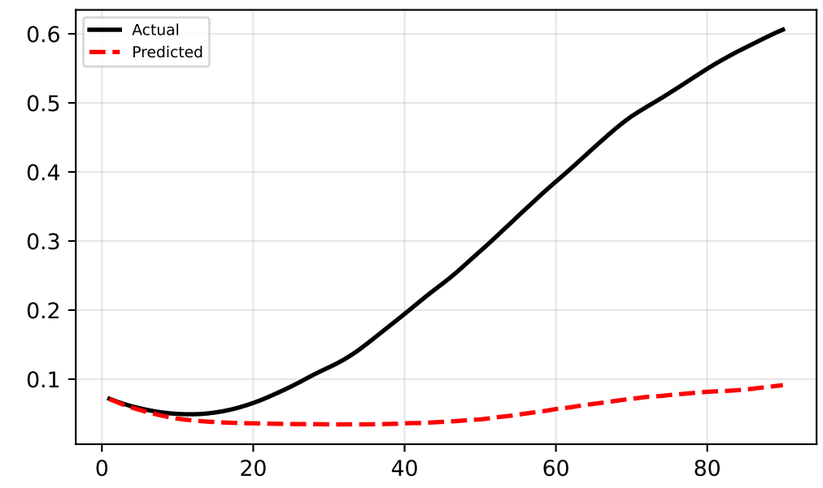


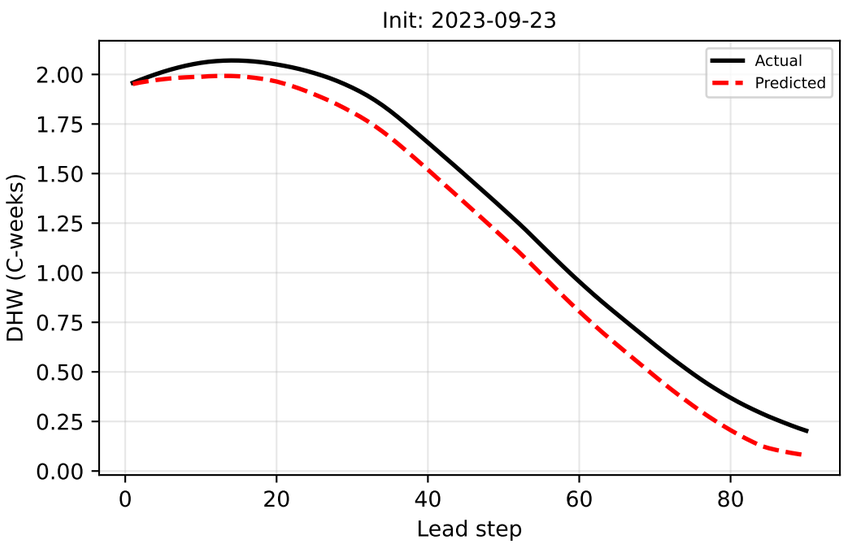




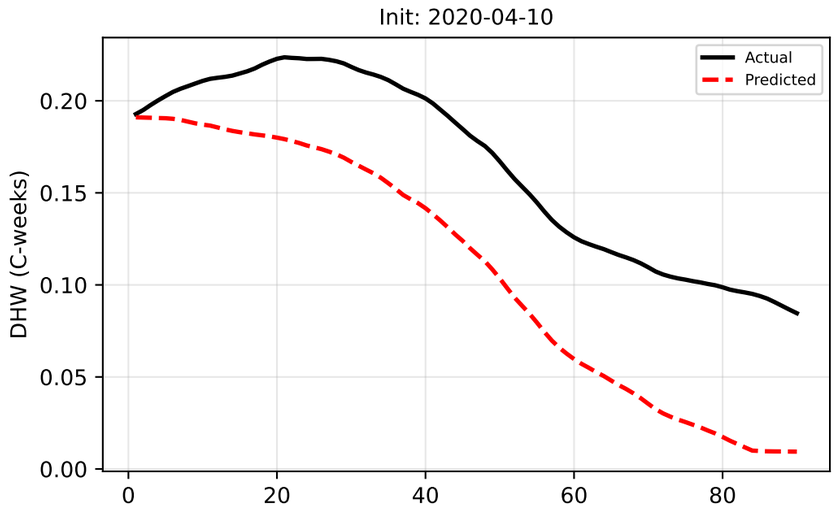


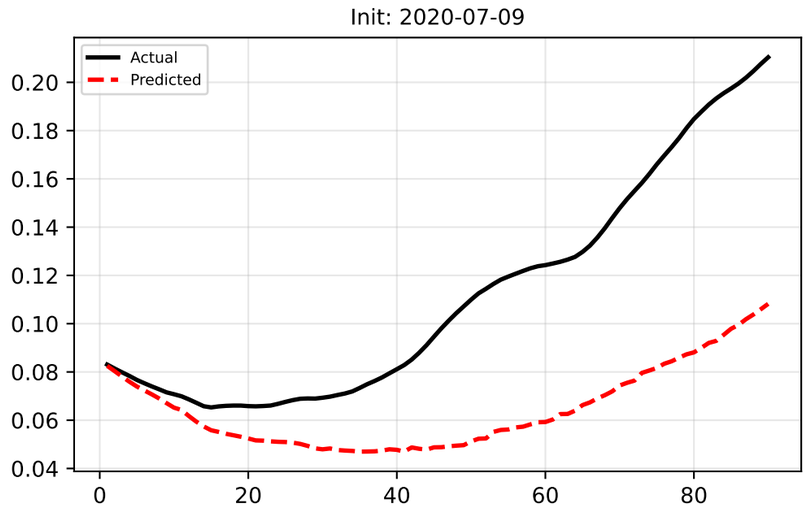


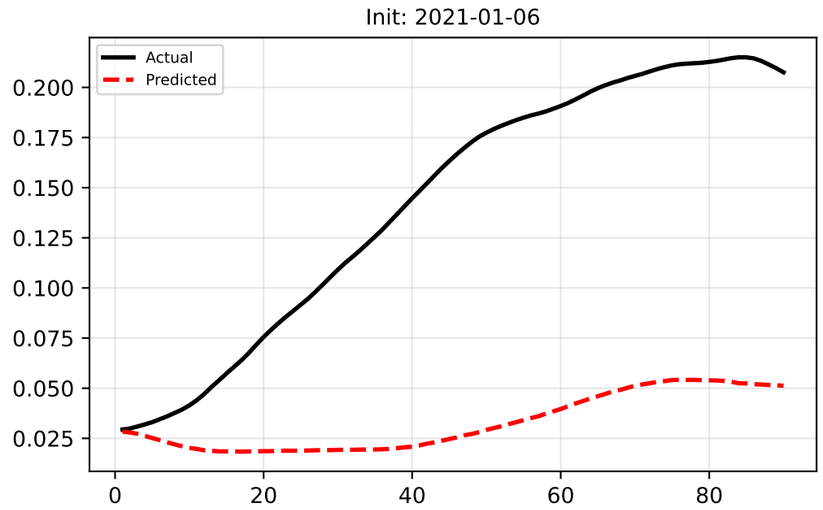


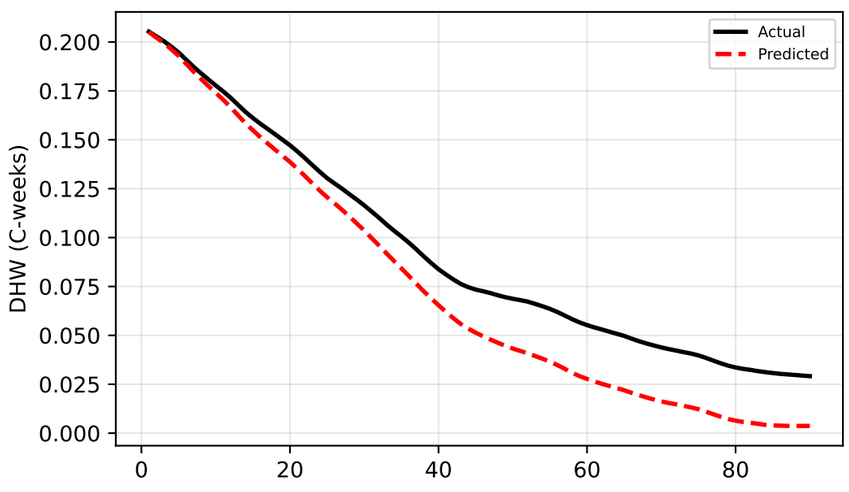


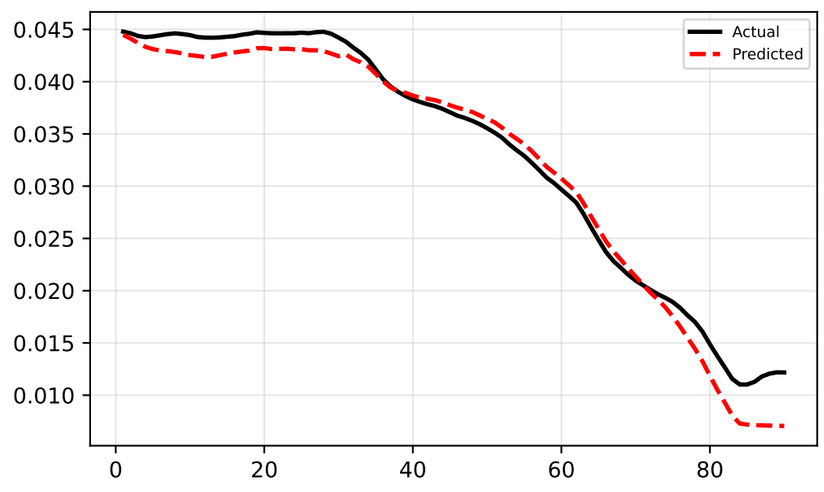
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

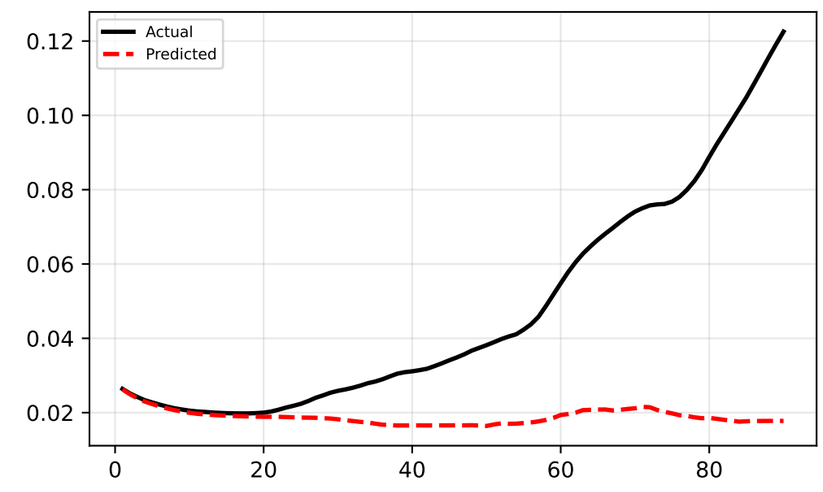


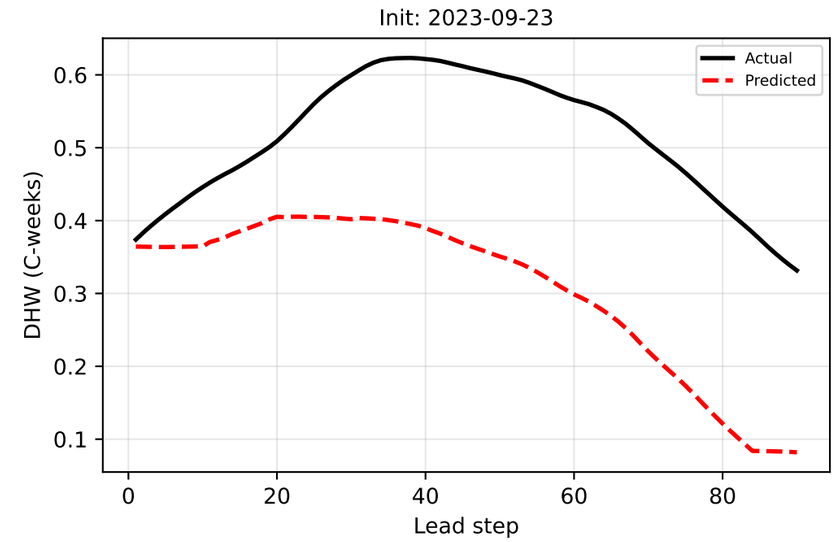


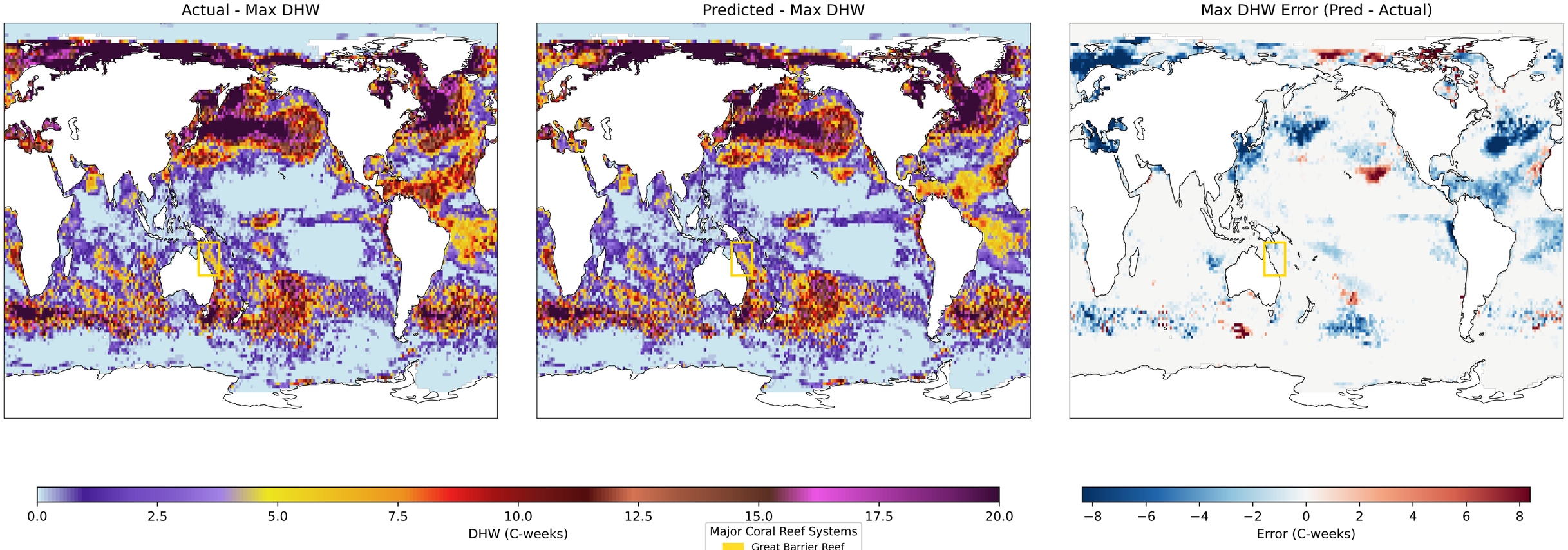


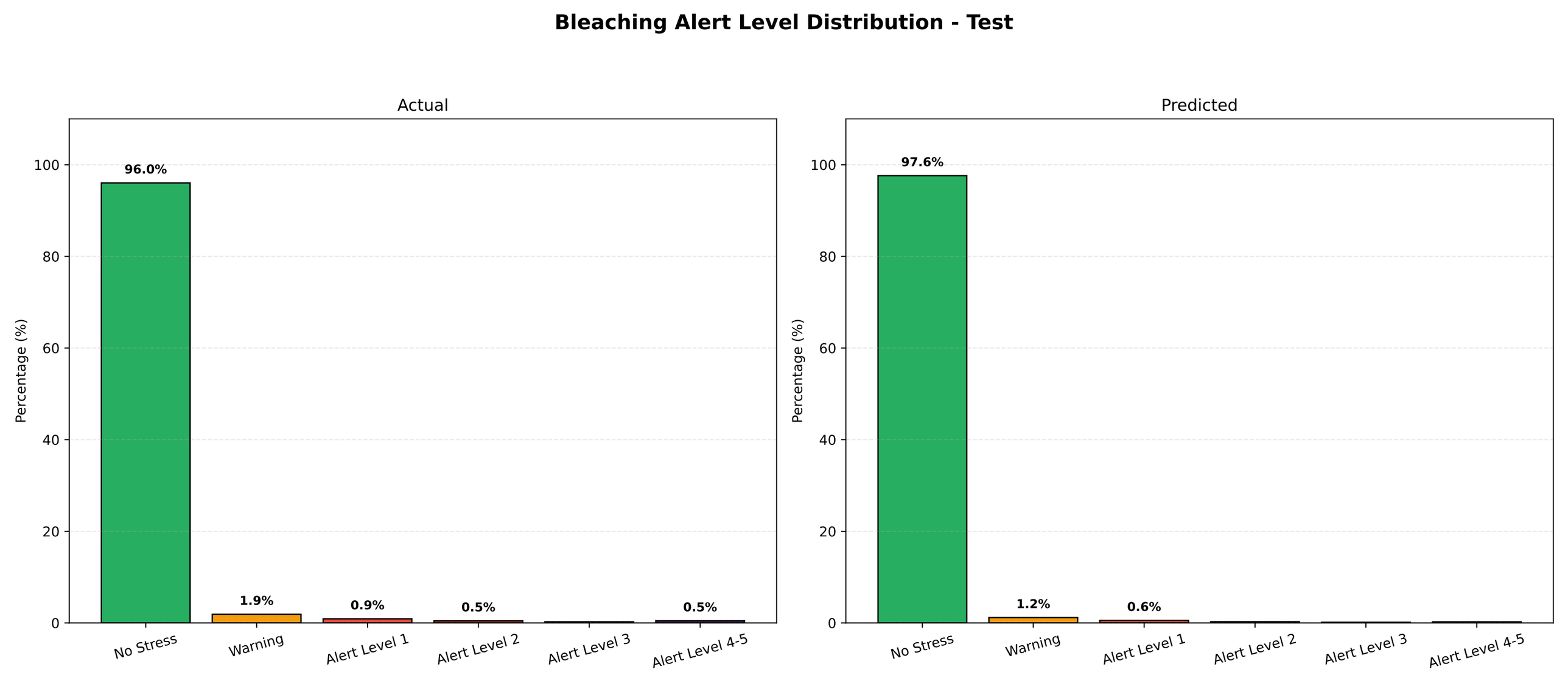






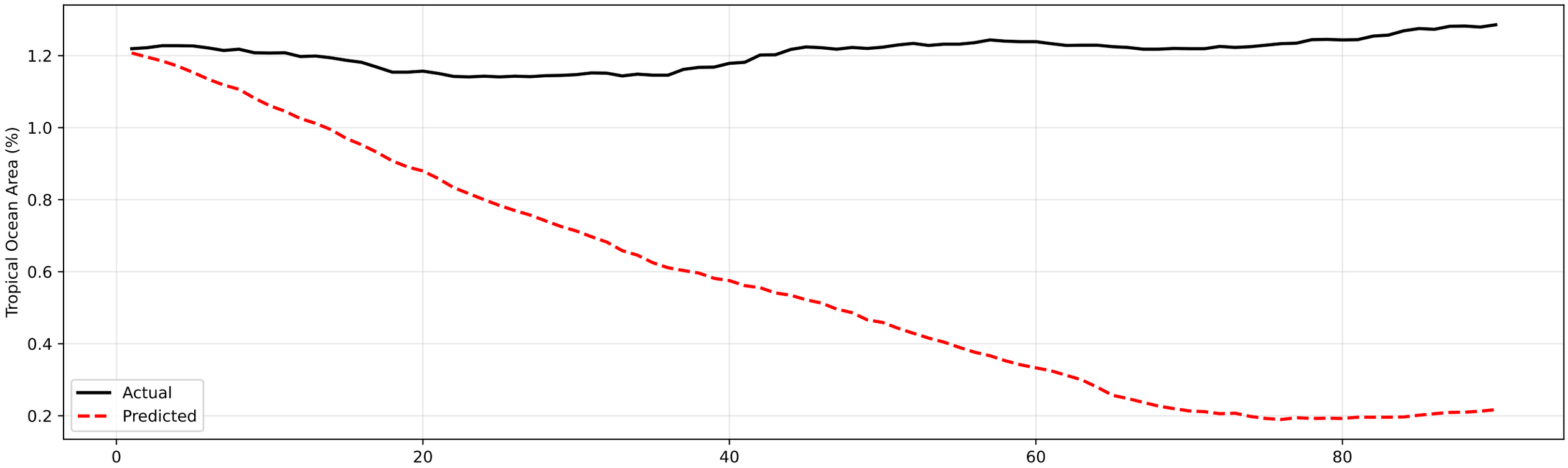



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)

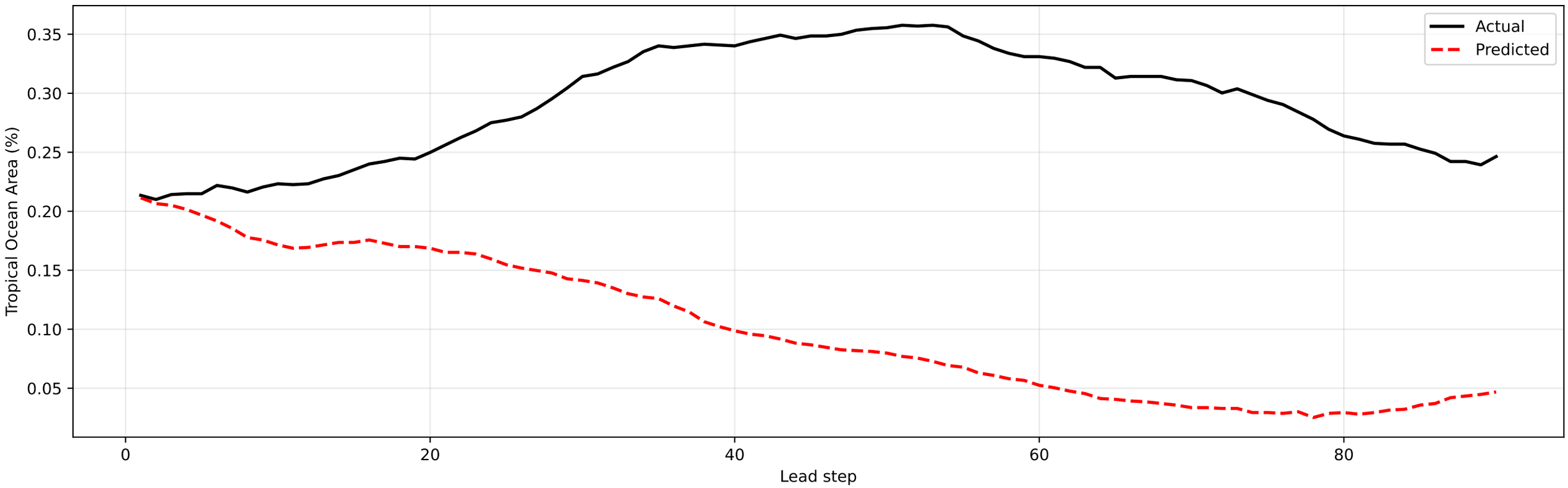

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.0%
1.9%
0.9%
0.5%
0.5%
97.6%
1.2%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



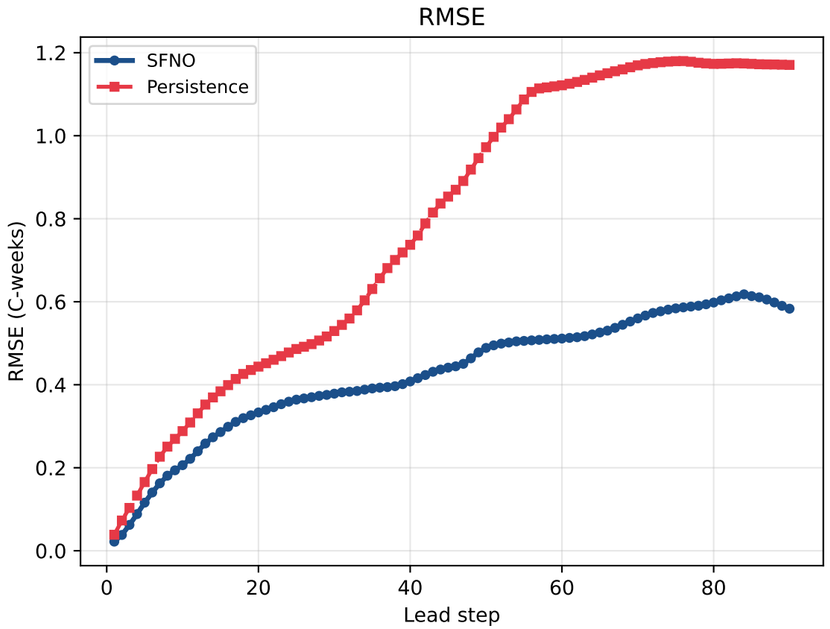
Great Barrier Reef Analysis - Test

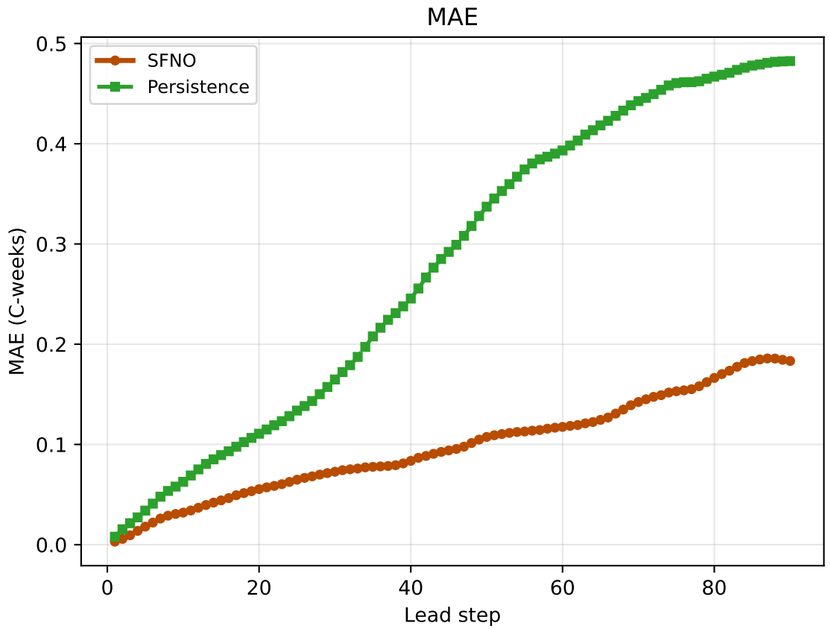
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

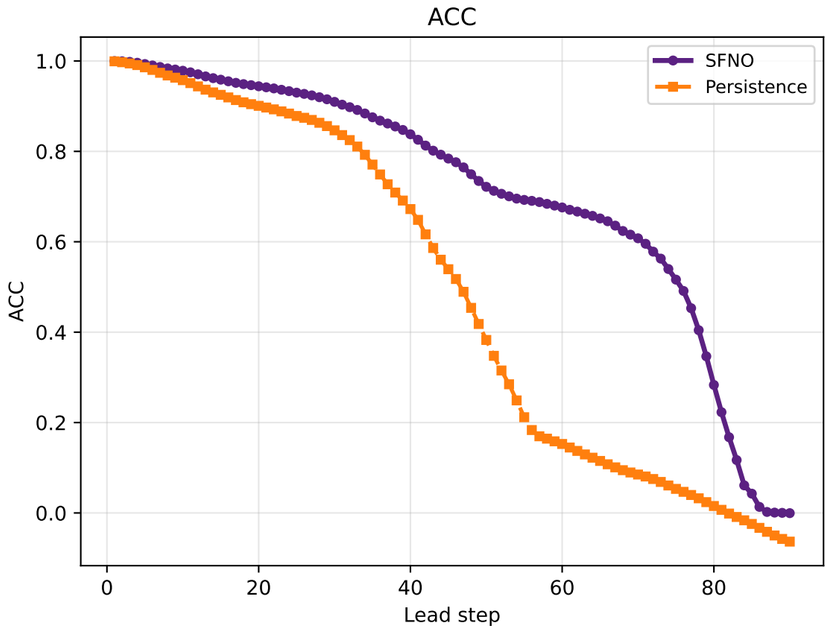
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

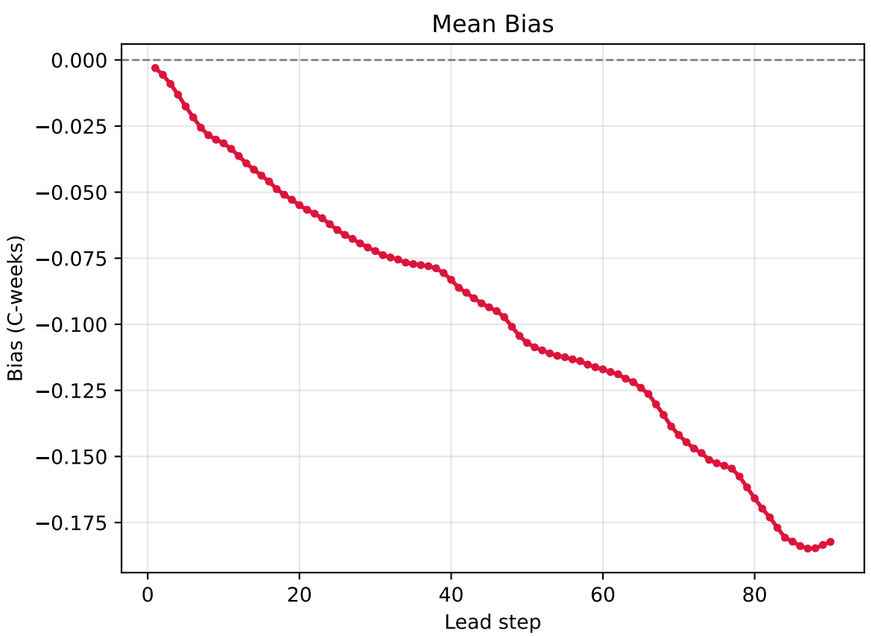
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0220	0.0386	43.0%	0.0030	0.0079	-0.0030	0.9998	0.9992	0.9995
2	0.0381	0.0729	47.7%	0.0058	0.0154	-0.0056	0.9993	0.9973	0.9986
3	0.0623	0.1032	39.6%	0.0094	0.0214	-0.0091	0.9980	0.9945	0.9962
4	0.0884	0.1329	33.5%	0.0138	0.0272	-0.0131	0.9960	0.9909	0.9923
5	0.1159	0.1655	30.0%	0.0180	0.0340	-0.0176	0.9932	0.9860	0.9869
6	0.1404	0.1968	28.7%	0.0221	0.0409	-0.0217	0.9900	0.9802	0.9808
7	0.1625	0.2266	28.3%	0.0261	0.0479	-0.0256	0.9867	0.9738	0.9744
8	0.1809	0.2509	27.9%	0.0289	0.0536	-0.0284	0.9835	0.9679	0.9683
9	0.1938	0.2696	28.1%	0.0306	0.0580	-0.0302	0.9810	0.9630	0.9636
10	0.2062	0.2883	28.5%	0.0320	0.0627	-0.0315	0.9784	0.9575	0.9586
11	0.2215	0.3090	28.3%	0.0341	0.0690	-0.0336	0.9749	0.9510	0.9519
12	0.2396	0.3310	27.6%	0.0368	0.0751	-0.0363	0.9706	0.9438	0.9437
13	0.2585	0.3523	26.6%	0.0396	0.0806	-0.0391	0.9658	0.9365	0.9349
14	0.2734	0.3694	26.0%	0.0420	0.0851	-0.0415	0.9620	0.9305	0.9276
15	0.2860	0.3842	25.6%	0.0442	0.0894	-0.0437	0.9586	0.9250	0.9212
16	0.2987	0.3991	25.2%	0.0465	0.0932	-0.0460	0.9550	0.9194	0.9146
17	0.3105	0.4140	25.0%	0.0493	0.0977	-0.0488	0.9516	0.9135	0.9082
18	0.3195	0.4260	25.0%	0.0515	0.1023	-0.0510	0.9488	0.9086	0.9030
19	0.3264	0.4355	25.0%	0.0534	0.1065	-0.0529	0.9465	0.9044	0.8987
20	0.3333	0.4437	24.9%	0.0554	0.1107	-0.0549	0.9440	0.9007	0.8942
21	0.3395	0.4518	24.9%	0.0572	0.1149	-0.0567	0.9418	0.8970	0.8900
22	0.3459	0.4602	24.8%	0.0586	0.1191	-0.0581	0.9393	0.8929	0.8856
23	0.3531	0.4690	24.7%	0.0603	0.1232	-0.0599	0.9363	0.8885	0.8801
24	0.3592	0.4777	24.8%	0.0626	0.1281	-0.0621	0.9333	0.8836	0.8748
25	0.3641	0.4861	25.1%	0.0648	0.1338	-0.0643	0.9299	0.8783	0.8689
26	0.3670	0.4916	25.3%	0.0667	0.1383	-0.0662	0.9272	0.8743	0.8641
27	0.3700	0.4979	25.7%	0.0682	0.1433	-0.0677	0.9239	0.8695	0.8584
28	0.3732	0.5066	26.3%	0.0699	0.1501	-0.0694	0.9198	0.8632	0.8512
29	0.3755	0.5163	27.3%	0.0714	0.1573	-0.0709	0.9152	0.8559	0.8433
30	0.3785	0.5294	28.5%	0.0728	0.1648	-0.0723	0.9095	0.8465	0.8334
31	0.3817	0.5442	29.9%	0.0743	0.1722	-0.0738	0.9033	0.8360	0.8229
32	0.3831	0.5596	31.5%	0.0752	0.1791	-0.0747	0.8978	0.8249	0.8134
33	0.3850	0.5792	33.5%	0.0760	0.1873	-0.0755	0.8916	0.8106	0.8029
34	0.3882	0.6035	35.7%	0.0771	0.1973	-0.0767	0.8838	0.7924	0.7899
35	0.3911	0.6307	38.0%	0.0777	0.2078	-0.0772	0.8754	0.7708	0.7760
36	0.3930	0.6566	40.1%	0.0781	0.2164	-0.0776	0.8680	0.7489	0.7639
37	0.3943	0.6811	42.1%	0.0785	0.2244	-0.0780	0.8616	0.7272	0.7536
38	0.3965	0.7006	43.4%	0.0793	0.2310	-0.0788	0.8551	0.7089	0.7432
39	0.4015	0.7187	44.1%	0.0811	0.2378	-0.0806	0.8474	0.6912	0.7309
40	0.4079	0.7372	44.7%	0.0836	0.2456	-0.0832	0.8378	0.6723	0.7157
41	0.4158	0.7596	45.3%	0.0867	0.2556	-0.0862	0.8257	0.6484	0.6966
42	0.4234	0.7886	46.3%	0.0885	0.2666	-0.0880	0.8128	0.6164	0.6760
43	0.4313	0.8150	47.1%	0.0907	0.2764	-0.0902	0.8015	0.5862	0.6577
44	0.4367	0.8367	47.8%	0.0925	0.2850	-0.0921	0.7927	0.5603	0.6431
45	0.4412	0.8533	48.3%	0.0941	0.2921	-0.0936	0.7841	0.5393	0.6290
46	0.4444	0.8698	48.9%	0.0955	0.2992	-0.0950	0.7759	0.5176	0.6156
47	0.4504	0.8909	49.4%	0.0978	0.3081	-0.0973	0.7645	0.4893	0.5974
48	0.4636	0.9184	49.5%	0.1014	0.3180	-0.1009	0.7494	0.4538	0.5735
49	0.4780	0.9460	49.5%	0.1049	0.3279	-0.1044	0.7344	0.4181	0.5500
50	0.4887	0.9723	49.7%	0.1075	0.3372	-0.1070	0.7216	0.3828	0.5298
51	0.4950	0.9973	50.4%	0.1092	0.3453	-0.1087	0.7127	0.3476	0.5154
52	0.4991	1.0196	51.1%	0.1103	0.3529	-0.1099	0.7061	0.3152	0.5042
53	0.5017	1.0399	51.7%	0.1115	0.3597	-0.1110	0.7005	0.2847	0.4946
54	0.5045	1.0634	52.6%	0.1124	0.3671	-0.1119	0.6955	0.2490	0.4856
55	0.5056	1.0873	53.5%	0.1129	0.3743	-0.1124	0.6926	0.2118	0.4796
56	0.5069	1.1055	54.1%	0.1137	0.3803	-0.1133	0.6905	0.1835	0.4754
57	0.5080	1.1139	54.4%	0.1144	0.3844	-0.1139	0.6879	0.1696	0.4708
58	0.5094	1.1164	54.4%	0.1157	0.3870	-0.1153	0.6839	0.1644	0.4655
59	0.5103	1.1190	54.4%	0.1167	0.3902	-0.1162	0.6800	0.1585	0.4596
60	0.5113	1.1215	54.4%	0.1175	0.3933	-0.1170	0.6759	0.1526	0.4532
61	0.5129	1.1253	54.4%	0.1185	0.3982	-0.1180	0.6707	0.1448	0.4447
62	0.5144	1.1295	54.5%	0.1194	0.4031	-0.1189	0.6667	0.1373	0.4379
63	0.5171	1.1345	54.4%	0.1210	0.4091	-0.1206	0.6620	0.1295	0.4314
64	0.5212	1.1400	54.3%	0.1224	0.4135	-0.1219	0.6573	0.1223	0.4250
65	0.5259	1.1455	54.1%	0.1245	0.4183	-0.1240	0.6516	0.1151	0.4179
66	0.5304	1.1506	53.9%	0.1269	0.4229	-0.1264	0.6454	0.1077	0.4098
67	0.5370	1.1553	53.5%	0.1308	0.4278	-0.1303	0.6357	0.1009	0.3988
68	0.5444	1.1599	53.1%	0.1348	0.4330	-0.1343	0.6238	0.0946	0.3866
69	0.5523	1.1649	52.6%	0.1391	0.4384	-0.1386	0.6157	0.0895	0.3783
70	0.5598	1.1696	52.1%	0.1424	0.4425	-0.1419	0.6077	0.0850	0.3698
71	0.5667	1.1728	51.7%	0.1451	0.4457	-0.1446	0.5956	0.0807	0.3582
72	0.5730	1.1751	51.2%	0.1475	0.4495	-0.1470	0.5781	0.0751	0.3410
73	0.5769	1.1771	51.0%	0.1492	0.4538	-0.1487	0.5627	0.0686	0.3249
74	0.5810	1.1786	50.7%	0.1518	0.4582	-0.1513	0.5397	0.0608	0.3030
75	0.5840	1.1796	50.5%	0.1531	0.4604	-0.1526	0.5162	0.0533	0.2815
76	0.5864	1.1798	50.3%	0.1540	0.4614	-0.1535	0.4911	0.0467	0.2603
77	0.5884	1.1783	50.1%	0.1551	0.4614	-0.1546	0.4530	0.0398	0.2299
78	0.5904	1.1759	49.8%	0.1581	0.4623	-0.1576	0.4044	0.0324	0.1935
79	0.5938	1.1743	49.4%	0.1622	0.4649	-0.1617	0.3465	0.0239	0.1522
80	0.5981	1.1731	49.0%	0.1663	0.4670	-0.1658	0.2832	0.0154	0.1085
81	0.6035	1.1734	48.6%	0.1702	0.4688	-0.1697	0.2230	0.0069	0.0673
82	0.6081	1.1740	48.2%	0.1736	0.4708	-0.1731	0.1676	-0.0013	0.0285
83	0.6130	1.1746	47.8%	0.1774	0.4737	-0.1770	0.1170	-0.0089	-0.0076
84	0.6179	1.1741	47.4%	0.1812	0.4758	-0.1807	0.0610	-0.0162	-0.0494
85	0.6136	1.1732	47.7%	0.1831	0.4780	-0.1822	0.0427	-0.0243	-0.0655
86	0.6106	1.1722	47.9%	0.1848	0.4791	-0.1839	0.0134	-0.0330	-0.0909
87	0.6053	1.1718	48.3%	0.1858	0.4807	-0.1848	0.0023	-0.0417	-0.1030
88	0.5982	1.1717	48.9%	0.1857	0.4817	-0.1847	0.0008	-0.0498	-0.1065
89	0.5902	1.1712	49.6%	0.1844	0.4821	-0.1835	0.0002	-0.0575	-0.1085
90	0.5832	1.1706	50.2%	0.1833	0.4825	-0.1823	-0.0004	-0.0634	-0.1099

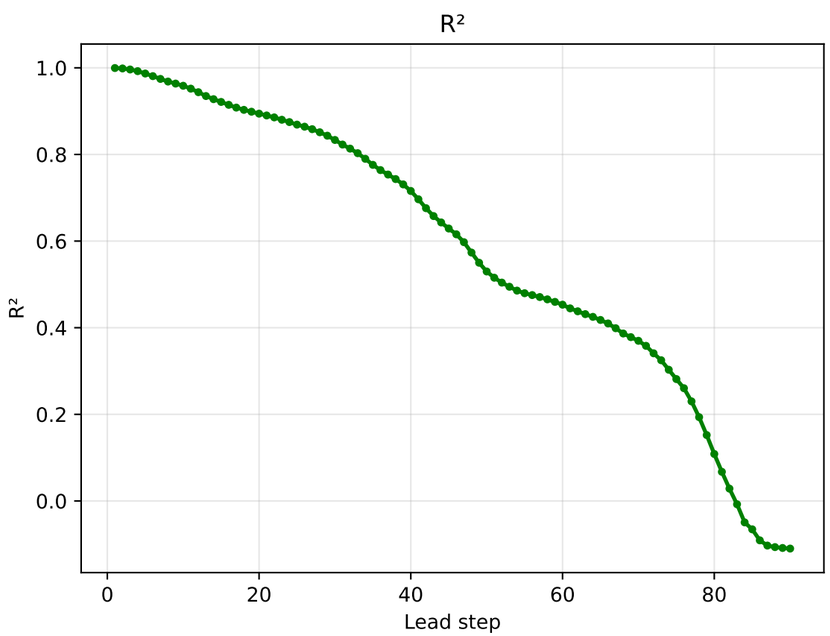
DHW Forecast Skill vs Lead Step - Test - GBR

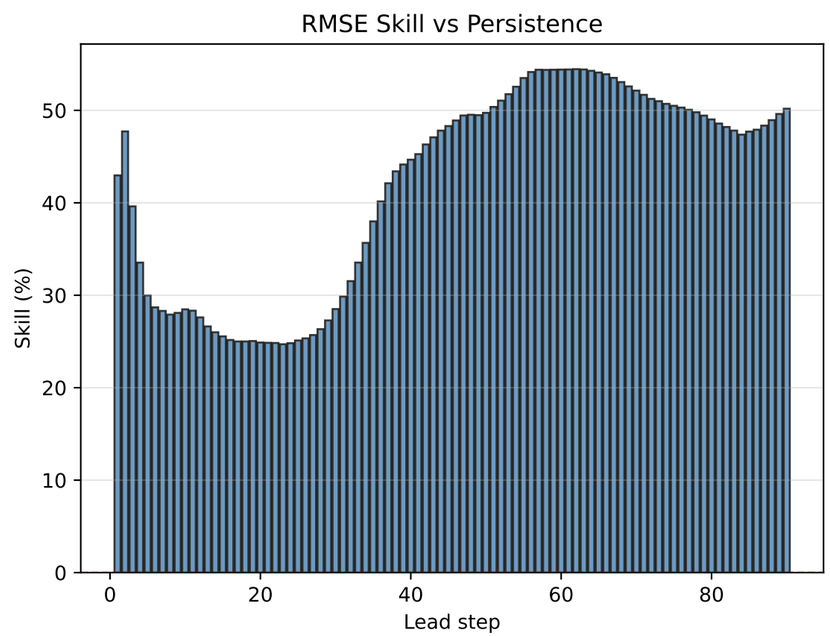


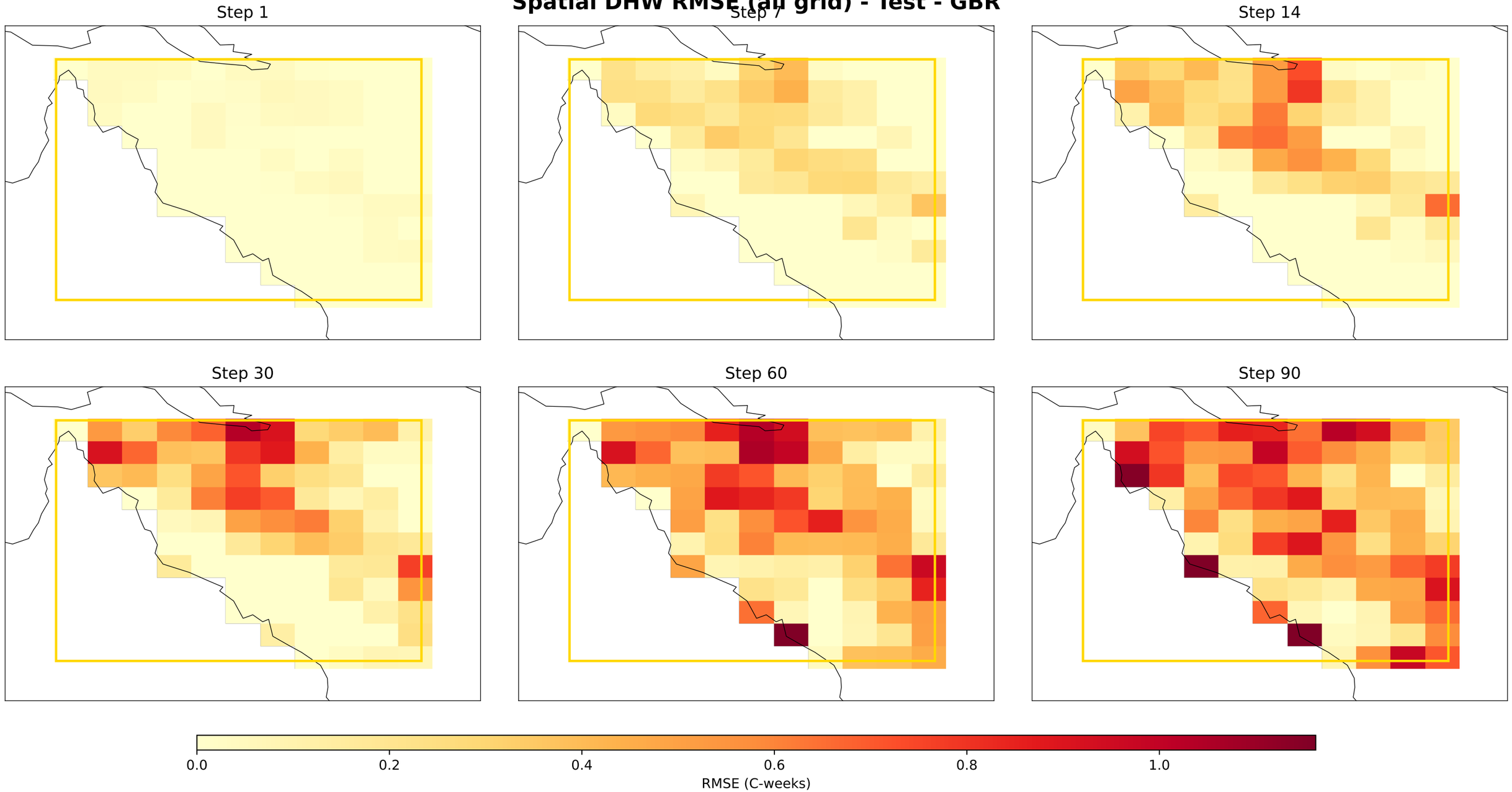


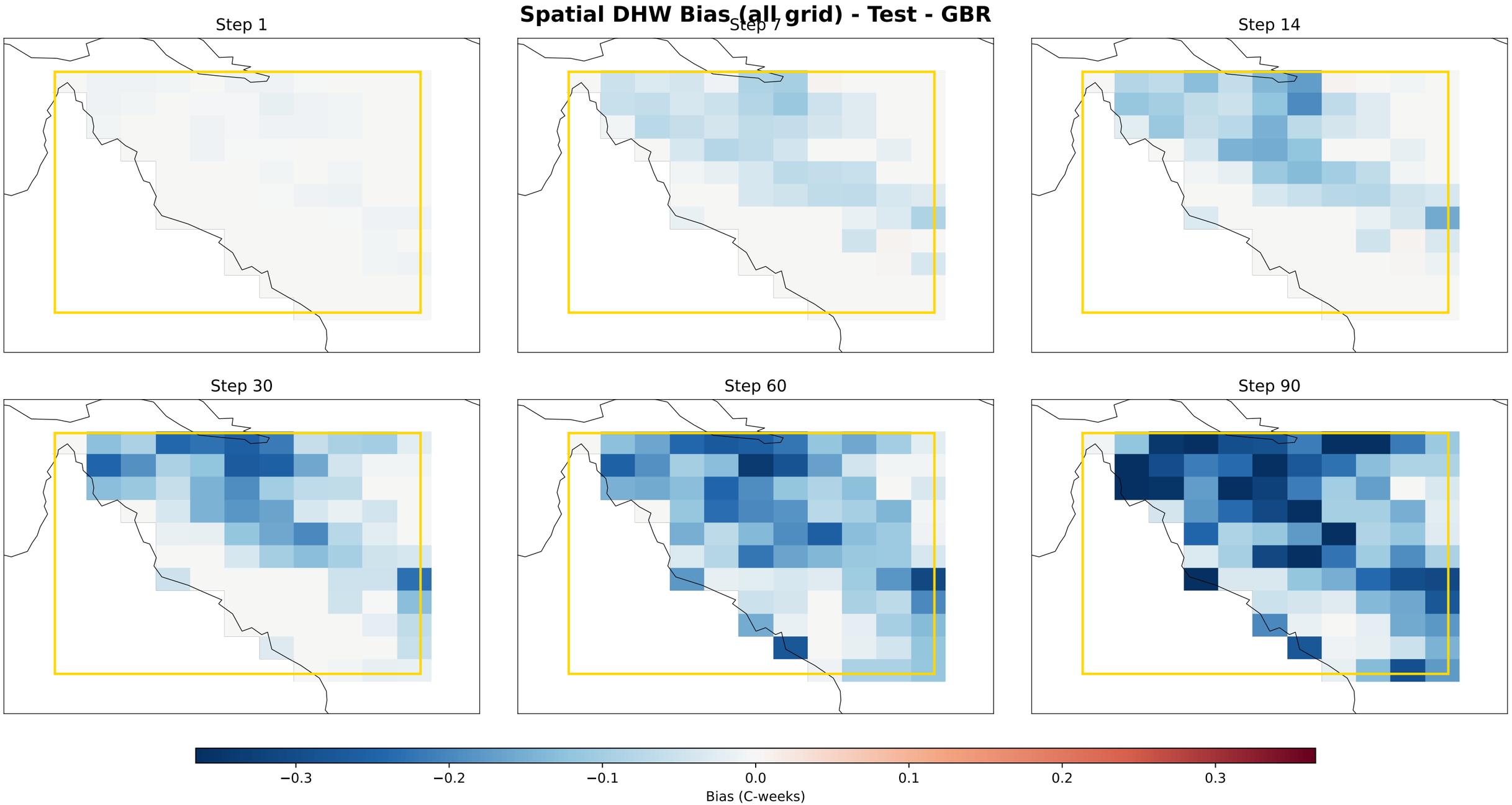


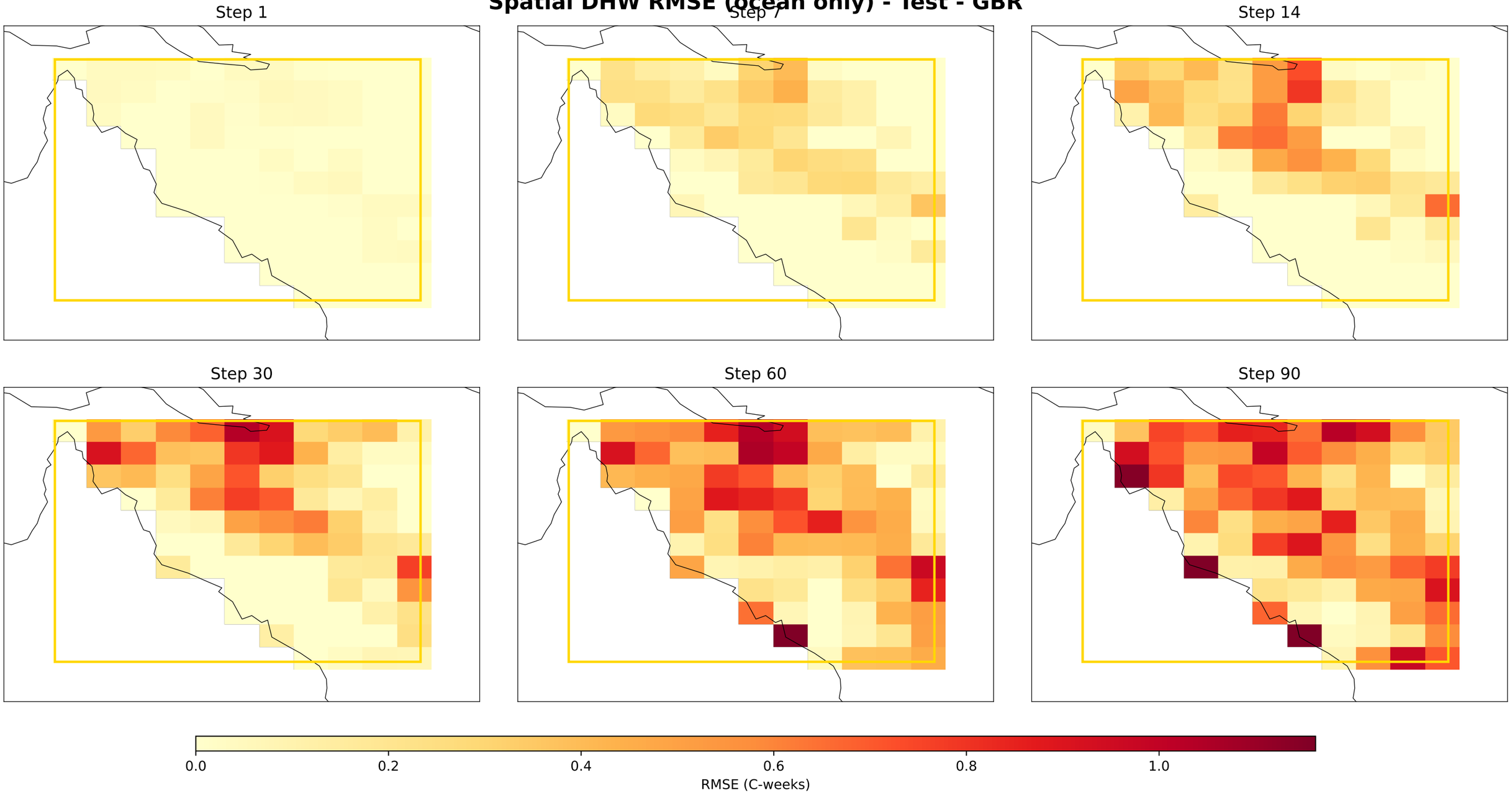


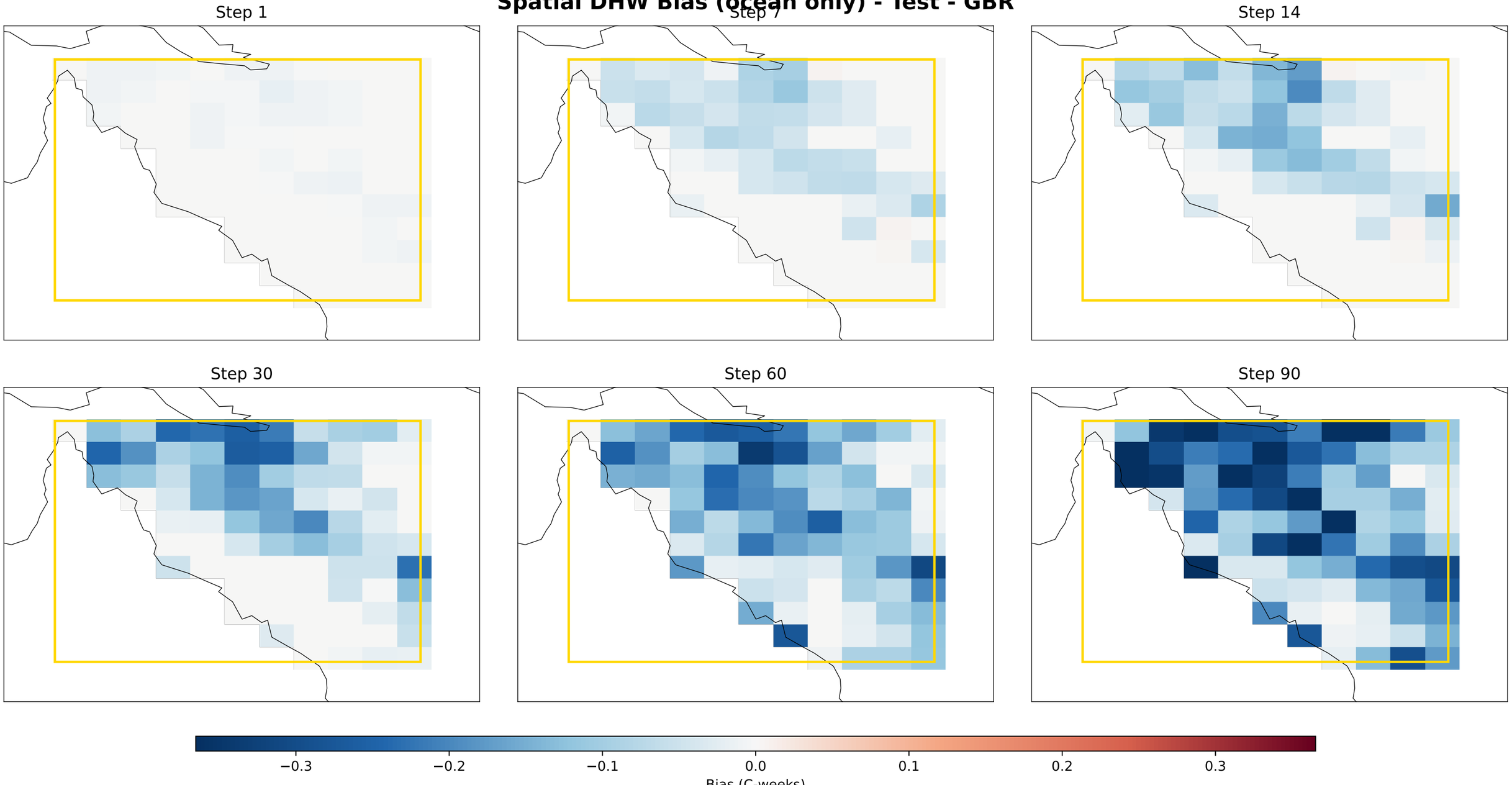





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

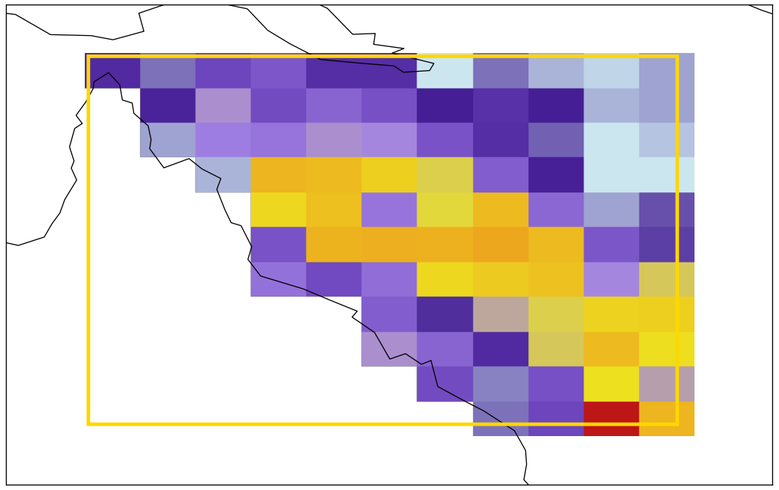

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

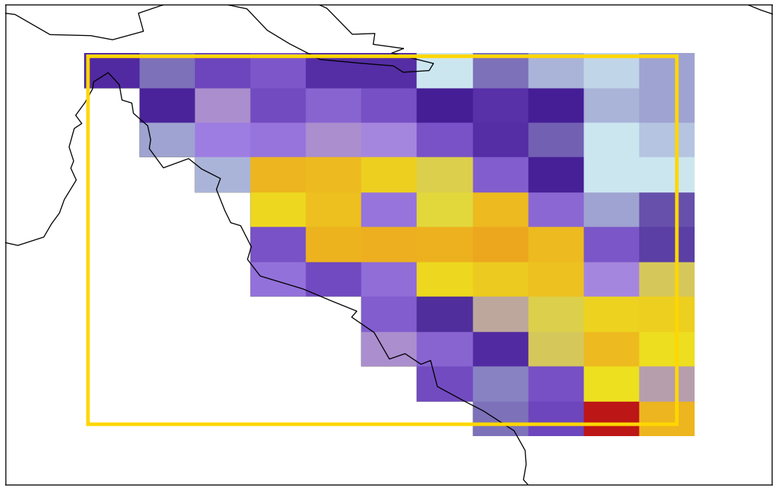

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

Actual DHW - Step 1



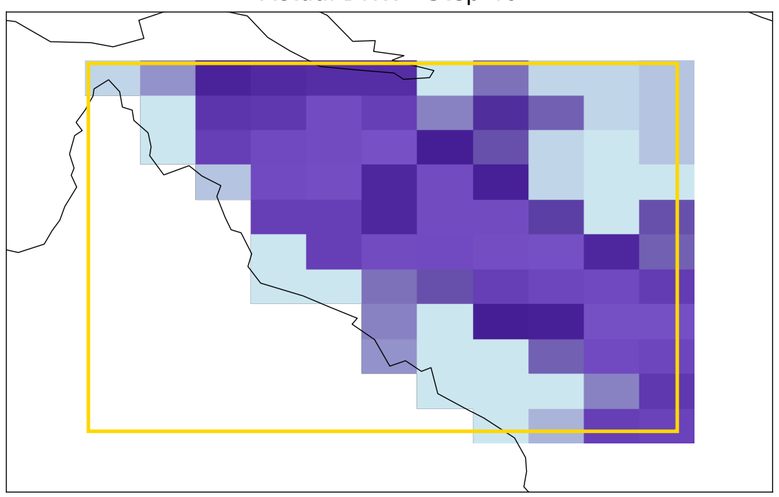
Predicted DHW - Step 1



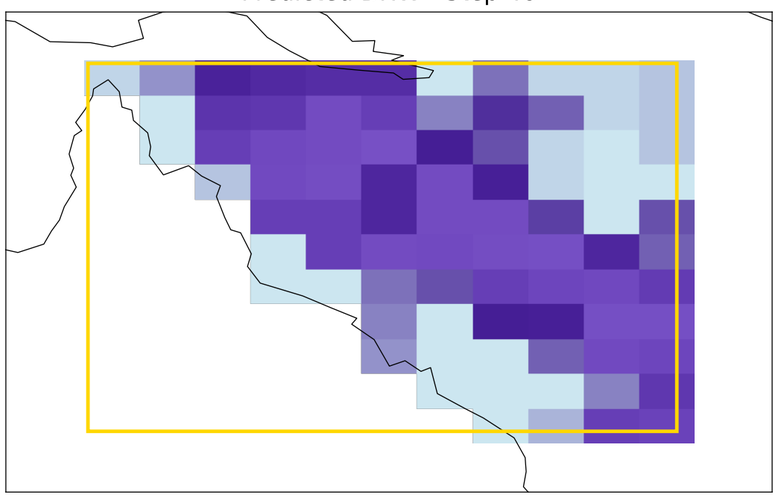
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



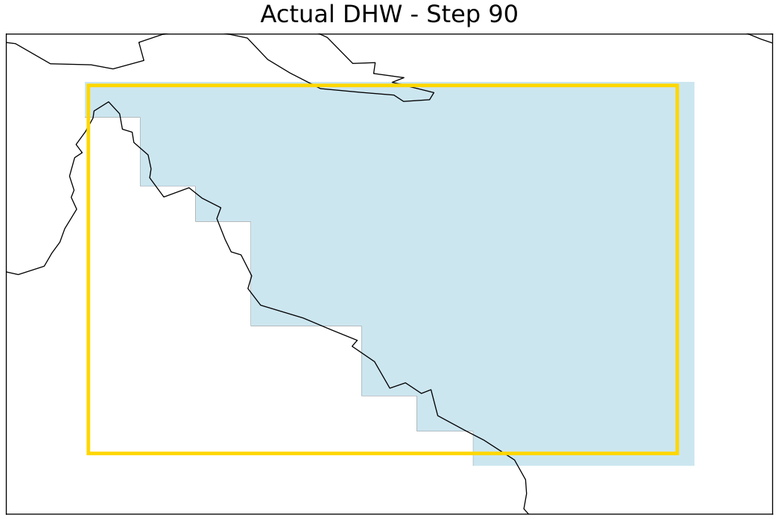
Predicted DHW - Step 46



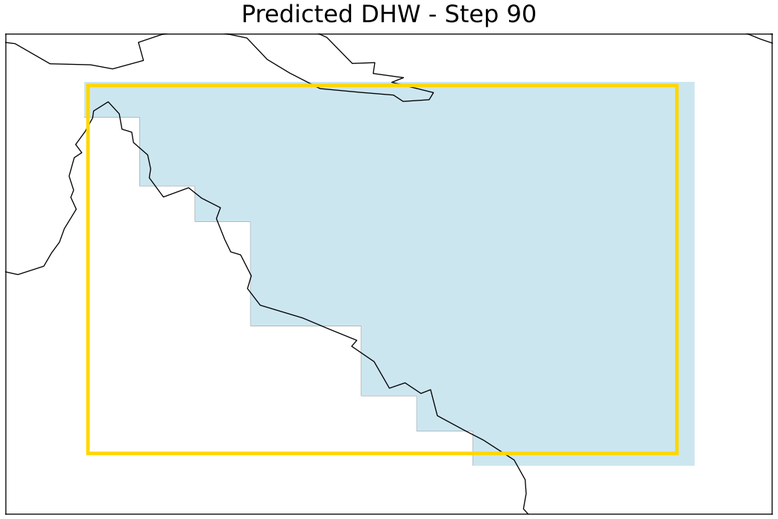
Error (Pred - Actual) - Step 46 [scale ±8]



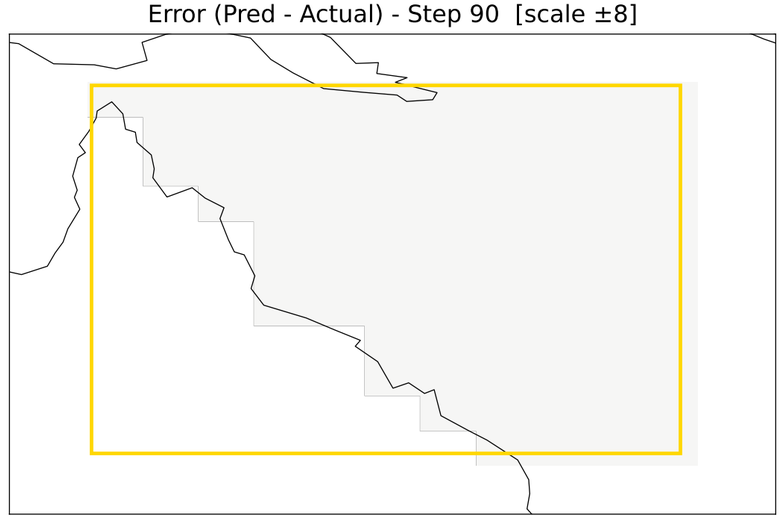
Actual DHW - Step 90

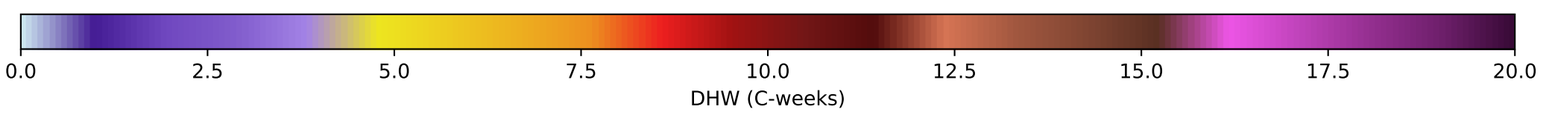


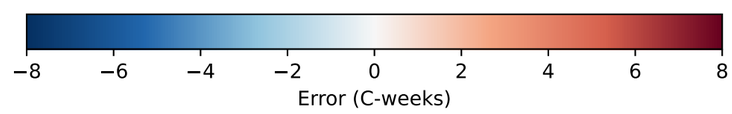
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

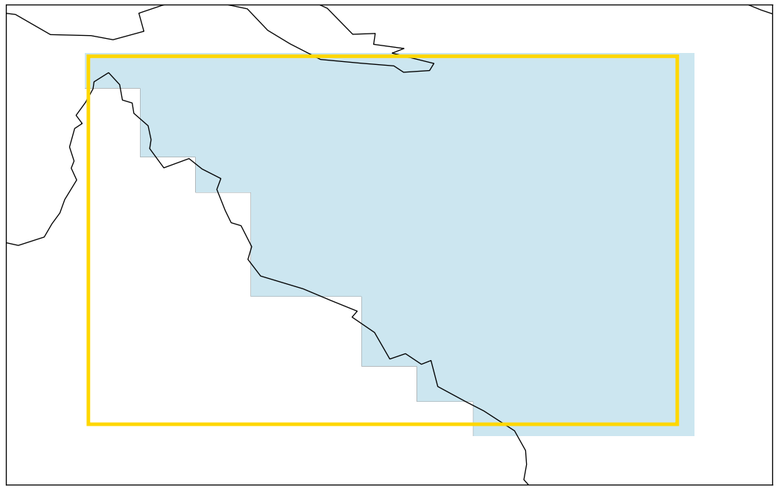




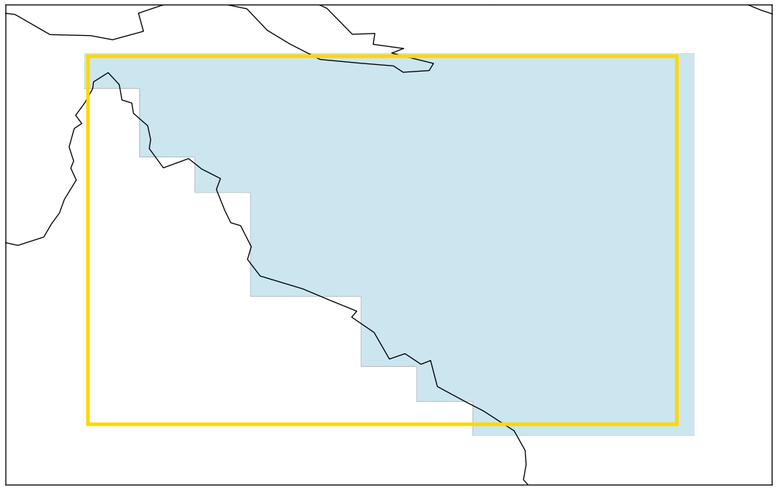


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

Actual DHW - Step 1



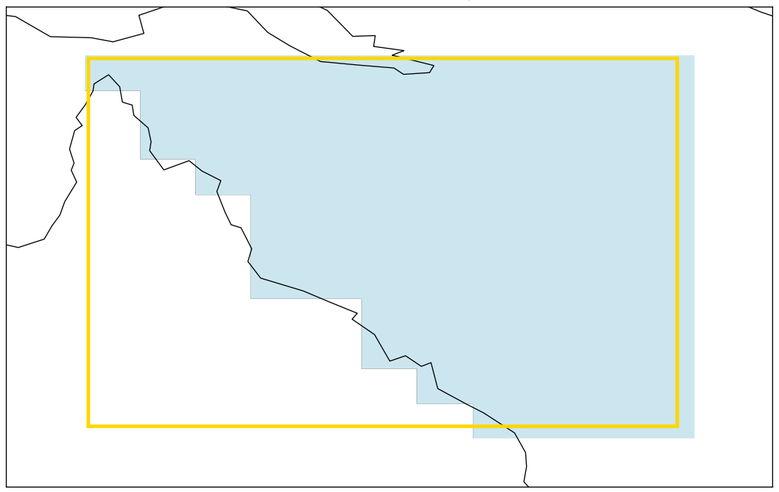
Predicted DHW - Step 1



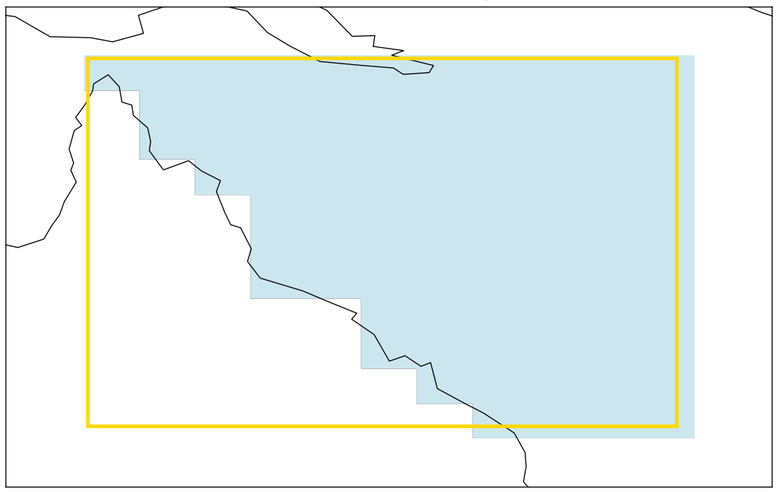
Error (Pred - Actual) - Step 1 [scale ±8]



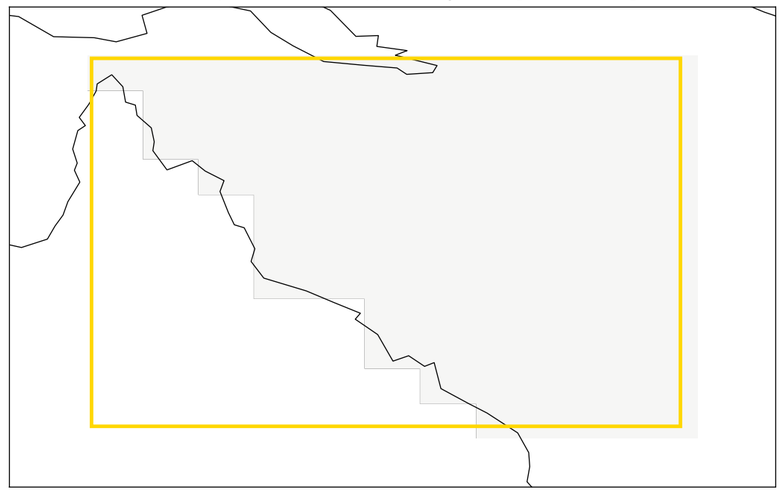
Actual DHW - Step 46



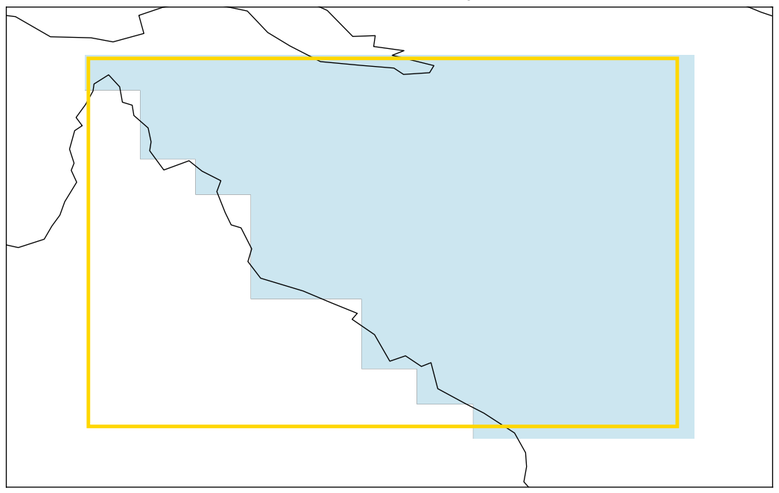
Predicted DHW - Step 46



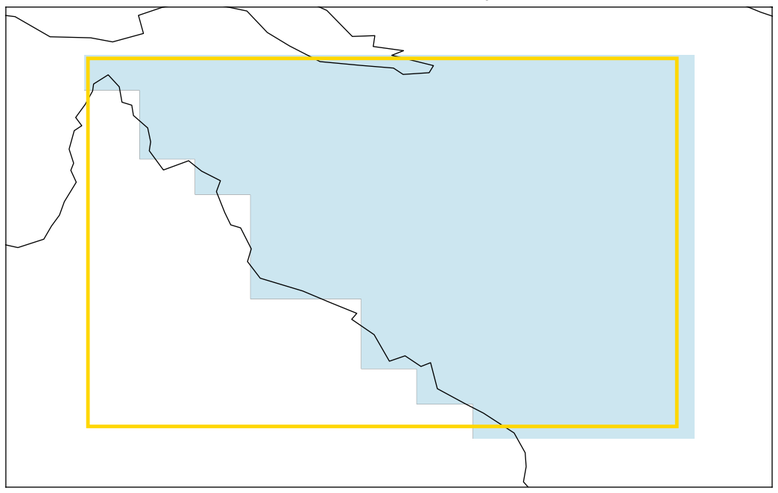
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

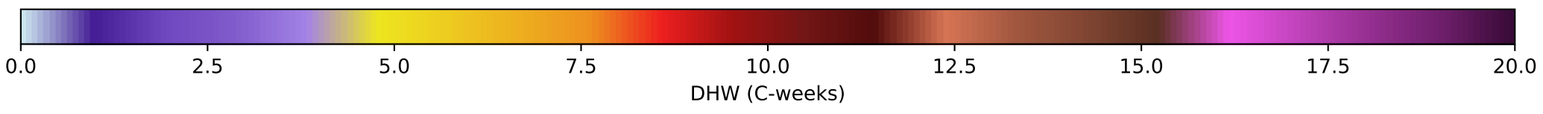


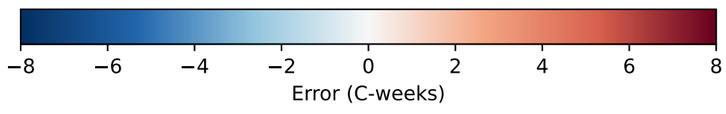
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

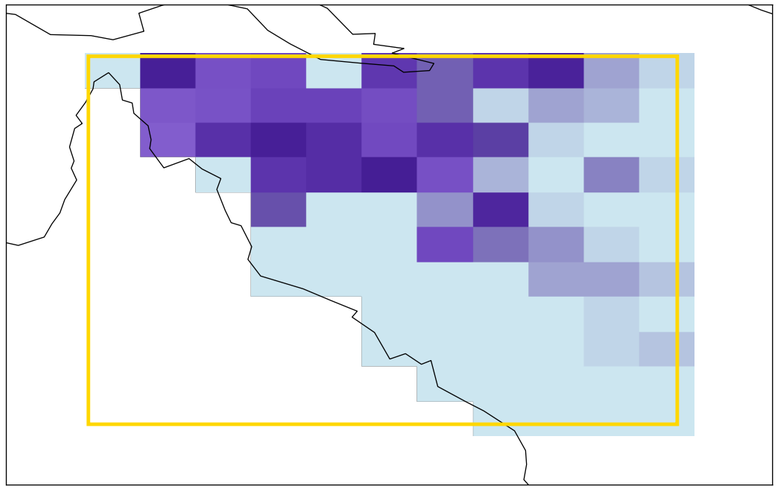




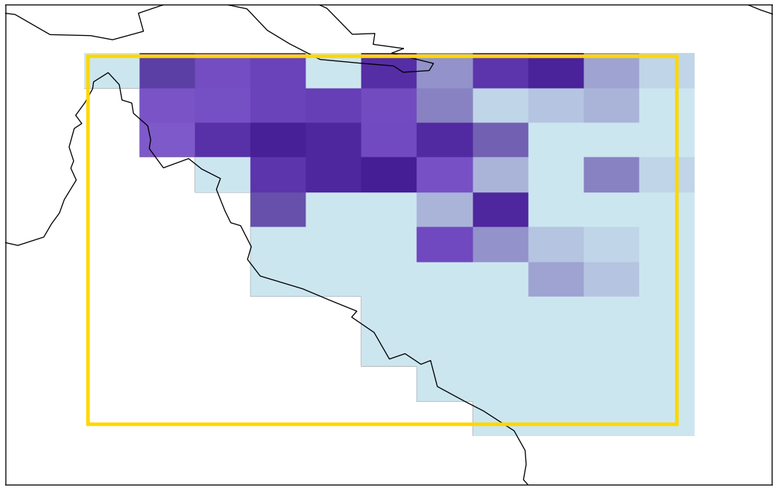


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

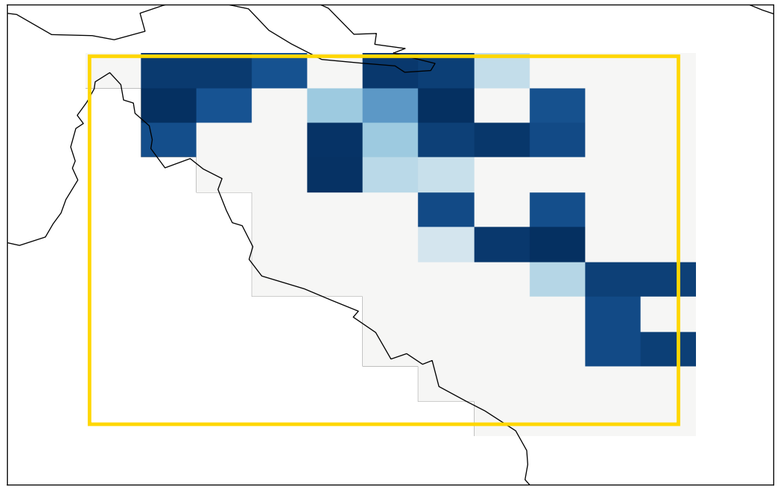
Actual DHW - Step 1



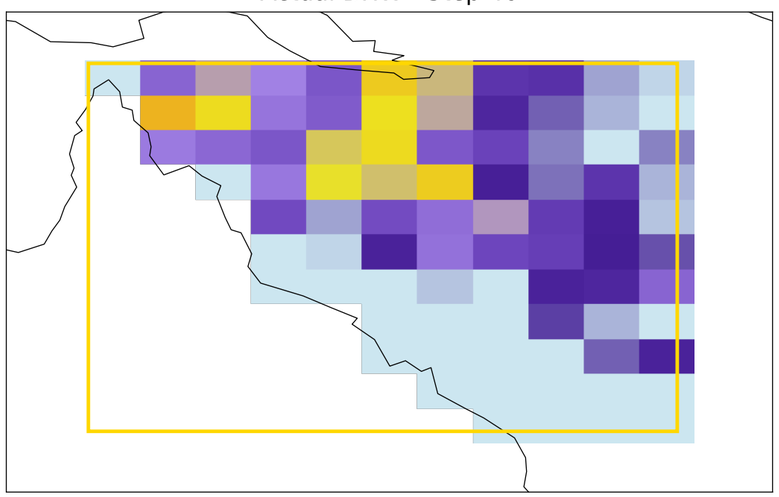
Predicted DHW - Step 1



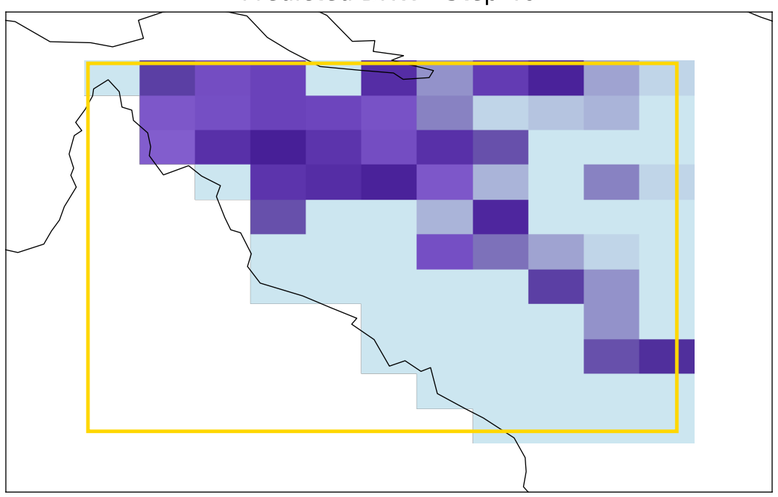
Error (Pred - Actual) - Step 1 [scale ±0.18]



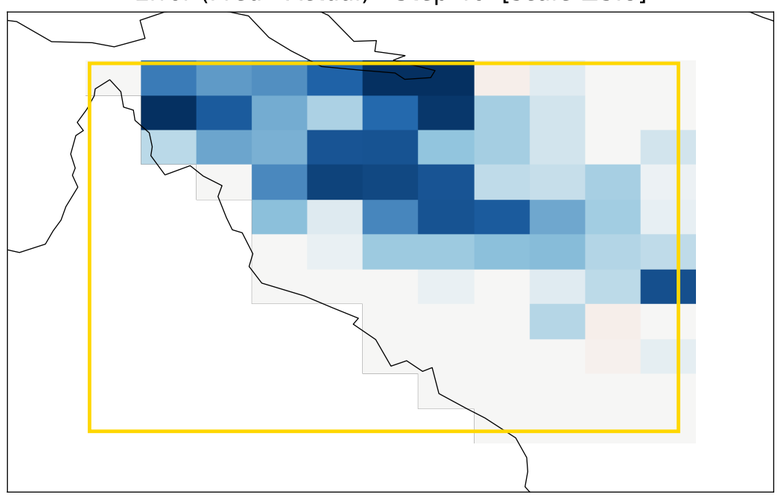
Actual DHW - Step 46



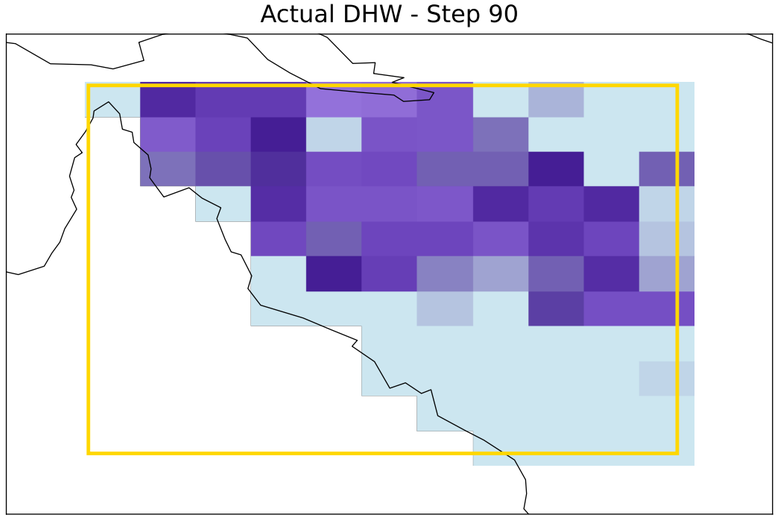
Predicted DHW - Step 46



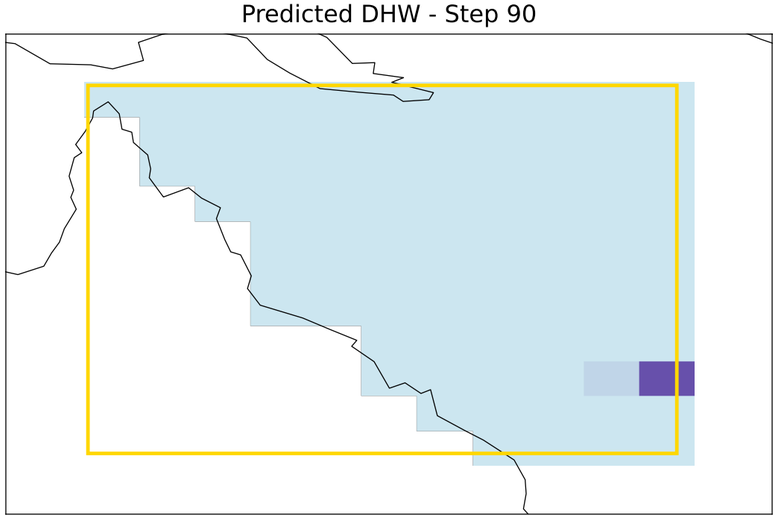
Error (Pred - Actual) - Step 46 [scale ±3.8]



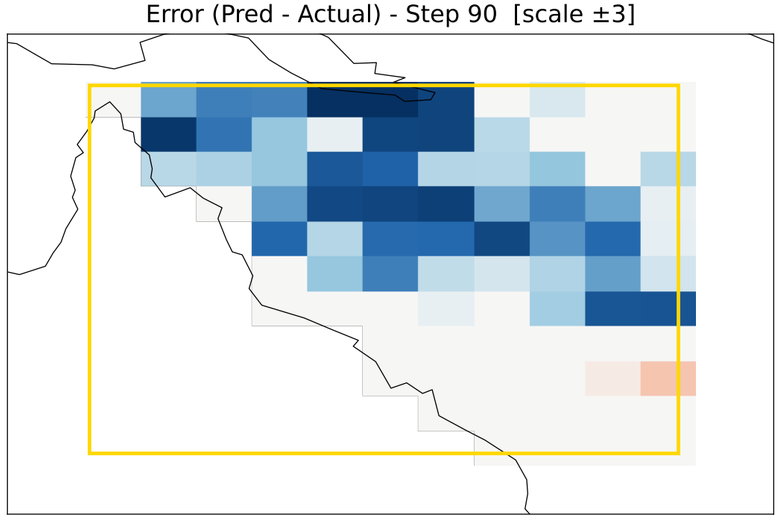
Actual DHW - Step 90

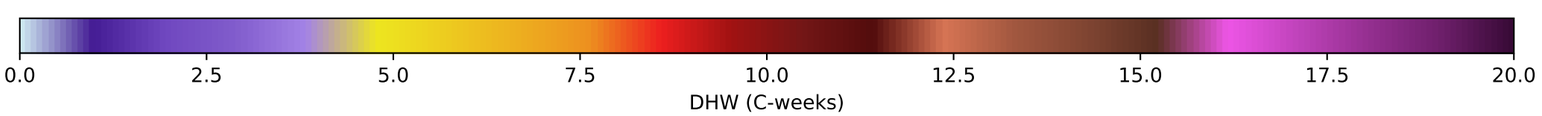


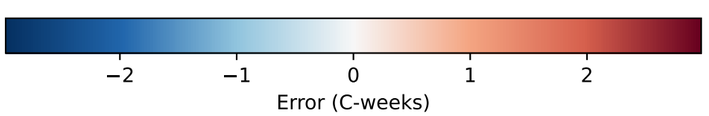
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

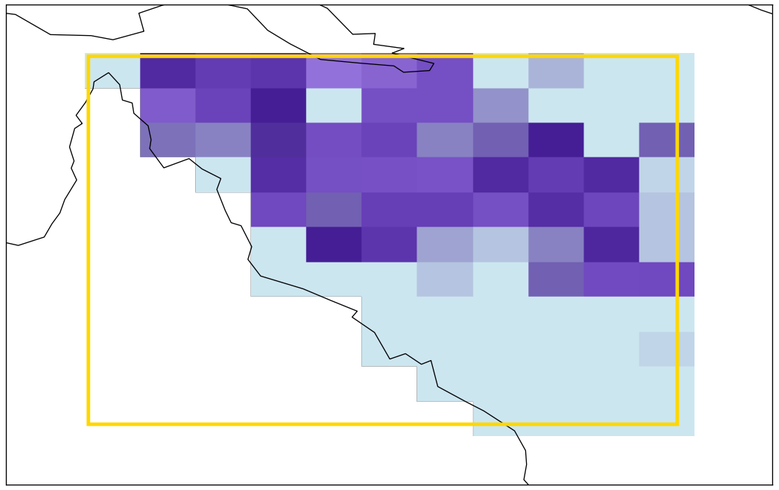




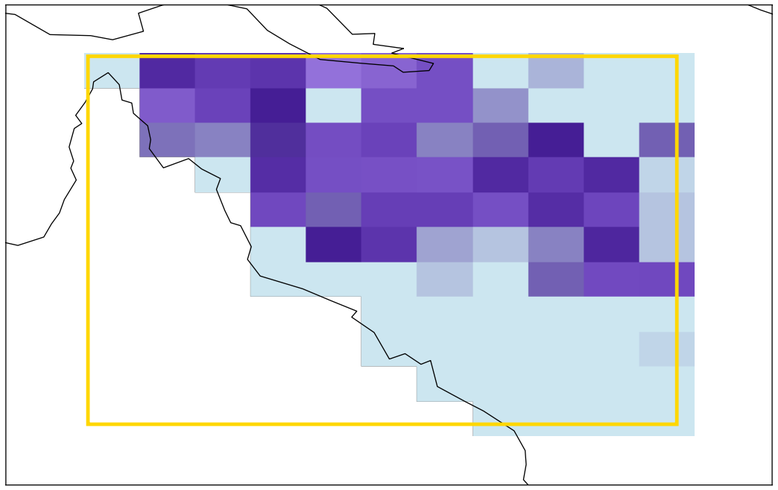


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

Actual DHW - Step 1



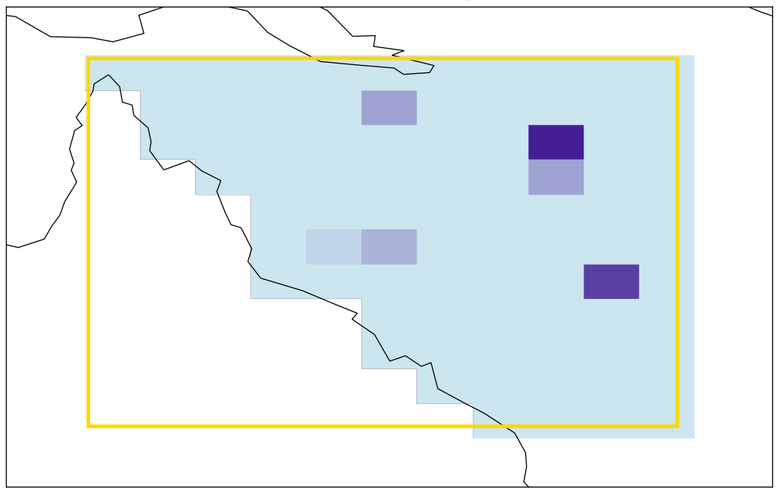
Predicted DHW - Step 1



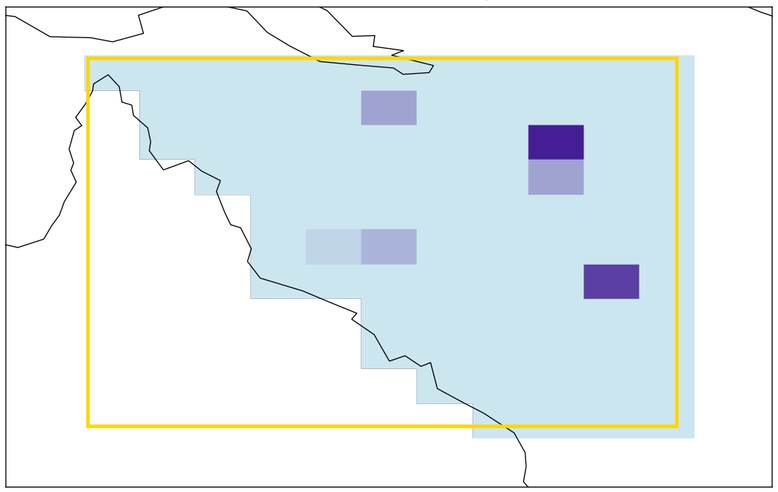
Error (Pred - Actual) - Step 1 [scale ±8]



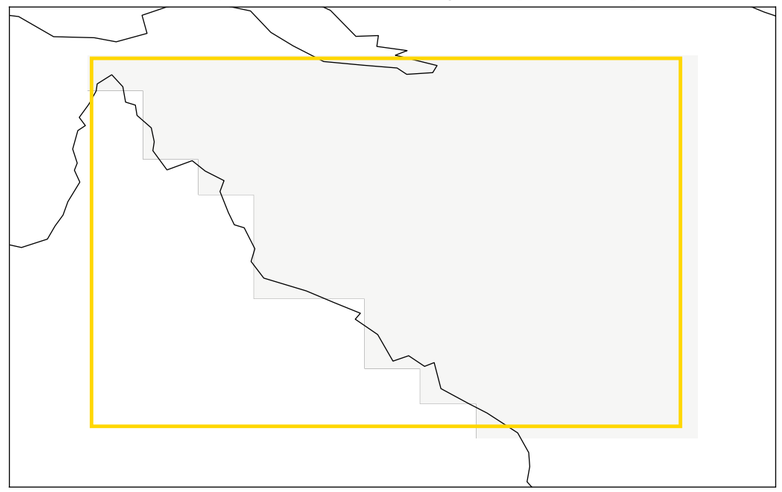
Actual DHW - Step 46



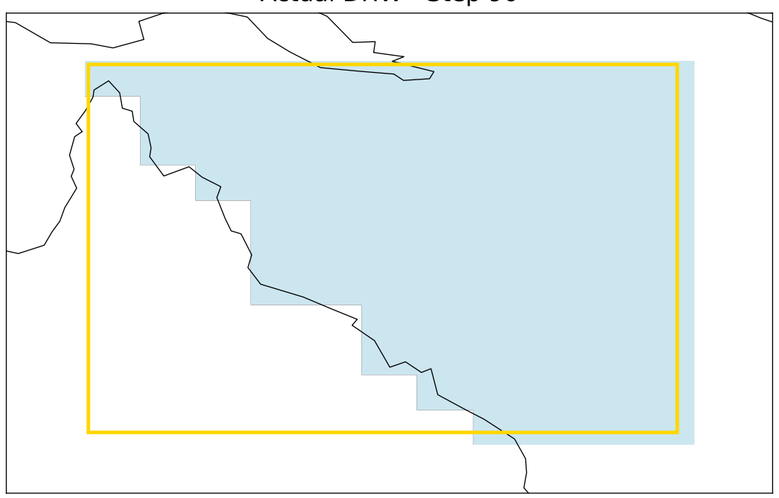
Predicted DHW - Step 46



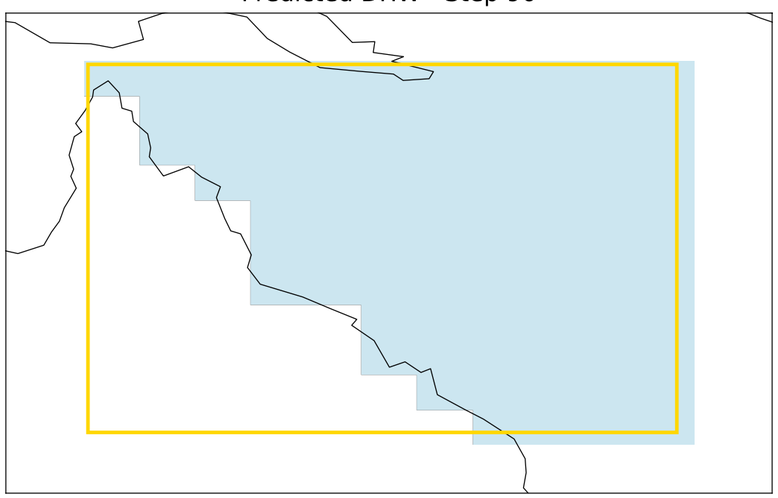
Error (Pred - Actual) - Step 46 [scale ±8]



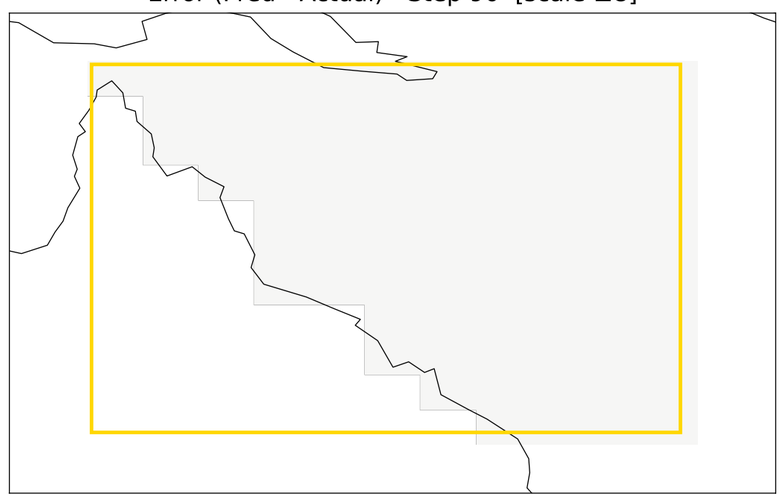
Actual DHW - Step 90

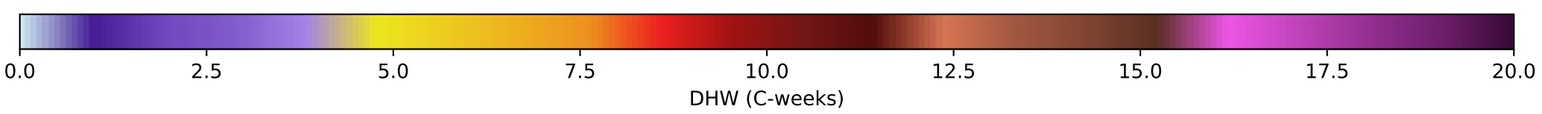


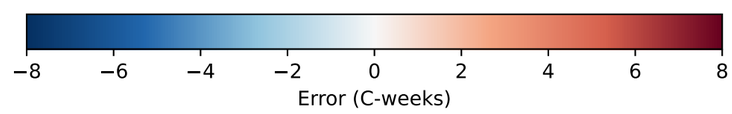
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

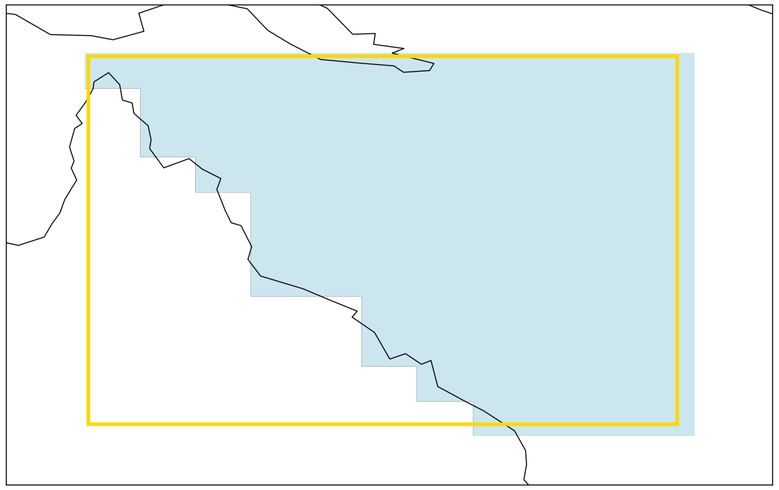




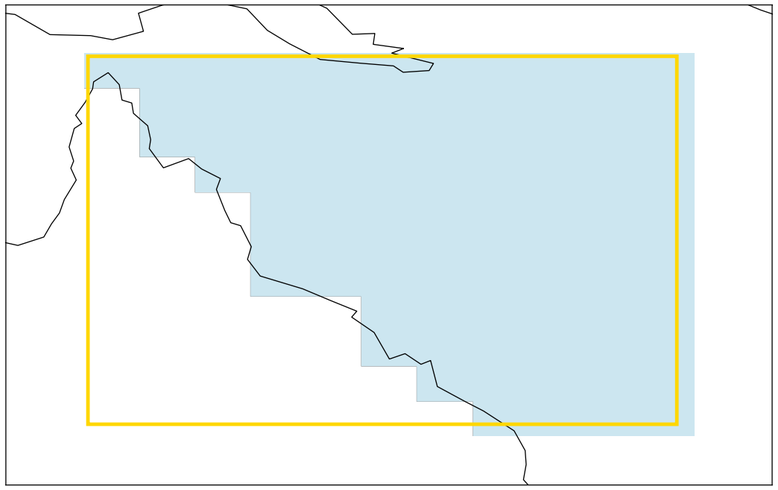


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

Actual DHW - Step 1



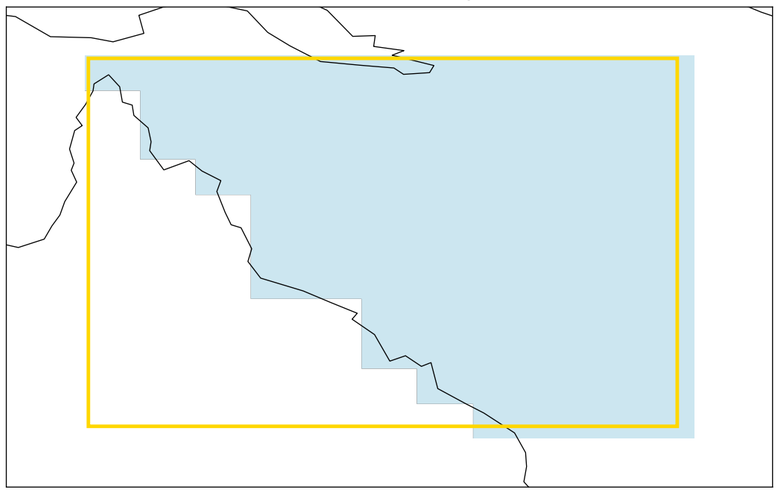
Predicted DHW - Step 1



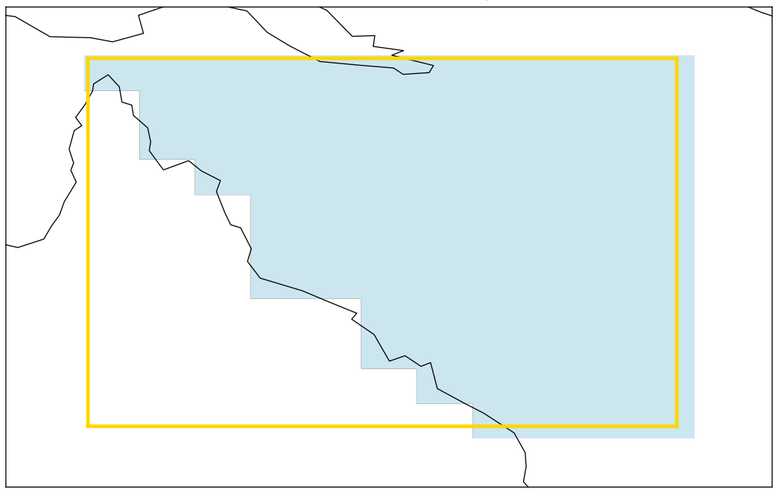
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



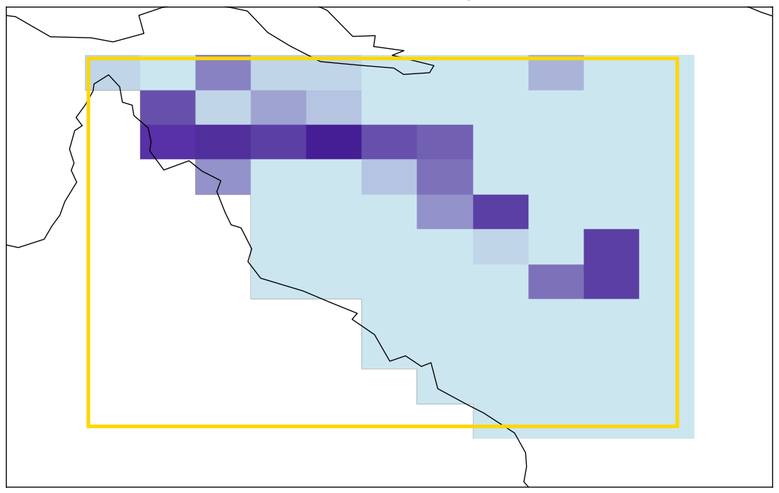
Predicted DHW - Step 46



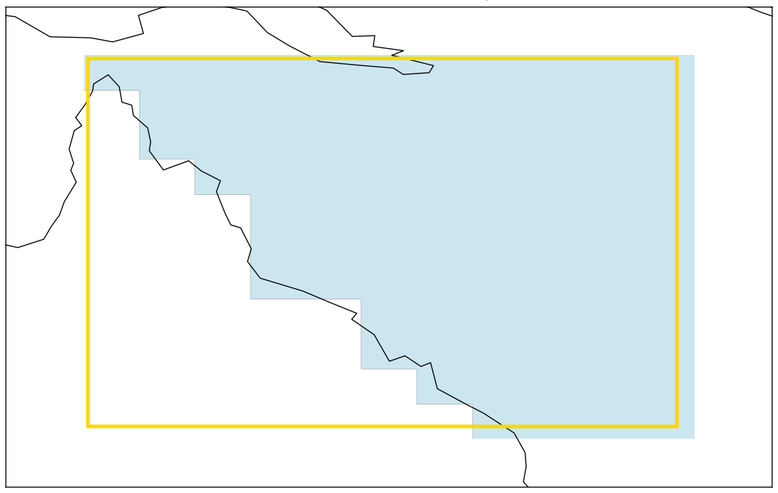
Error (Pred - Actual) - Step 46 [scale ±8]



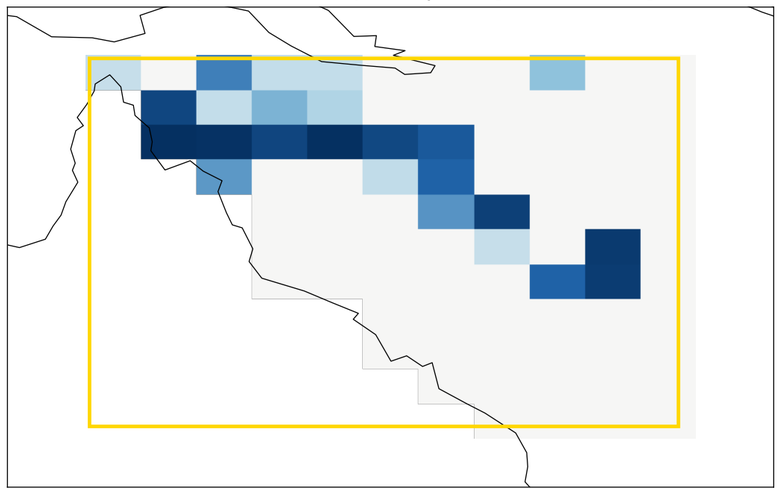
Actual DHW - Step 90

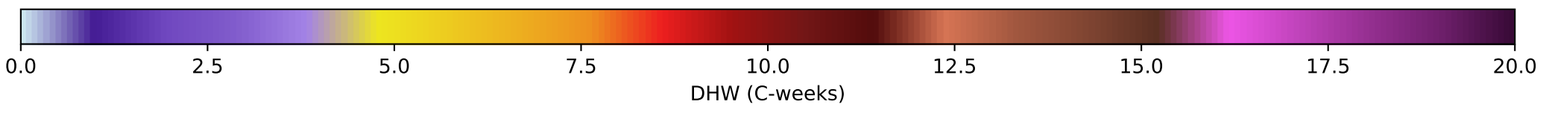


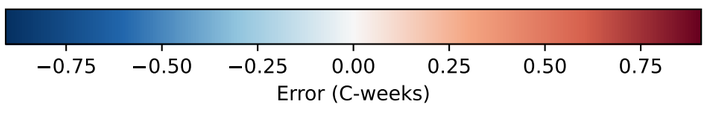
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.91]

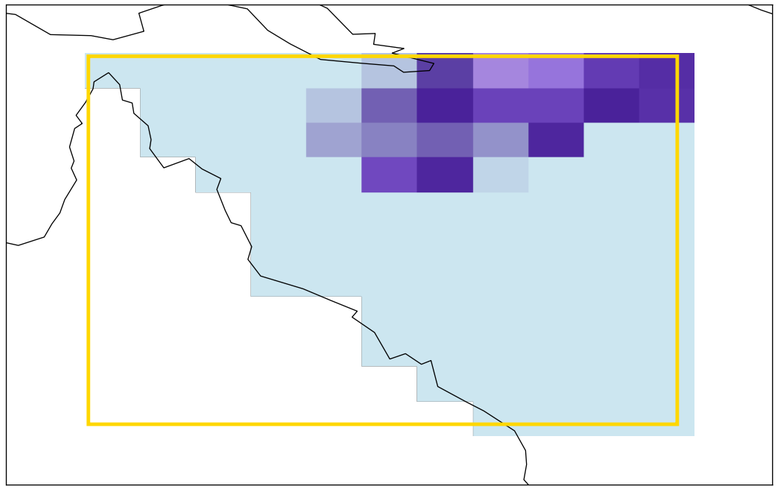




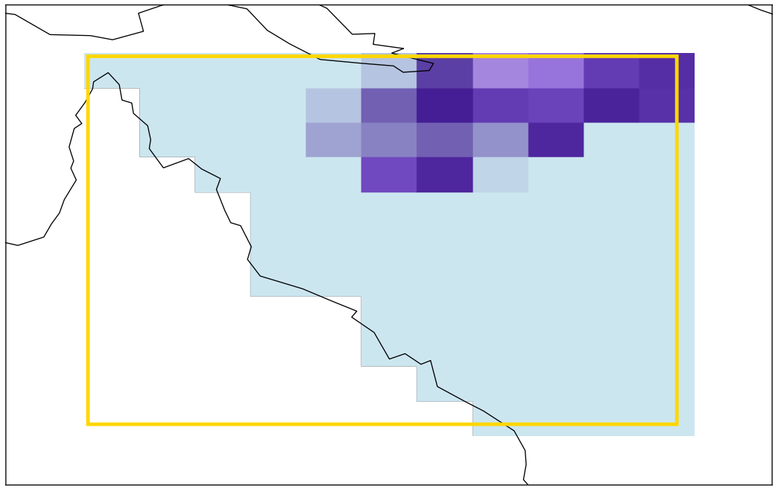


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

Actual DHW - Step 1



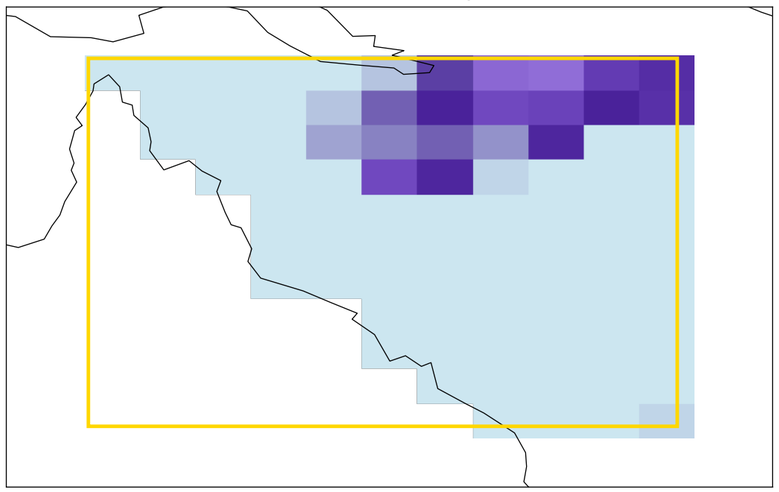
Predicted DHW - Step 1



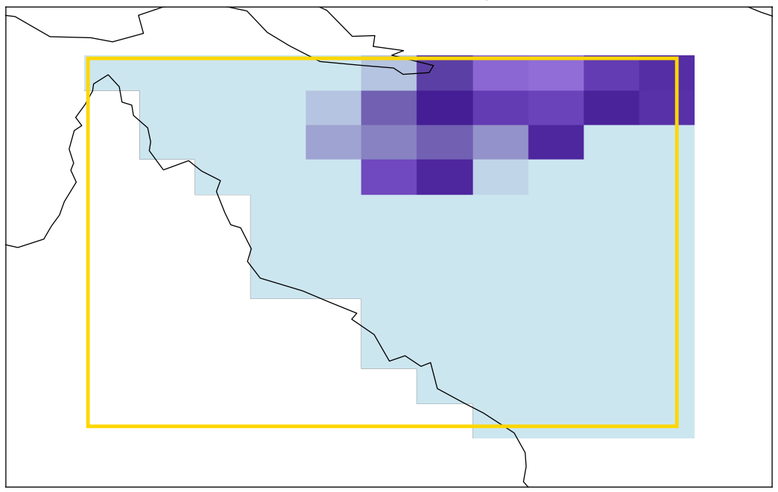
Error (Pred - Actual) - Step 1 [scale ±0.051]



Actual DHW - Step 46



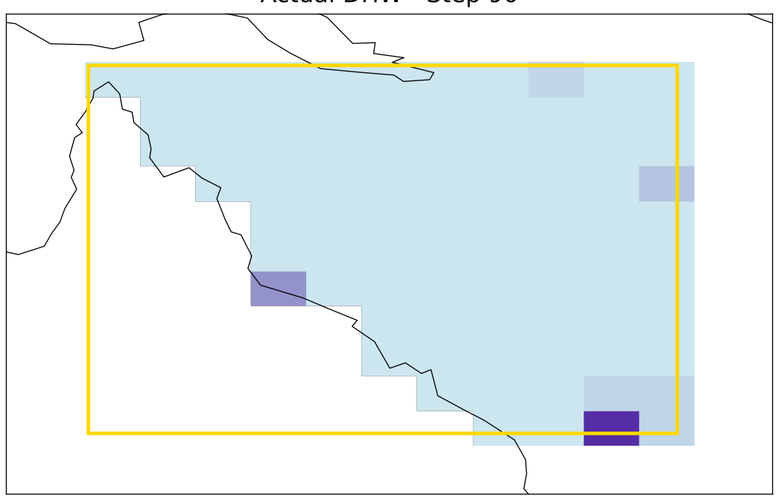
Predicted DHW - Step 46



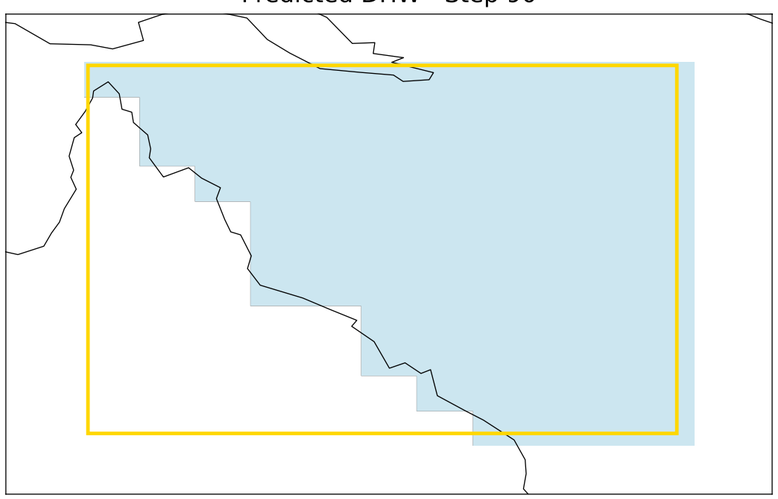
Error (Pred - Actual) - Step 46 [scale ±0.15]



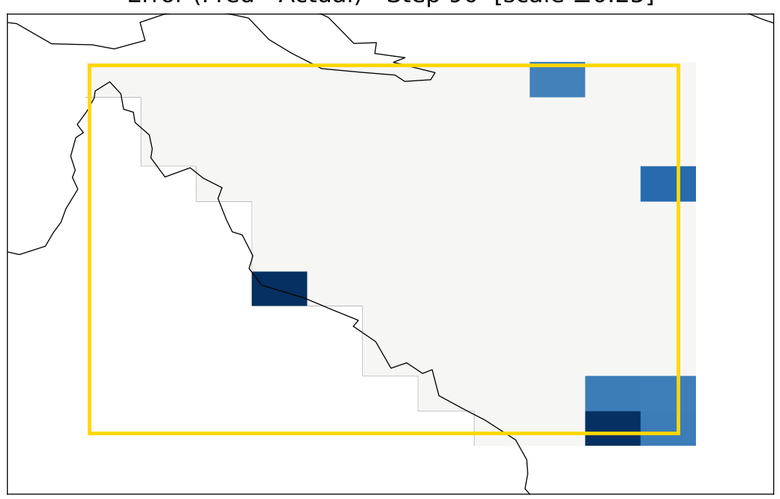
Actual DHW - Step 90

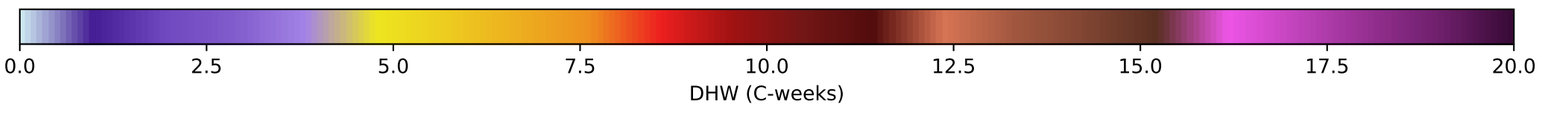


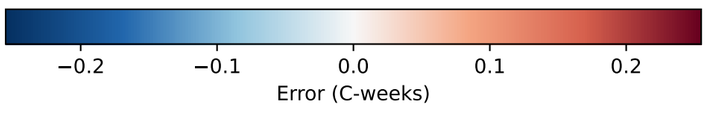
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.25]

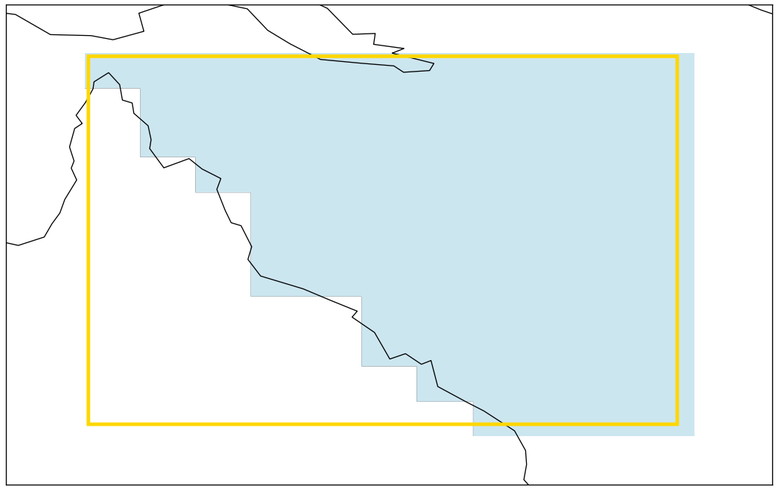




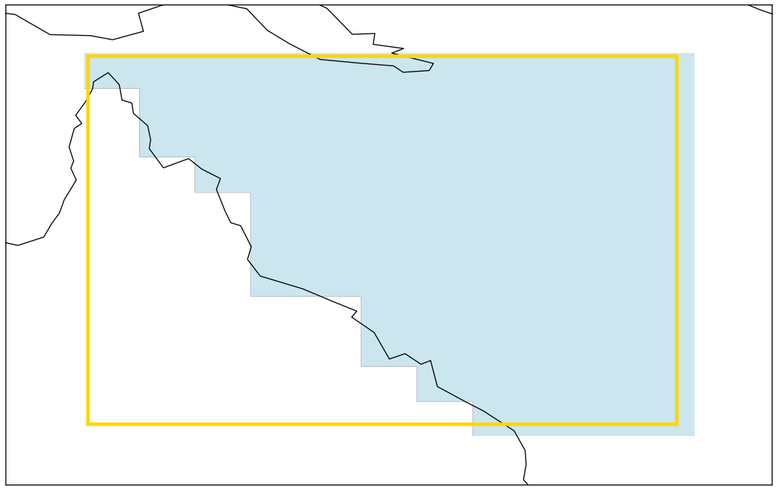


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

Actual DHW - Step 1



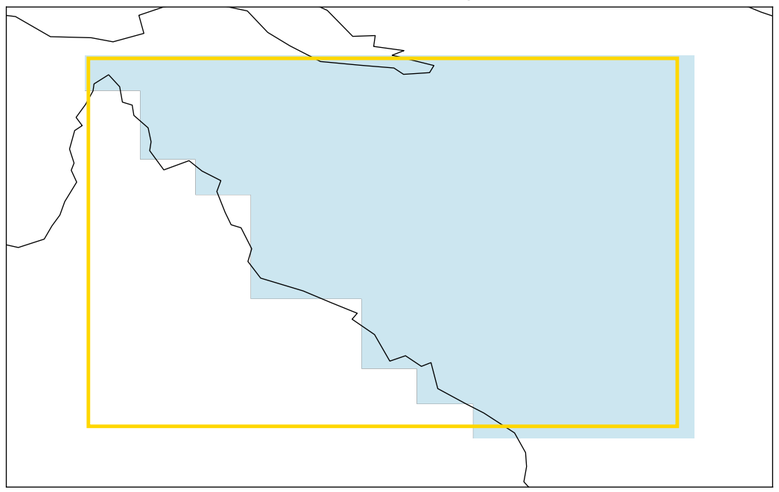
Predicted DHW - Step 1



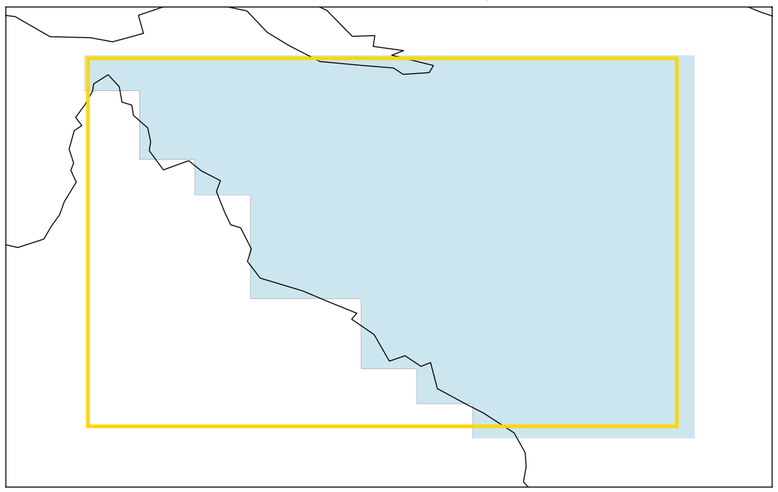
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



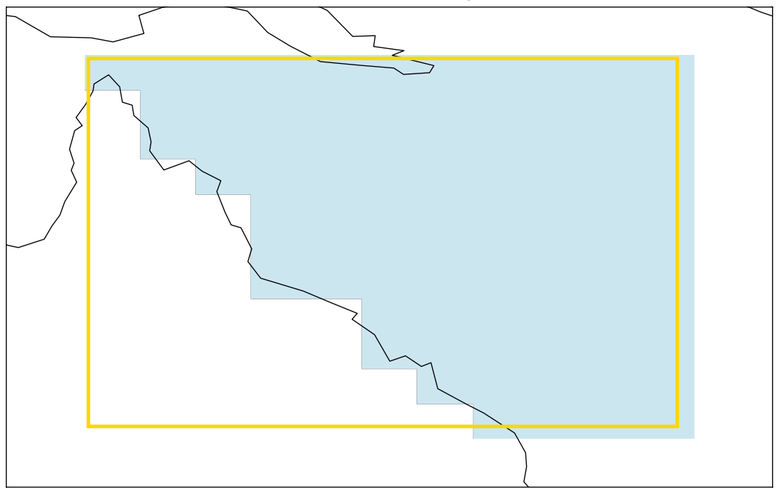
Predicted DHW - Step 46



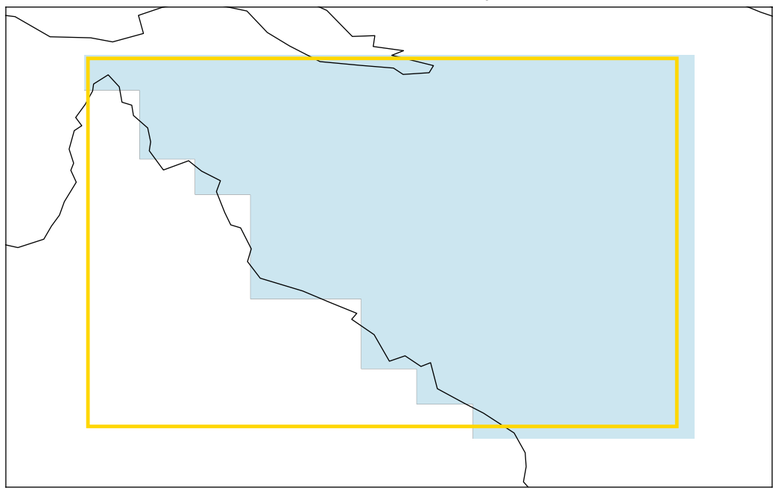
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

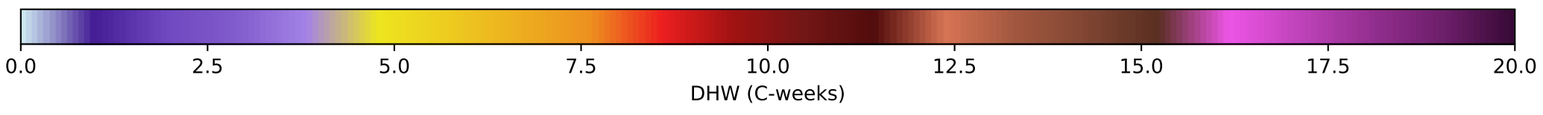


Predicted DHW - Step 90



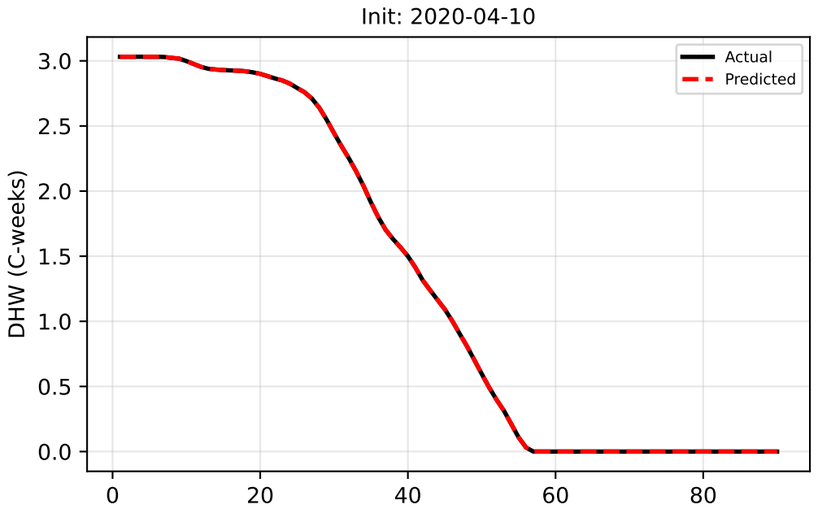
Error (Pred - Actual) - Step 90 [scale ±8]

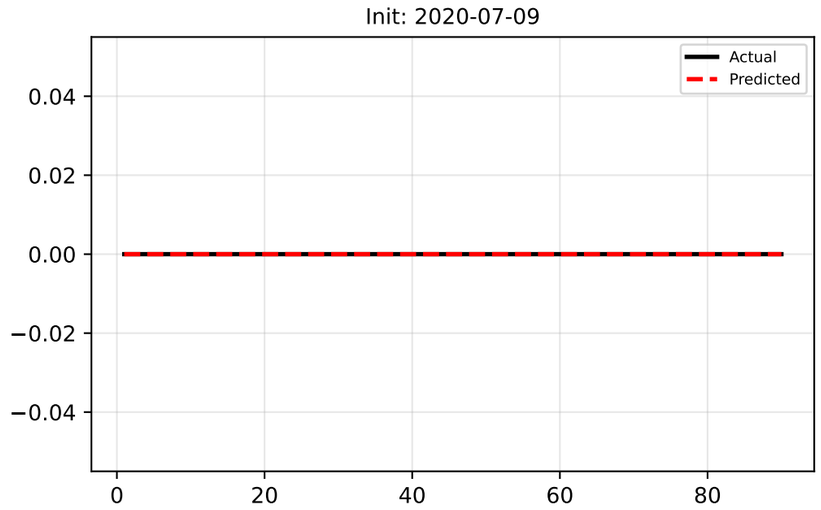


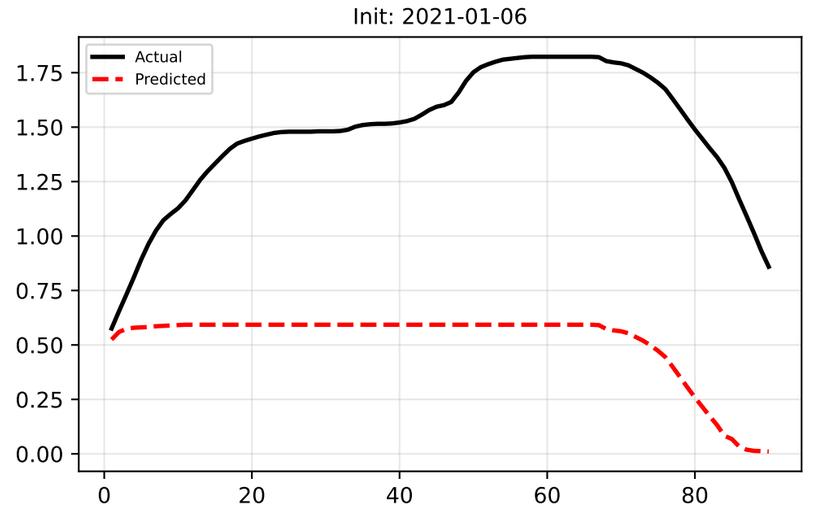


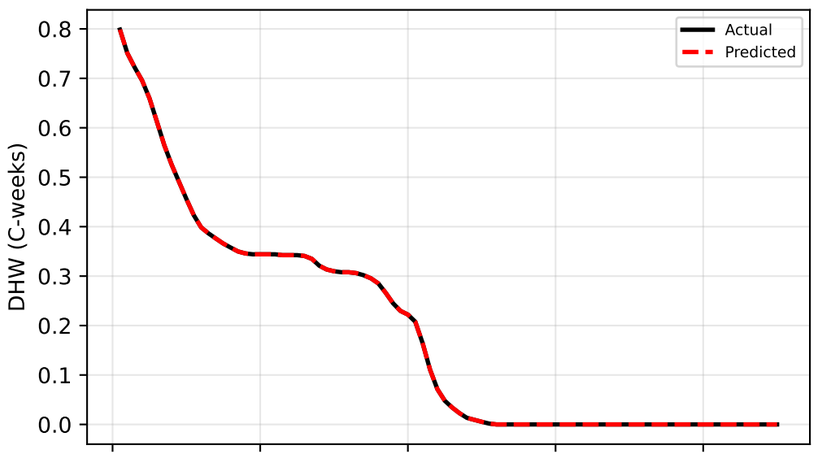


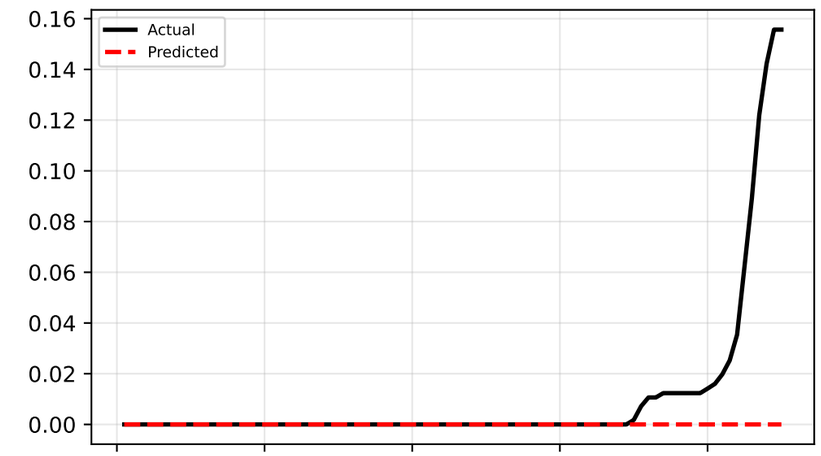
Global-Mean DHW - Test - GBR (C-weeks)

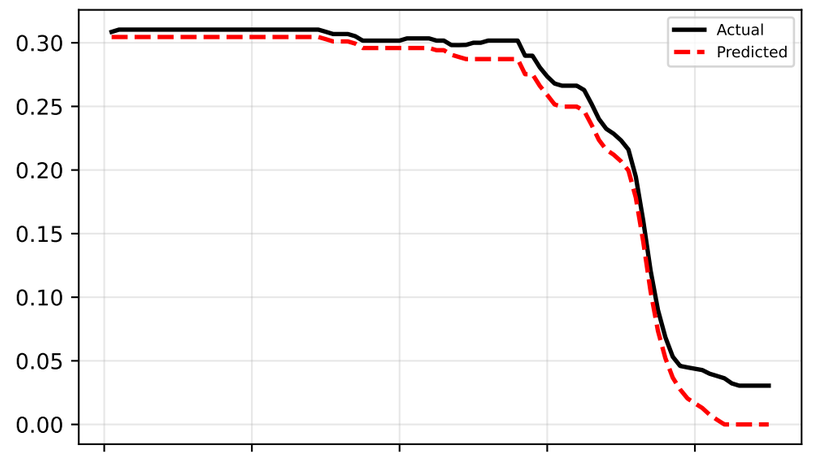


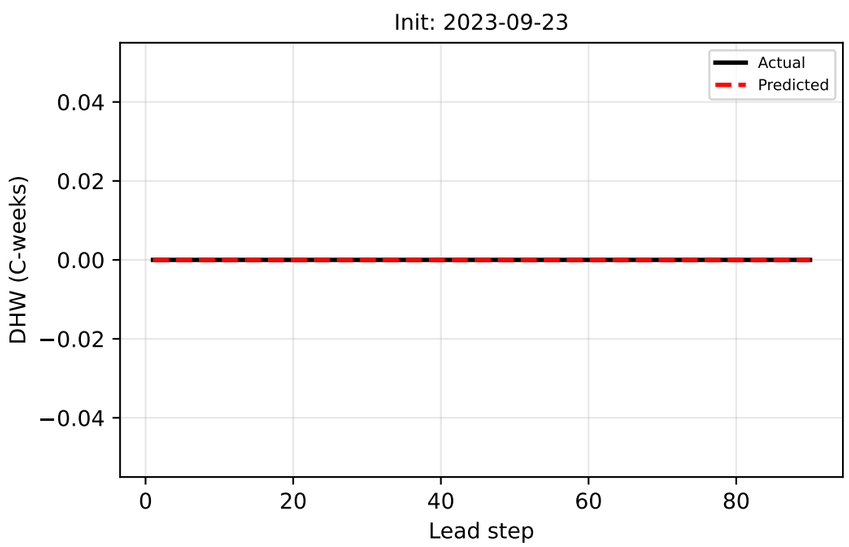


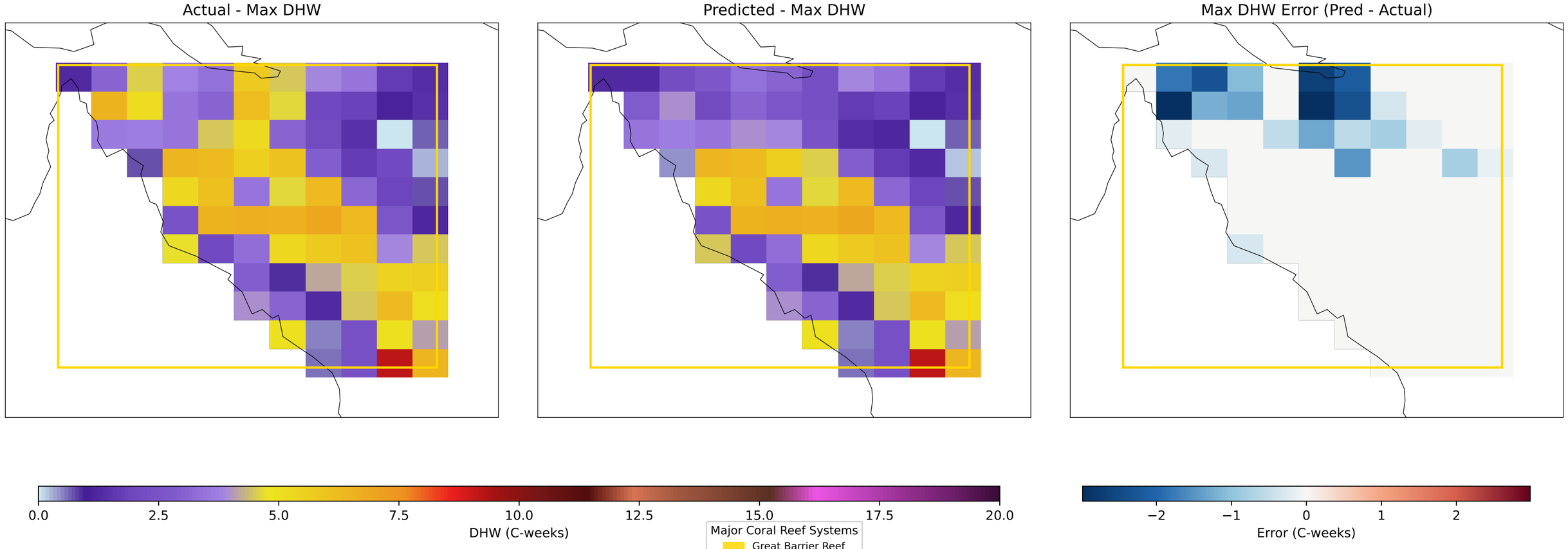


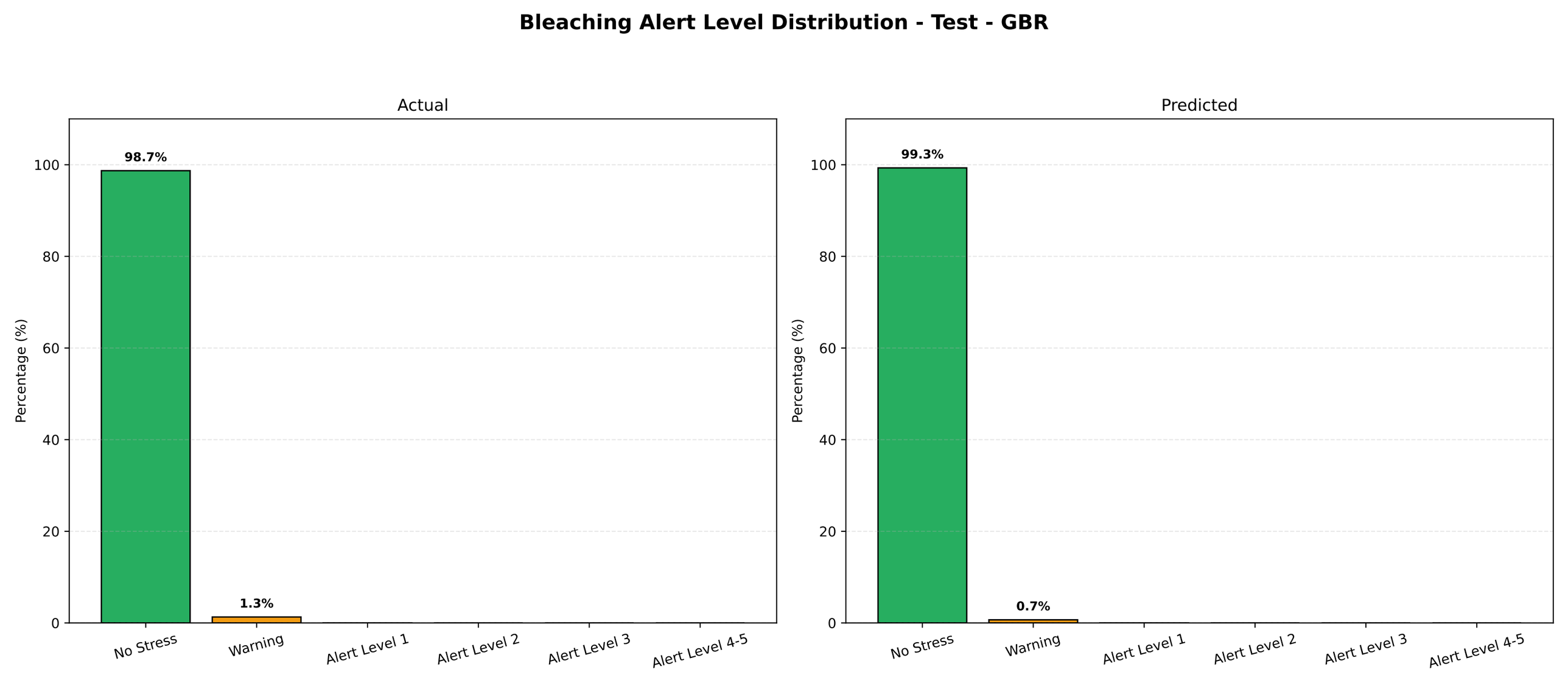






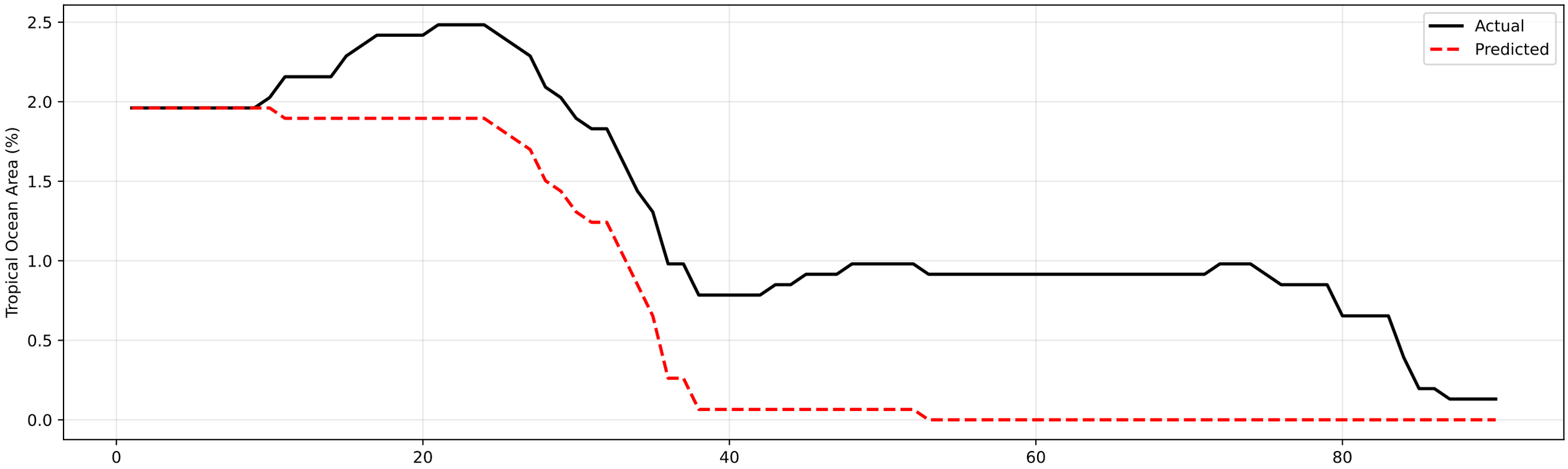



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)

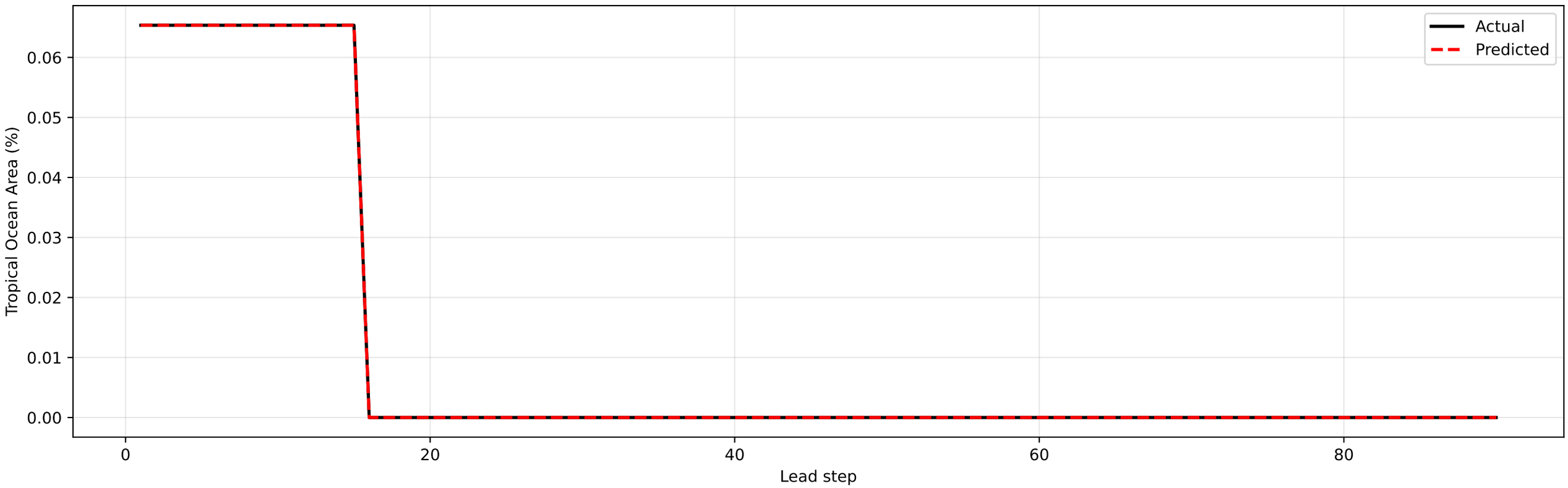

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.7%
1.3%
99.3%
0.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



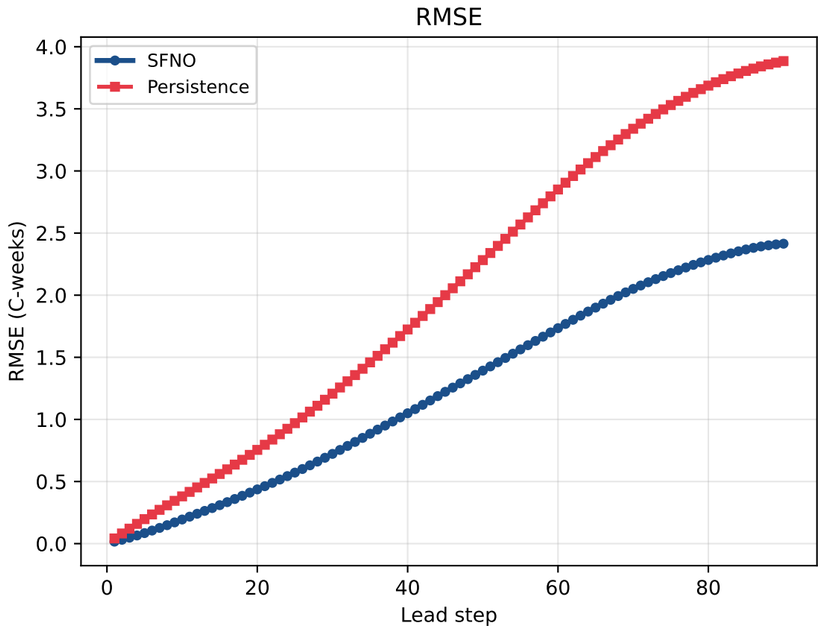
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

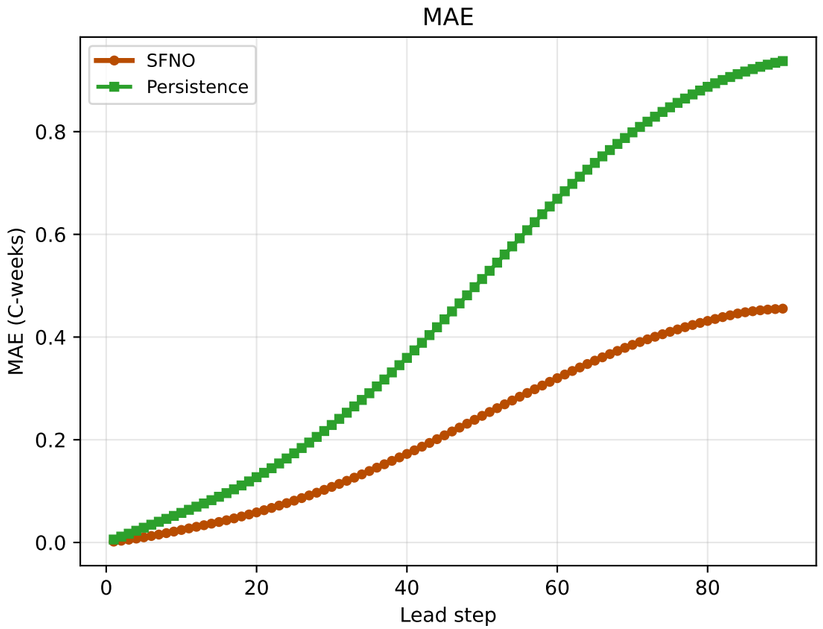


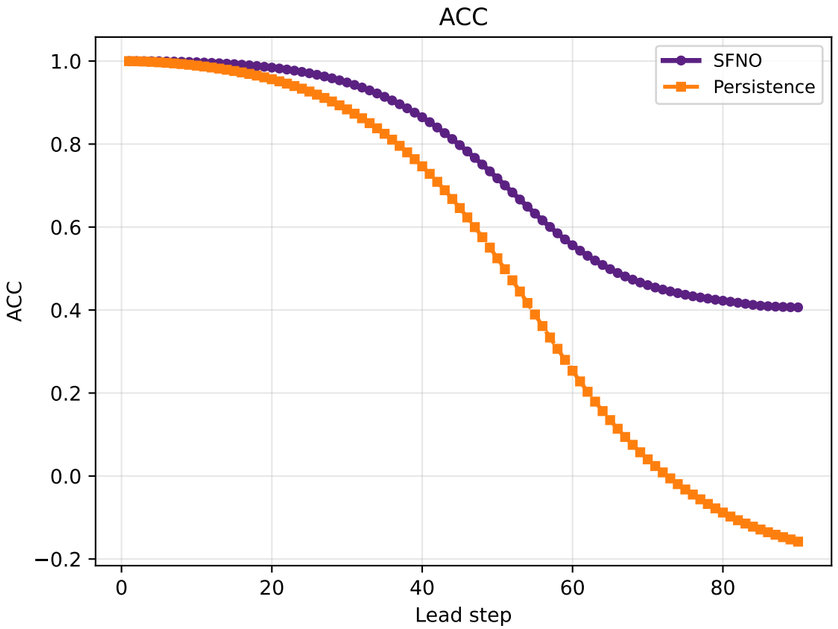
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

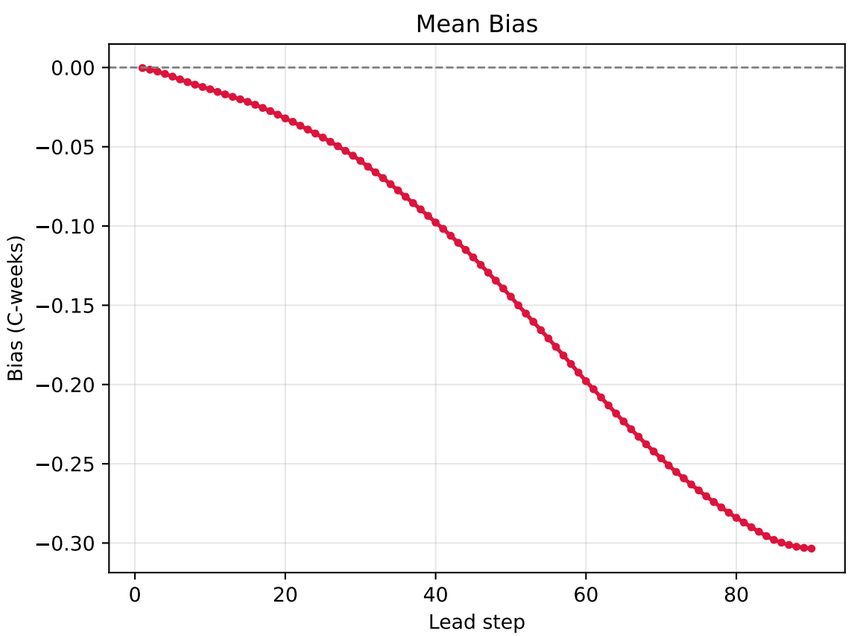
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0161	0.0418	61.4%	0.0016	0.0057	-0.0003	1.0000	0.9999	1.0000
2	0.0312	0.0818	61.9%	0.0034	0.0114	-0.0014	0.9999	0.9995	0.9999
3	0.0479	0.1205	60.3%	0.0054	0.0171	-0.0026	0.9998	0.9989	0.9997
4	0.0657	0.1589	58.6%	0.0076	0.0228	-0.0040	0.9997	0.9981	0.9995
5	0.0848	0.1972	57.0%	0.0100	0.0287	-0.0057	0.9995	0.9971	0.9991
6	0.1051	0.2350	55.3%	0.0126	0.0346	-0.0075	0.9992	0.9959	0.9986
7	0.1265	0.2720	53.5%	0.0154	0.0403	-0.0092	0.9988	0.9944	0.9980
8	0.1485	0.3084	51.9%	0.0182	0.0460	-0.0108	0.9984	0.9928	0.9973
9	0.1709	0.3444	50.4%	0.0212	0.0517	-0.0123	0.9978	0.9911	0.9964
10	0.1941	0.3804	49.0%	0.0243	0.0576	-0.0137	0.9972	0.9891	0.9953
11	0.2171	0.4166	47.9%	0.0274	0.0636	-0.0154	0.9965	0.9869	0.9941
12	0.2405	0.4526	46.9%	0.0305	0.0697	-0.0169	0.9956	0.9845	0.9928
13	0.2634	0.4885	46.1%	0.0336	0.0760	-0.0185	0.9947	0.9819	0.9913
14	0.2864	0.5245	45.4%	0.0367	0.0824	-0.0201	0.9937	0.9791	0.9896
15	0.3097	0.5608	44.8%	0.0399	0.0890	-0.0217	0.9926	0.9760	0.9878
16	0.3340	0.5981	44.2%	0.0433	0.0961	-0.0235	0.9913	0.9727	0.9857
17	0.3592	0.6362	43.5%	0.0469	0.1034	-0.0255	0.9898	0.9691	0.9833
18	0.3849	0.6749	43.0%	0.0506	0.1111	-0.0275	0.9881	0.9651	0.9807
19	0.4106	0.7145	42.5%	0.0545	0.1189	-0.0297	0.9863	0.9608	0.9778
20	0.4366	0.7547	42.2%	0.0586	0.1271	-0.0321	0.9843	0.9562	0.9746
21	0.4624	0.7957	41.9%	0.0628	0.1356	-0.0343	0.9821	0.9512	0.9712
22	0.4890	0.8377	41.6%	0.0673	0.1446	-0.0367	0.9796	0.9457	0.9675
23	0.5159	0.8806	41.4%	0.0718	0.1539	-0.0391	0.9769	0.9399	0.9634
24	0.5436	0.9245	41.2%	0.0766	0.1636	-0.0416	0.9739	0.9335	0.9588
25	0.5718	0.9697	41.0%	0.0814	0.1736	-0.0442	0.9706	0.9266	0.9538
26	0.6006	1.0157	40.9%	0.0864	0.1839	-0.0469	0.9670	0.9191	0.9483
27	0.6304	1.0625	40.7%	0.0916	0.1945	-0.0497	0.9629	0.9112	0.9423
28	0.6606	1.1099	40.5%	0.0969	0.2055	-0.0526	0.9585	0.9026	0.9358
29	0.6912	1.1584	40.3%	0.1024	0.2170	-0.0556	0.9536	0.8934	0.9287
30	0.7224	1.2074	40.2%	0.1082	0.2288	-0.0589	0.9483	0.8836	0.9210
31	0.7542	1.2569	40.0%	0.1140	0.2407	-0.0626	0.9425	0.8732	0.9127
32	0.7864	1.3067	39.8%	0.1200	0.2527	-0.0661	0.9362	0.8621	0.9037
33	0.8186	1.3569	39.7%	0.1262	0.2650	-0.0698	0.9293	0.8504	0.8942
34	0.8511	1.4076	39.5%	0.1325	0.2776	-0.0736	0.9219	0.8380	0.8839
35	0.8846	1.4593	39.4%	0.1391	0.2906	-0.0776	0.9138	0.8247	0.8728
36	0.9178	1.5117	39.3%	0.1457	0.3038	-0.0815	0.9051	0.8106	0.8610
37	0.9506	1.5646	39.2%	0.1523	0.3173	-0.0855	0.8960	0.7957	0.8487
38	0.9832	1.6177	39.2%	0.1589	0.3312	-0.0895	0.8862	0.7800	0.8359
39	1.0163	1.6708	39.2%	0.1656	0.3452	-0.0936	0.8758	0.7636	0.8222
40	1.0495	1.7241	39.1%	0.1724	0.3594	-0.0978	0.8646	0.7464	0.8079
41	1.0830	1.7780	39.1%	0.1794	0.3740	-0.1018	0.8527	0.7282	0.7928
42	1.1174	1.8327	39.0%	0.1865	0.3888	-0.1061	0.8400	0.7090	0.7768
43	1.1525	1.8883	39.0%	0.1938	0.4038	-0.1106	0.8263	0.6888	0.7599
44	1.1875	1.9443	38.9%	0.2012	0.4191	-0.1151	0.8121	0.6676	0.7424
45	1.2221	1.9999	38.9%	0.2086	0.4345	-0.1198	0.7974	0.6458	0.7246
46	1.2560	2.0555	38.9%	0.2161	0.4499	-0.1246	0.7823	0.6233	0.7066
47	1.2900	2.1115	38.9%	0.2237	0.4656	-0.1294	0.7667	0.5999	0.6882
48	1.3241	2.1682	38.9%	0.2313	0.4813	-0.1345	0.7506	0.5756	0.6695
49	1.3583	2.2251	39.0%	0.2389	0.4971	-0.1395	0.7342	0.5505	0.6505
50	1.3925	2.2823	39.0%	0.2466	0.5130	-0.1446	0.7175	0.5248	0.6314
51	1.4266	2.3396	39.0%	0.2540	0.5290	-0.1501	0.7005	0.4984	0.6121
52	1.4605	2.3967	39.1%	0.2615	0.5449	-0.1552	0.6834	0.4717	0.5928
53	1.4946	2.4537	39.1%	0.2690	0.5607	-0.1604	0.6664	0.4446	0.5737
54	1.5290	2.5113	39.1%	0.2764	0.5766	-0.1657	0.6492	0.4170	0.5546
55	1.5634	2.5689	39.1%	0.2838	0.5923	-0.1709	0.6325	0.3893	0.5360
56	1.5975	2.6262	39.2%	0.2911	0.6080	-0.1762	0.6161	0.3615	0.5180
57	1.6317	2.6833	39.2%	0.2983	0.6237	-0.1817	0.6003	0.3339	0.5006
58	1.6660	2.7398	39.2%	0.3056	0.6392	-0.1871	0.5850	0.3066	0.4838
59	1.7002	2.7954	39.2%	0.3128	0.6543	-0.1925	0.5702	0.2799	0.4676
60	1.7345	2.8507	39.2%	0.3199	0.6692	-0.1979	0.5562	0.2536	0.4522
61	1.7685	2.9054	39.1%	0.3270	0.6839	-0.2029	0.5431	0.2280	0.4377
62	1.8020	2.9593	39.1%	0.3339	0.6983	-0.2081	0.5308	0.2031	0.4241
63	1.8354	3.0117	39.1%	0.3408	0.7123	-0.2132	0.5193	0.1793	0.4111
64	1.8681	3.0626	39.0%	0.3475	0.7259	-0.2183	0.5086	0.1565	0.3989
65	1.9000	3.1120	38.9%	0.3541	0.7391	-0.2233	0.4986	0.1348	0.3874
66	1.9316	3.1601	38.9%	0.3605	0.7519	-0.2282	0.4893	0.1140	0.3764
67	1.9627	3.2071	38.8%	0.3668	0.7641	-0.2330	0.4810	0.0941	0.3663
68	1.9931	3.2530	38.7%	0.3730	0.7761	-0.2377	0.4733	0.0752	0.3569
69	2.0225	3.2971	38.7%	0.3790	0.7876	-0.2422	0.4663	0.0572	0.3482
70	2.0507	3.3397	38.6%	0.3848	0.7986	-0.2466	0.4601	0.0402	0.3402
71	2.0779	3.3806	38.5%	0.3903	0.8092	-0.2510	0.4543	0.0240	0.3328
72	2.1042	3.4203	38.5%	0.3956	0.8194	-0.2552	0.4494	0.0088	0.3261
73	2.1294	3.4585	38.4%	0.4006	0.8291	-0.2591	0.4449	-0.0057	0.3200
74	2.1540	3.4954	38.4%	0.4055	0.8384	-0.2631	0.4408	-0.0195	0.3142
75	2.1779	3.5305	38.3%	0.4102	0.8474	-0.2668	0.4368	-0.0325	0.3086
76	2.2007	3.5641	38.3%	0.4148	0.8560	-0.2705	0.4332	-0.0446	0.3034
77	2.2227	3.5969	38.2%	0.4193	0.8644	-0.2741	0.4303	-0.0562	0.2990
78	2.2438	3.6284	38.2%	0.4235	0.8724	-0.2775	0.4276	-0.0673	0.2948
79	2.2640	3.6583	38.1%	0.4275	0.8801	-0.2808	0.4249	-0.0779	0.2905
80	2.2834	3.6871	38.1%	0.4314	0.8873	-0.2841	0.4224	-0.0880	0.2864
81	2.3017	3.7142	38.0%	0.4352	0.8942	-0.2871	0.4201	-0.0976	0.2824
82	2.3193	3.7392	38.0%	0.4389	0.9006	-0.2901	0.4176	-0.1065	0.2784
83	2.3364	3.7626	37.9%	0.4424	0.9064	-0.2929	0.4151	-0.1145	0.2743
84	2.3532	3.7845	37.8%	0.4458	0.9119	-0.2956	0.4125	-0.1220	0.2703
85	2.3680	3.8050	37.8%	0.4483	0.9171	-0.2980	0.4104	-0.1289	0.2672
86	2.3809	3.8241	37.7%	0.4503	0.9218	-0.2997	0.4092	-0.1354	0.2648
87	2.3922	3.8419	37.7%	0.4521	0.9263	-0.3012	0.4083	-0.1415	0.2630
88	2.4015	3.8579	37.8%	0.4535	0.9304	-0.3023	0.4075	-0.1473	0.2614
89	2.4088	3.8719	37.8%	0.4545	0.9341	-0.3030	0.4069	-0.1528	0.2600
90	2.4143	3.8845	37.8%	0.4553	0.9374	-0.3035	0.4064	-0.1581	0.2588

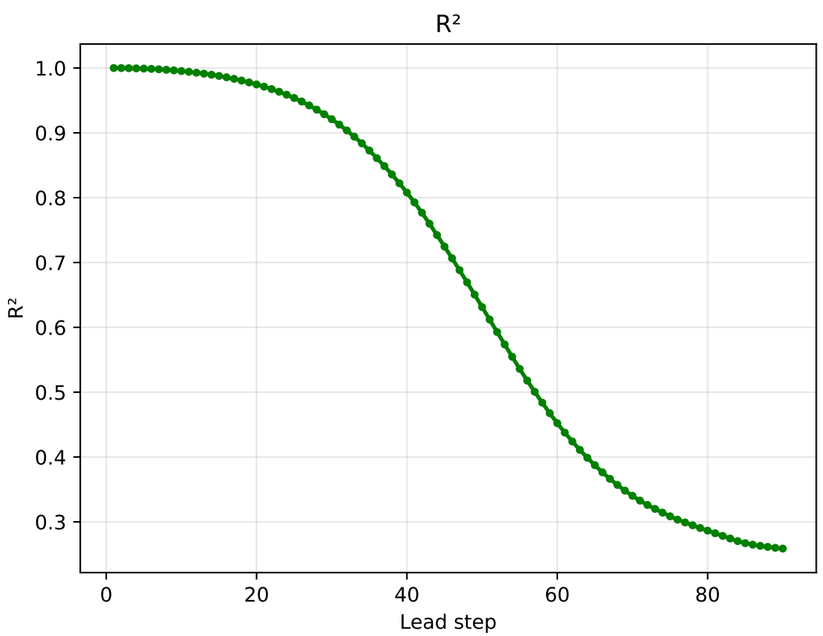
DHW Forecast Skill vs Lead Step - Val

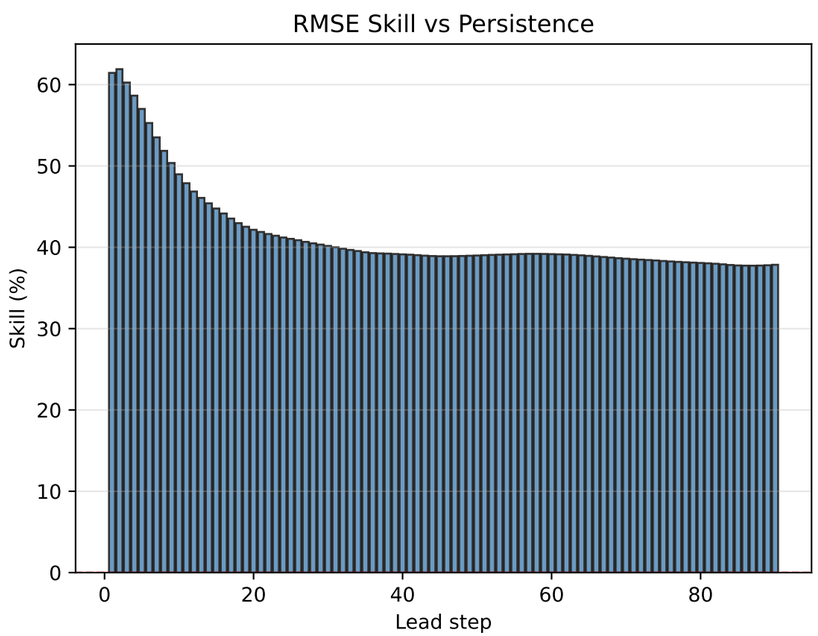


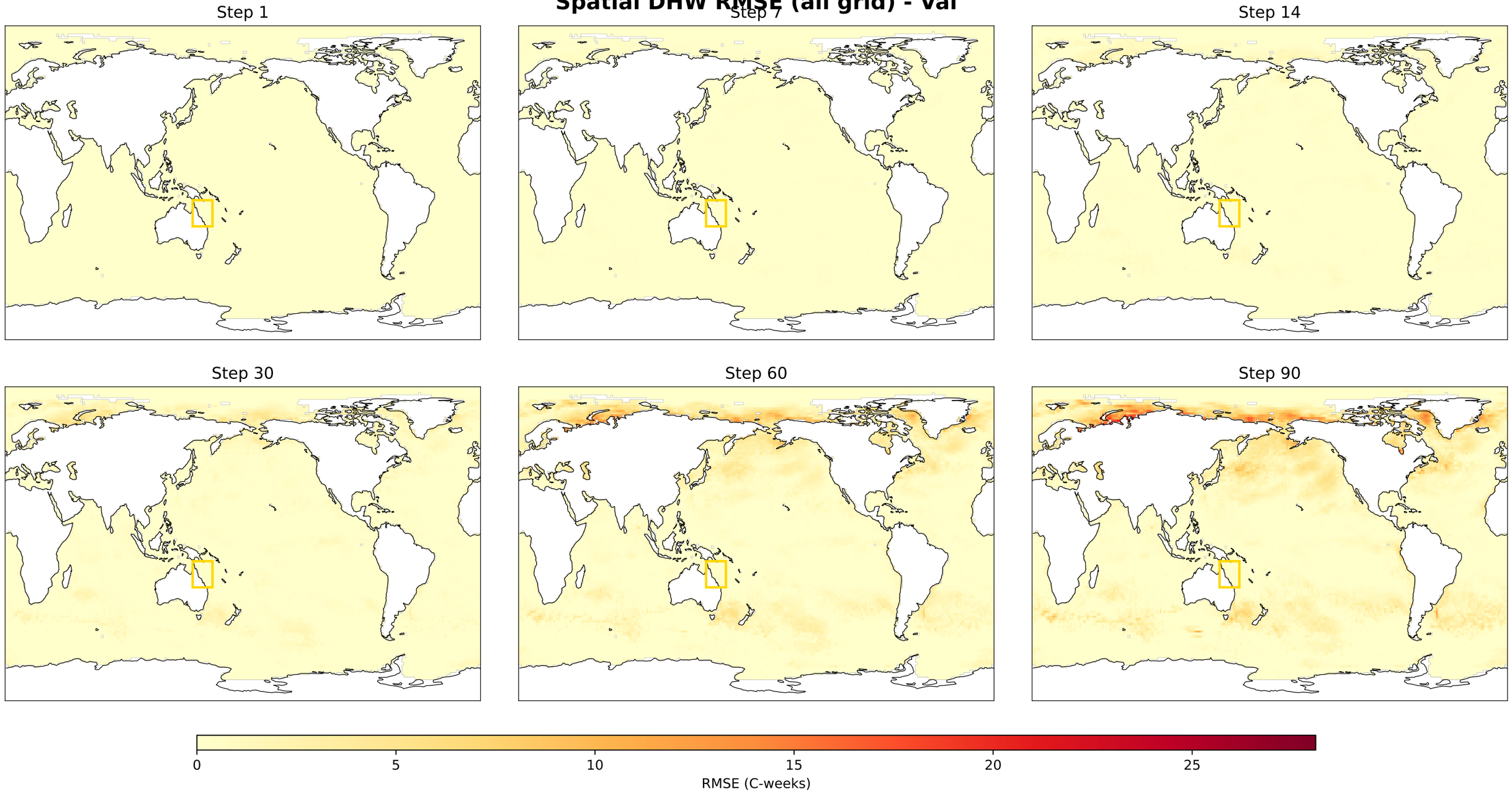


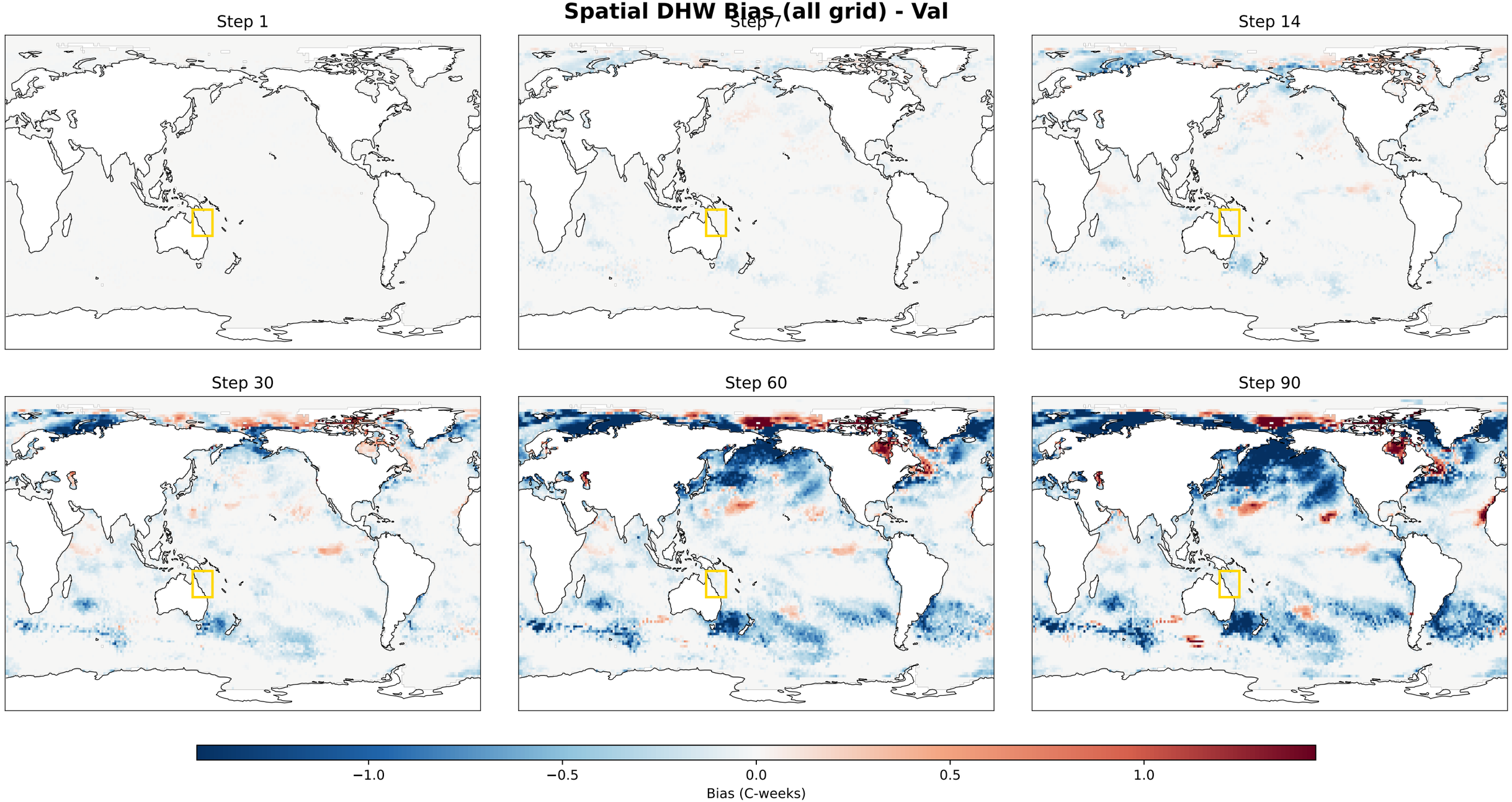


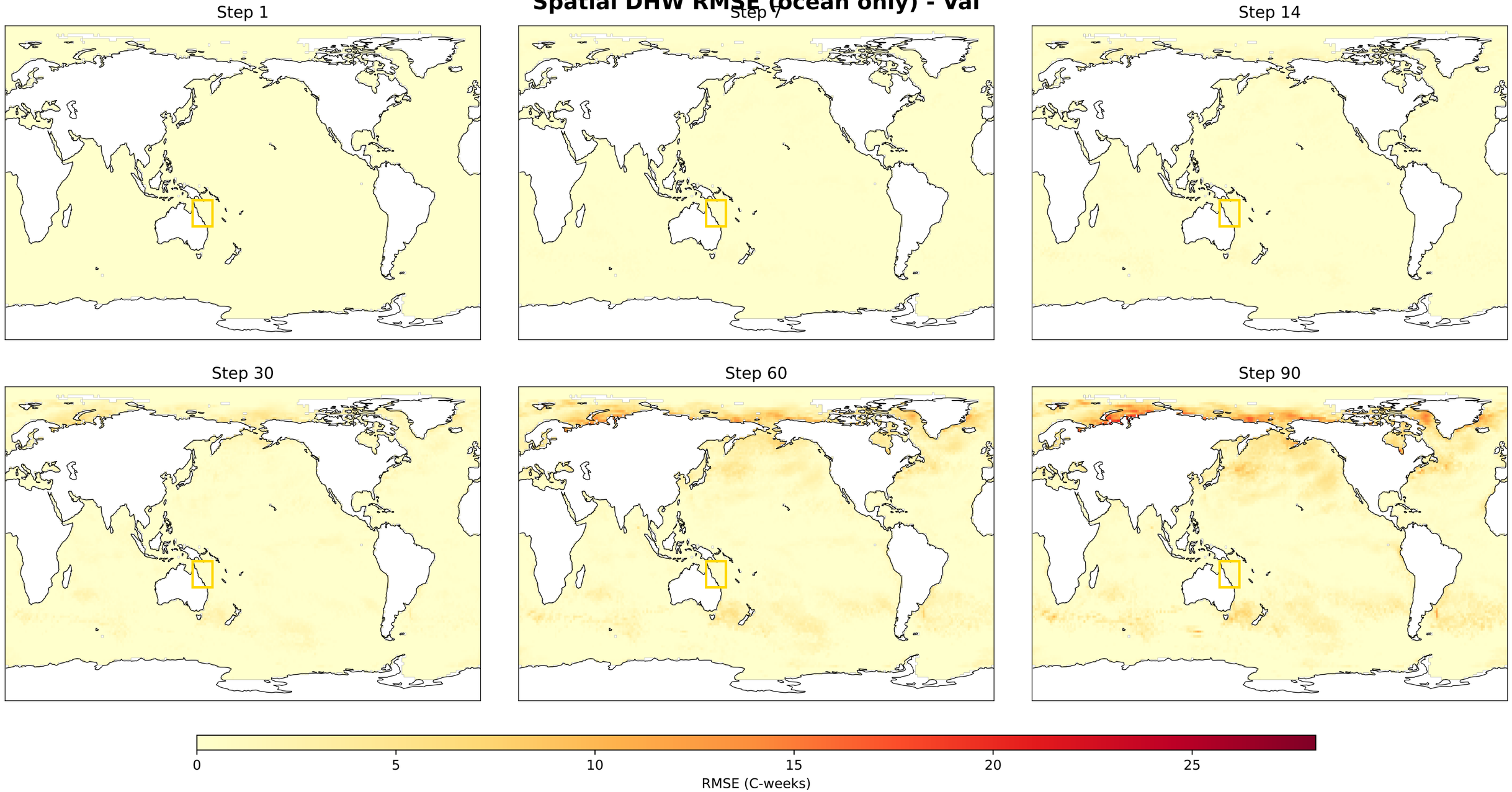


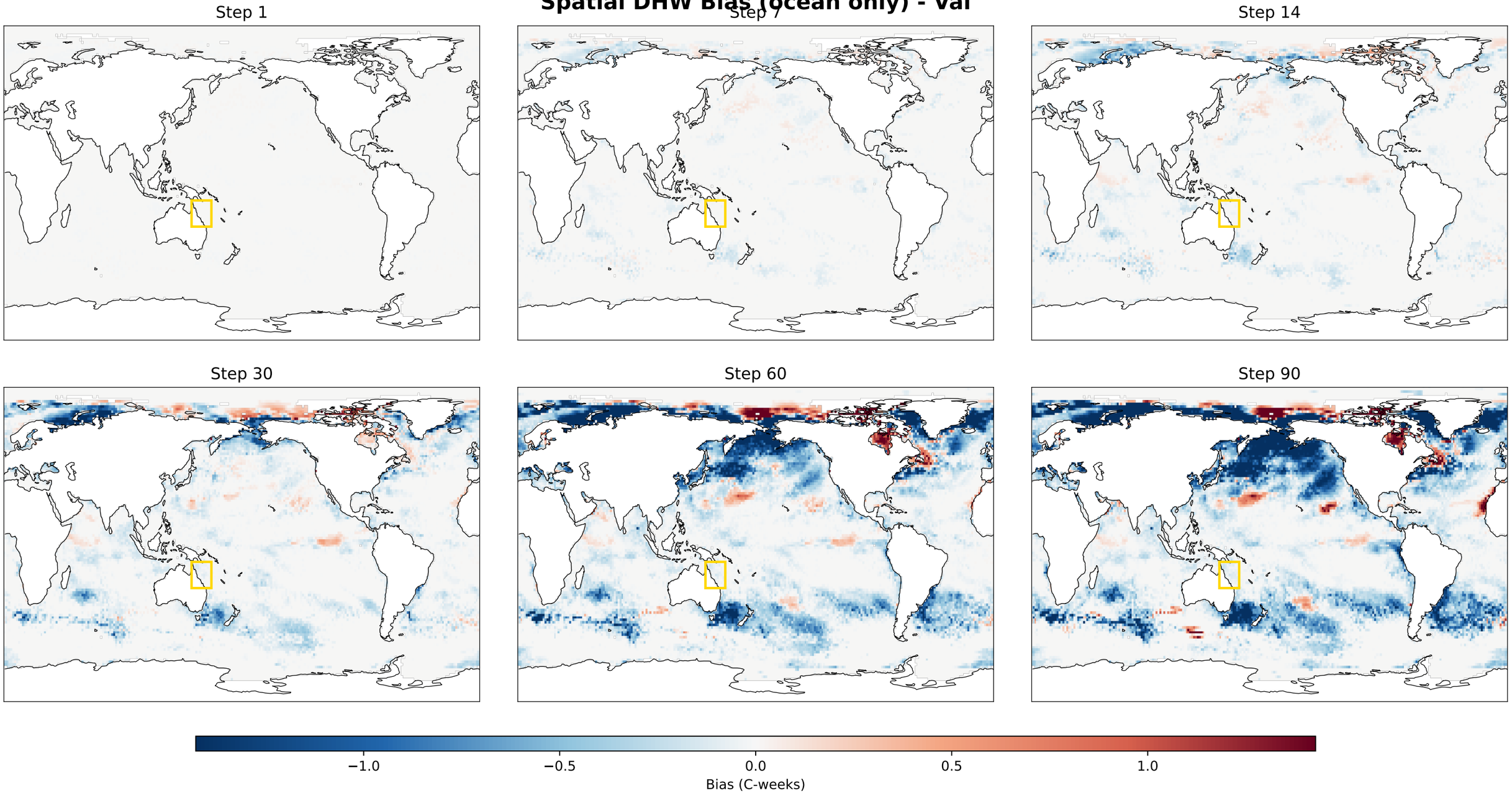





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

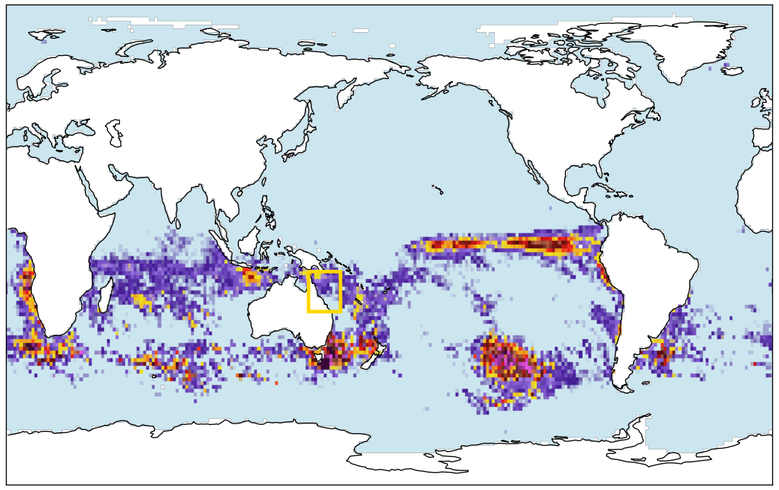

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

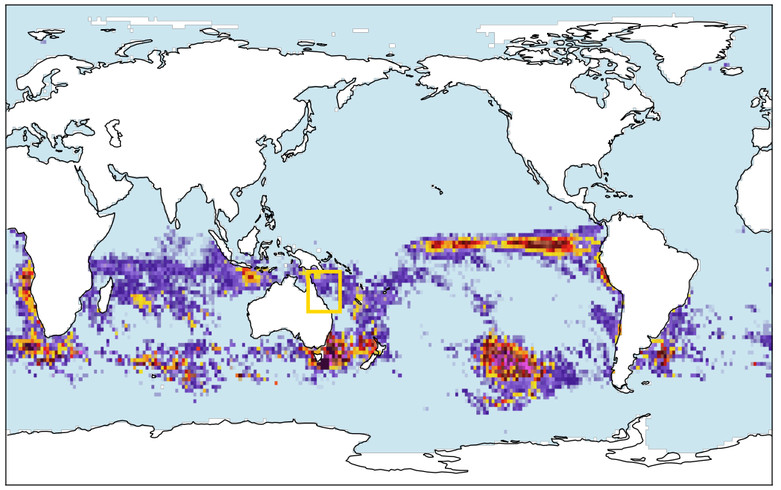

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

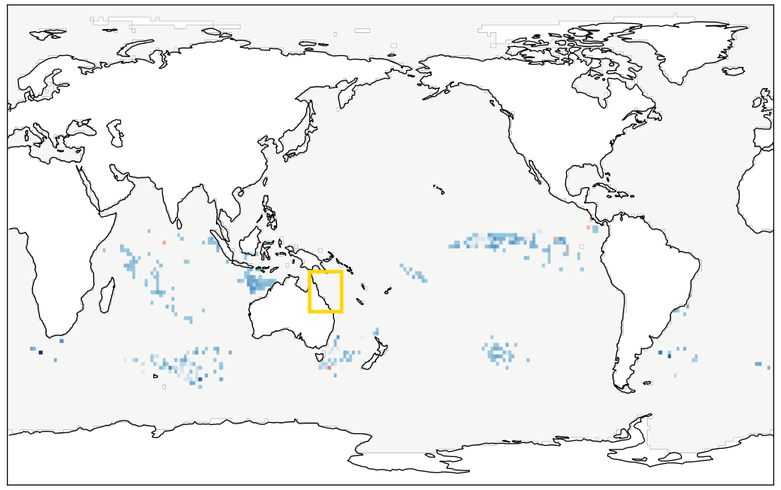
Actual DHW - Step 1



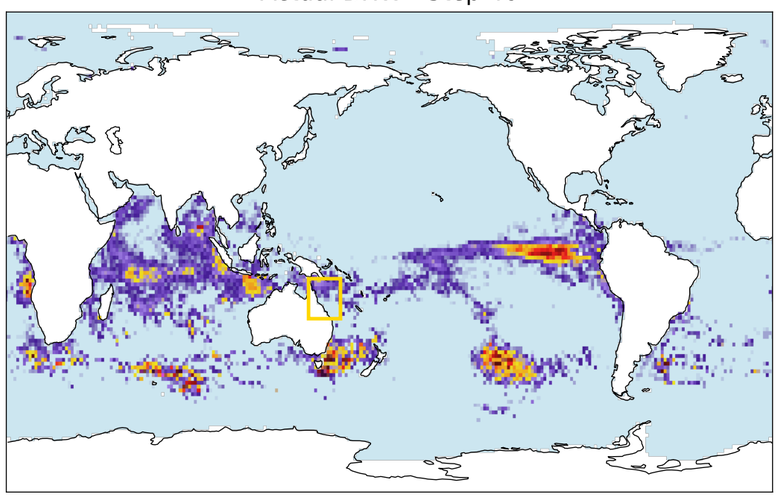
Predicted DHW - Step 1



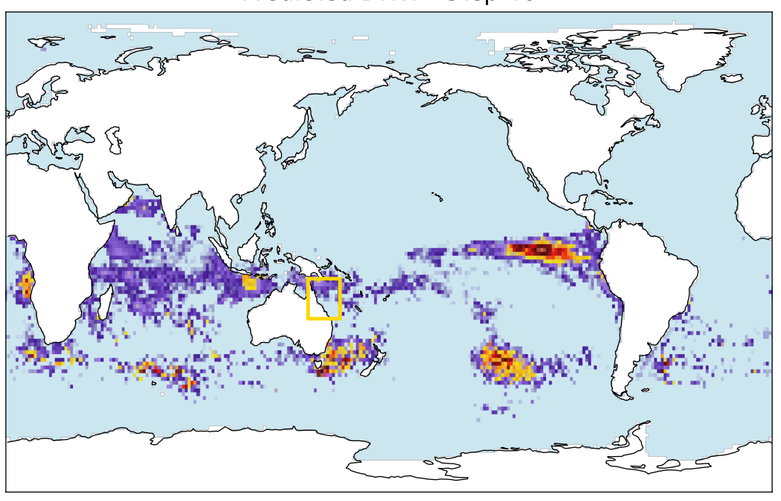
Error (Pred - Actual) - Step 1 [scale ±0.48]



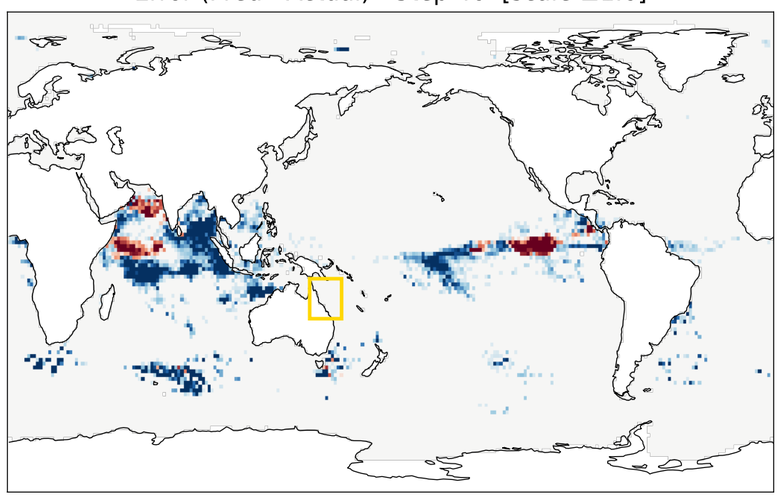
Actual DHW - Step 46



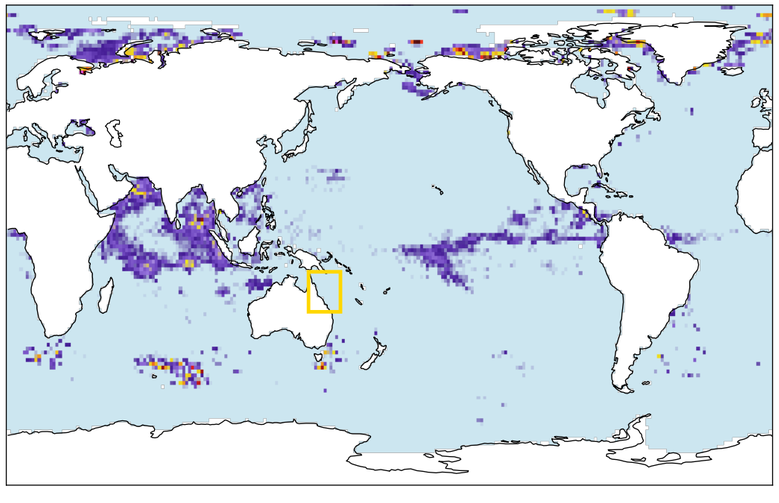
Predicted DHW - Step 46



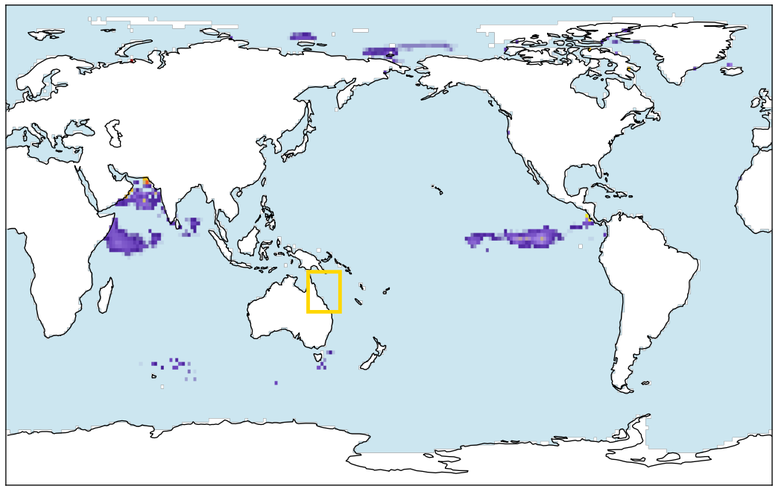
Error (Pred - Actual) - Step 46 [scale ±1.6]



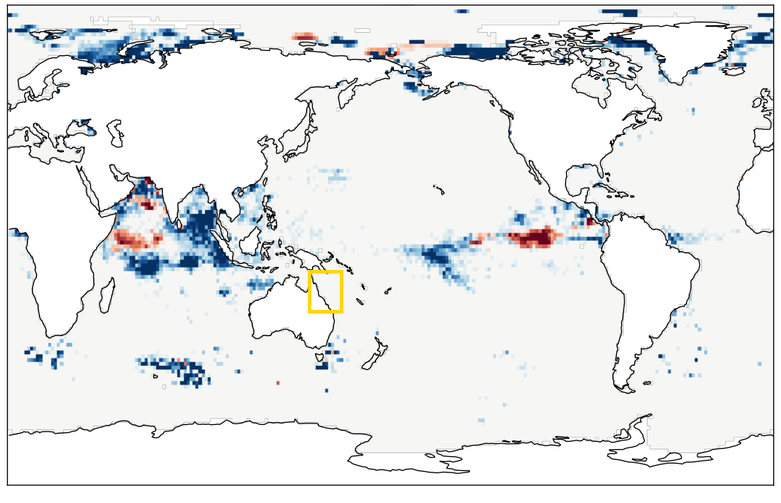
Actual DHW - Step 90

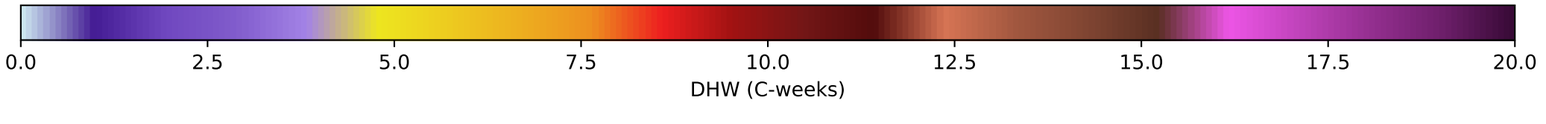


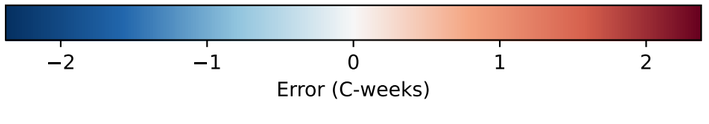
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.4]

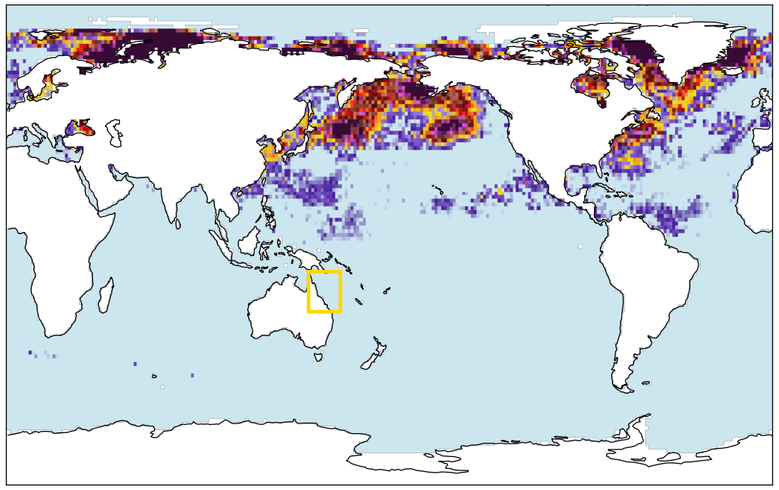




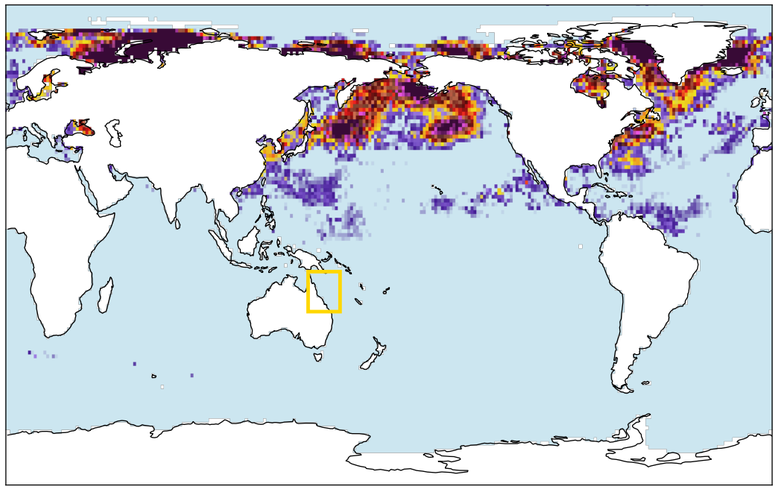


DHW Chain init: 2016-10-07 - Val (C-weeks)

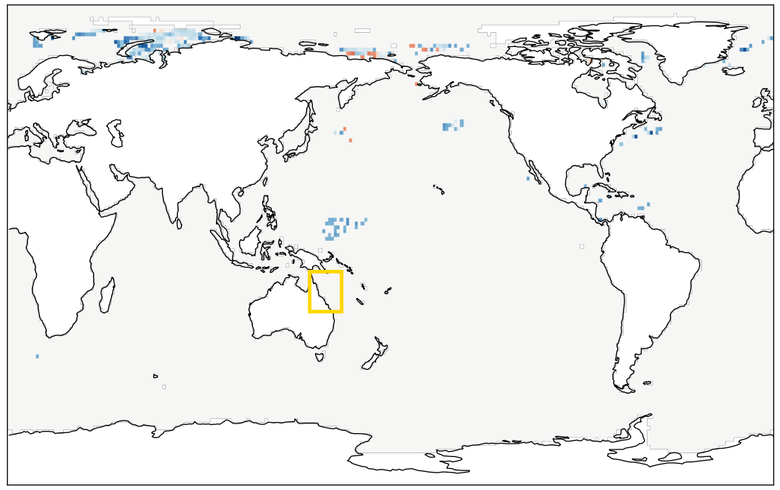
Actual DHW - Step 1



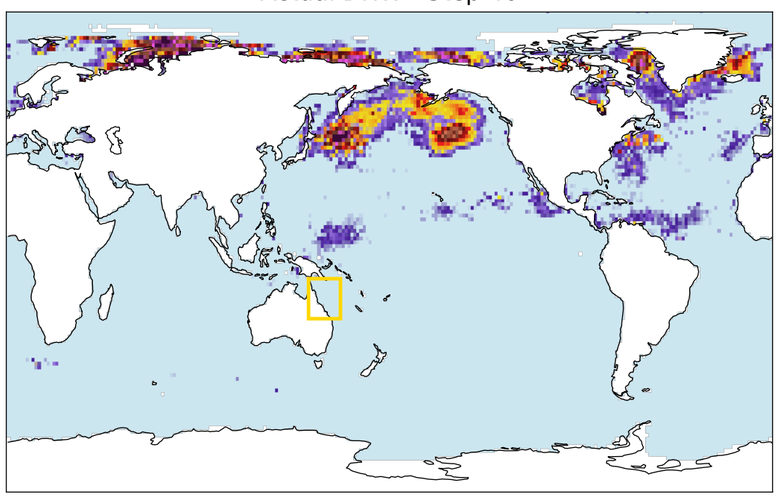
Predicted DHW - Step 1



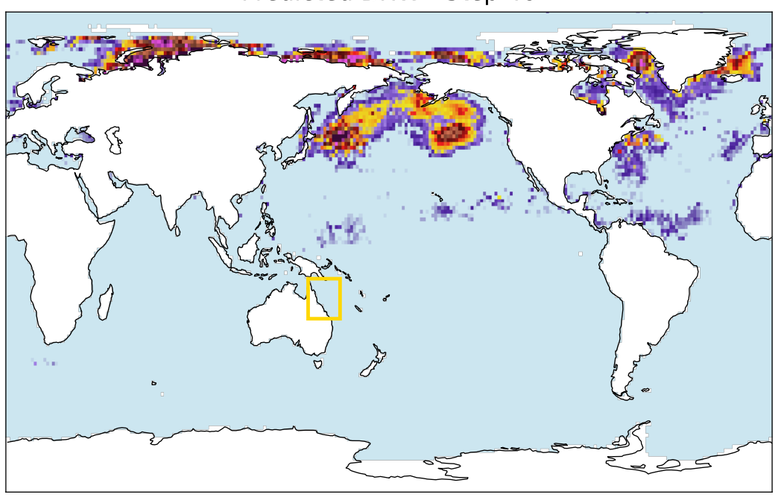
Error (Pred - Actual) - Step 1 [scale ±0.37]



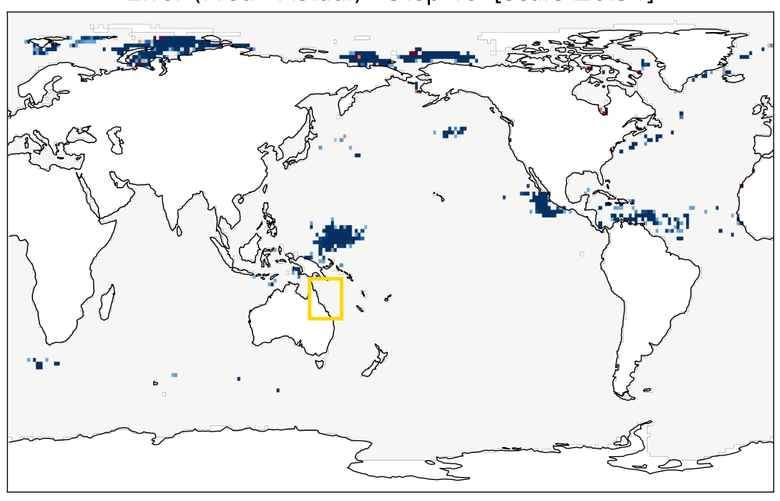
Actual DHW - Step 46



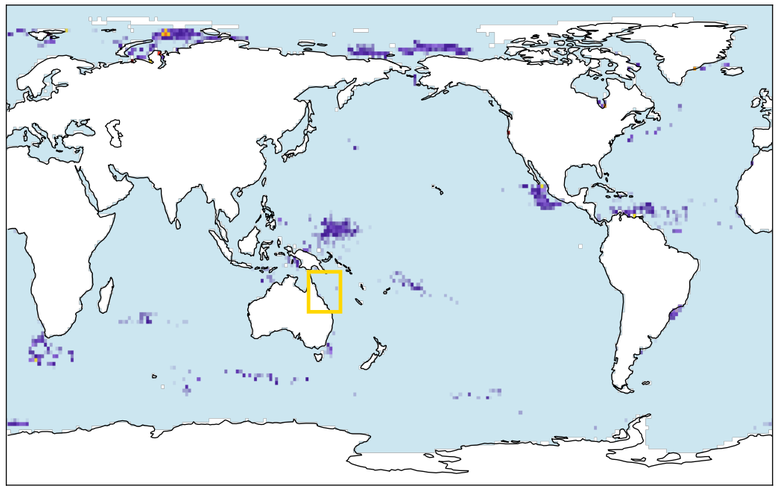
Predicted DHW - Step 46



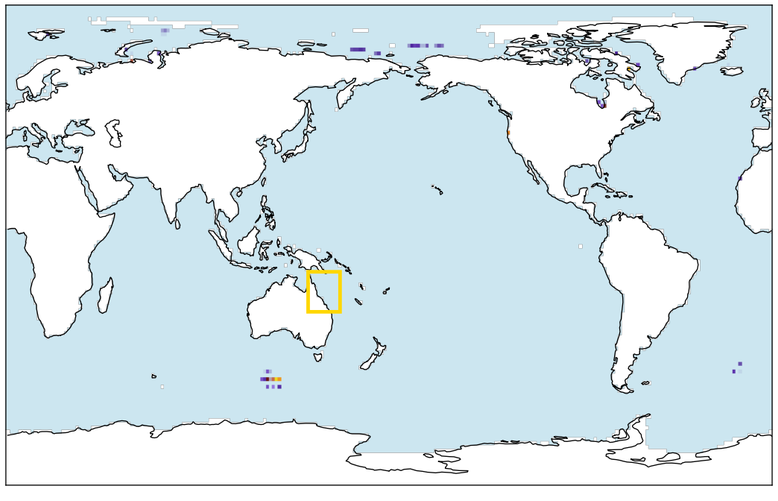
Error (Pred - Actual) - Step 46 [scale ±0.34]



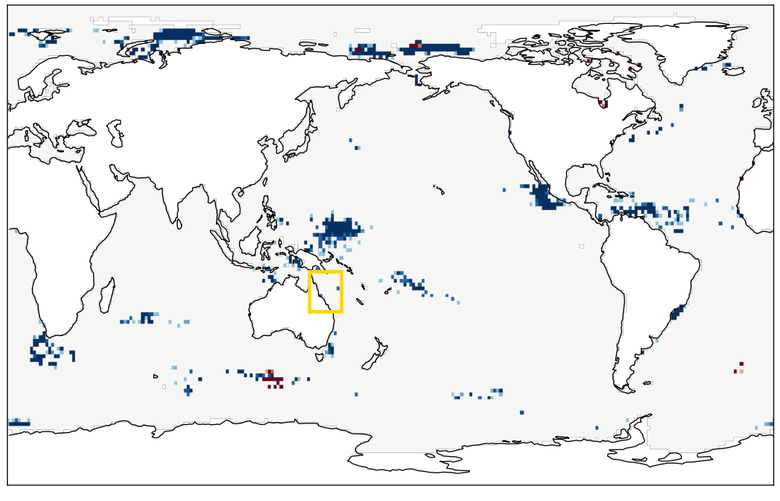
Actual DHW - Step 90

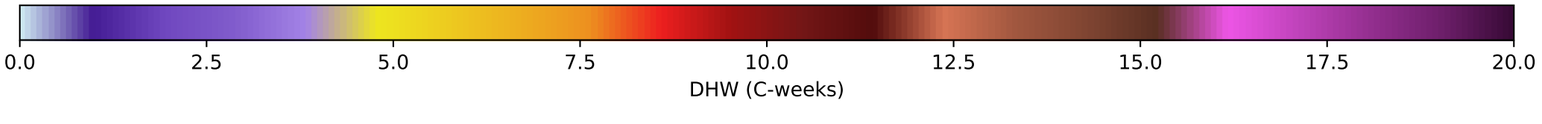


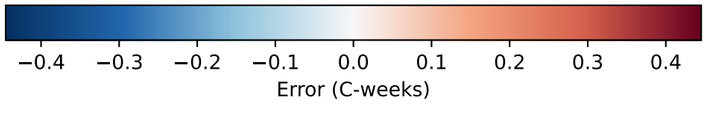
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.45]

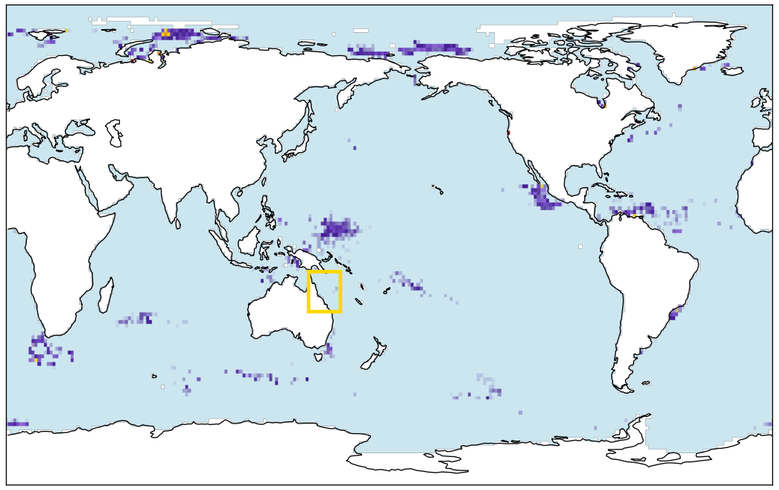




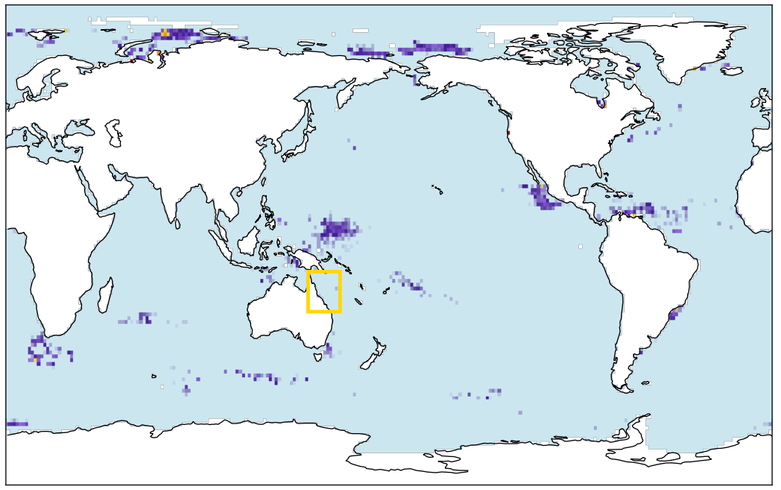


DHW Chain init: 2017-01-05 - Val (C-weeks)

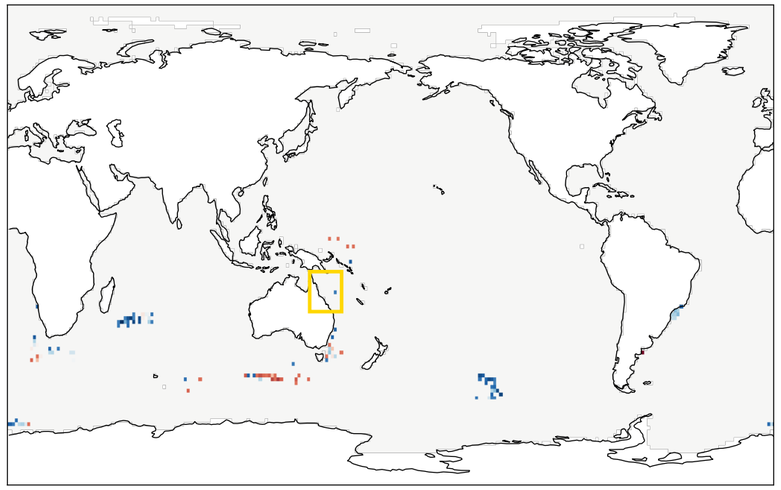
Actual DHW - Step 1



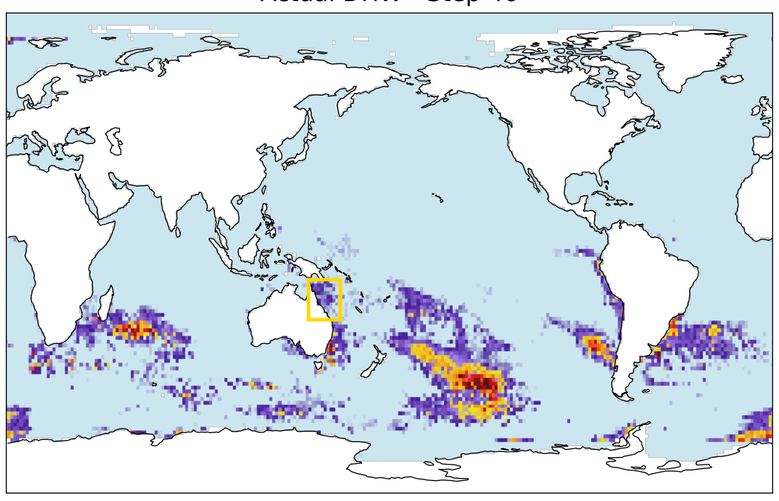
Predicted DHW - Step 1



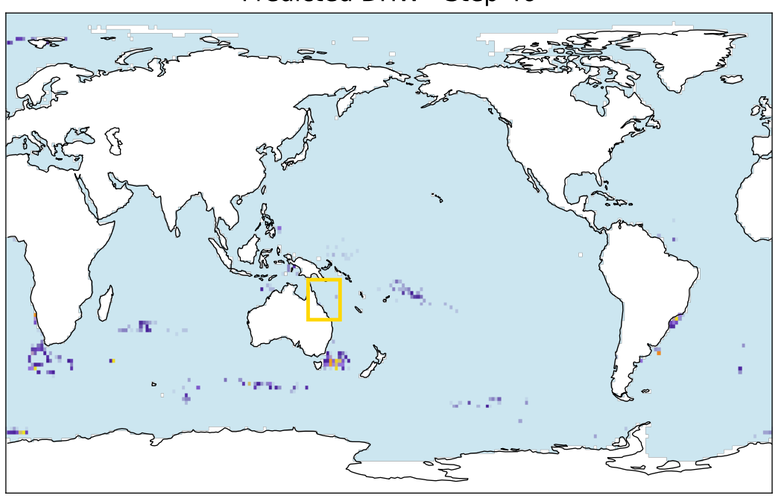
Error (Pred - Actual) - Step 1 [scale ±0.26]



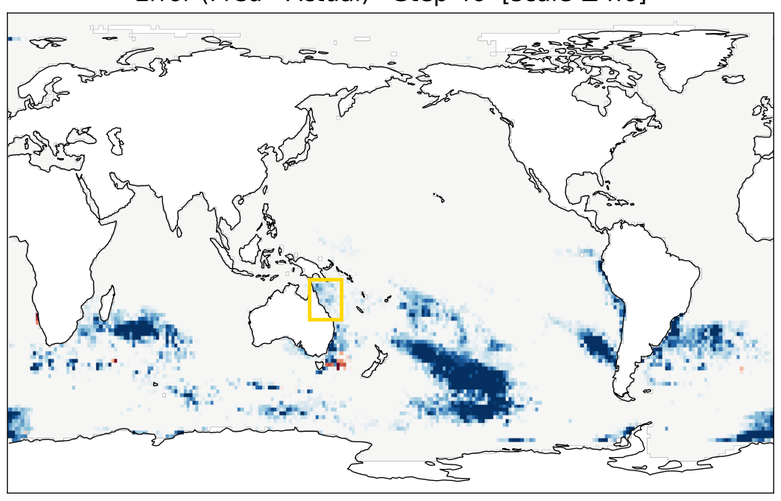
Actual DHW - Step 46



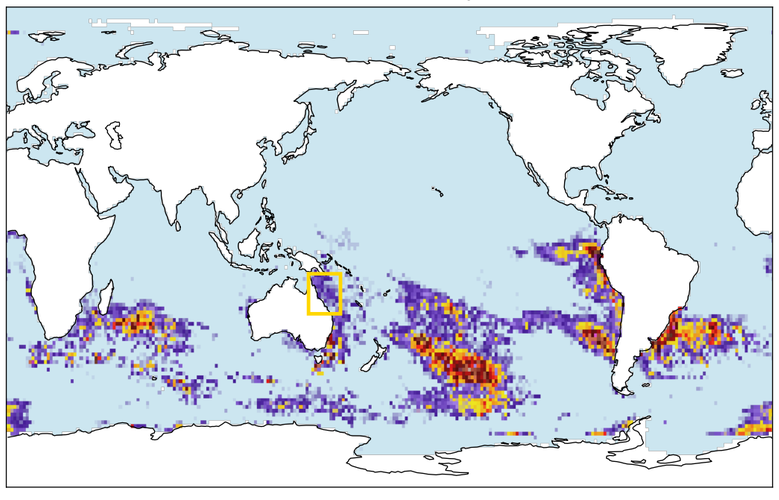
Predicted DHW - Step 46



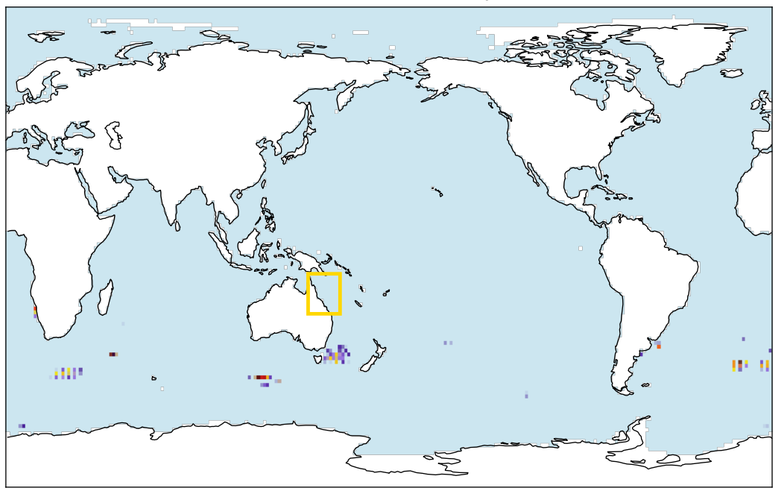
Error (Pred - Actual) - Step 46 [scale ±4.6]



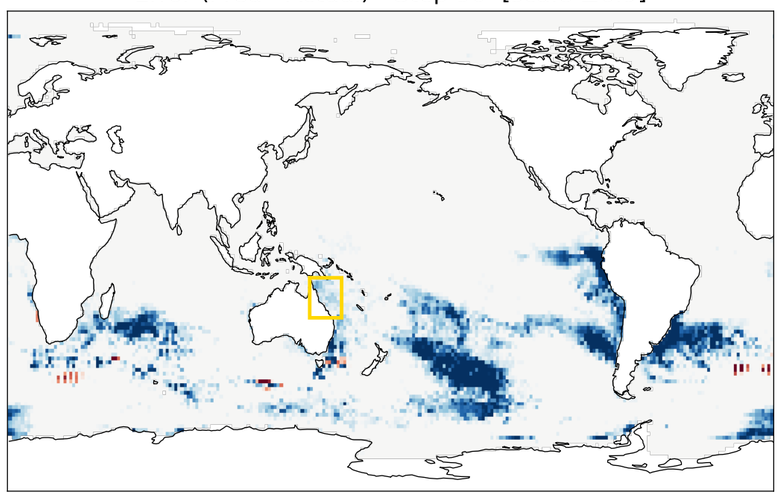
Actual DHW - Step 90

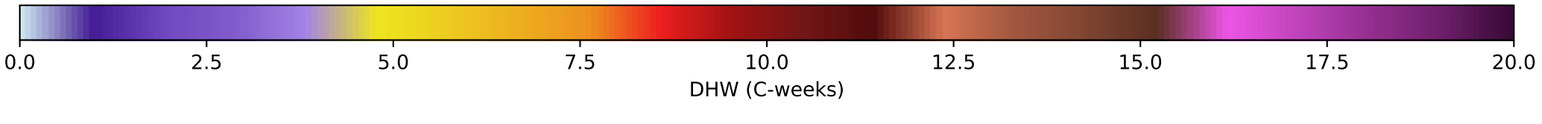


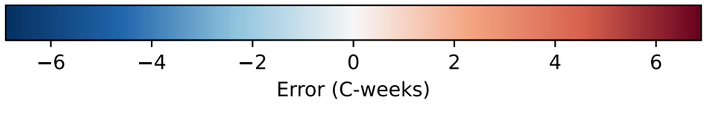
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.9]

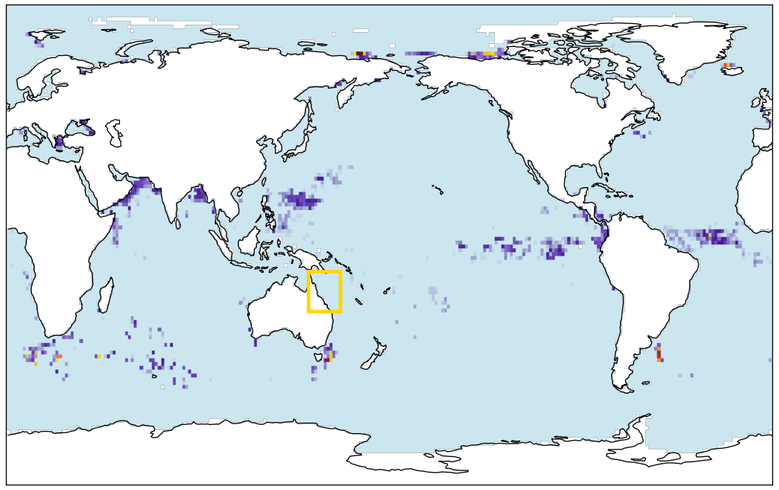




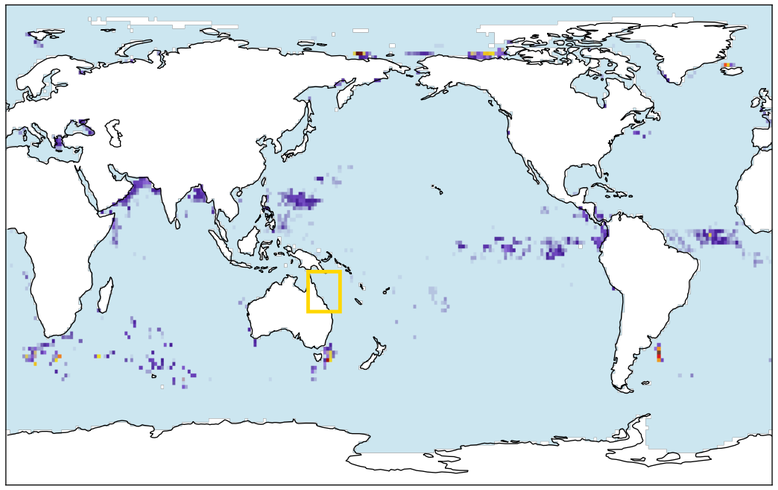


DHW Chain init: 2017-07-04 - Val (C-weeks)

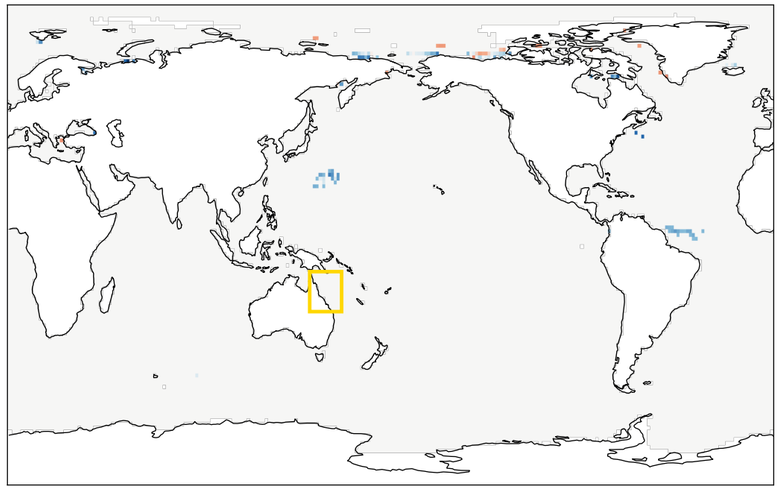
Actual DHW - Step 1



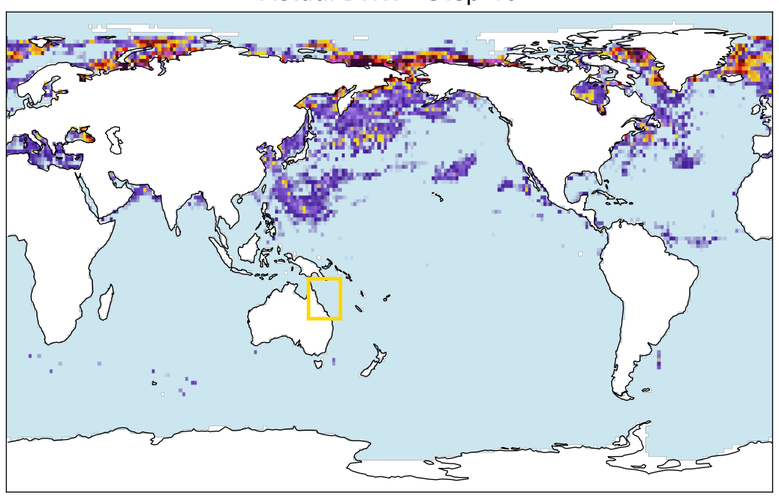
Predicted DHW - Step 1



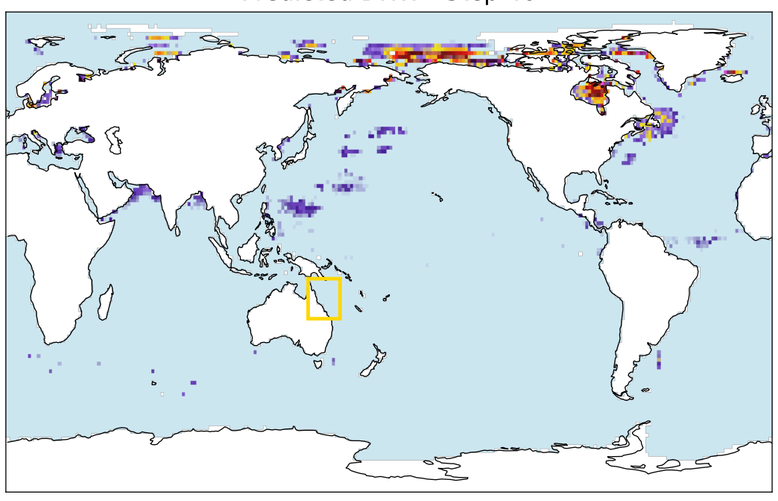
Error (Pred - Actual) - Step 1 [scale ±0.4]



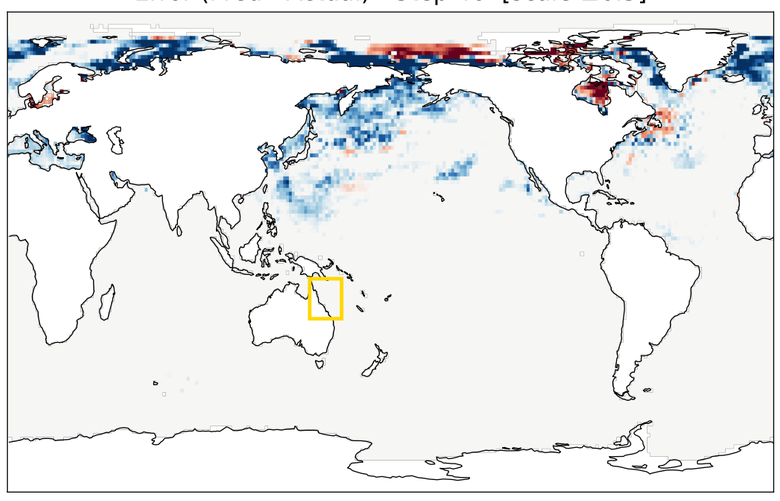
Actual DHW - Step 46



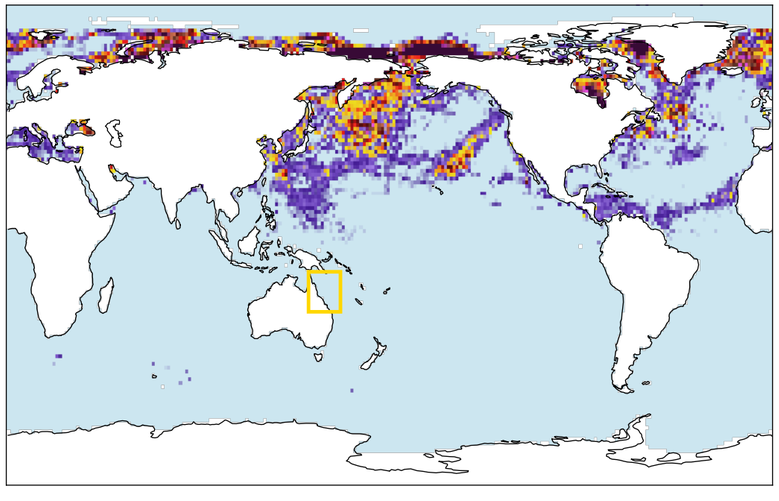
Predicted DHW - Step 46



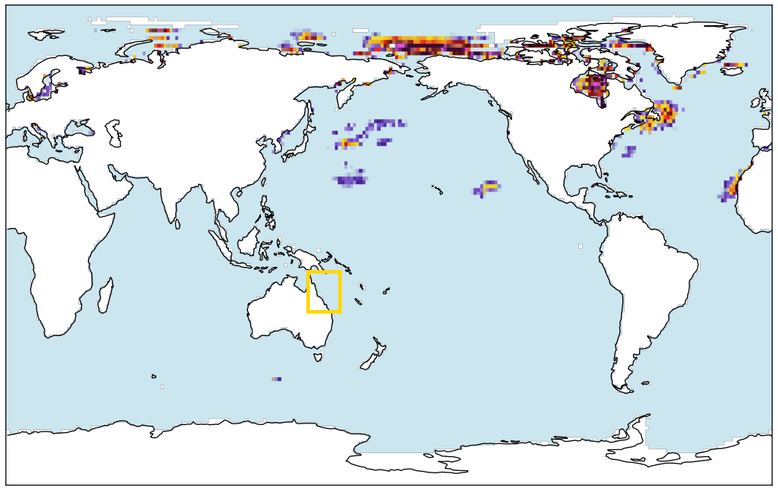
Error (Pred - Actual) - Step 46 [scale ±6.5]



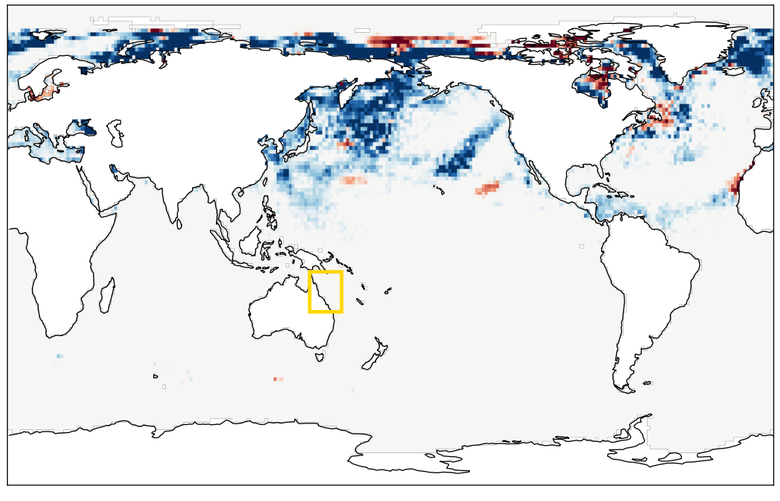
Actual DHW - Step 90

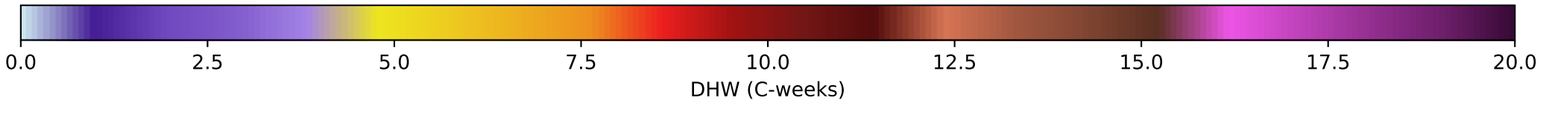


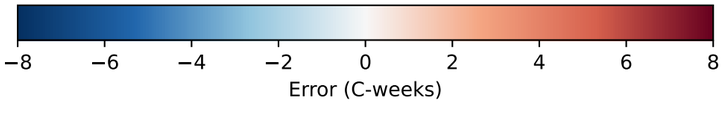
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

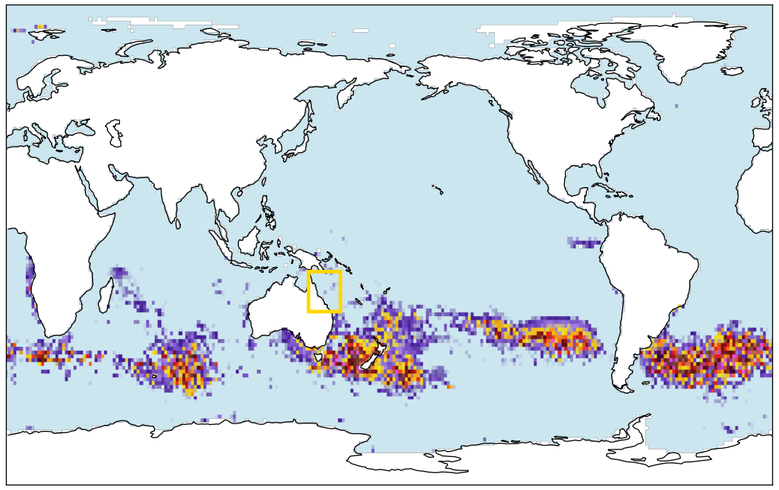




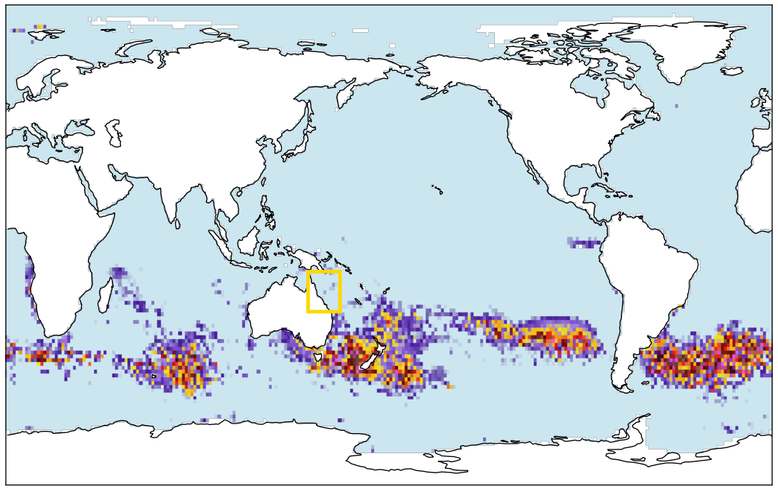


DHW Chain init: 2018-03-31 - Val (C-weeks)

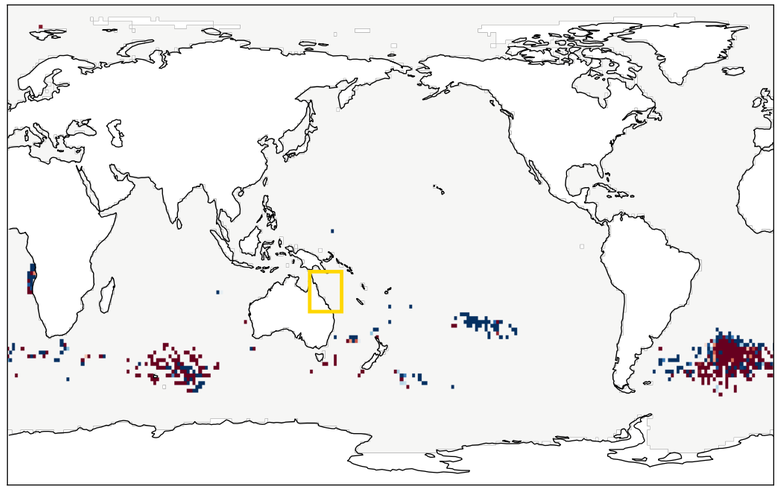
Actual DHW - Step 1



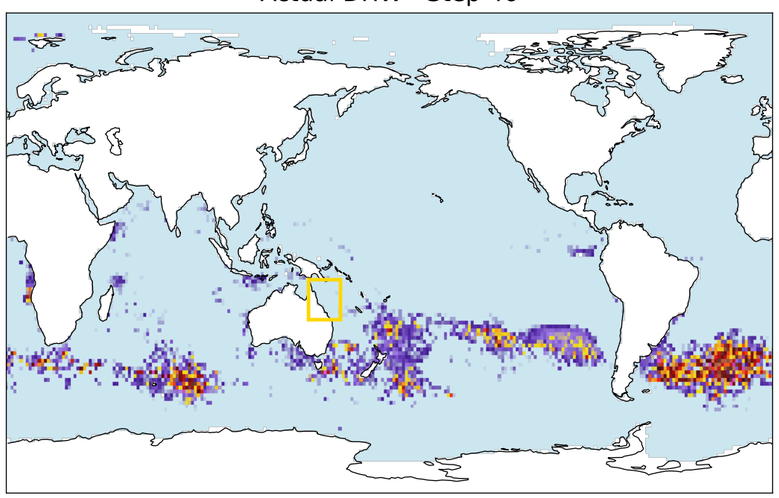
Predicted DHW - Step 1



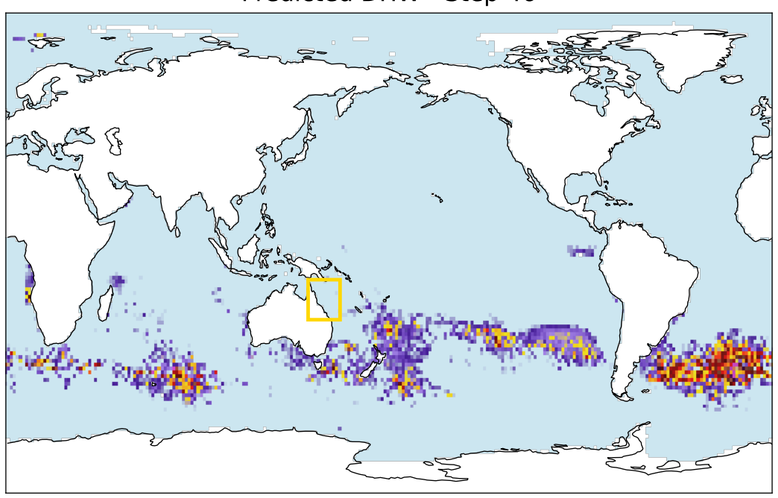
Error (Pred - Actual) - Step 1 [scale ±0.0076]



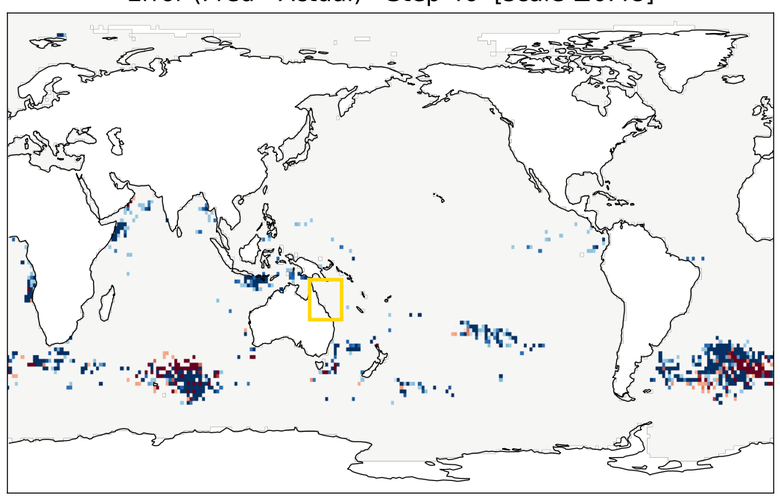
Actual DHW - Step 46



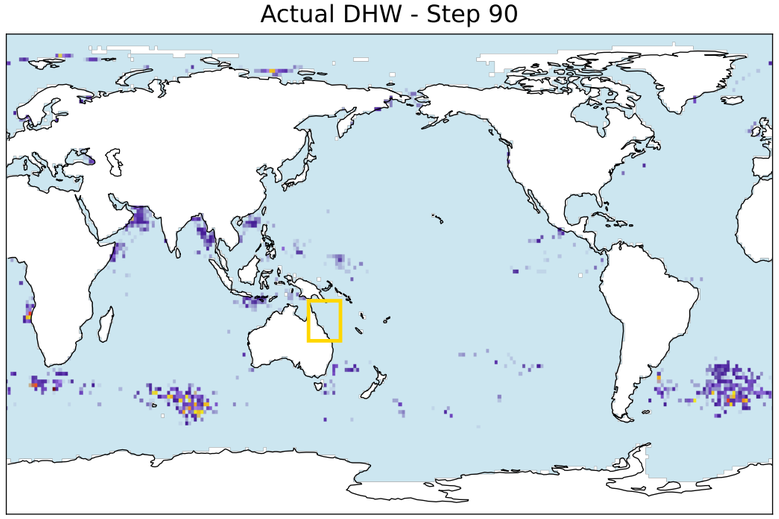
Predicted DHW - Step 46



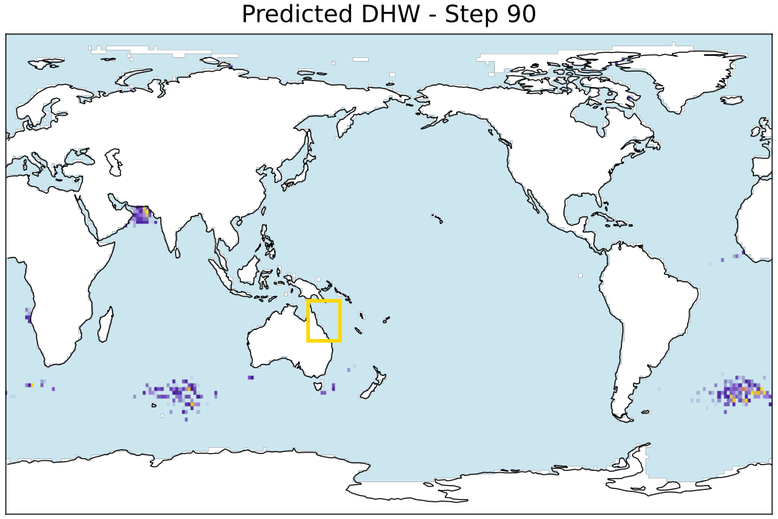
Error (Pred - Actual) - Step 46 [scale ±0.48]



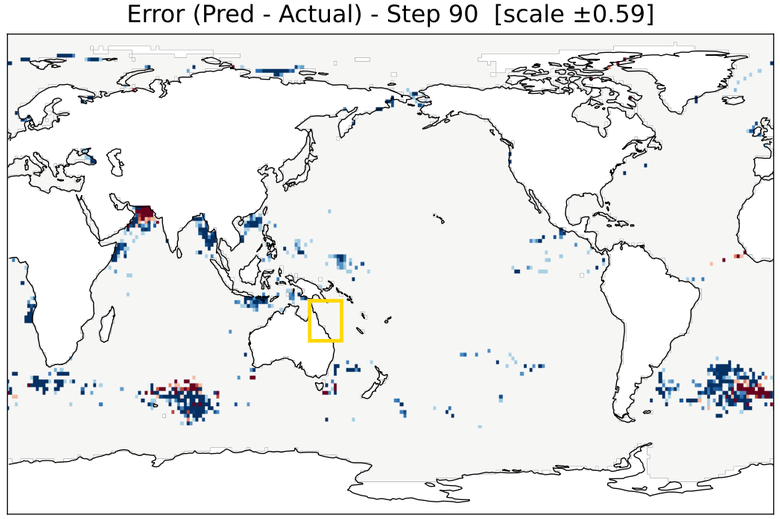
Actual DHW - Step 90

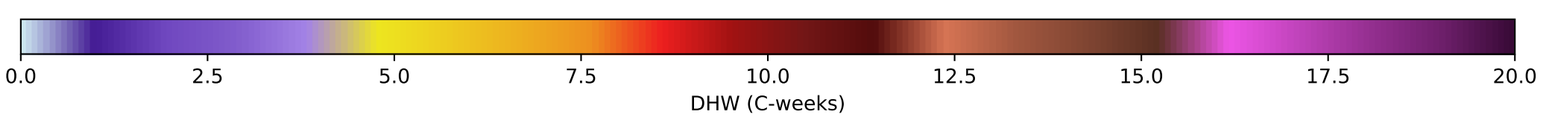


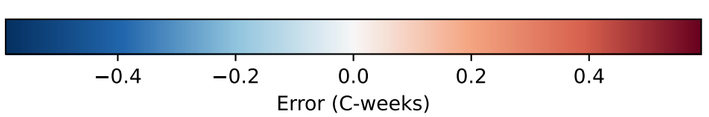
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.59]

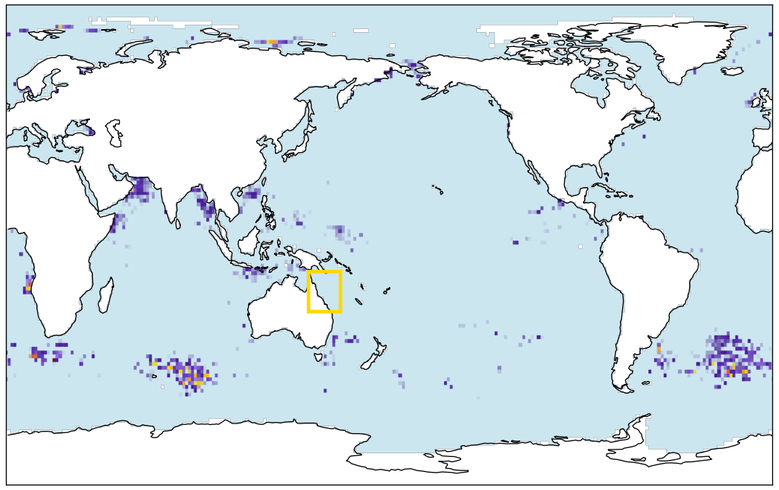




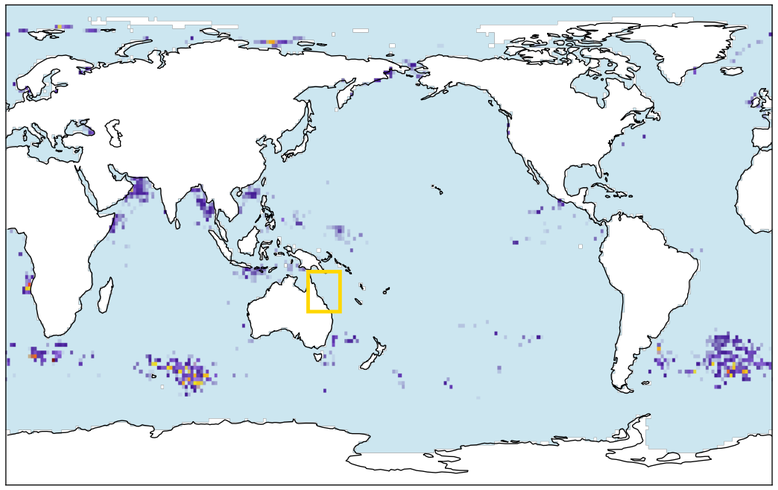


DHW Chain init: 2018-06-29 - Val (C-weeks)

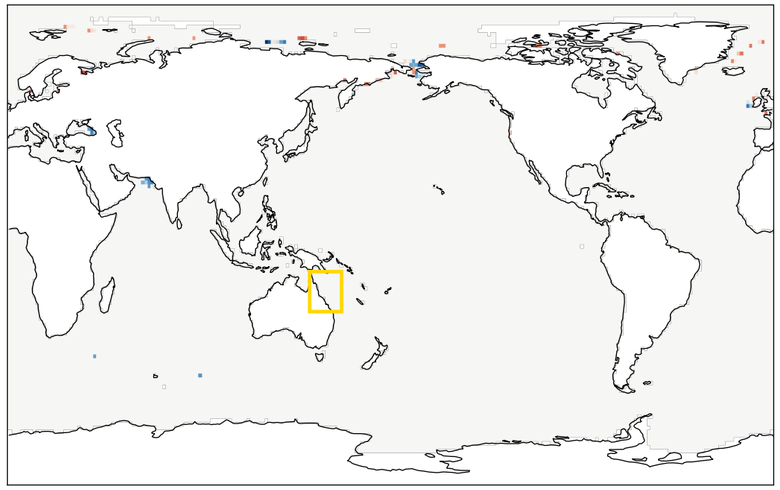
Actual DHW - Step 1



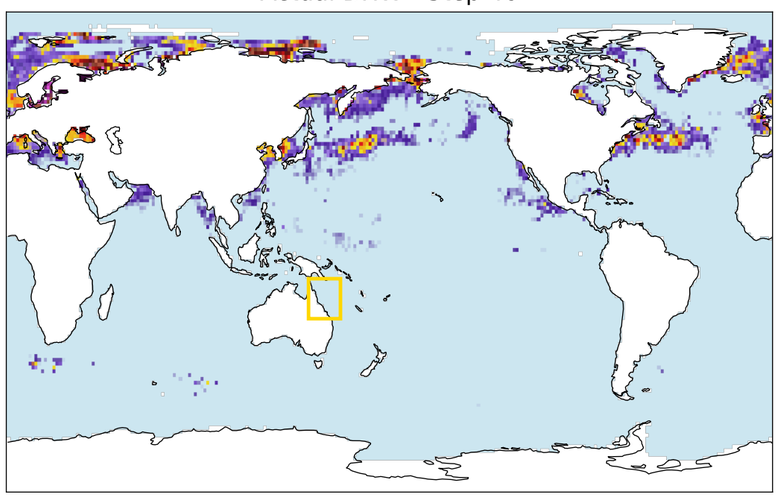
Predicted DHW - Step 1



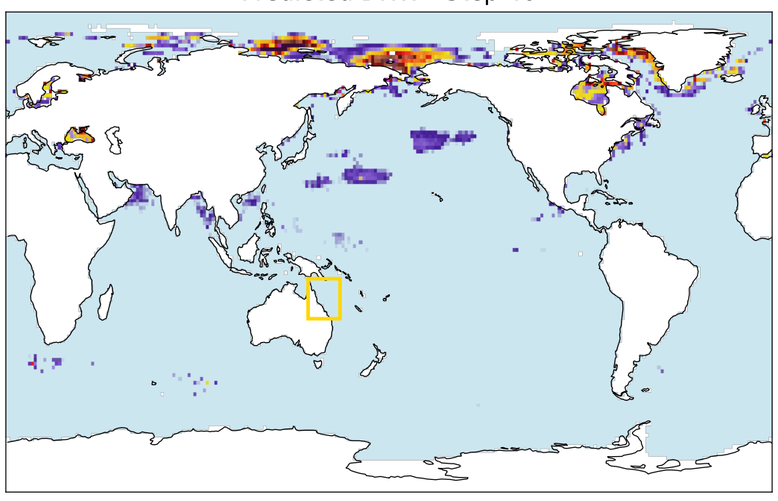
Error (Pred - Actual) - Step 1 [scale ±0.36]



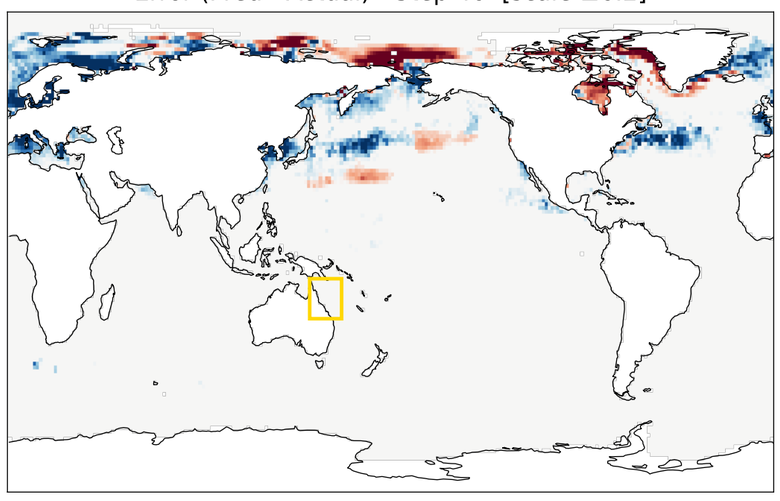
Actual DHW - Step 46



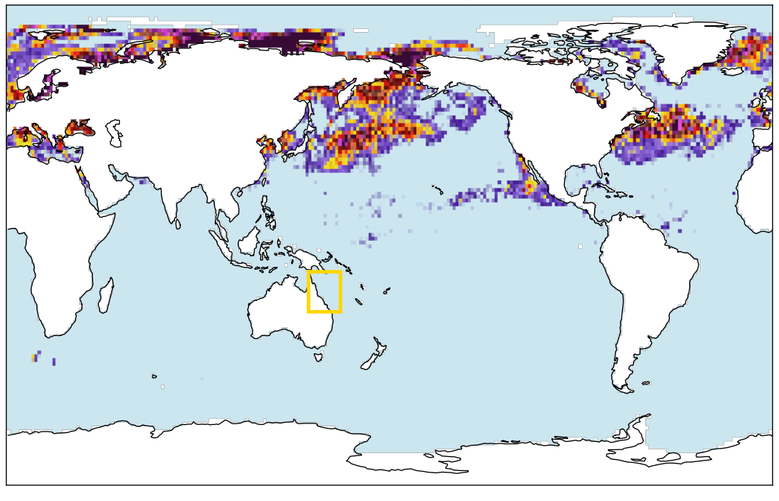
Predicted DHW - Step 46



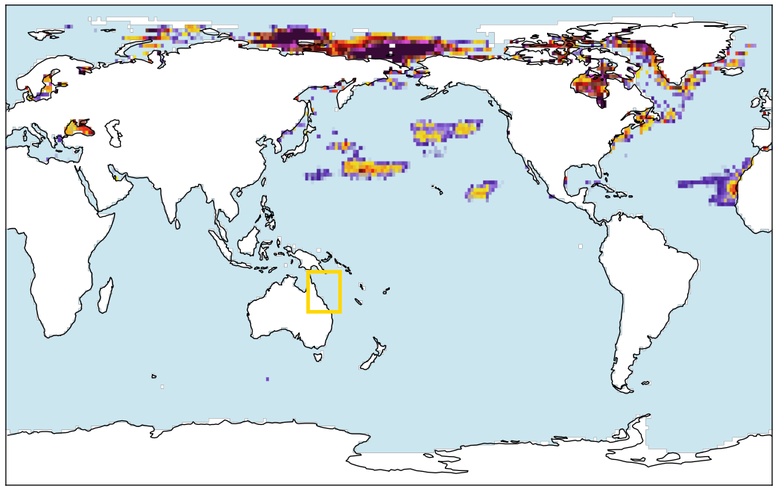
Error (Pred - Actual) - Step 46 [scale ±6.2]



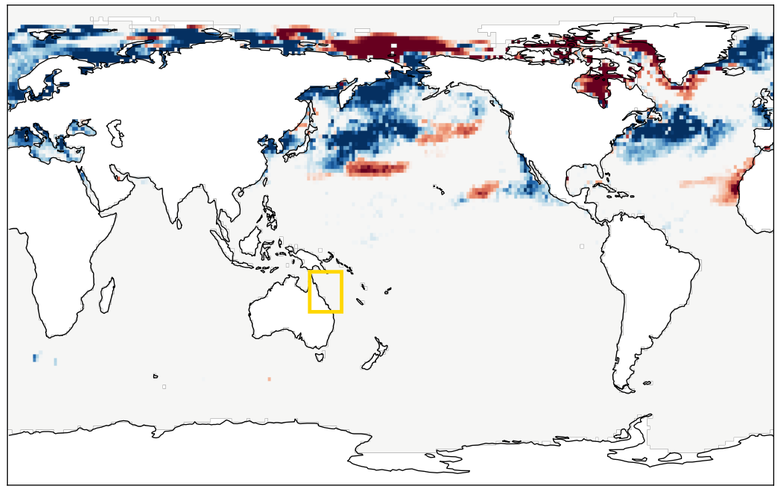
Actual DHW - Step 90

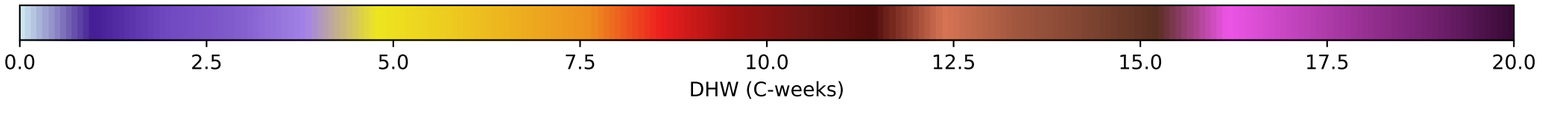


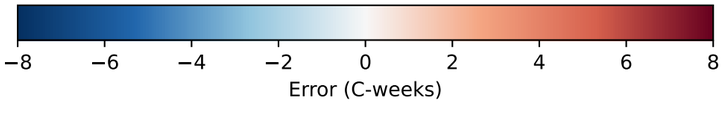
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

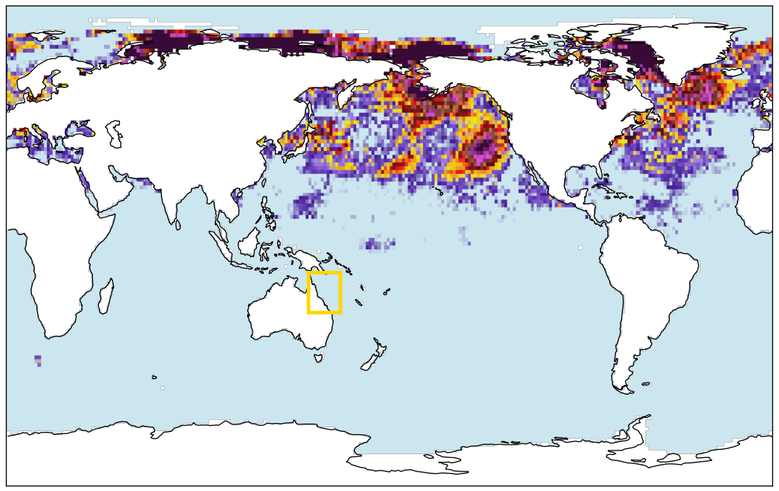




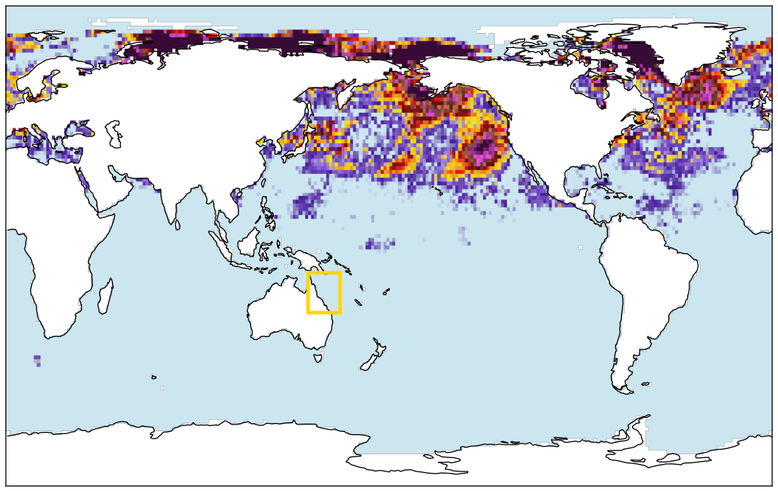


DHW Chain init: 2019-09-22 - Val (C-weeks)

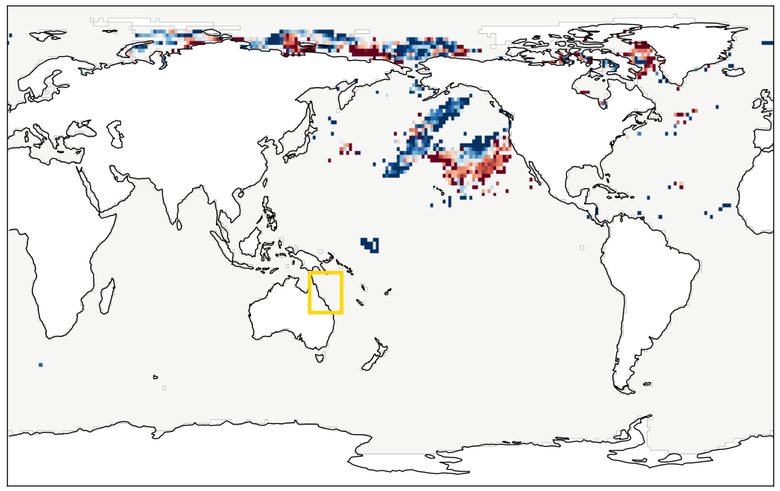
Actual DHW - Step 1



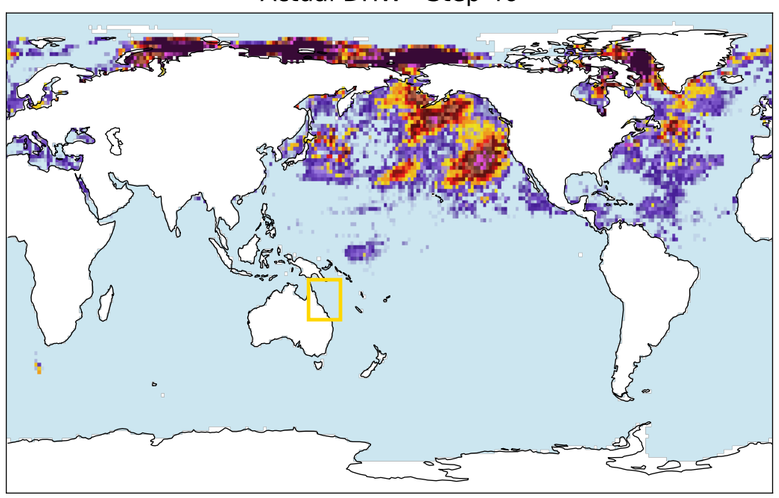
Predicted DHW - Step 1



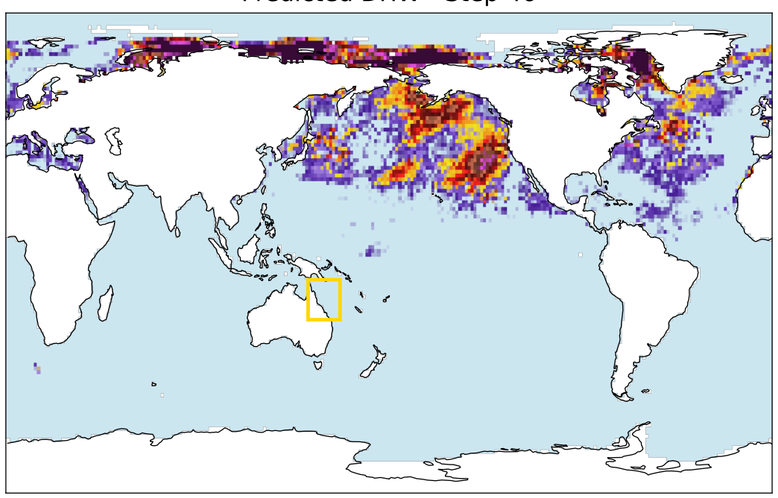
Error (Pred - Actual) - Step 1 [scale ±0.058]



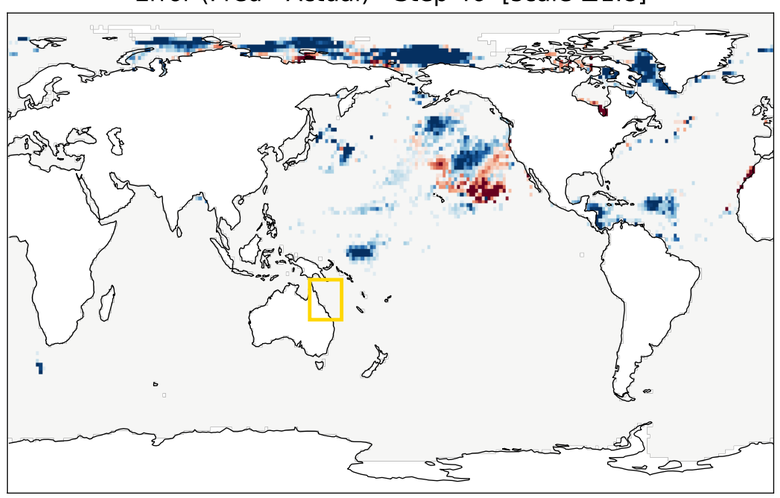
Actual DHW - Step 46



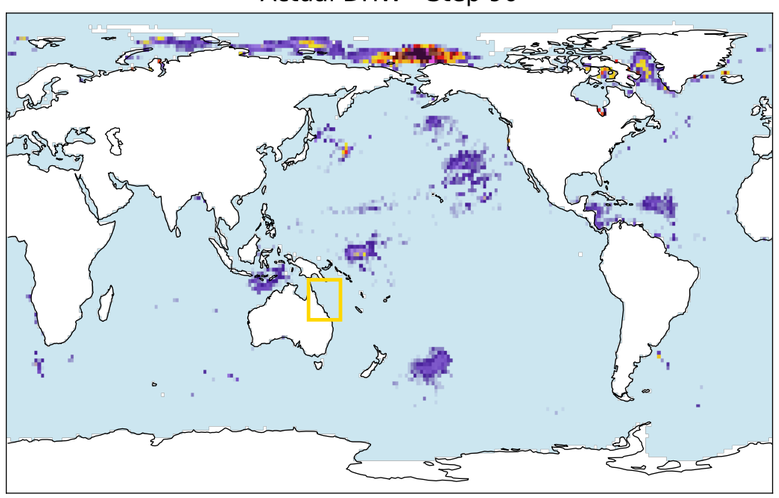
Predicted DHW - Step 46



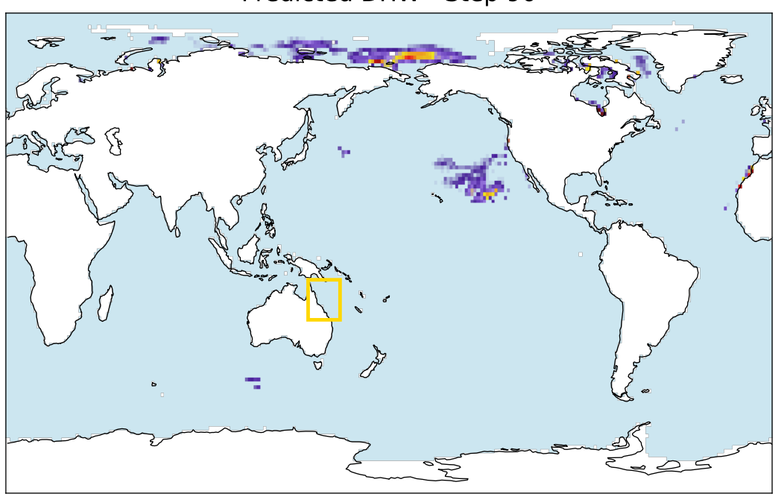
Error (Pred - Actual) - Step 46 [scale ±1.8]



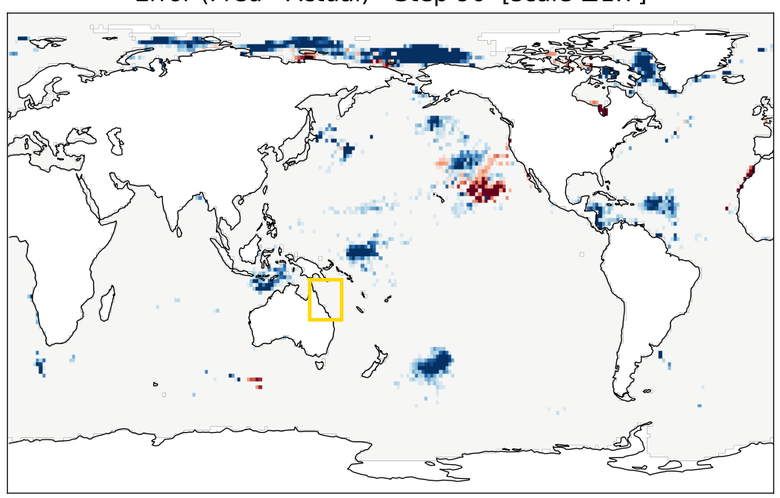
Actual DHW - Step 90

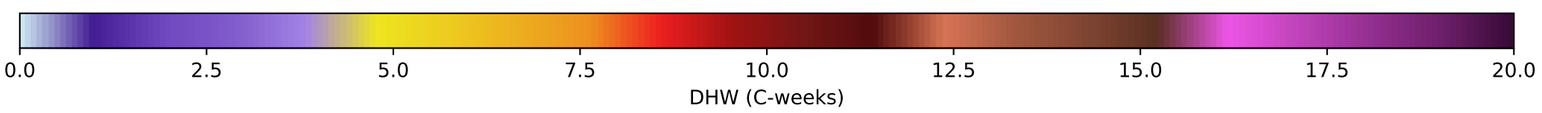


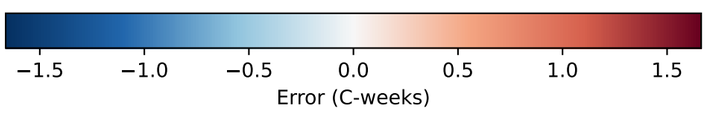
Predicted DHW - Step 90



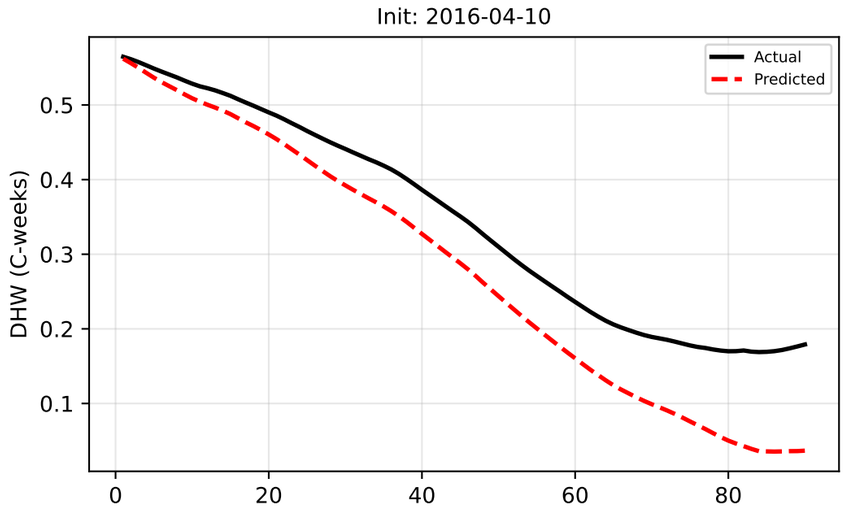
Error (Pred - Actual) - Step 90 [scale ±1.7]

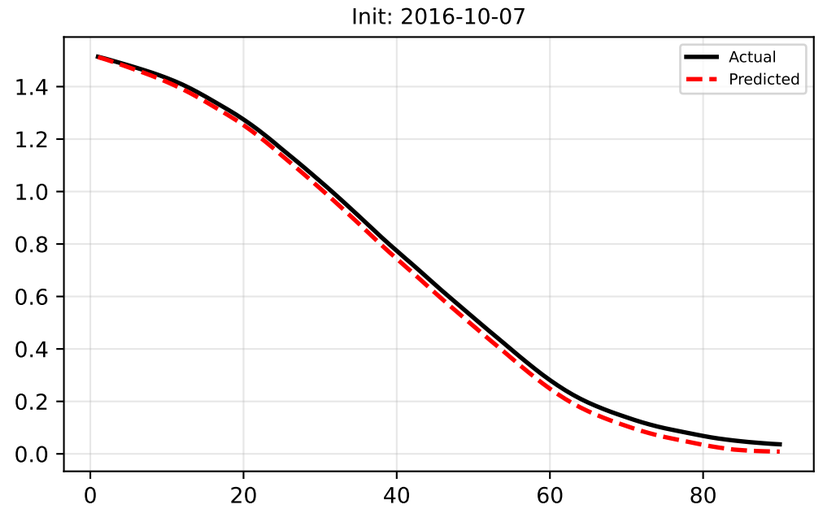


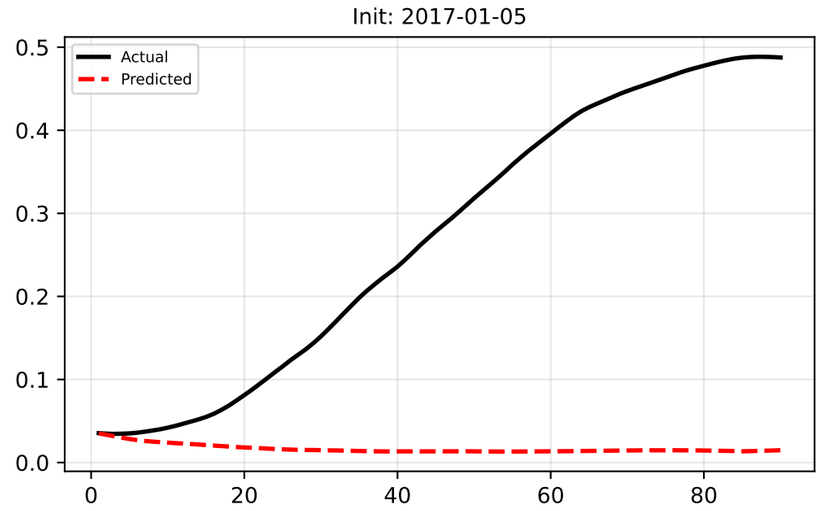


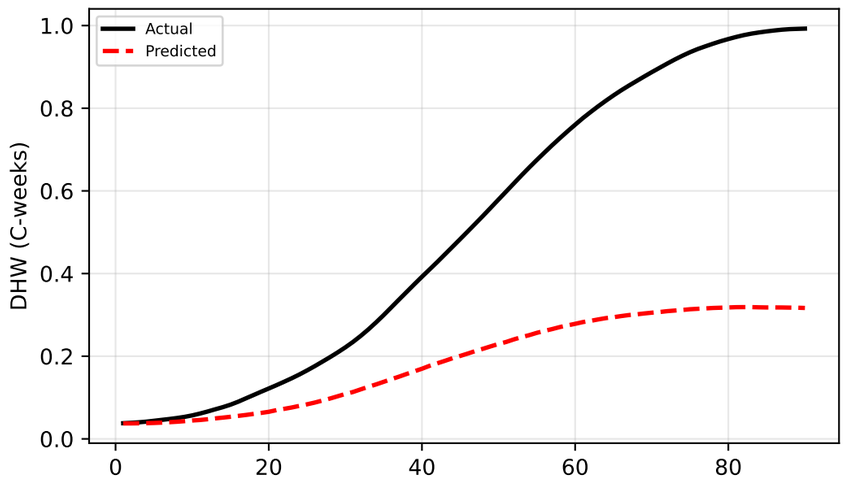


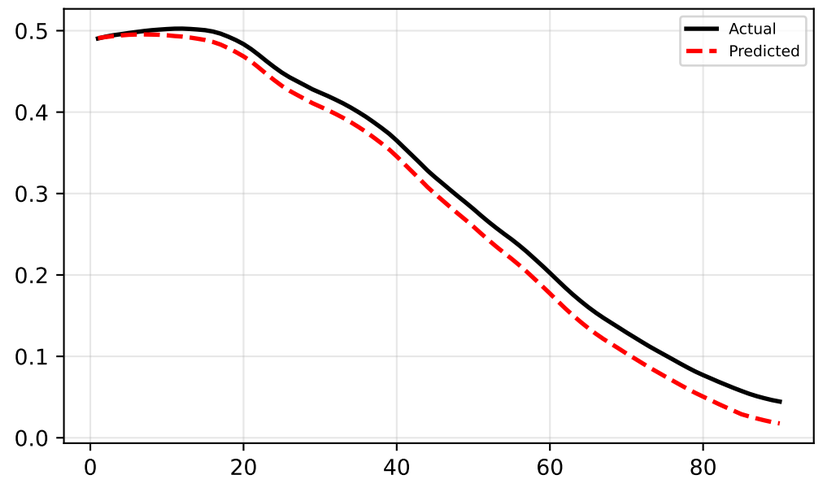
Global-Mean DHW - Val (C-weeks)

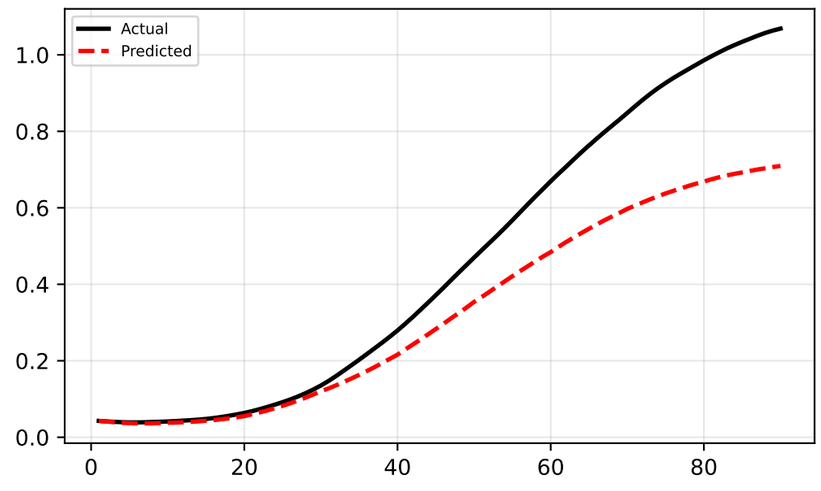


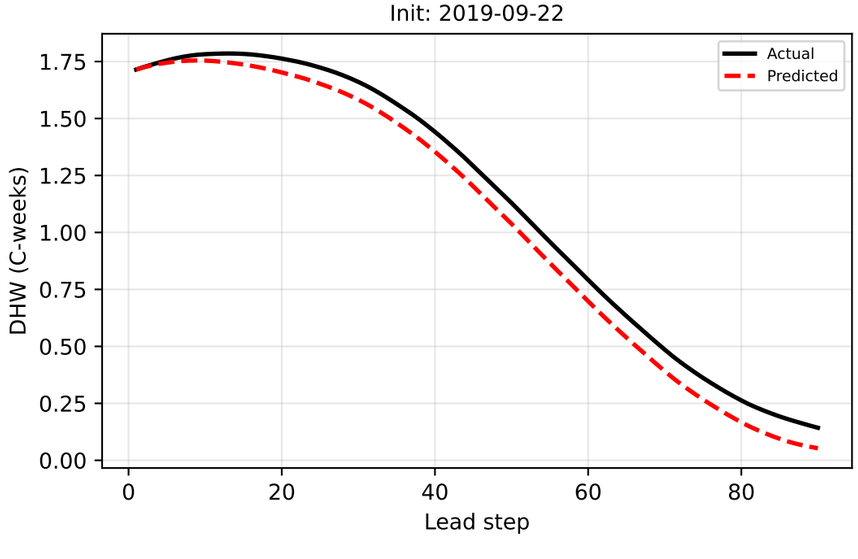




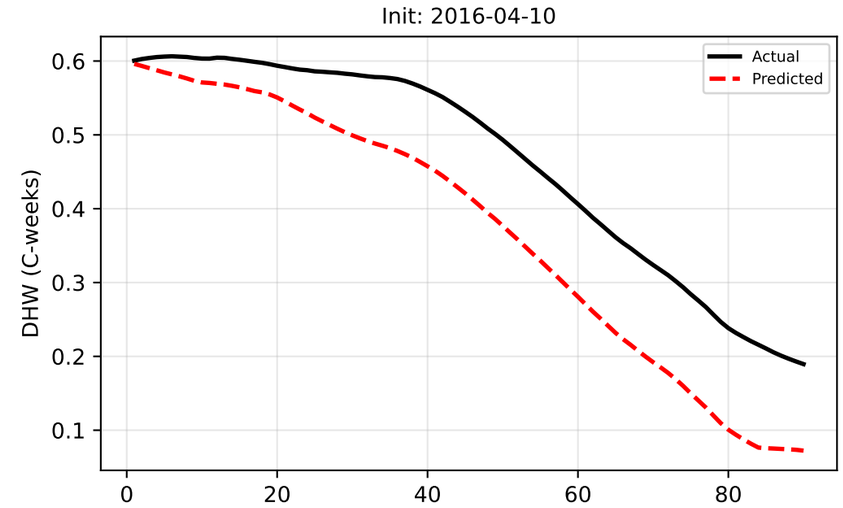


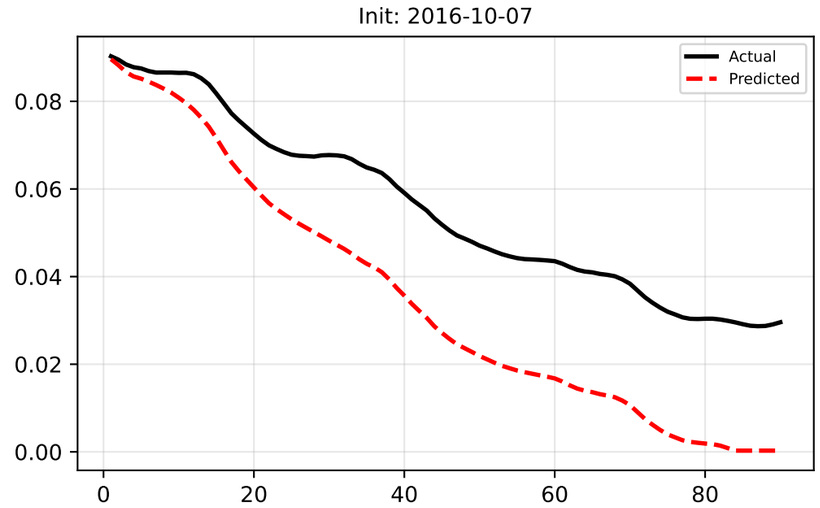


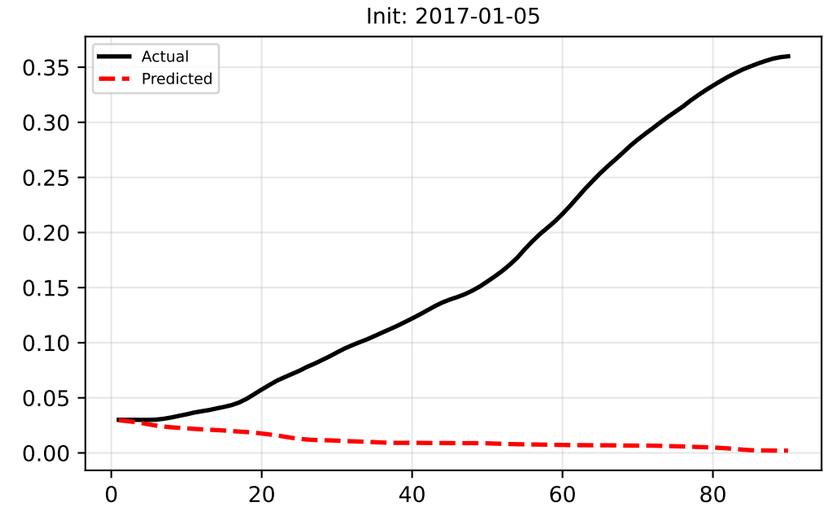


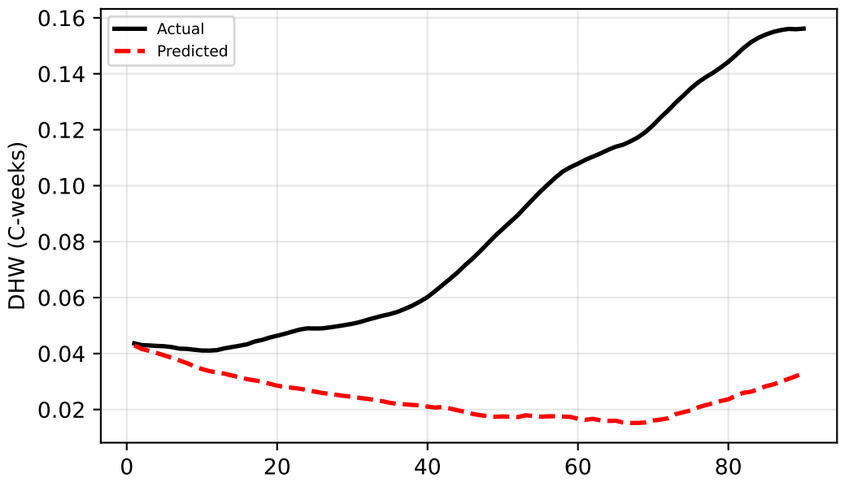


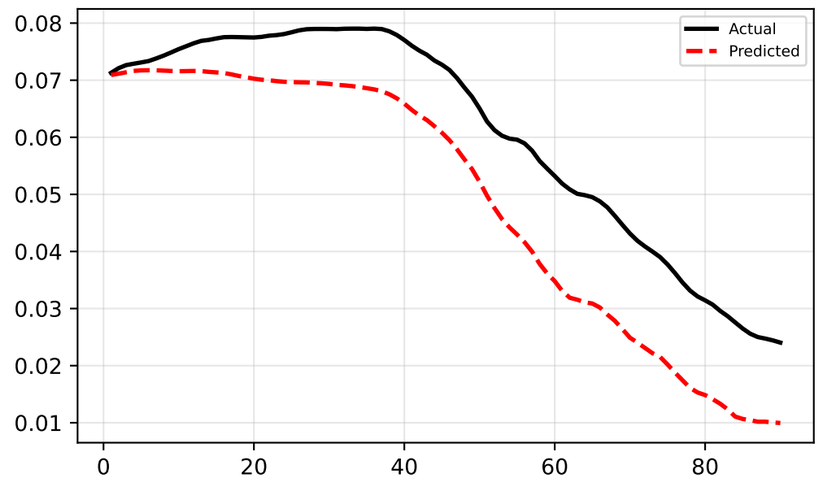
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

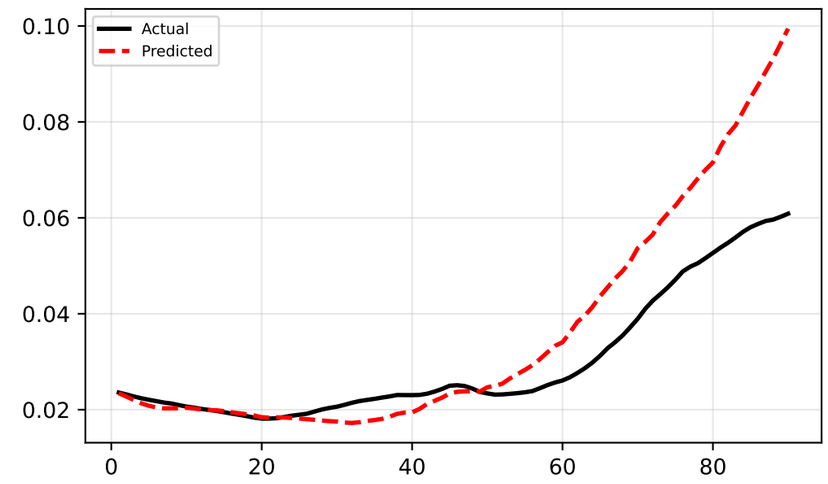


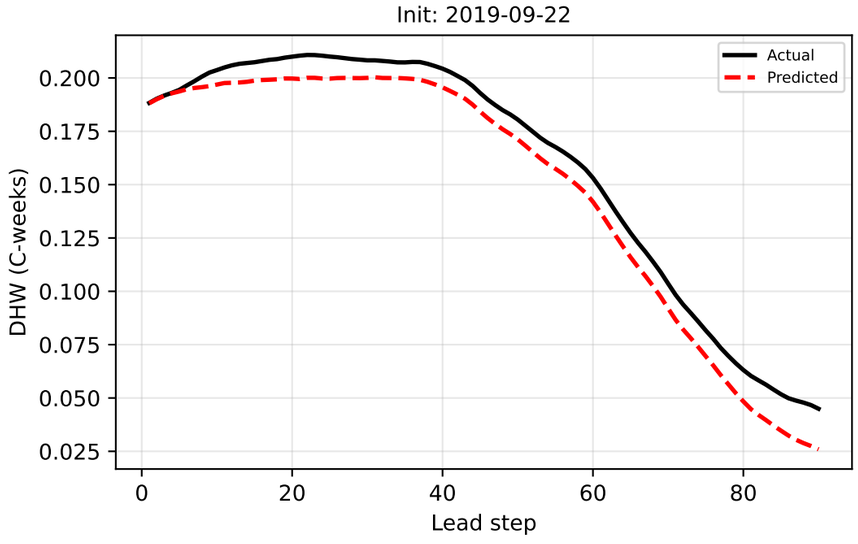




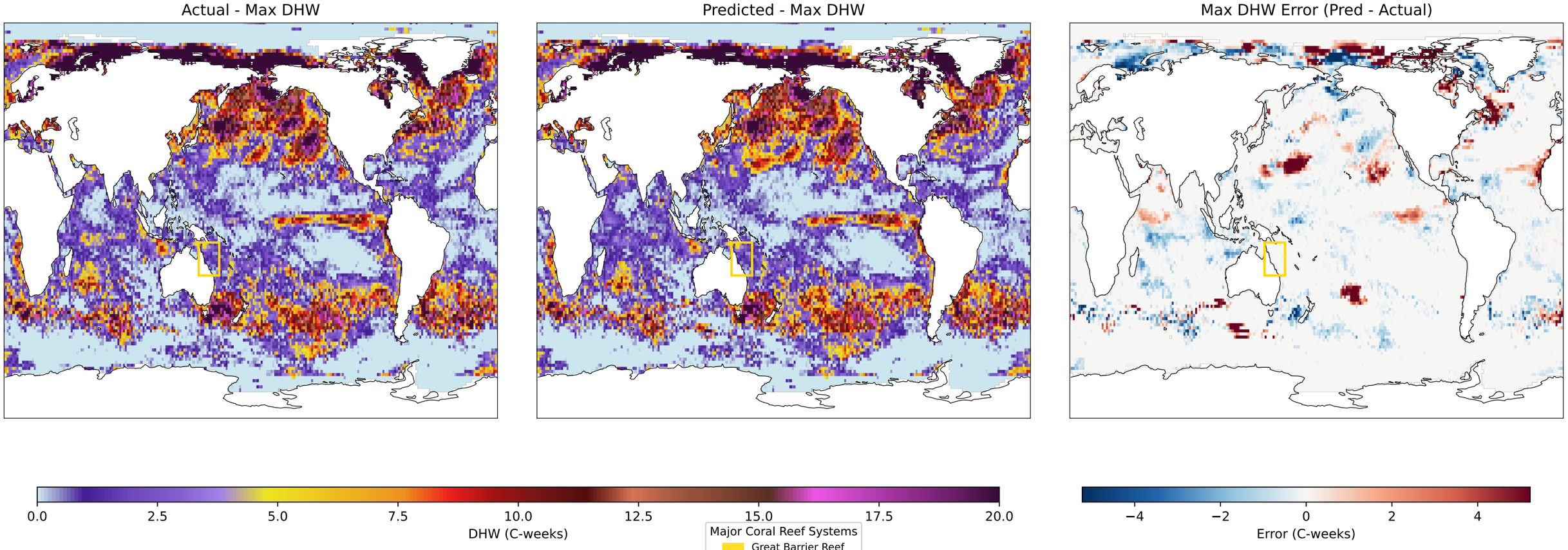


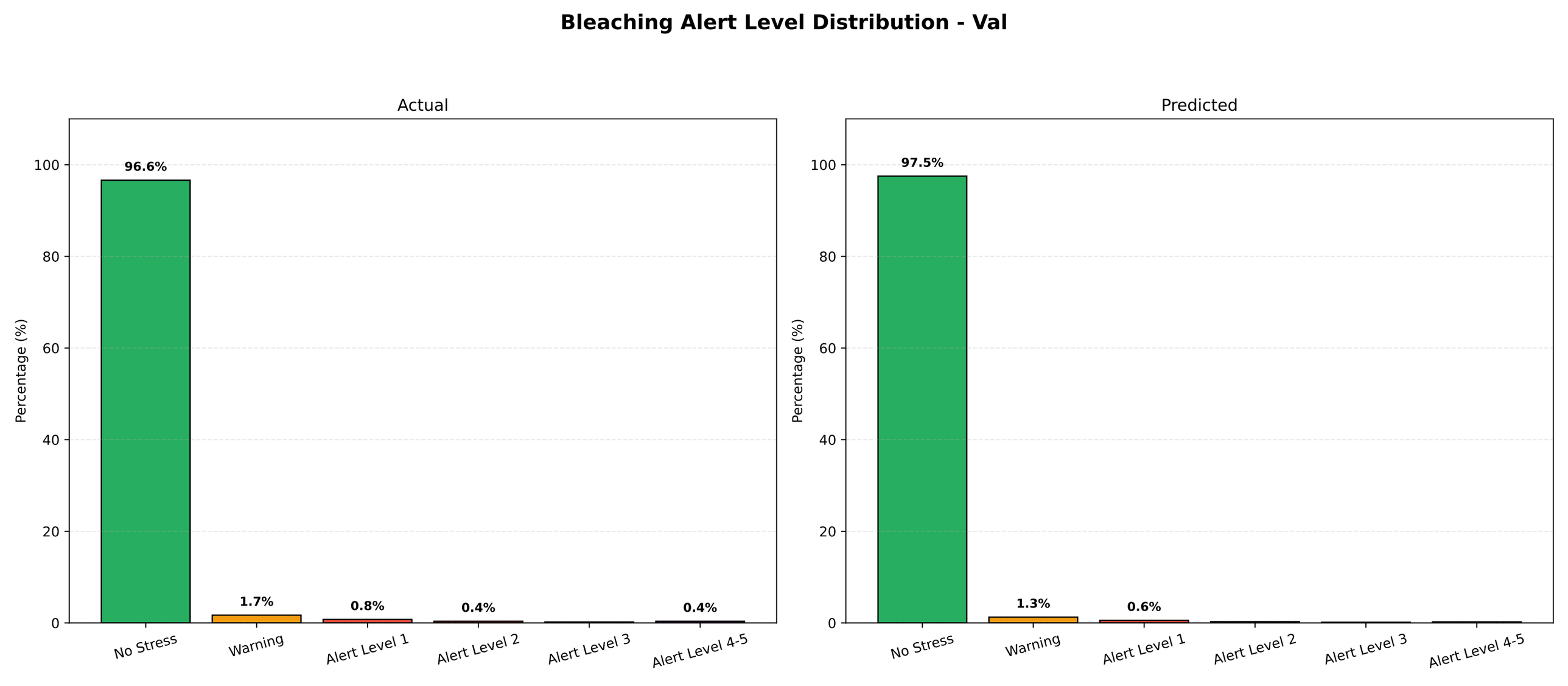




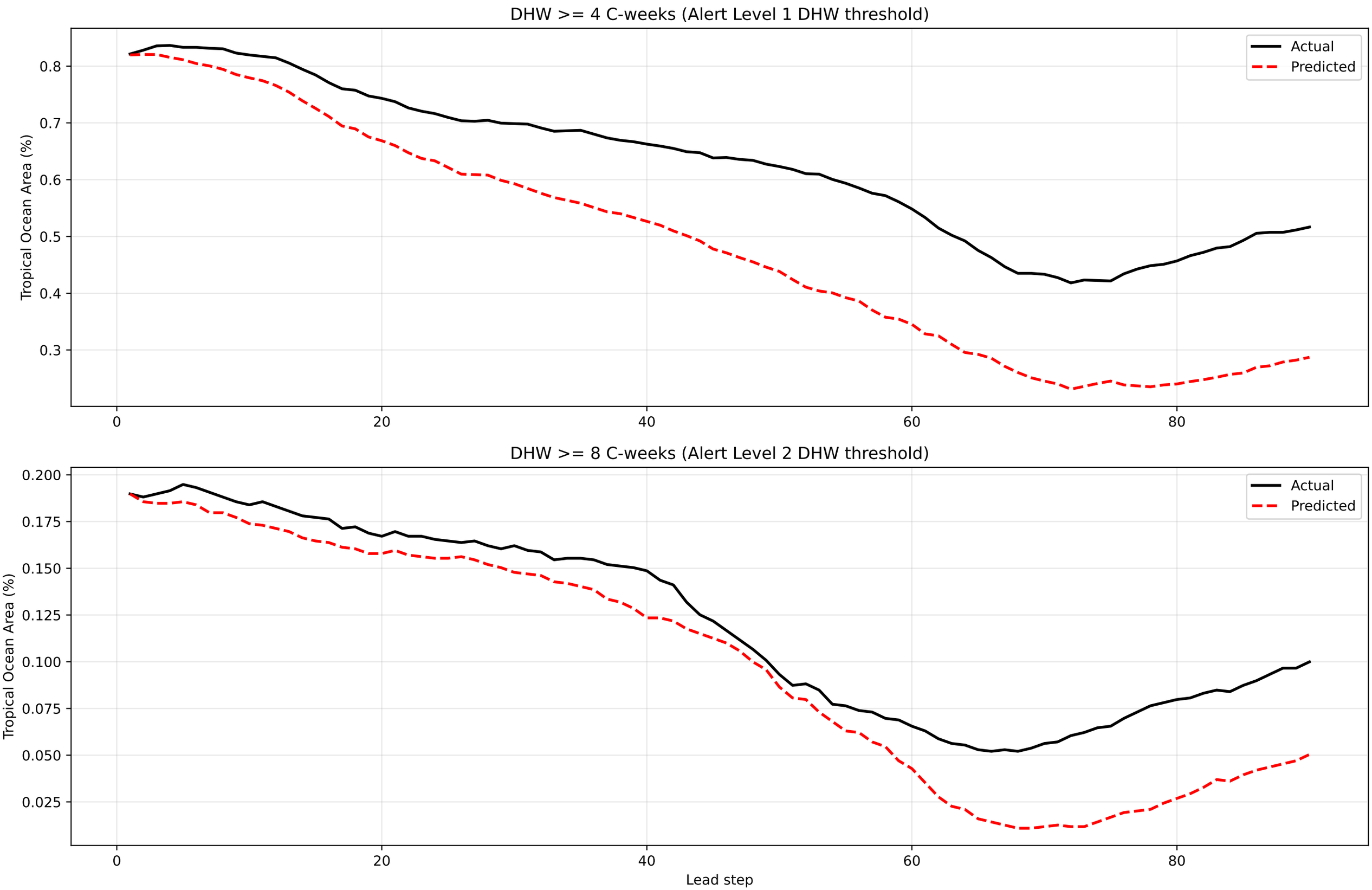


Peak DHW (max over all chains & leads) - Val (C-weeks)




Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.8%
0.4%
0.4%
97.5%
1.3%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val



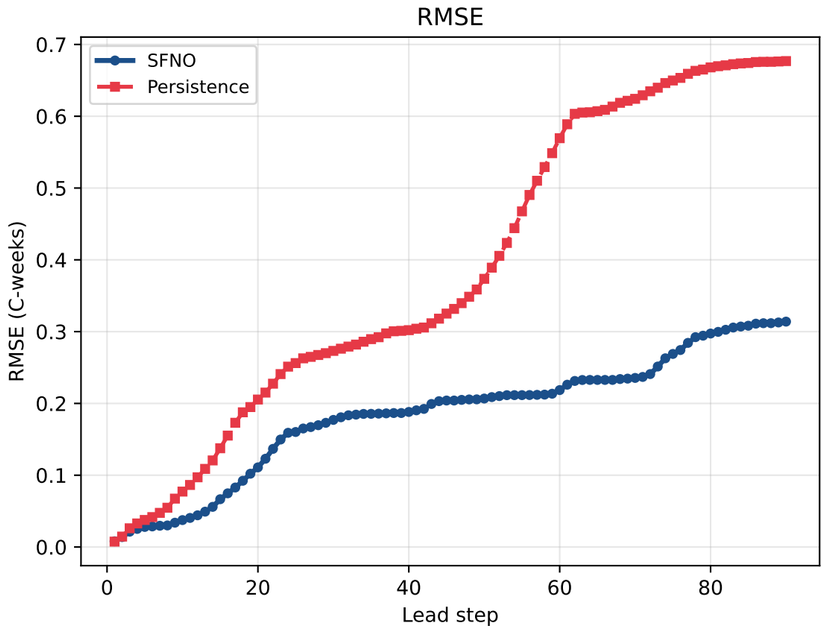
Great Barrier Reef Analysis - Val

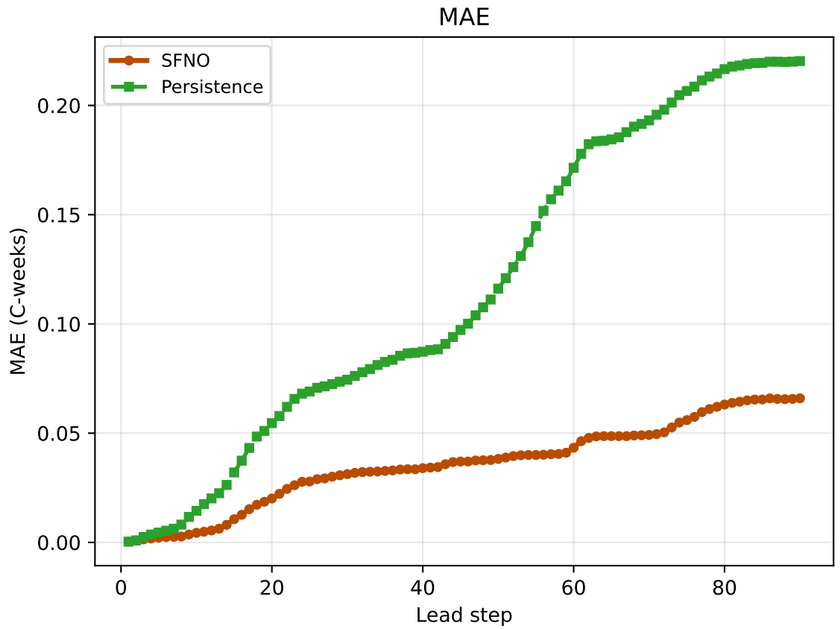
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

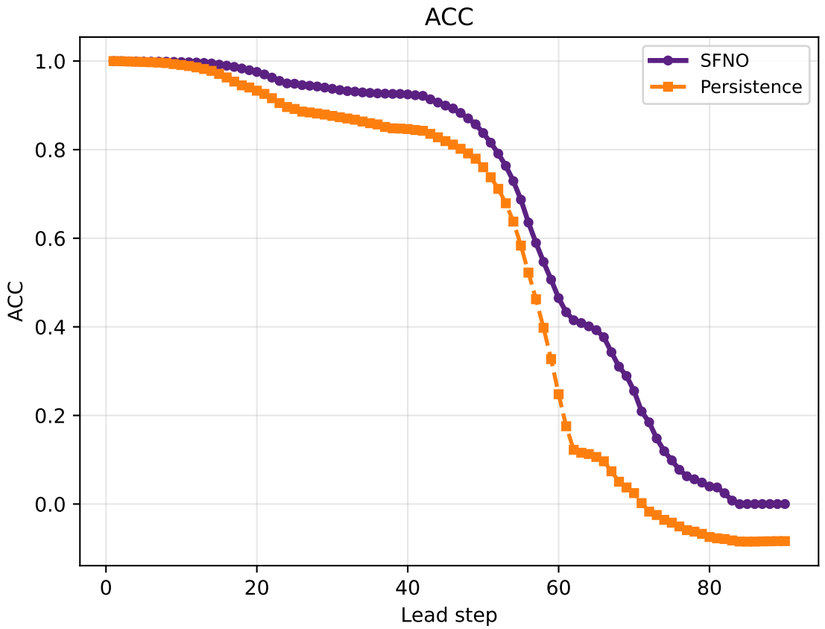
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

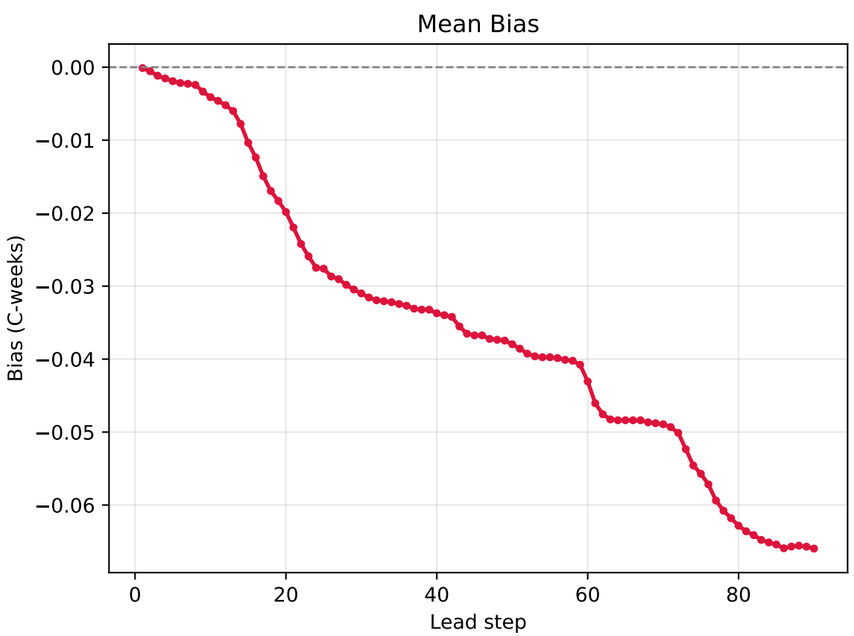
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0079	0.0075	-5.8%	0.0004	0.0004	-0.0001	0.9999	0.9999	0.9998
2	0.0138	0.0145	4.7%	0.0008	0.0009	-0.0006	0.9997	0.9997	0.9994
3	0.0213	0.0261	18.2%	0.0014	0.0025	-0.0012	0.9993	0.9989	0.9986
4	0.0252	0.0328	23.2%	0.0018	0.0036	-0.0015	0.9990	0.9983	0.9981
5	0.0279	0.0377	25.8%	0.0022	0.0045	-0.0019	0.9988	0.9978	0.9977
6	0.0287	0.0417	31.2%	0.0024	0.0053	-0.0022	0.9987	0.9973	0.9975
7	0.0296	0.0476	37.8%	0.0025	0.0062	-0.0023	0.9986	0.9965	0.9973
8	0.0300	0.0547	45.1%	0.0027	0.0082	-0.0024	0.9985	0.9954	0.9972
9	0.0339	0.0674	49.7%	0.0036	0.0117	-0.0033	0.9981	0.9931	0.9964
10	0.0375	0.0772	51.4%	0.0044	0.0144	-0.0041	0.9976	0.9910	0.9955
11	0.0405	0.0864	53.1%	0.0049	0.0175	-0.0046	0.9972	0.9887	0.9947
12	0.0442	0.0971	54.5%	0.0055	0.0201	-0.0052	0.9966	0.9857	0.9936
13	0.0492	0.1088	54.8%	0.0063	0.0225	-0.0060	0.9958	0.9820	0.9920
14	0.0559	0.1206	53.6%	0.0080	0.0263	-0.0078	0.9945	0.9778	0.9895
15	0.0666	0.1375	51.6%	0.0106	0.0320	-0.0104	0.9920	0.9710	0.9848
16	0.0748	0.1552	51.8%	0.0126	0.0374	-0.0124	0.9896	0.9629	0.9803
17	0.0829	0.1730	52.1%	0.0152	0.0432	-0.0149	0.9868	0.9536	0.9750
18	0.0923	0.1876	50.8%	0.0172	0.0484	-0.0170	0.9832	0.9451	0.9683
19	0.1021	0.1949	47.6%	0.0186	0.0510	-0.0183	0.9793	0.9403	0.9611
20	0.1109	0.2054	46.0%	0.0201	0.0546	-0.0198	0.9755	0.9330	0.9539
21	0.1229	0.2151	42.8%	0.0222	0.0578	-0.0220	0.9698	0.9258	0.9434
22	0.1367	0.2275	39.9%	0.0245	0.0621	-0.0242	0.9626	0.9161	0.9303
23	0.1497	0.2409	37.8%	0.0262	0.0657	-0.0259	0.9552	0.9051	0.9168
24	0.1591	0.2512	36.7%	0.0278	0.0681	-0.0275	0.9495	0.8962	0.9066
25	0.1602	0.2562	37.5%	0.0279	0.0691	-0.0276	0.9488	0.8919	0.9052
26	0.1651	0.2628	37.2%	0.0289	0.0708	-0.0287	0.9457	0.8859	0.8998
27	0.1671	0.2649	36.9%	0.0293	0.0714	-0.0290	0.9443	0.8840	0.8973
28	0.1697	0.2675	36.6%	0.0301	0.0725	-0.0298	0.9426	0.8816	0.8942
29	0.1730	0.2700	35.9%	0.0307	0.0736	-0.0305	0.9401	0.8792	0.8899
30	0.1770	0.2732	35.2%	0.0313	0.0745	-0.0310	0.9374	0.8762	0.8851
31	0.1807	0.2762	34.6%	0.0318	0.0762	-0.0316	0.9345	0.8732	0.8801
32	0.1835	0.2792	34.3%	0.0322	0.0779	-0.0319	0.9322	0.8703	0.8762
33	0.1844	0.2819	34.6%	0.0323	0.0794	-0.0321	0.9310	0.8676	0.8742
34	0.1854	0.2859	35.2%	0.0325	0.0812	-0.0322	0.9292	0.8635	0.8711
35	0.1855	0.2895	35.9%	0.0327	0.0826	-0.0324	0.9280	0.8597	0.8689
36	0.1858	0.2922	36.4%	0.0330	0.0836	-0.0327	0.9270	0.8569	0.8671
37	0.1862	0.2976	37.4%	0.0334	0.0854	-0.0331	0.9264	0.8512	0.8661
38	0.1866	0.3005	37.9%	0.0335	0.0866	-0.0332	0.9258	0.8481	0.8651
39	0.1866	0.3011	38.0%	0.0335	0.0868	-0.0332	0.9258	0.8475	0.8650
40	0.1881	0.3020	37.7%	0.0340	0.0873	-0.0337	0.9247	0.8465	0.8631
41	0.1904	0.3041	37.4%	0.0343	0.0880	-0.0340	0.9229	0.8443	0.8600
42	0.1924	0.3057	37.1%	0.0345	0.0884	-0.0342	0.9214	0.8425	0.8574
43	0.1992	0.3116	36.1%	0.0358	0.0909	-0.0355	0.9136	0.8357	0.8437
44	0.2031	0.3180	36.1%	0.0368	0.0940	-0.0365	0.9063	0.8279	0.8311
45	0.2040	0.3251	37.2%	0.0370	0.0972	-0.0367	0.8995	0.8193	0.8195
46	0.2040	0.3317	38.5%	0.0370	0.1001	-0.0367	0.8929	0.8114	0.8084
47	0.2049	0.3397	39.7%	0.0375	0.1039	-0.0372	0.8828	0.8019	0.7913
48	0.2056	0.3486	41.0%	0.0376	0.1076	-0.0373	0.8705	0.7915	0.7708
49	0.2057	0.3587	42.7%	0.0377	0.1112	-0.0375	0.8571	0.7796	0.7488
50	0.2068	0.3735	44.6%	0.0382	0.1161	-0.0380	0.8375	0.7603	0.7171
51	0.2088	0.3891	46.3%	0.0388	0.1209	-0.0386	0.8157	0.7376	0.6827
52	0.2102	0.4055	48.2%	0.0395	0.1260	-0.0393	0.7909	0.7115	0.6449
53	0.2115	0.4235	50.1%	0.0399	0.1311	-0.0396	0.7633	0.6790	0.6041
54	0.2115	0.4441	52.4%	0.0400	0.1374	-0.0397	0.7291	0.6376	0.5561
55	0.2115	0.4675	54.7%	0.0400	0.1448	-0.0397	0.6870	0.5834	0.5005
56	0.2117	0.4904	56.8%	0.0401	0.1517	-0.0399	0.6354	0.5223	0.4356
57	0.2119	0.5101	58.5%	0.0404	0.1570	-0.0401	0.5894	0.4623	0.3798
58	0.2123	0.5292	59.9%	0.0405	0.1610	-0.0402	0.5467	0.3978	0.3305
59	0.2135	0.5485	61.1%	0.0410	0.1653	-0.0408	0.5065	0.3272	0.2865
60	0.2186	0.5694	61.6%	0.0433	0.1715	-0.0431	0.4649	0.2479	0.2424
61	0.2261	0.5888	61.6%	0.0463	0.1779	-0.0461	0.4331	0.1755	0.2097
62	0.2313	0.6033	61.7%	0.0478	0.1822	-0.0476	0.4148	0.1224	0.1927
63	0.2326	0.6051	61.6%	0.0485	0.1836	-0.0483	0.4085	0.1157	0.1874
64	0.2327	0.6055	61.6%	0.0486	0.1838	-0.0484	0.4011	0.1129	0.1808
65	0.2327	0.6070	61.7%	0.0486	0.1845	-0.0484	0.3926	0.1062	0.1727
66	0.2327	0.6089	61.8%	0.0486	0.1854	-0.0484	0.3765	0.0965	0.1589
67	0.2327	0.6135	62.1%	0.0486	0.1878	-0.0484	0.3427	0.0739	0.1317
68	0.2340	0.6187	62.2%	0.0489	0.1903	-0.0487	0.3101	0.0504	0.1069
69	0.2345	0.6215	62.3%	0.0491	0.1916	-0.0488	0.2890	0.0374	0.0917
70	0.2355	0.6242	62.3%	0.0492	0.1932	-0.0489	0.2552	0.0247	0.0693
71	0.2369	0.6292	62.4%	0.0496	0.1958	-0.0493	0.2090	0.0020	0.0414
72	0.2409	0.6347	62.1%	0.0504	0.1981	-0.0501	0.1845	-0.0171	0.0286
73	0.2514	0.6397	60.7%	0.0526	0.2013	-0.0523	0.1481	-0.0248	0.0121
74	0.2629	0.6462	59.3%	0.0548	0.2047	-0.0546	0.1192	-0.0357	0.0004
75	0.2689	0.6498	58.6%	0.0560	0.2066	-0.0557	0.0986	-0.0421	-0.0075
76	0.2744	0.6534	58.0%	0.0574	0.2086	-0.0572	0.0774	-0.0506	-0.0164
77	0.2843	0.6591	56.9%	0.0597	0.2115	-0.0594	0.0627	-0.0589	-0.0219
78	0.2923	0.6632	55.9%	0.0610	0.2132	-0.0608	0.0558	-0.0623	-0.0239
79	0.2943	0.6651	55.7%	0.0621	0.2146	-0.0618	0.0484	-0.0673	-0.0274
80	0.2973	0.6680	55.5%	0.0631	0.2166	-0.0628	0.0396	-0.0744	-0.0306
81	0.2997	0.6697	55.3%	0.0639	0.2178	-0.0636	0.0374	-0.0774	-0.0314
82	0.3026	0.6709	54.9%	0.0644	0.2183	-0.0641	0.0244	-0.0790	-0.0369
83	0.3057	0.6725	54.5%	0.0650	0.2190	-0.0648	0.0078	-0.0822	-0.0441
84	0.3071	0.6736	54.4%	0.0654	0.2194	-0.0651	-0.0003	-0.0846	-0.0476
85	0.3082	0.6741	54.3%	0.0654	0.2195	-0.0654	-0.0000	-0.0849	-0.0471
86	0.3111	0.6755	53.9%	0.0659	0.2200	-0.0659	0.0000	-0.0848	-0.0470
87	0.3117	0.6758	53.9%	0.0657	0.2200	-0.0657	0.0000	-0.0843	-0.0465
88	0.3118	0.6758	53.9%	0.0656	0.2199	-0.0656	0.0000	-0.0841	-0.0463
89	0.3128	0.6763	53.8%	0.0657	0.2200	-0.0657	-0.0000	-0.0838	-0.0462
90	0.3139	0.6768	53.6%	0.0660	0.2203	-0.0660	0.0000	-0.0838	-0.0462

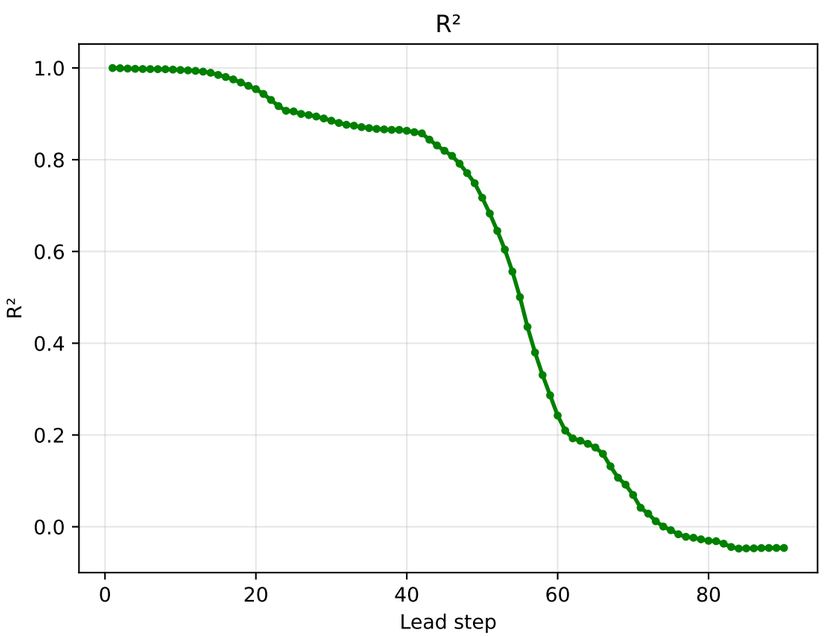
DHW Forecast Skill vs Lead Step - Val - GBR

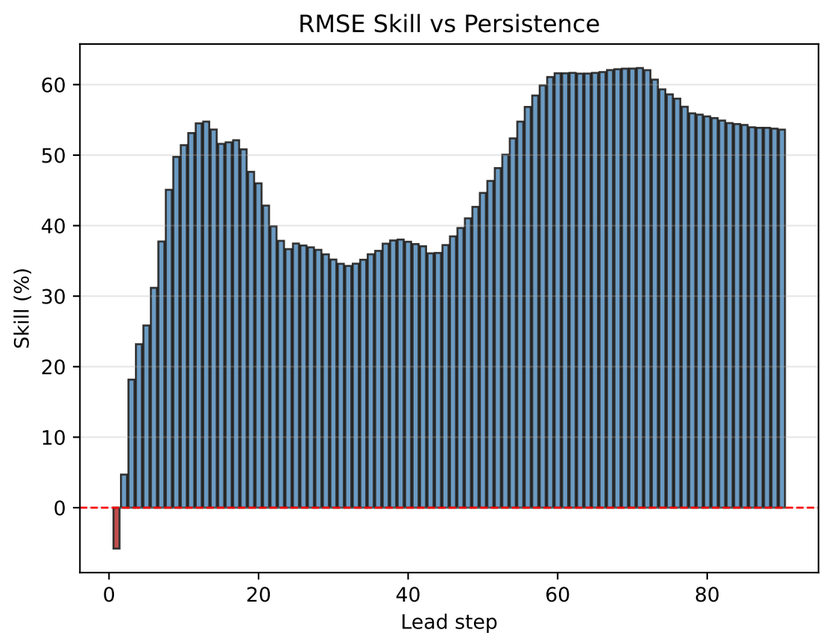


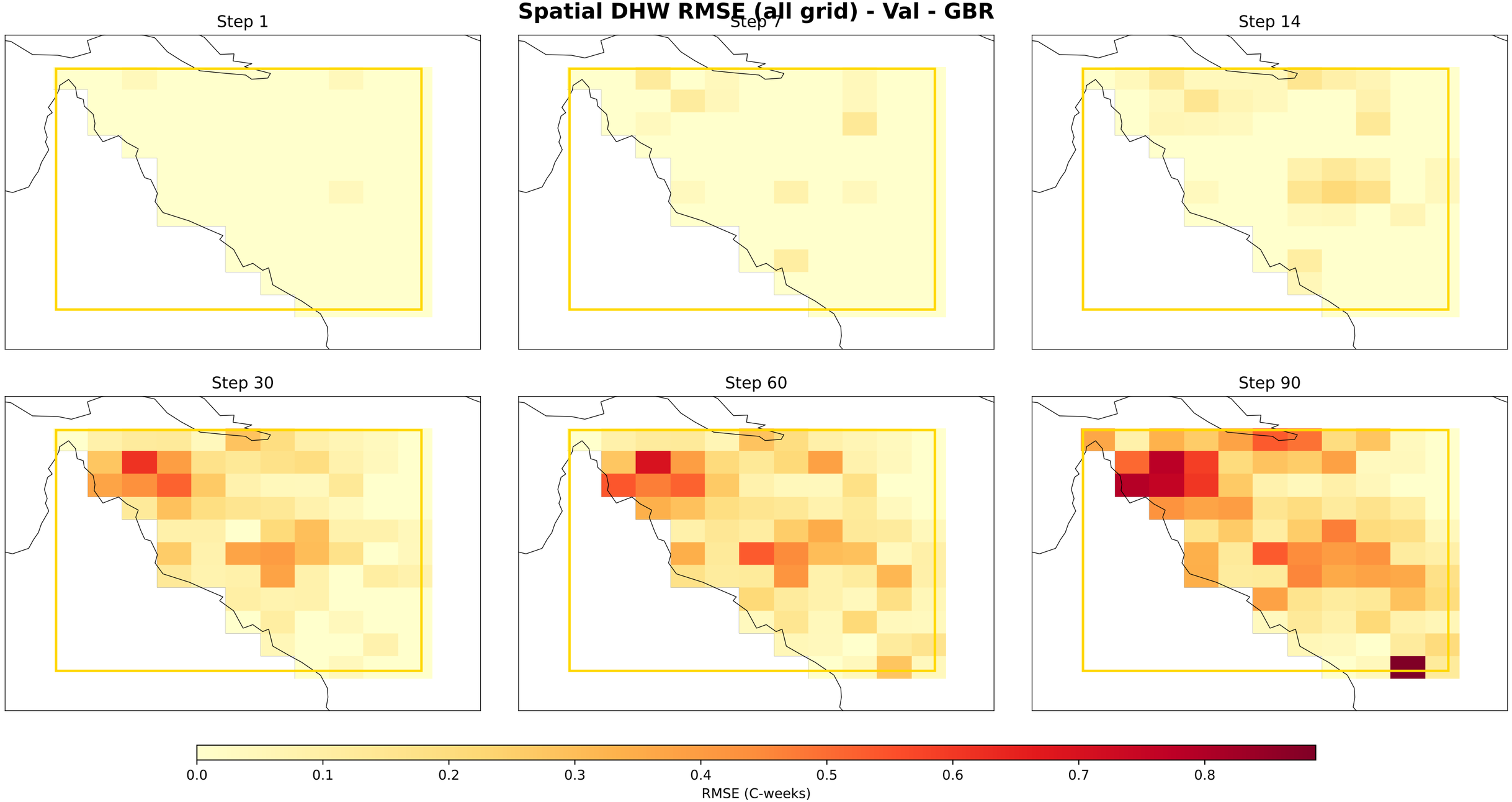


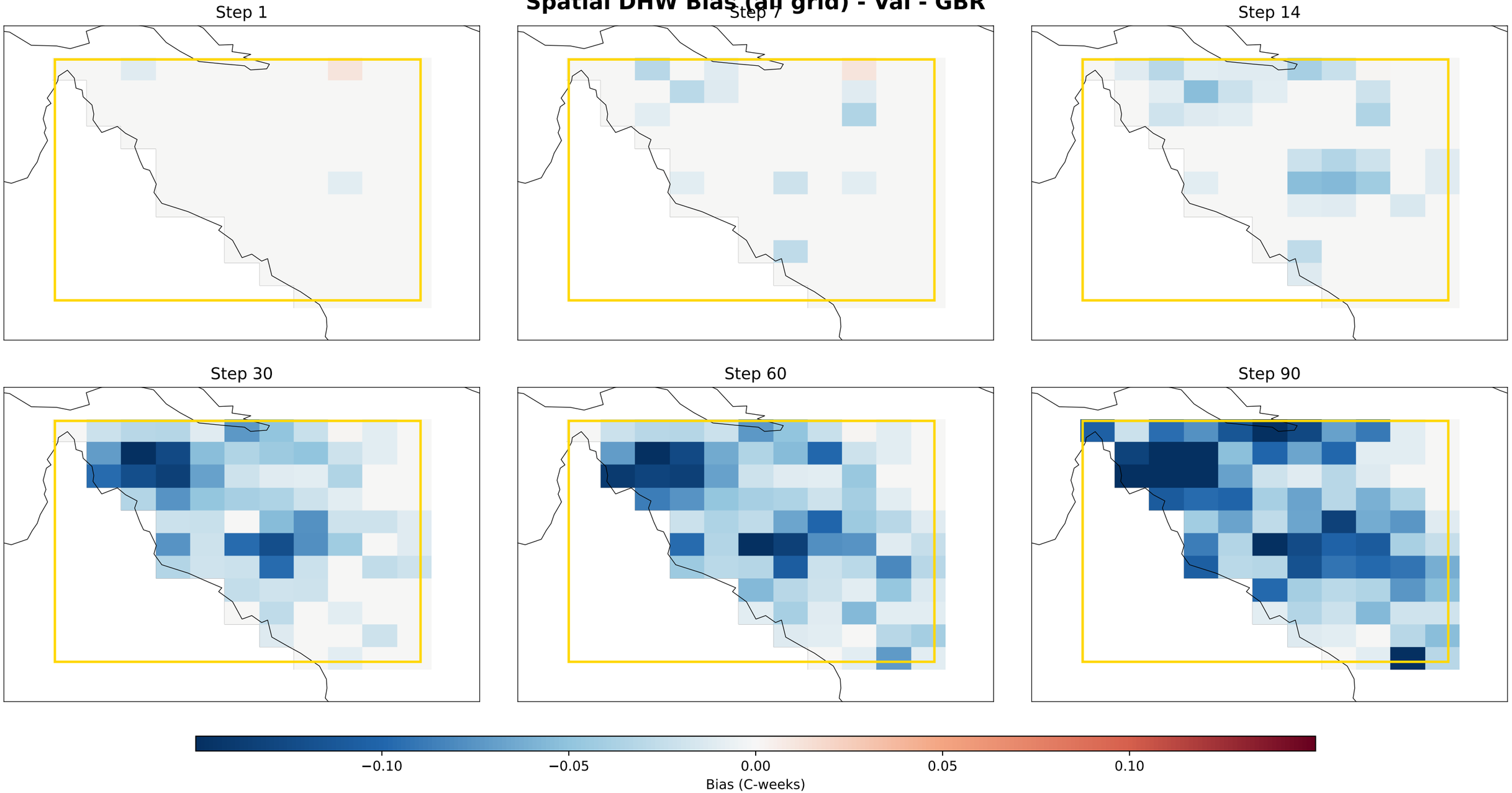


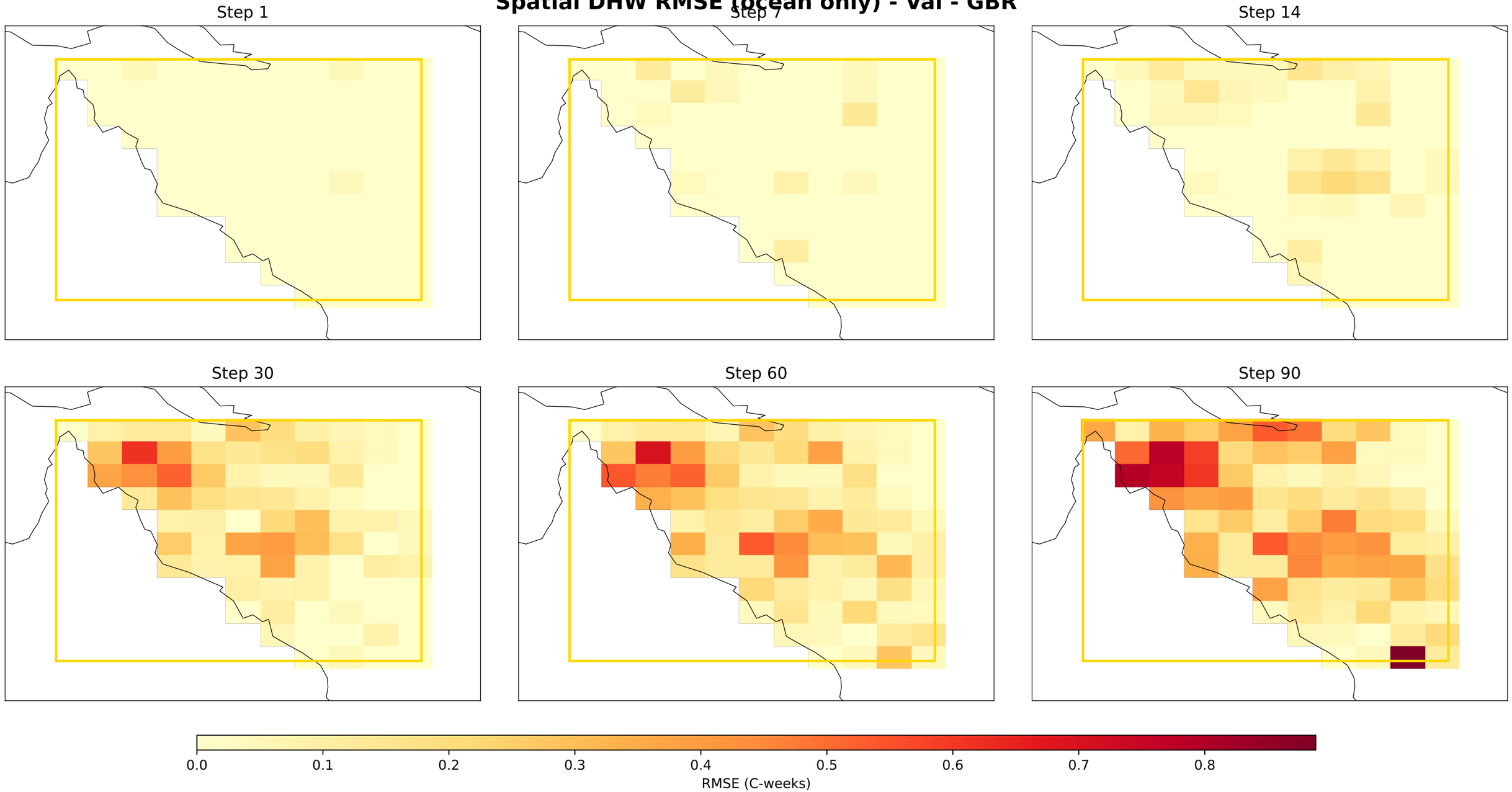


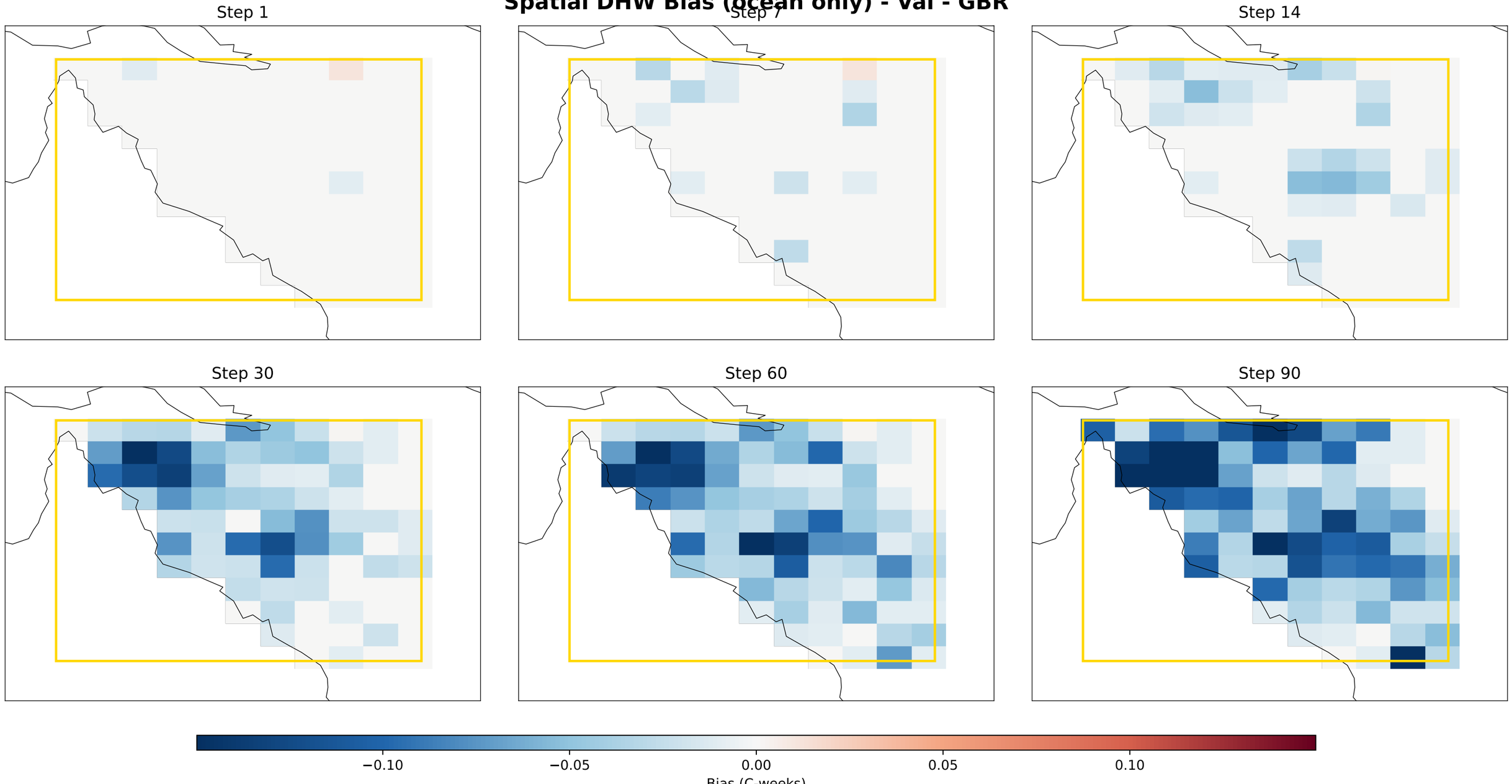





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

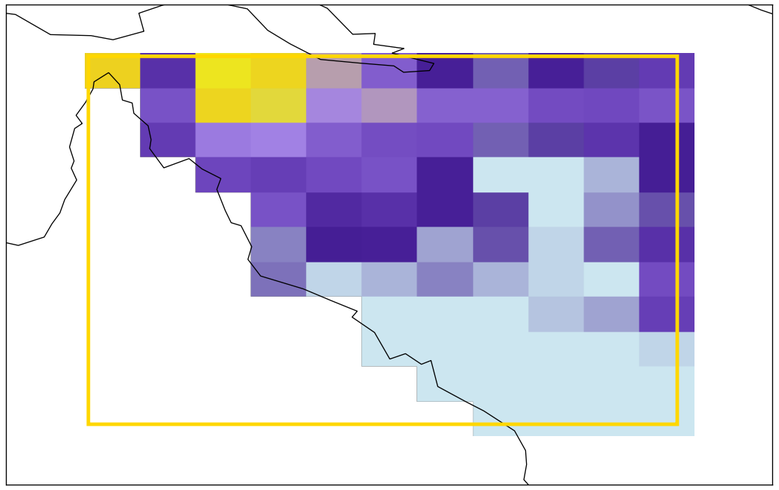

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

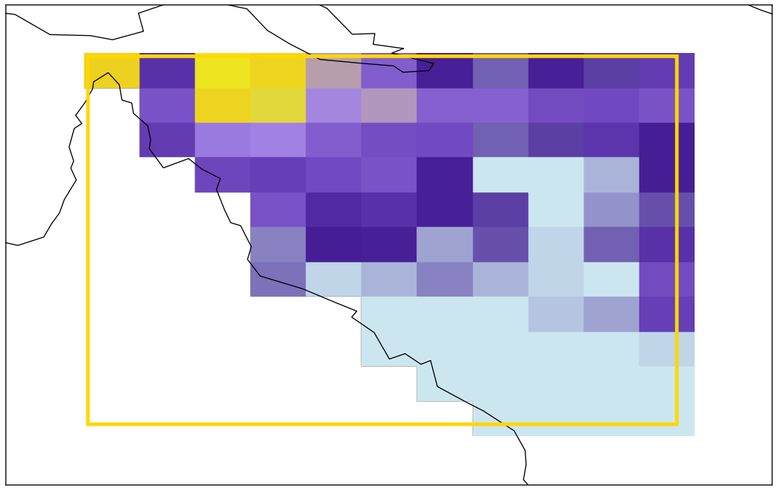

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

Actual DHW - Step 1



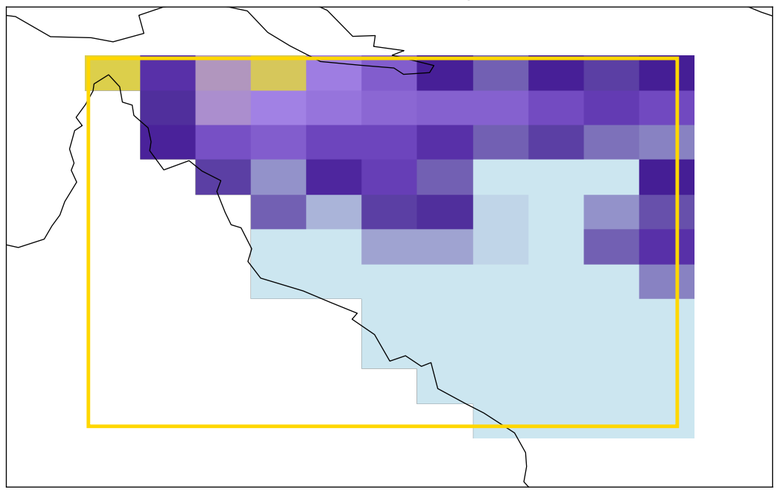
Predicted DHW - Step 1



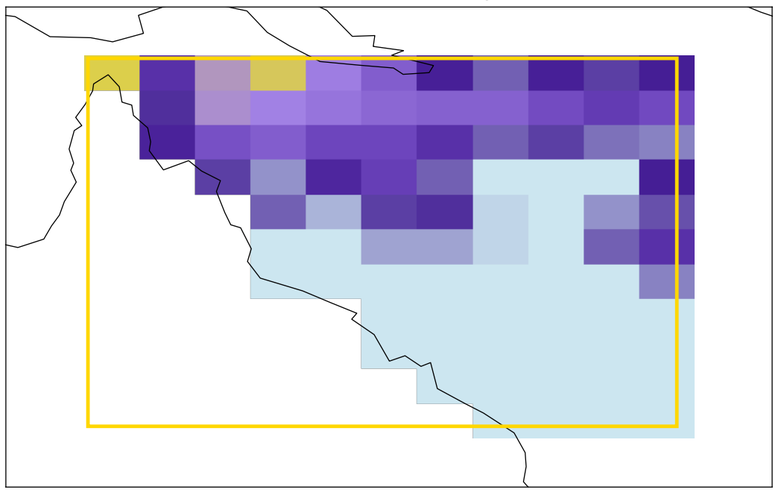
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



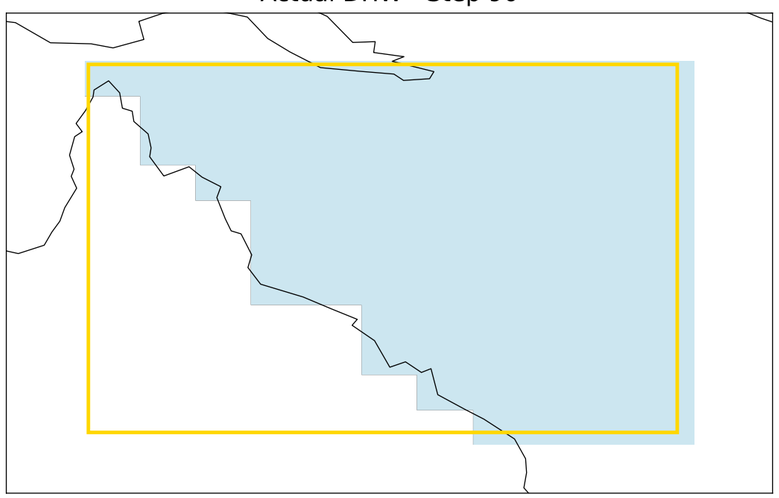
Predicted DHW - Step 46



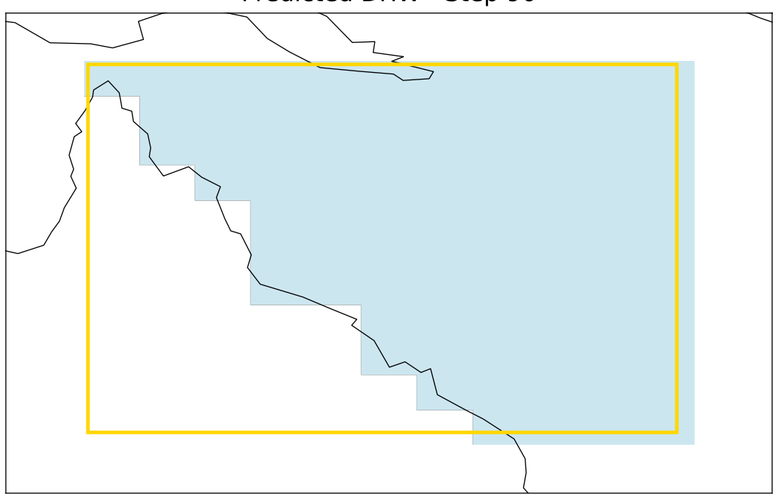
Error (Pred - Actual) - Step 46 [scale ±8]



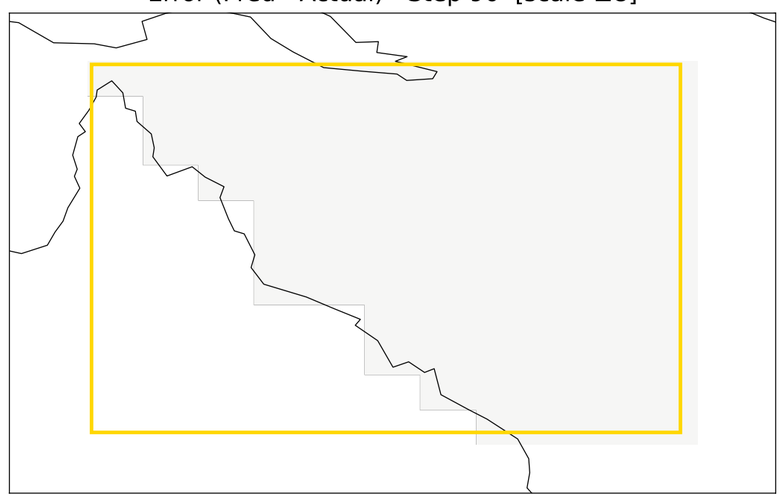
Actual DHW - Step 90

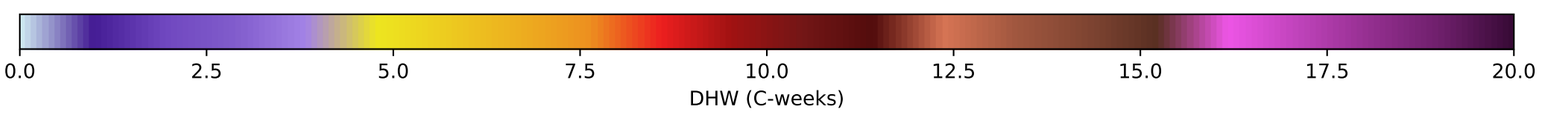


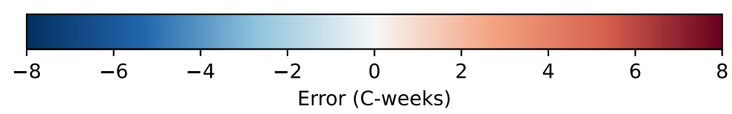
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

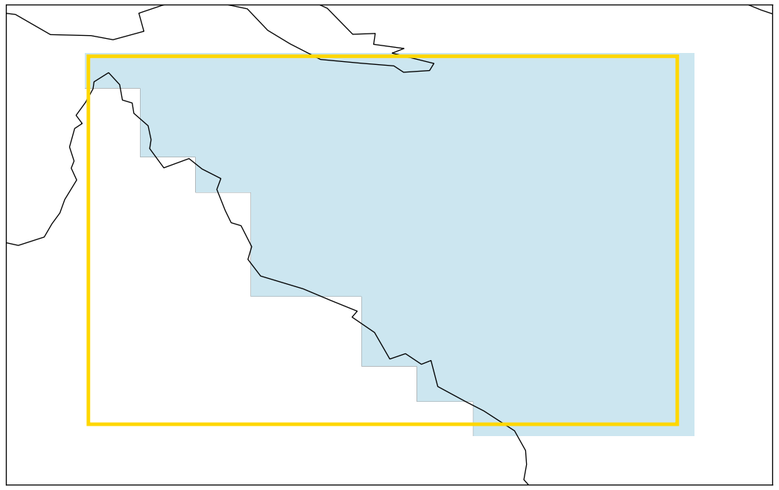




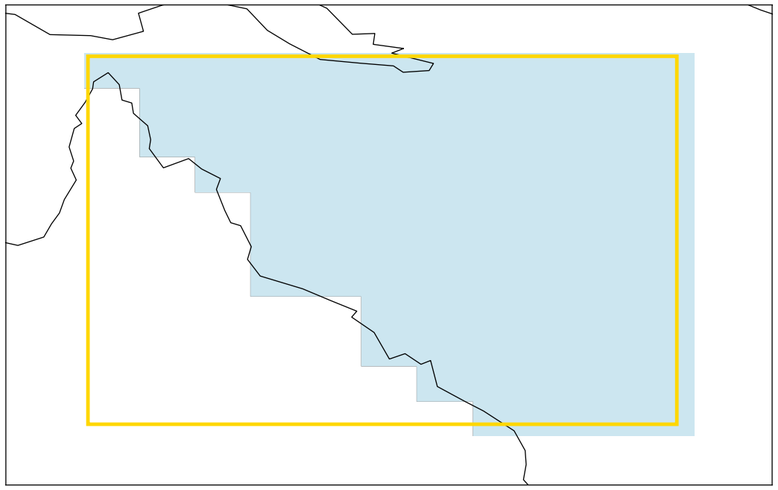


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

Actual DHW - Step 1



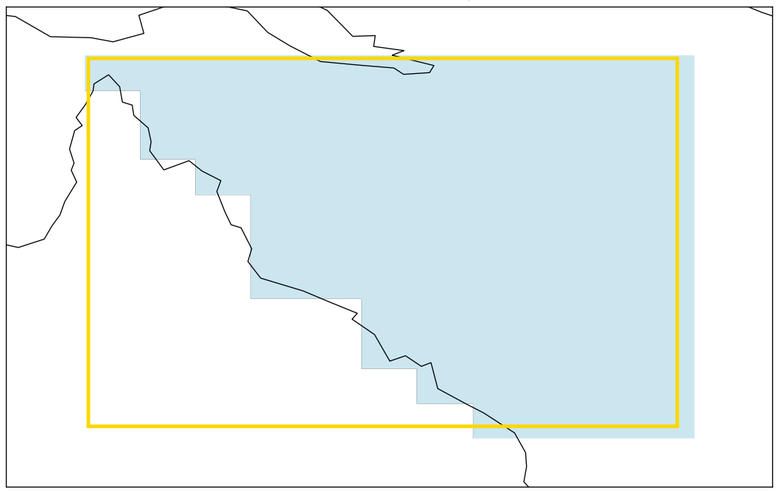
Predicted DHW - Step 1



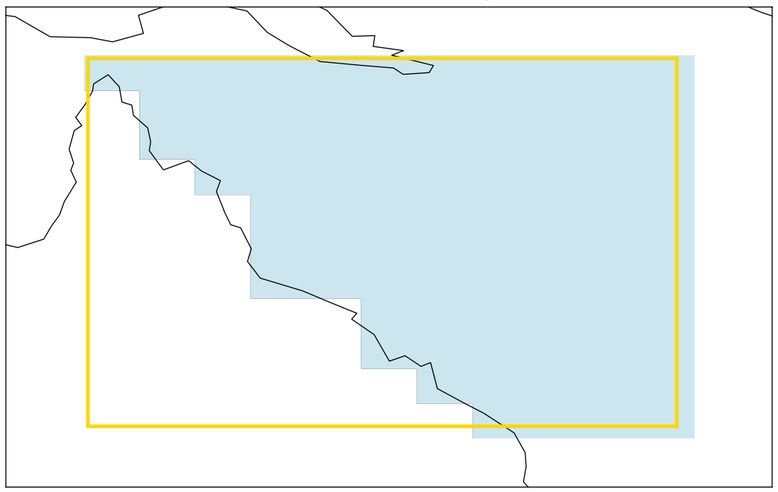
Error (Pred - Actual) - Step 1 [scale ±8]



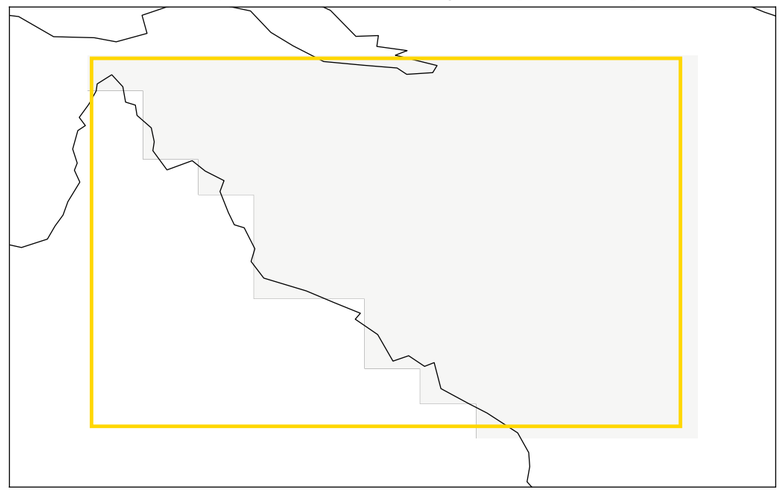
Actual DHW - Step 46



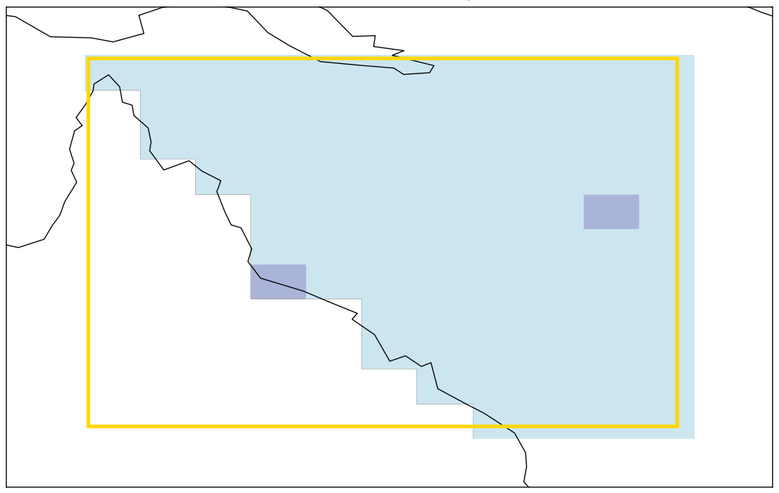
Predicted DHW - Step 46



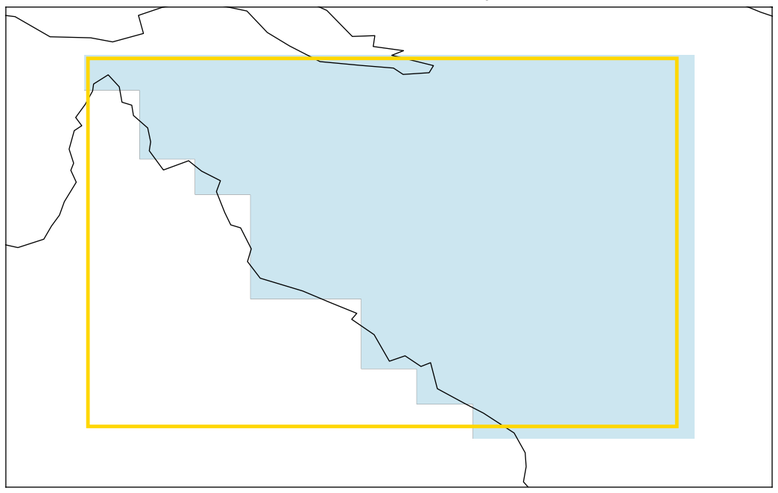
Error (Pred - Actual) - Step 46 [scale ±8]



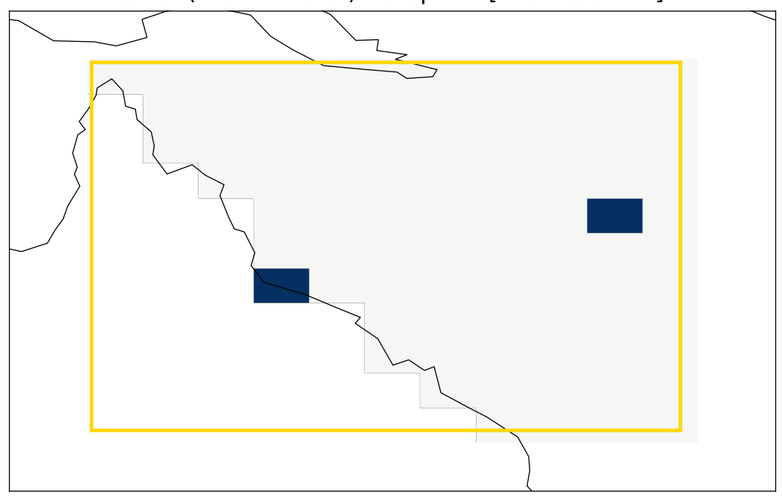
Actual DHW - Step 90

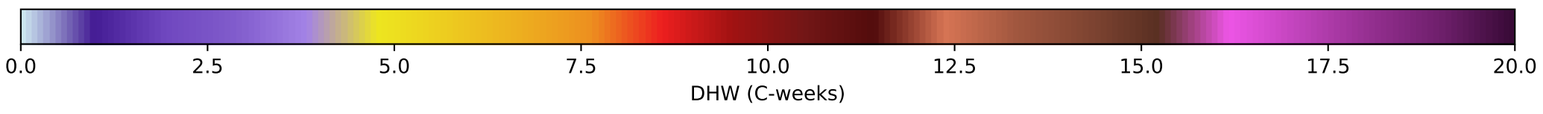


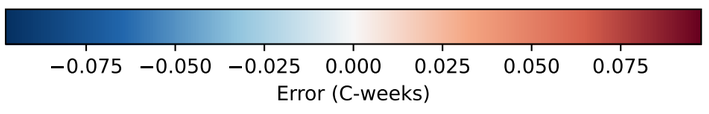
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.098]

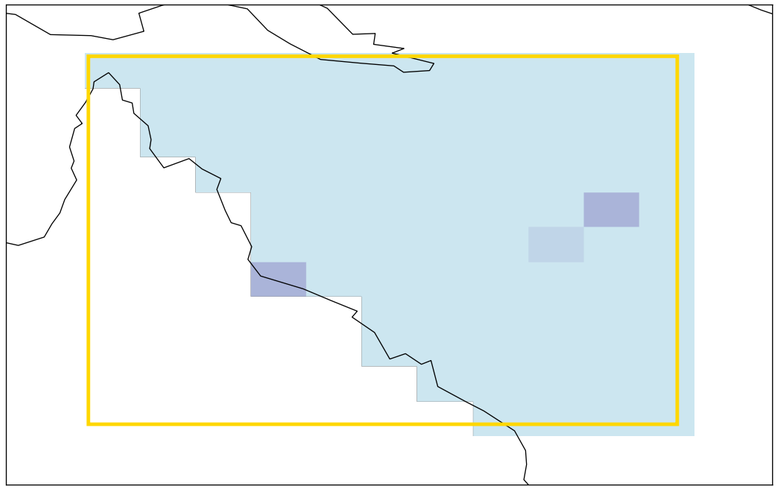




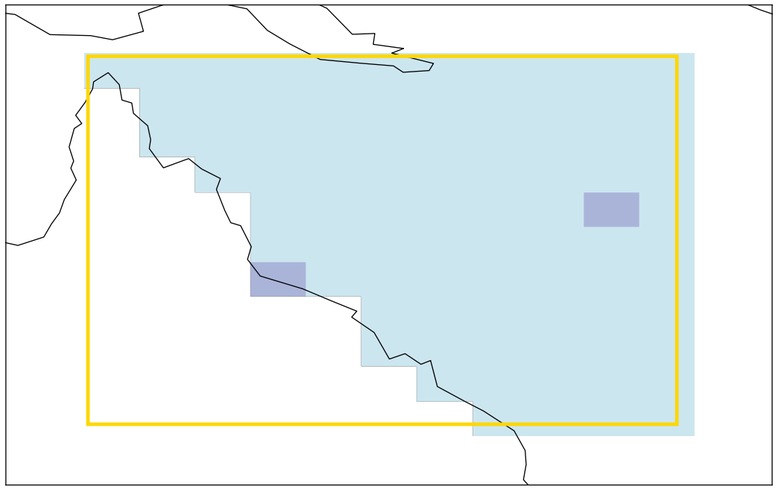


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

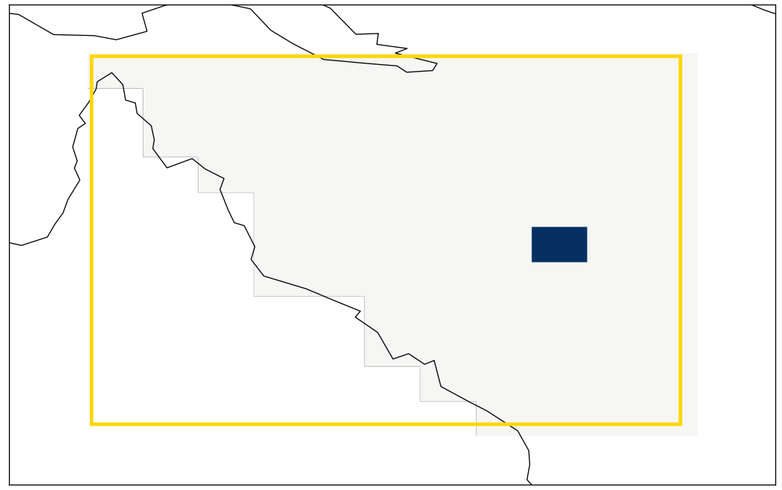
Actual DHW - Step 1



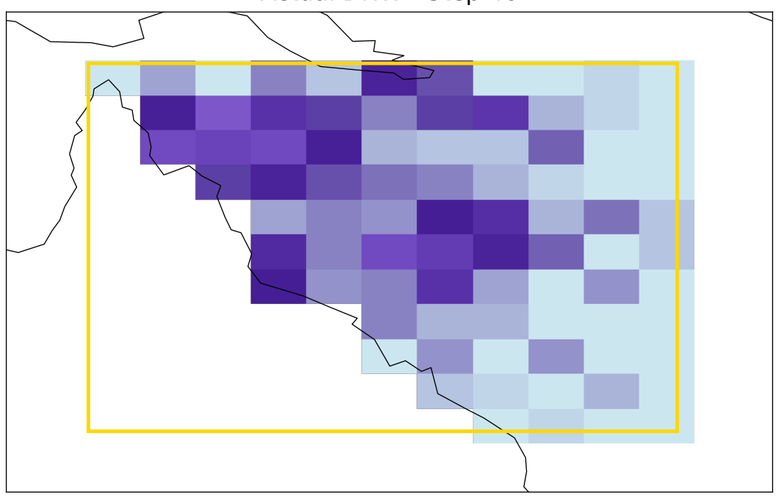
Predicted DHW - Step 1



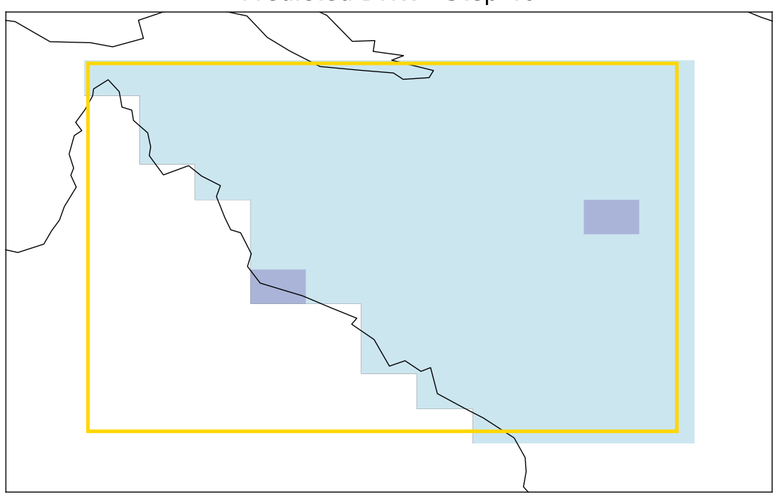
Error (Pred - Actual) - Step 1 [scale ±0.15]



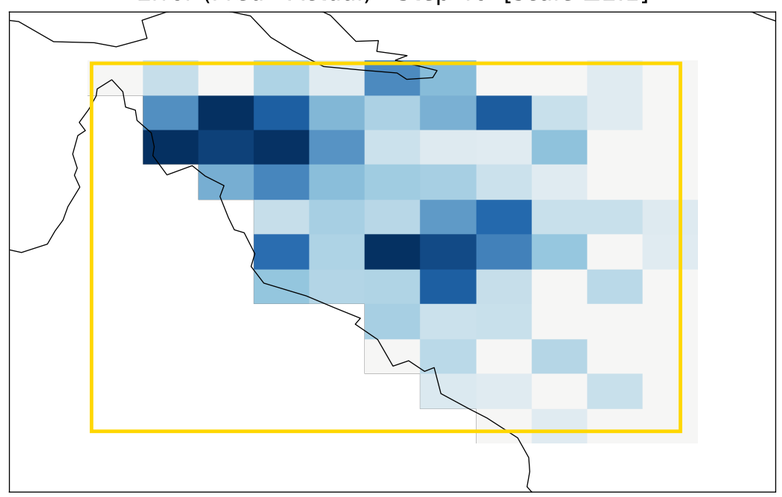
Actual DHW - Step 46



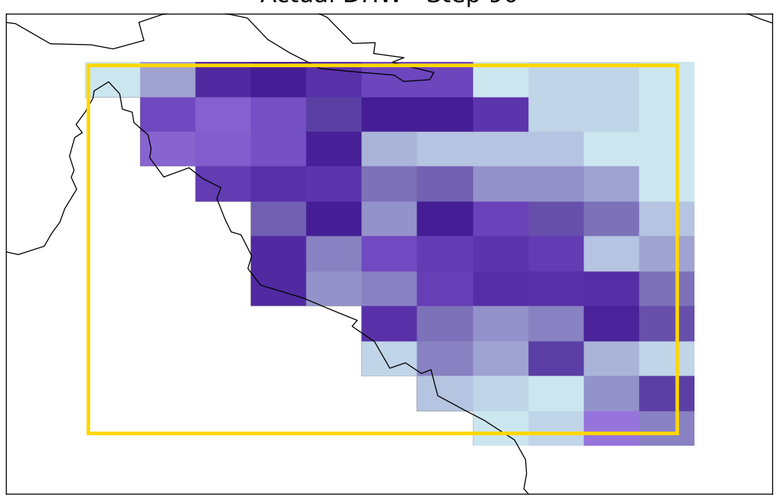
Predicted DHW - Step 46



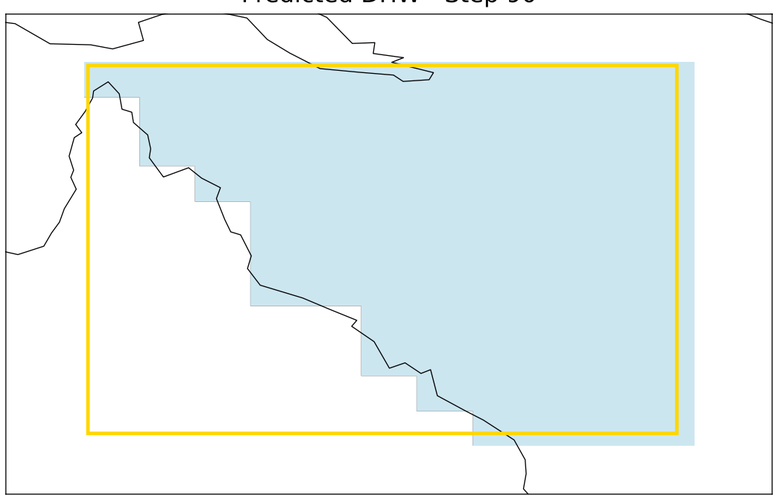
Error (Pred - Actual) - Step 46 [scale ±2.1]



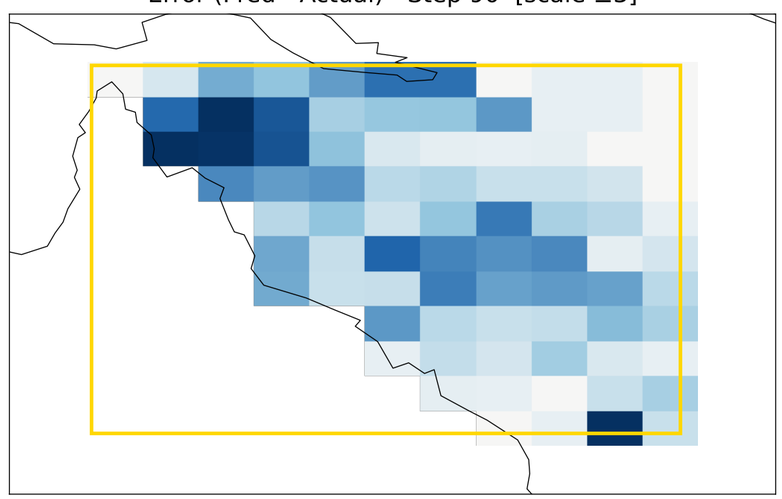
Actual DHW - Step 90

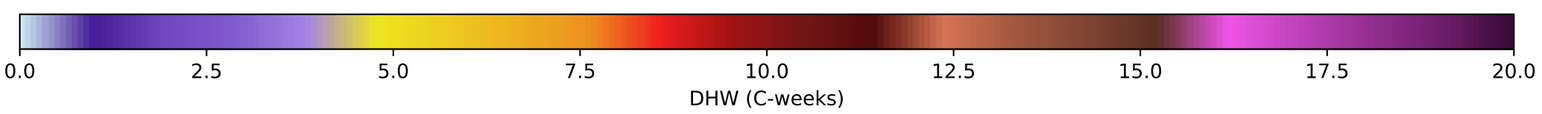


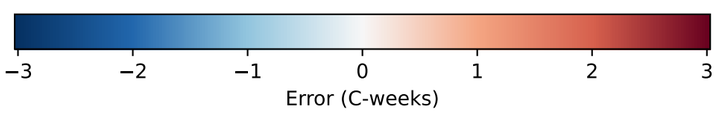
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

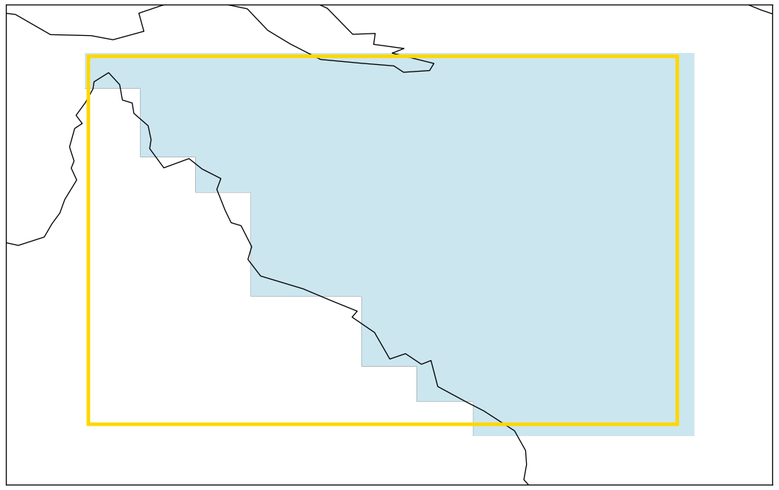




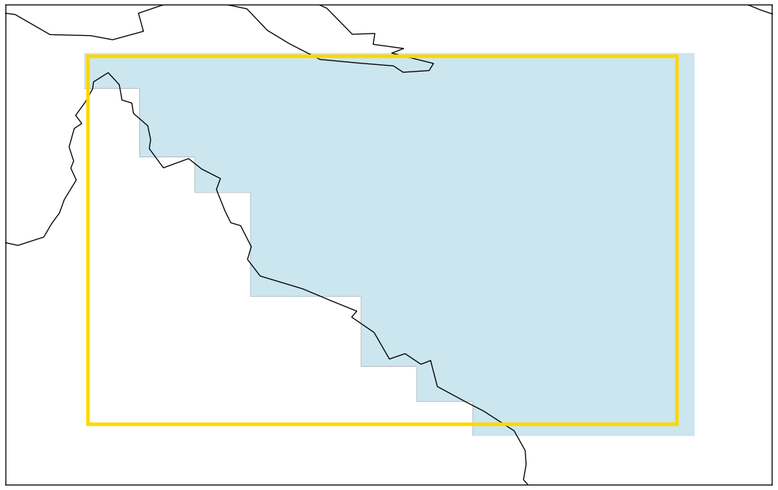


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

Actual DHW - Step 1



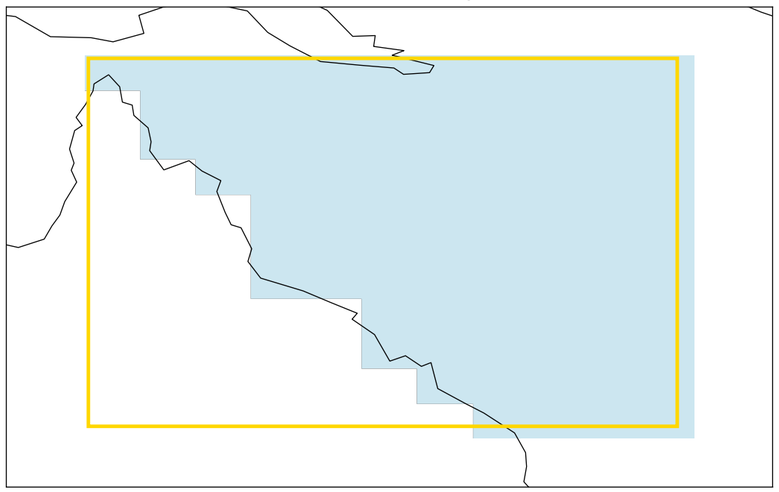
Predicted DHW - Step 1



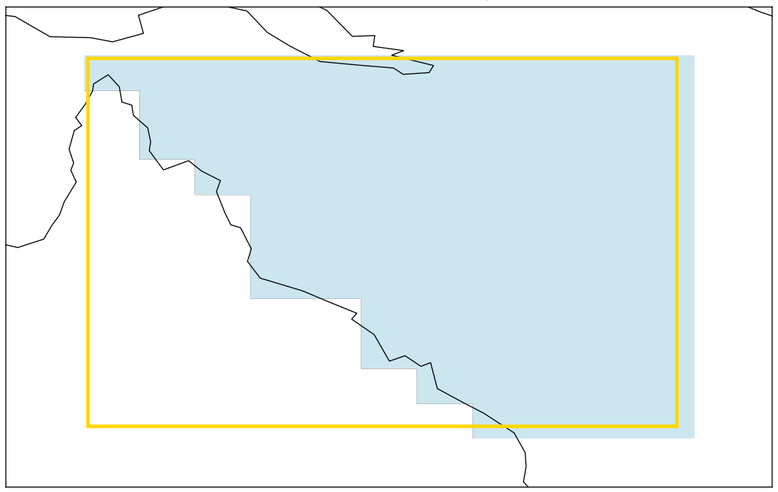
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



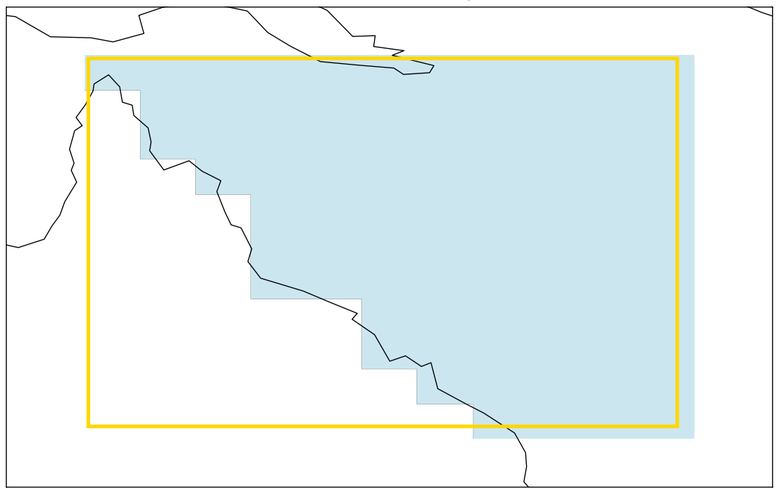
Predicted DHW - Step 46



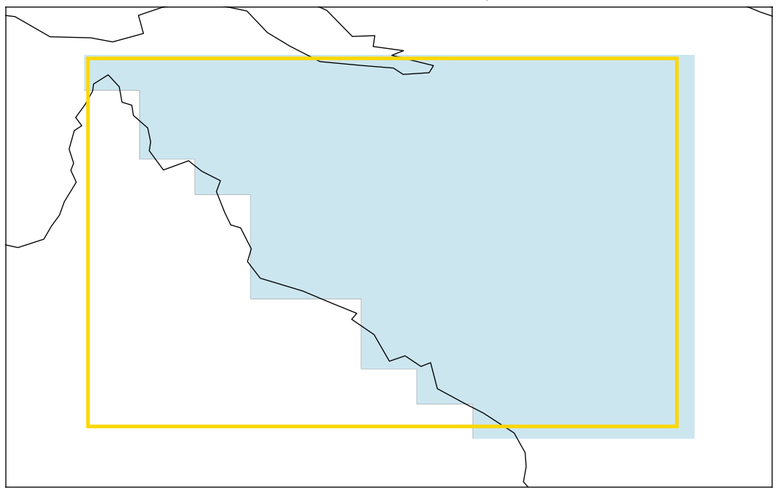
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

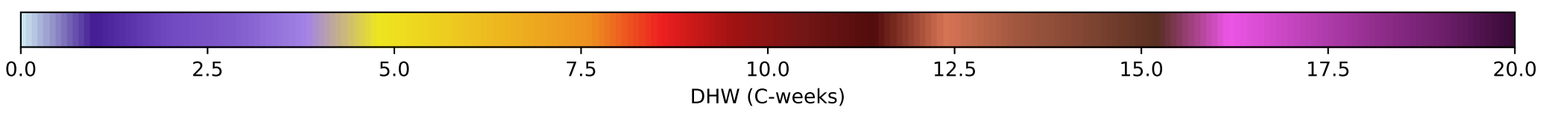


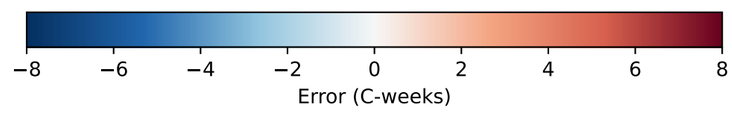
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

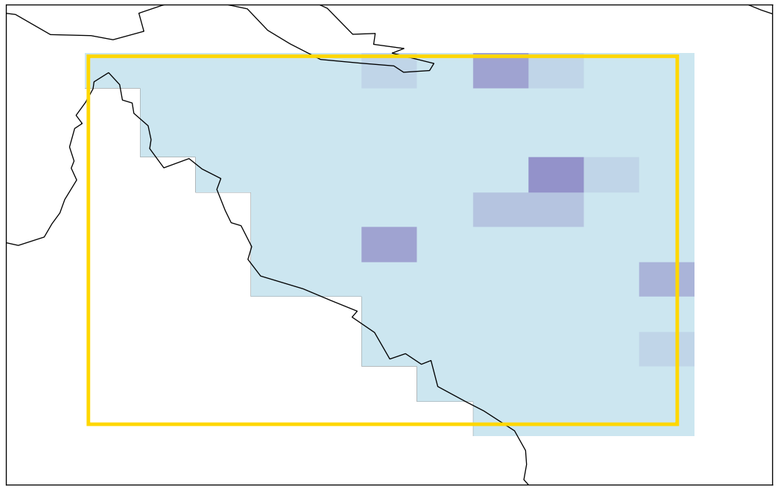




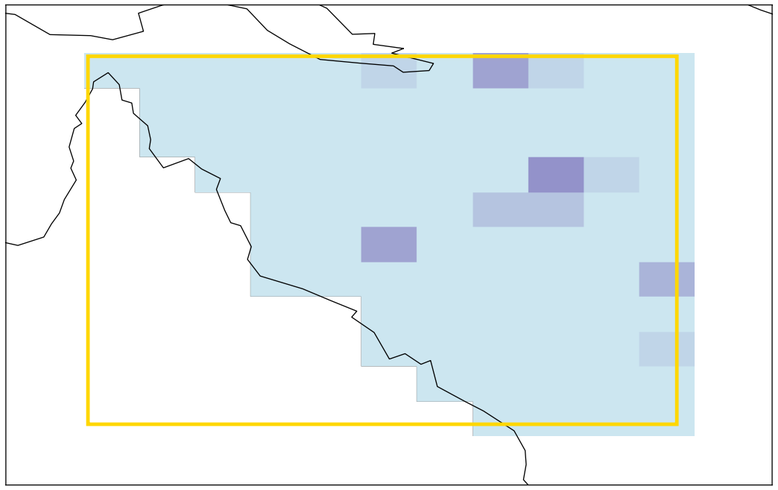


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

Actual DHW - Step 1



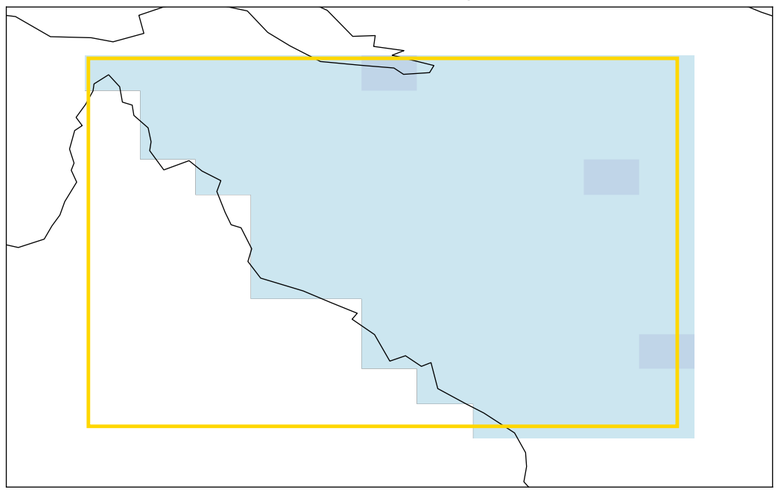
Predicted DHW - Step 1



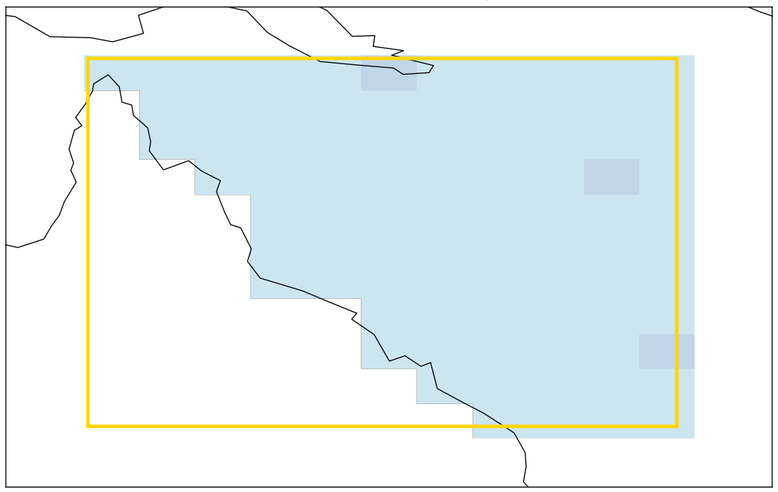
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



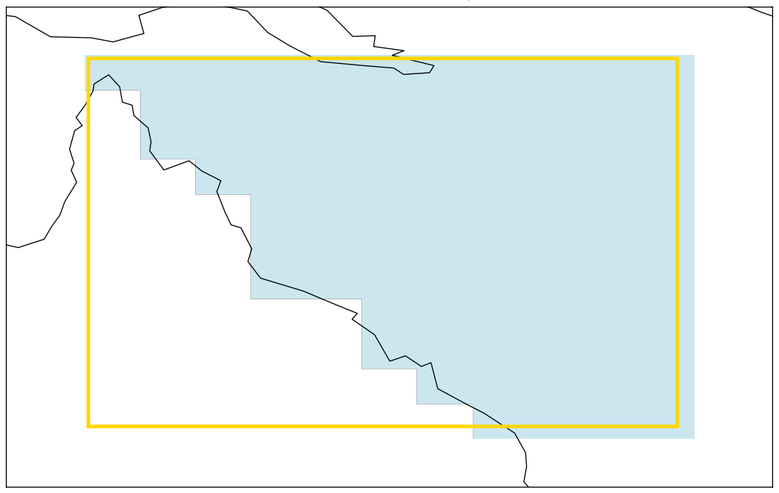
Predicted DHW - Step 46



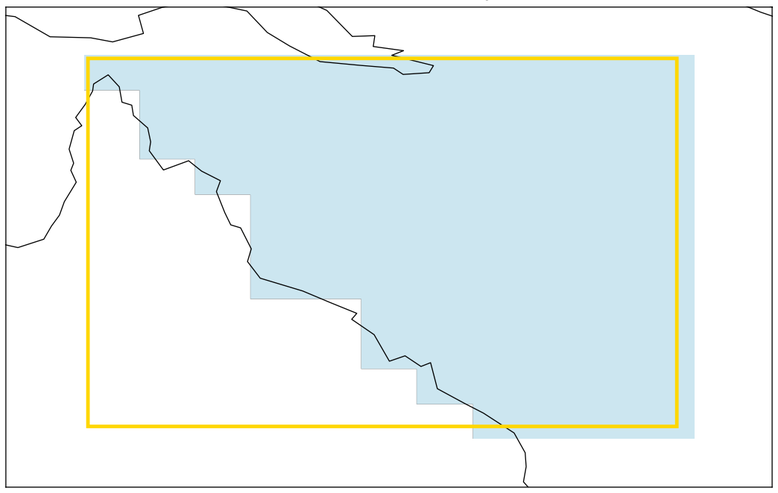
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

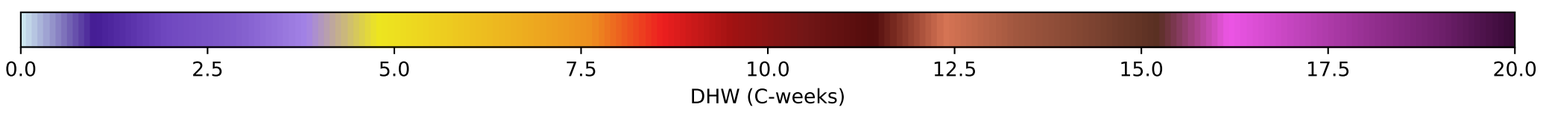


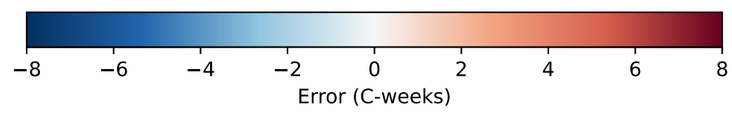
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

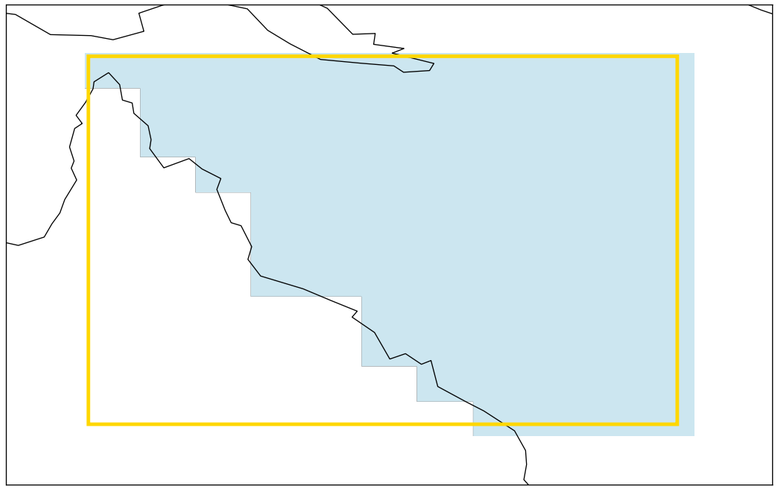




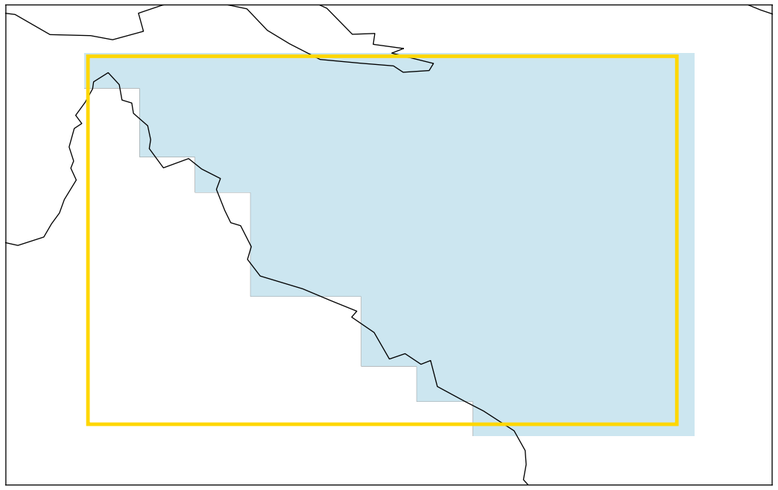


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

Actual DHW - Step 1



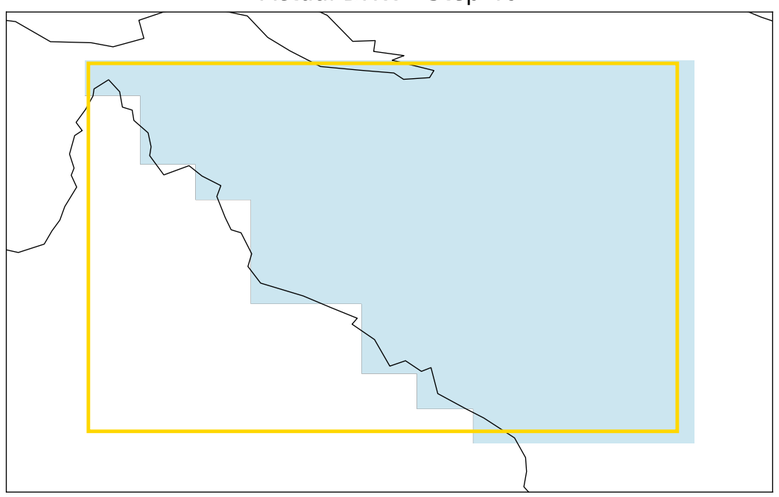
Predicted DHW - Step 1



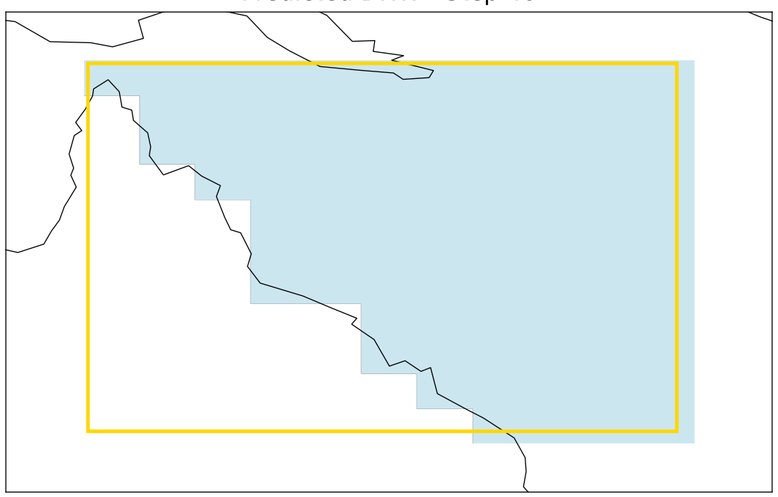
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



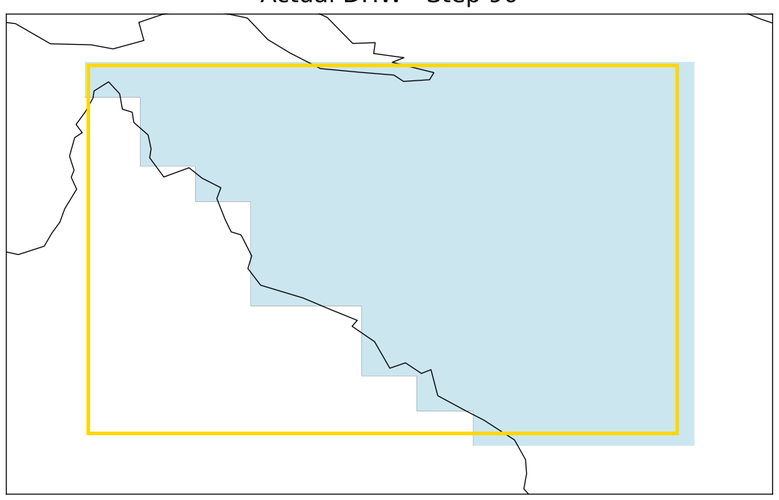
Predicted DHW - Step 46



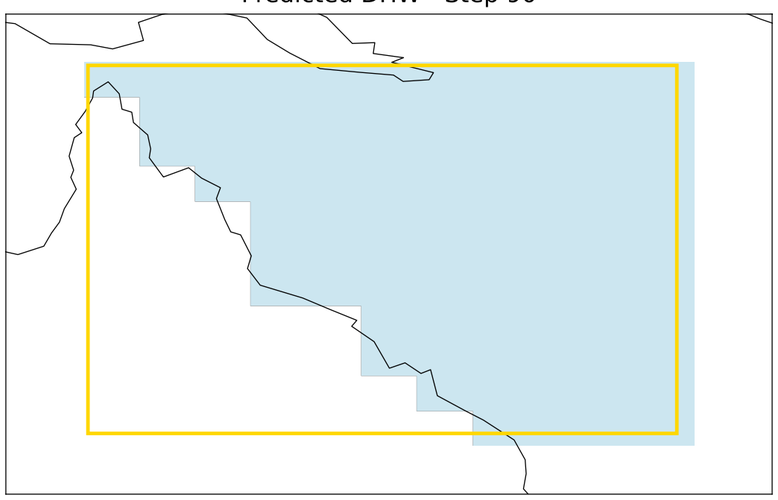
Error (Pred - Actual) - Step 46 [scale ±8]



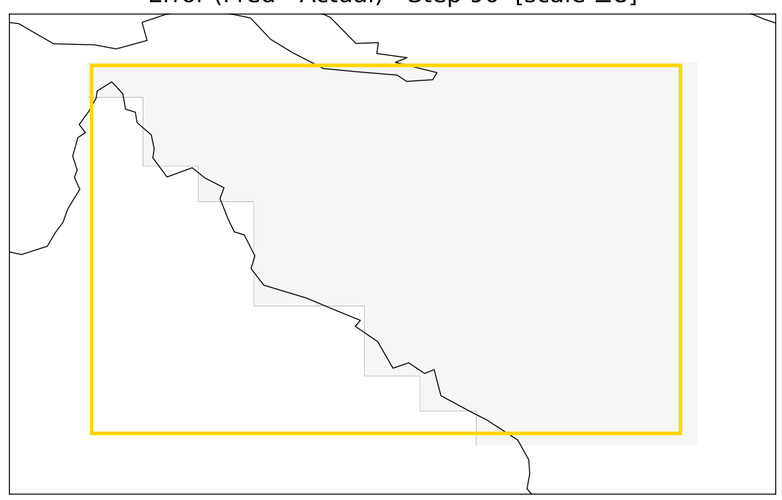
Actual DHW - Step 90

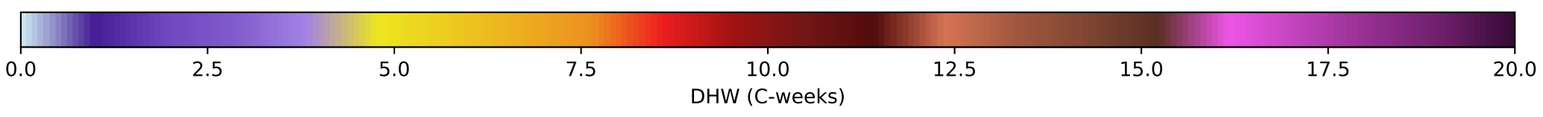


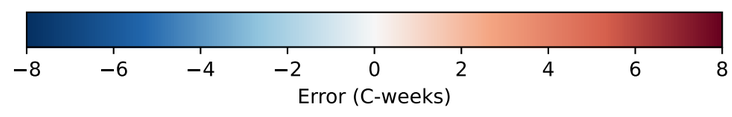
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

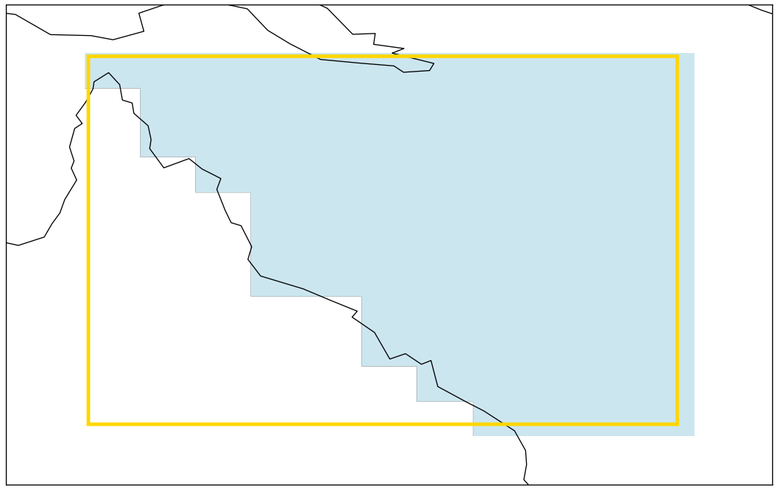




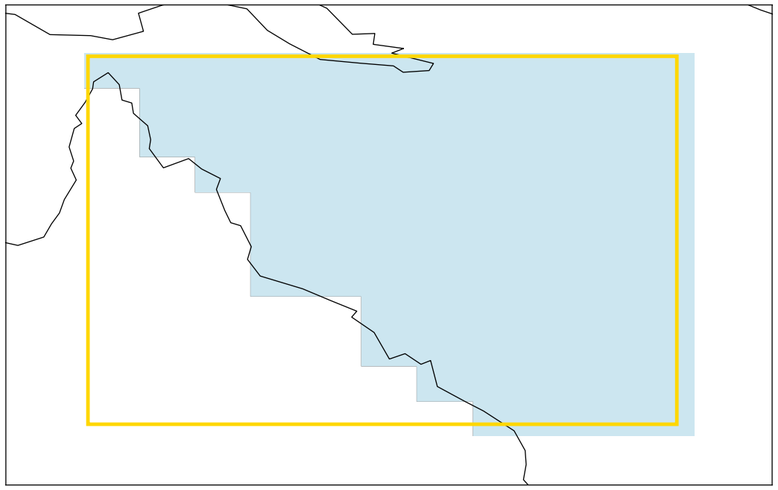


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

Actual DHW - Step 1



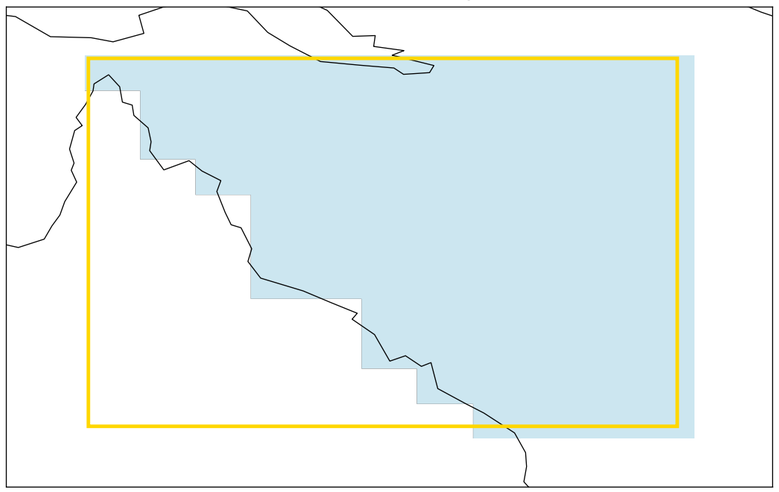
Predicted DHW - Step 1



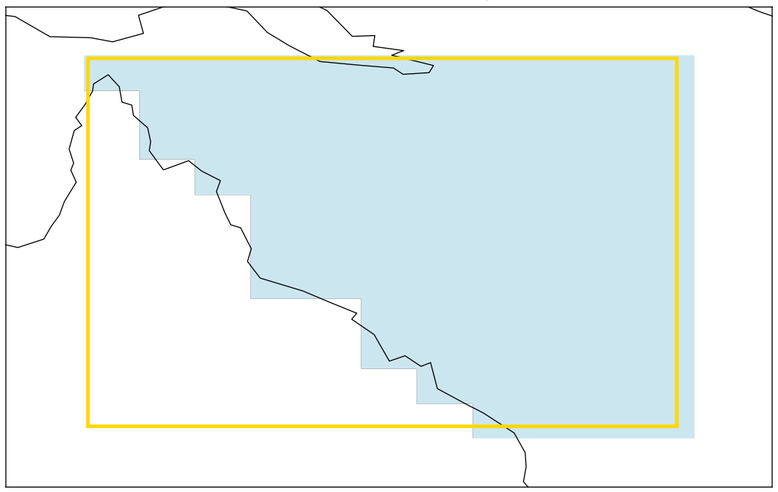
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



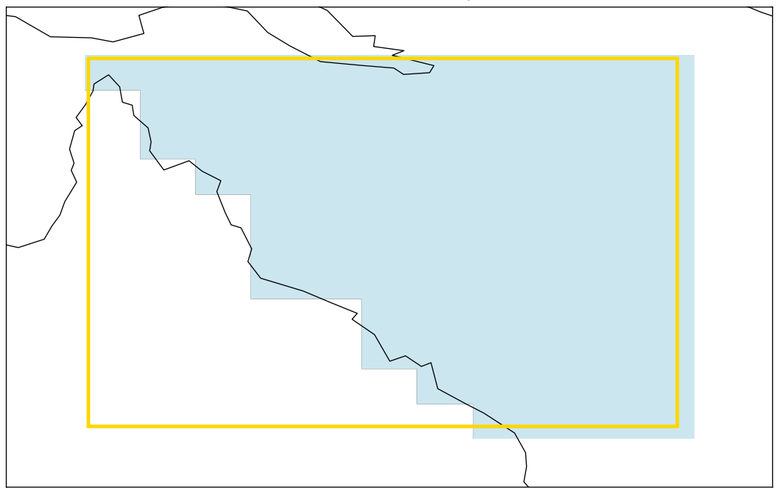
Predicted DHW - Step 46



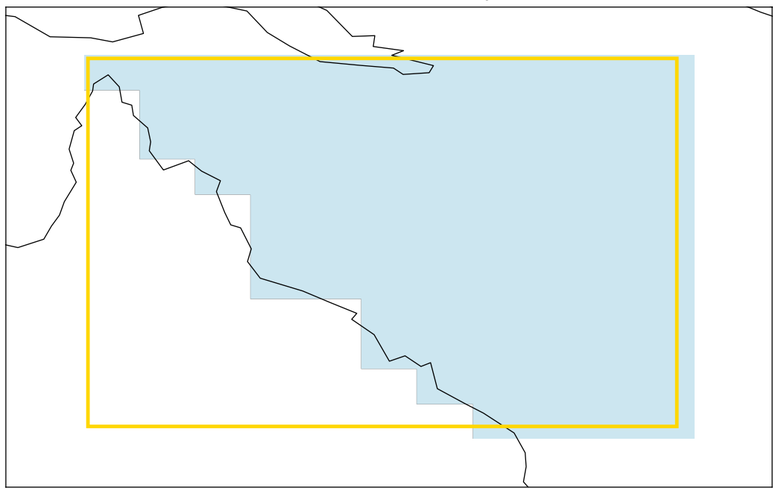
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

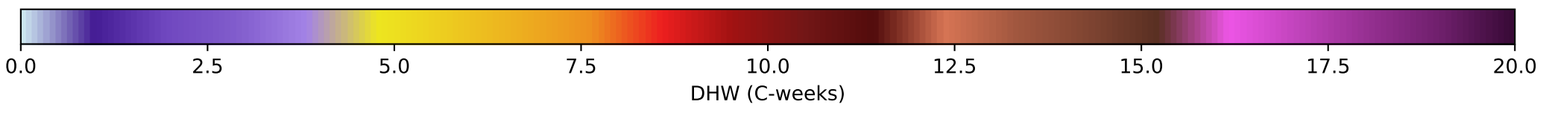


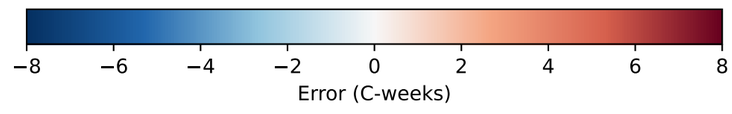
Predicted DHW - Step 90



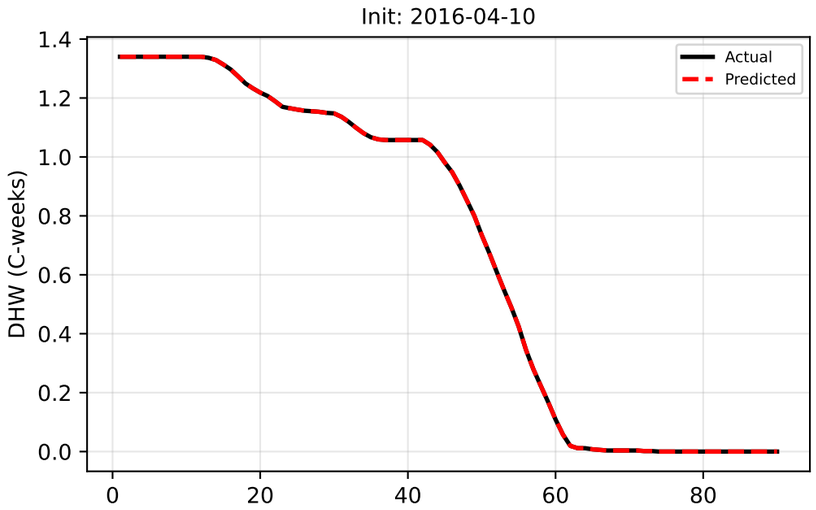
Error (Pred - Actual) - Step 90 [scale ±8]

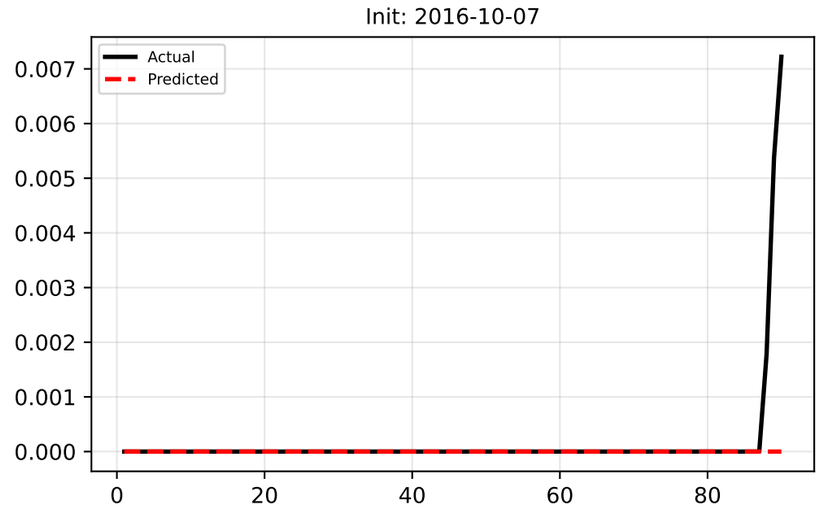


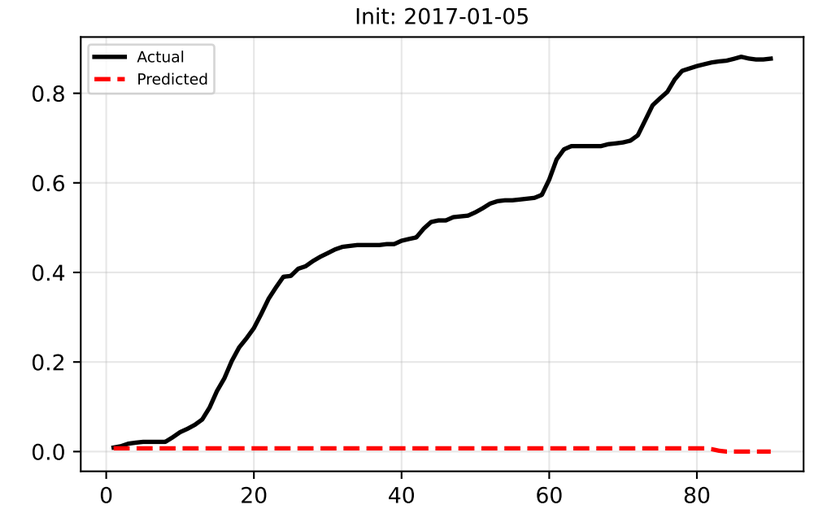


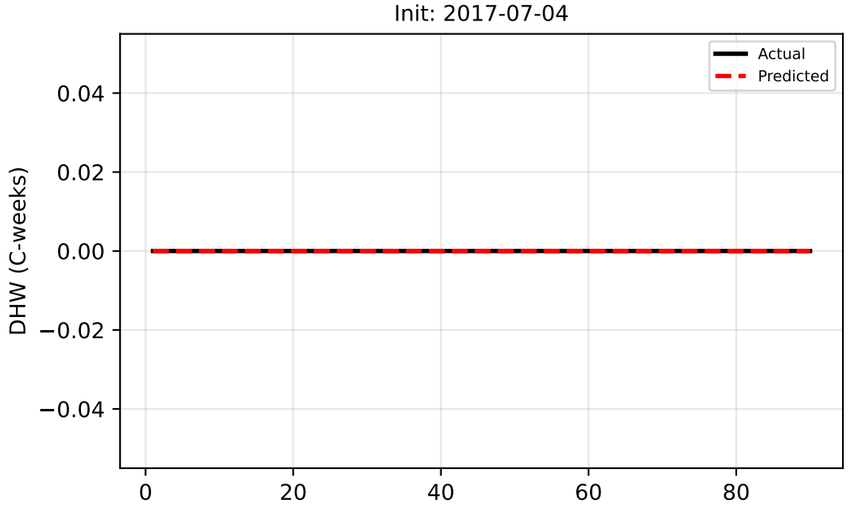


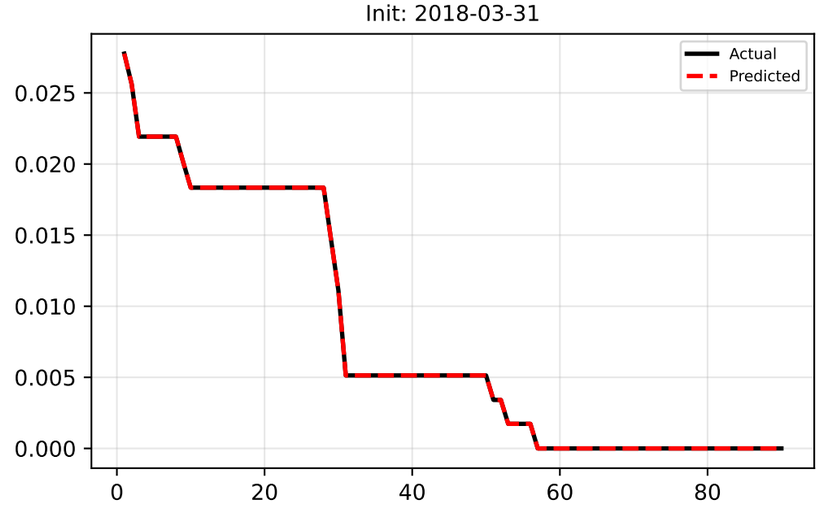
Global-Mean DHW - Val - GBR (C-weeks)

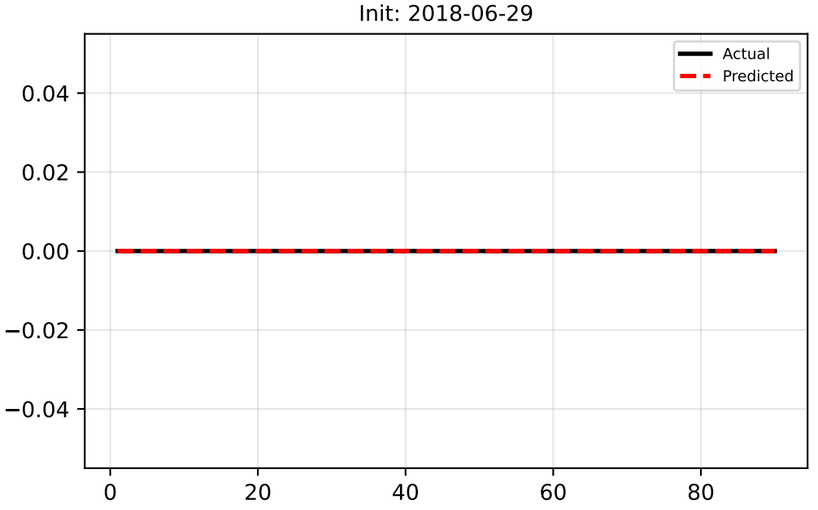


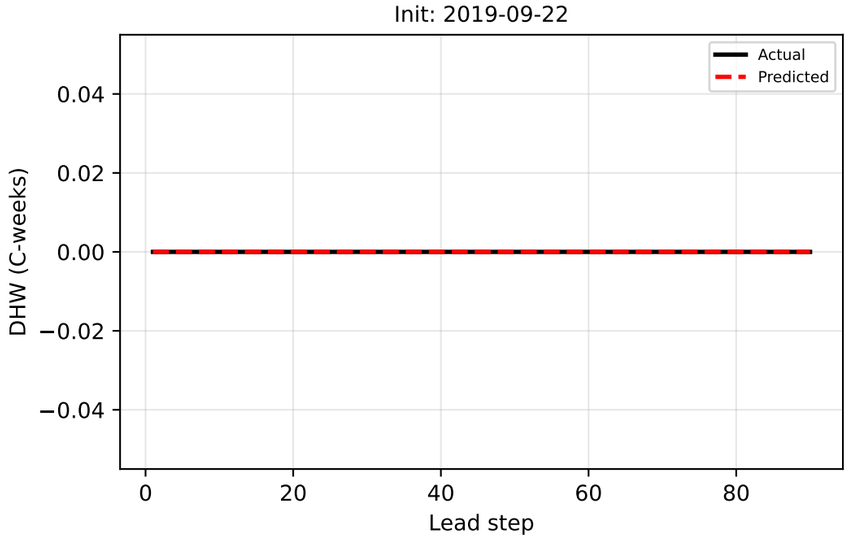


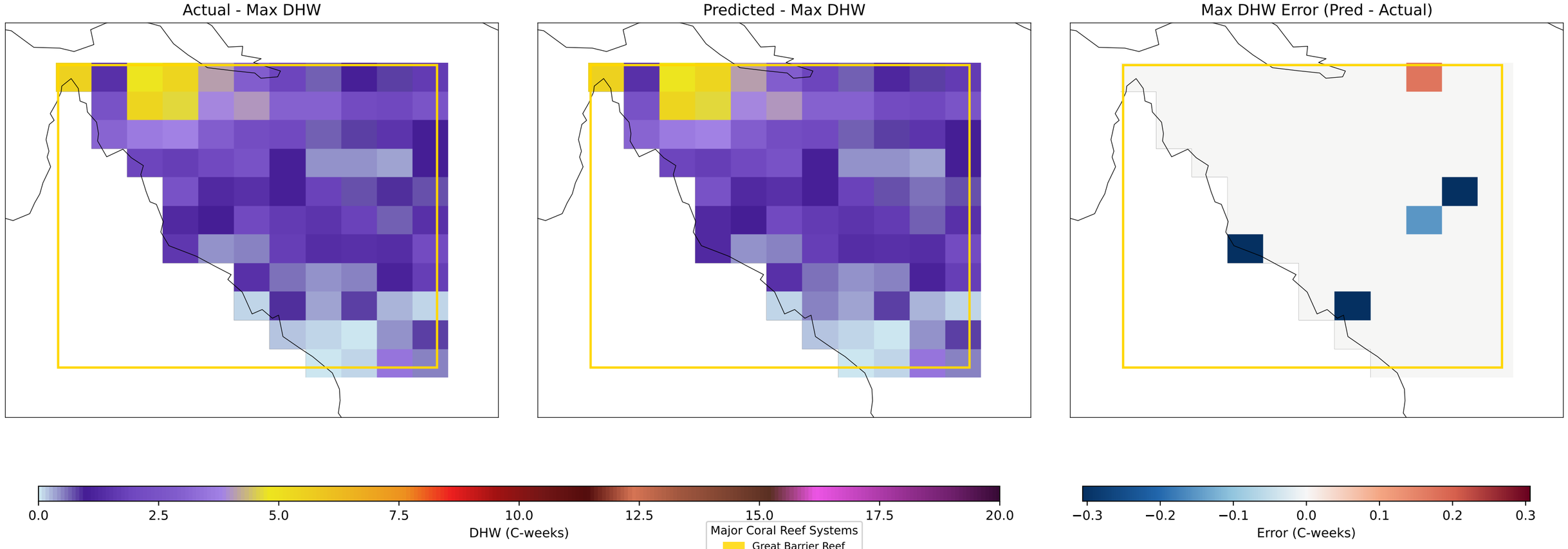


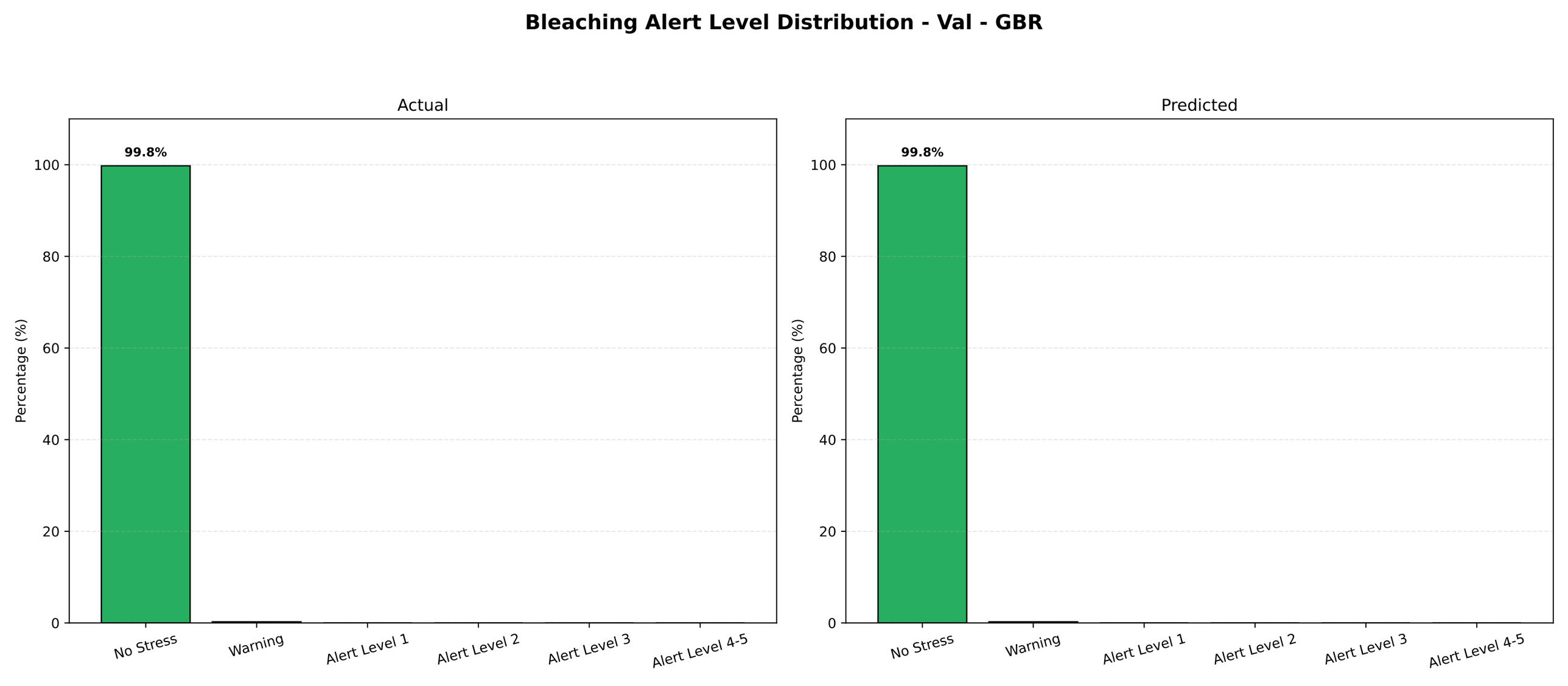






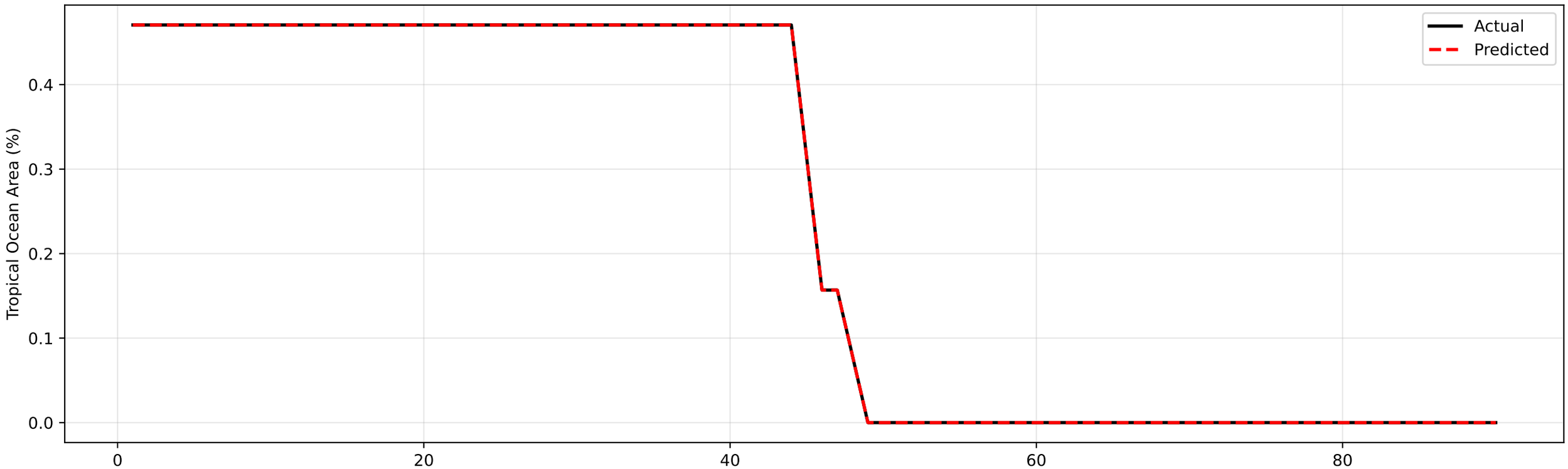



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

